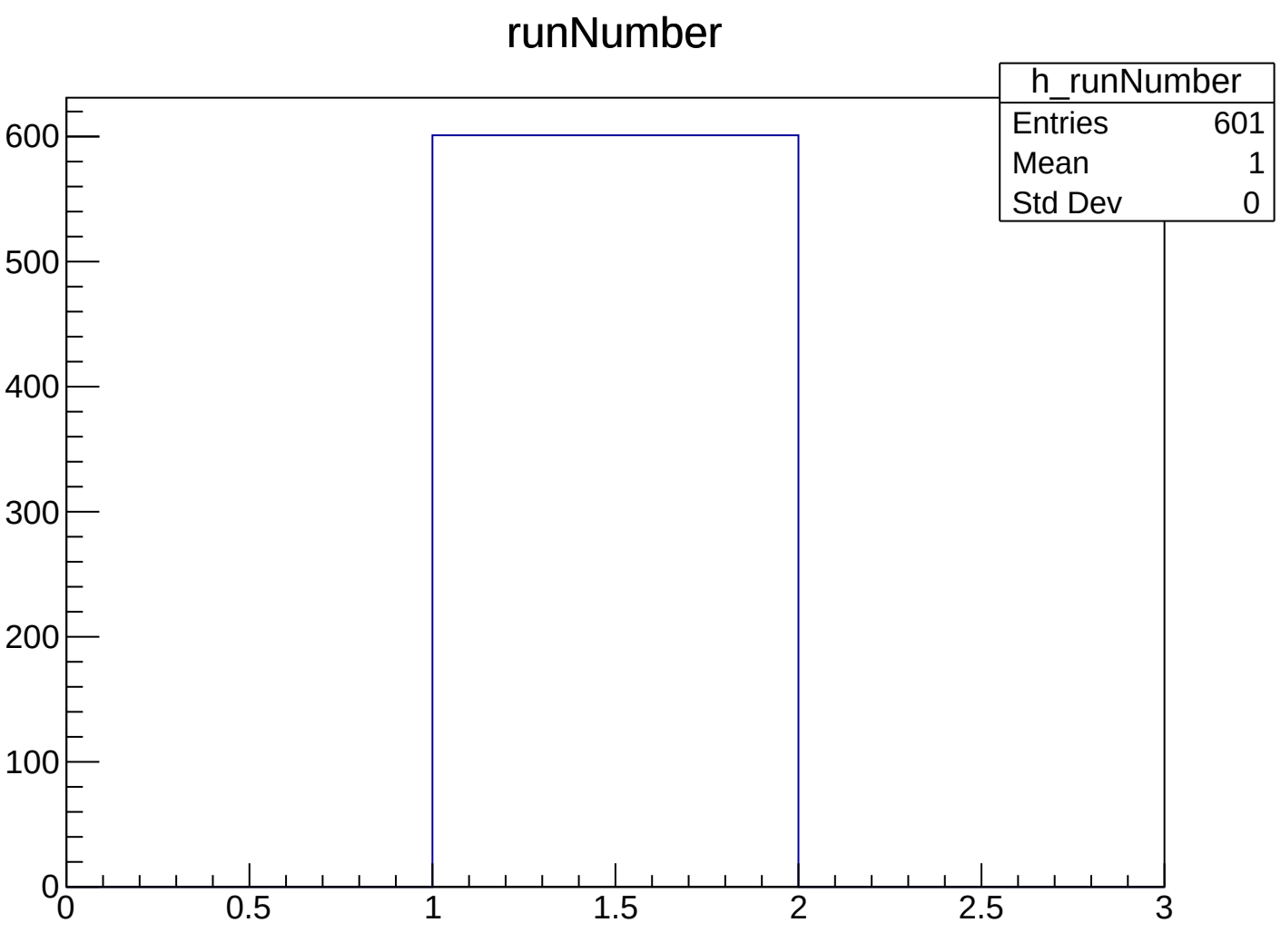
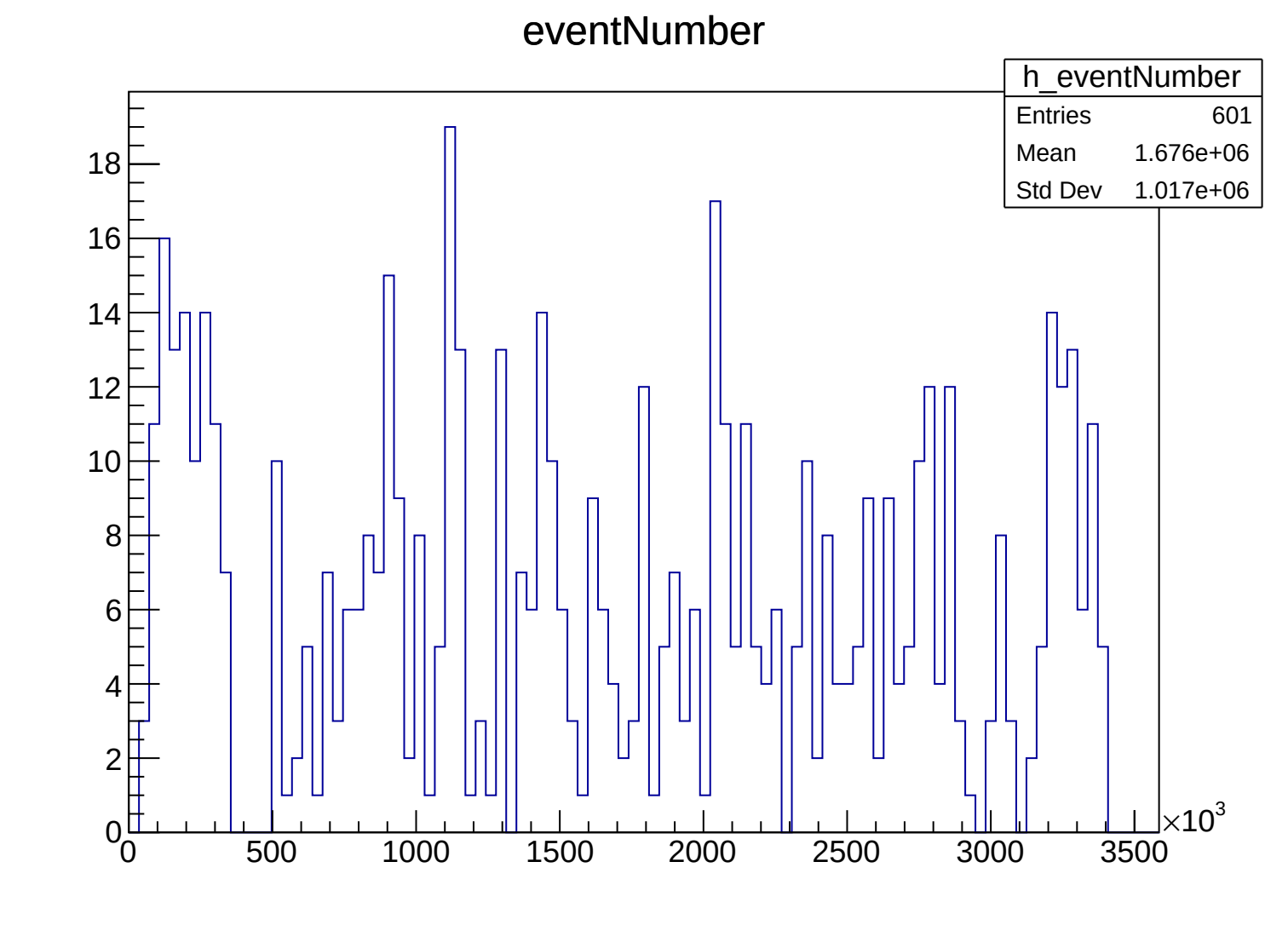
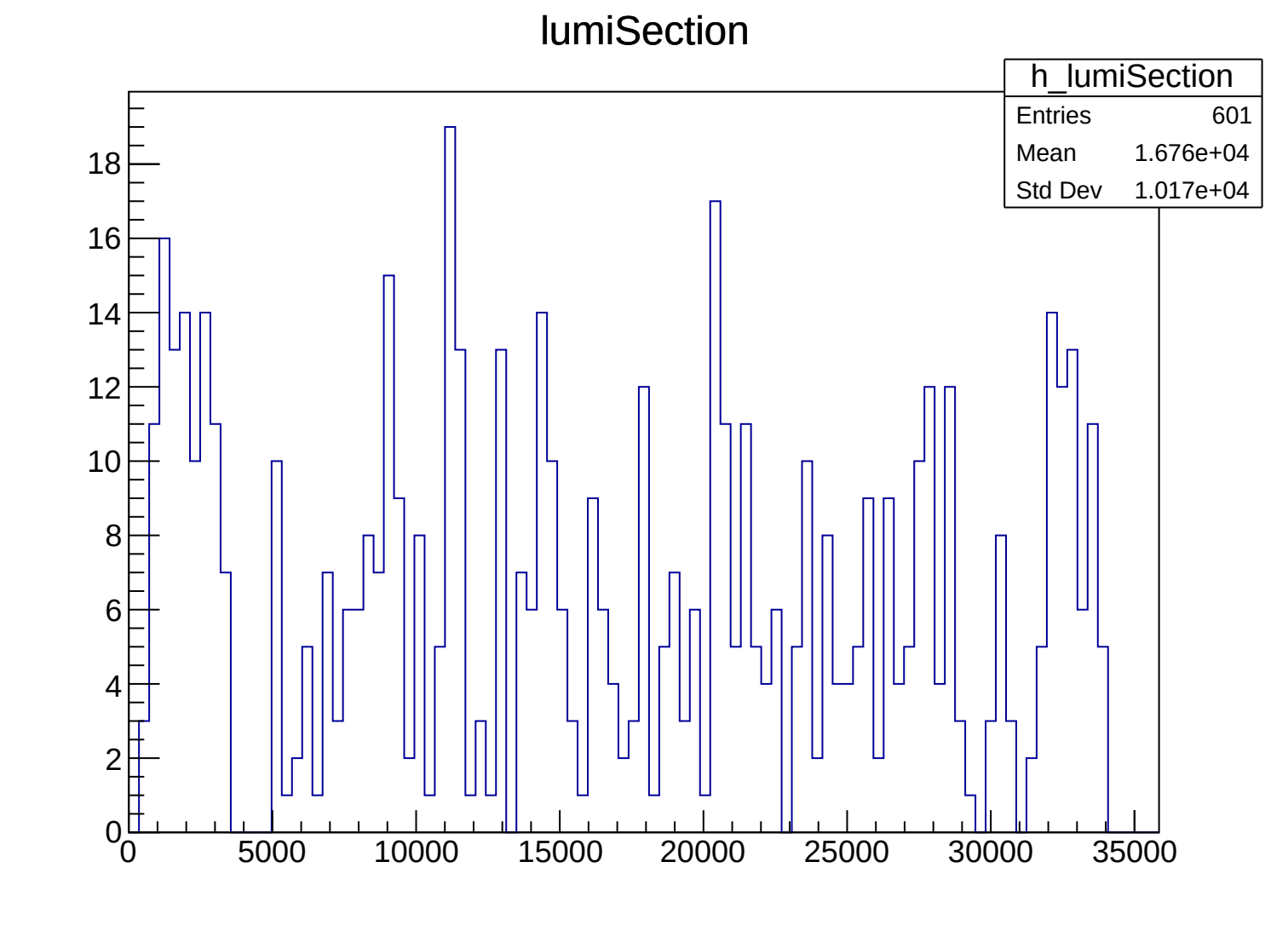
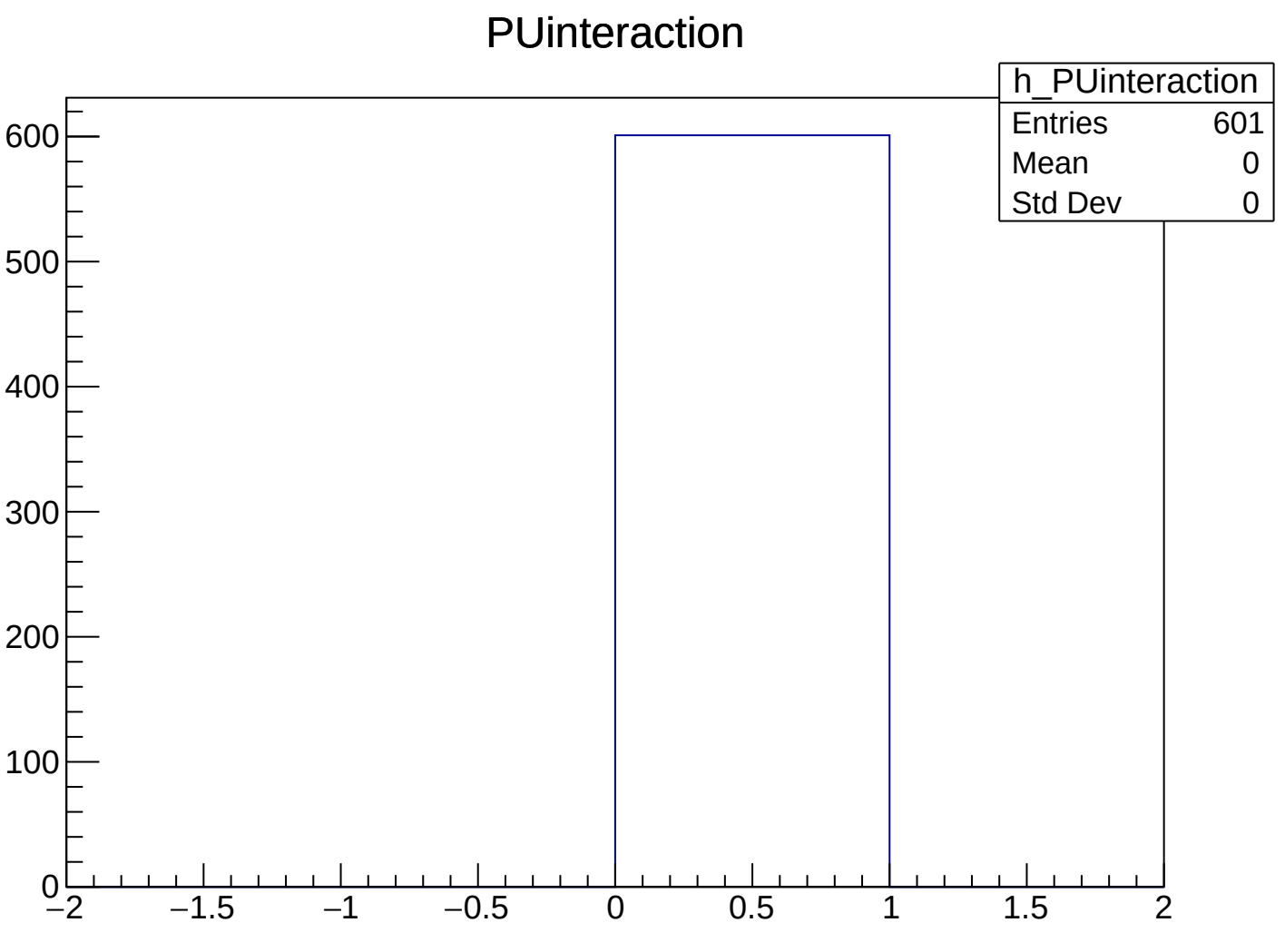
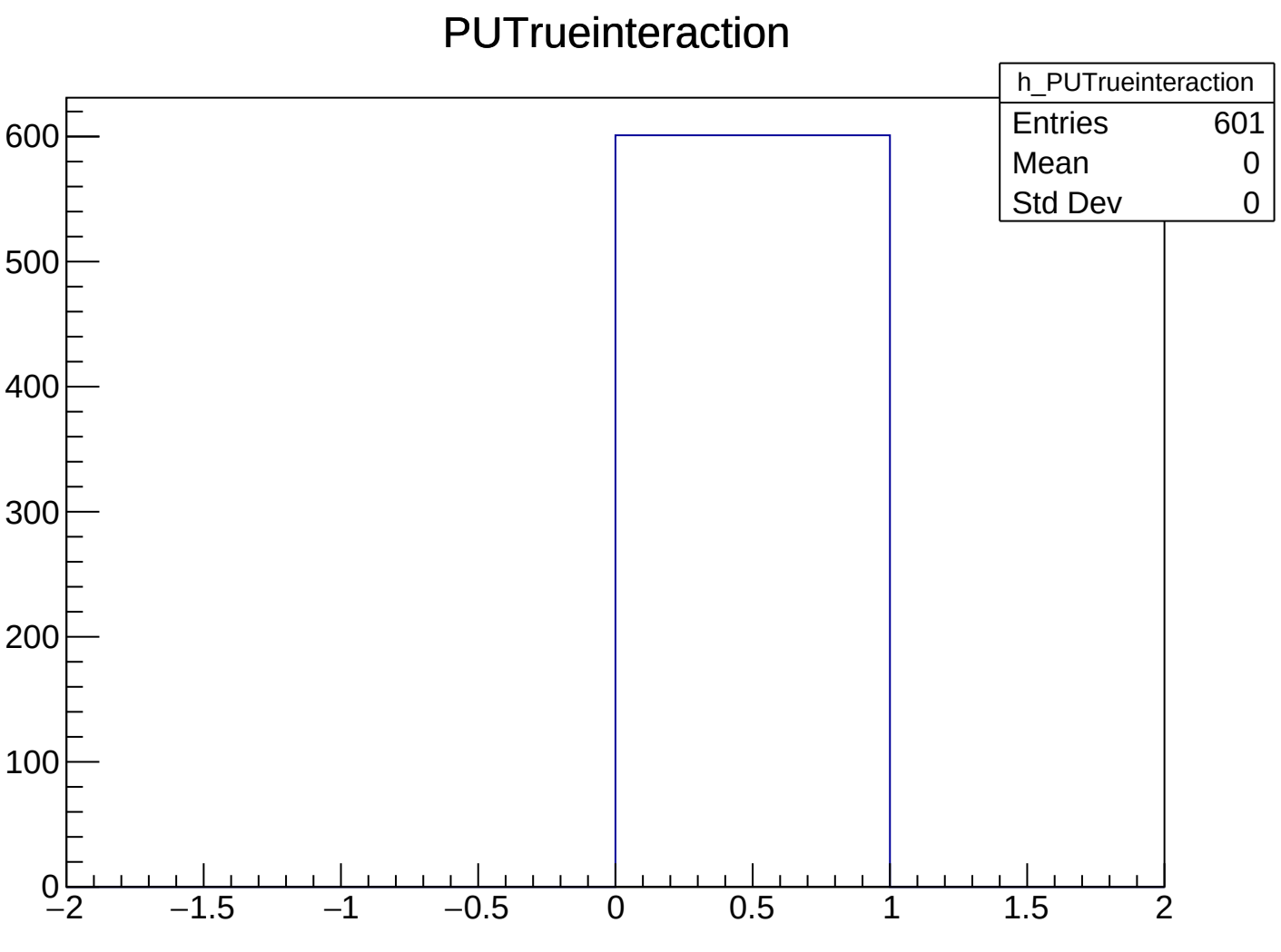
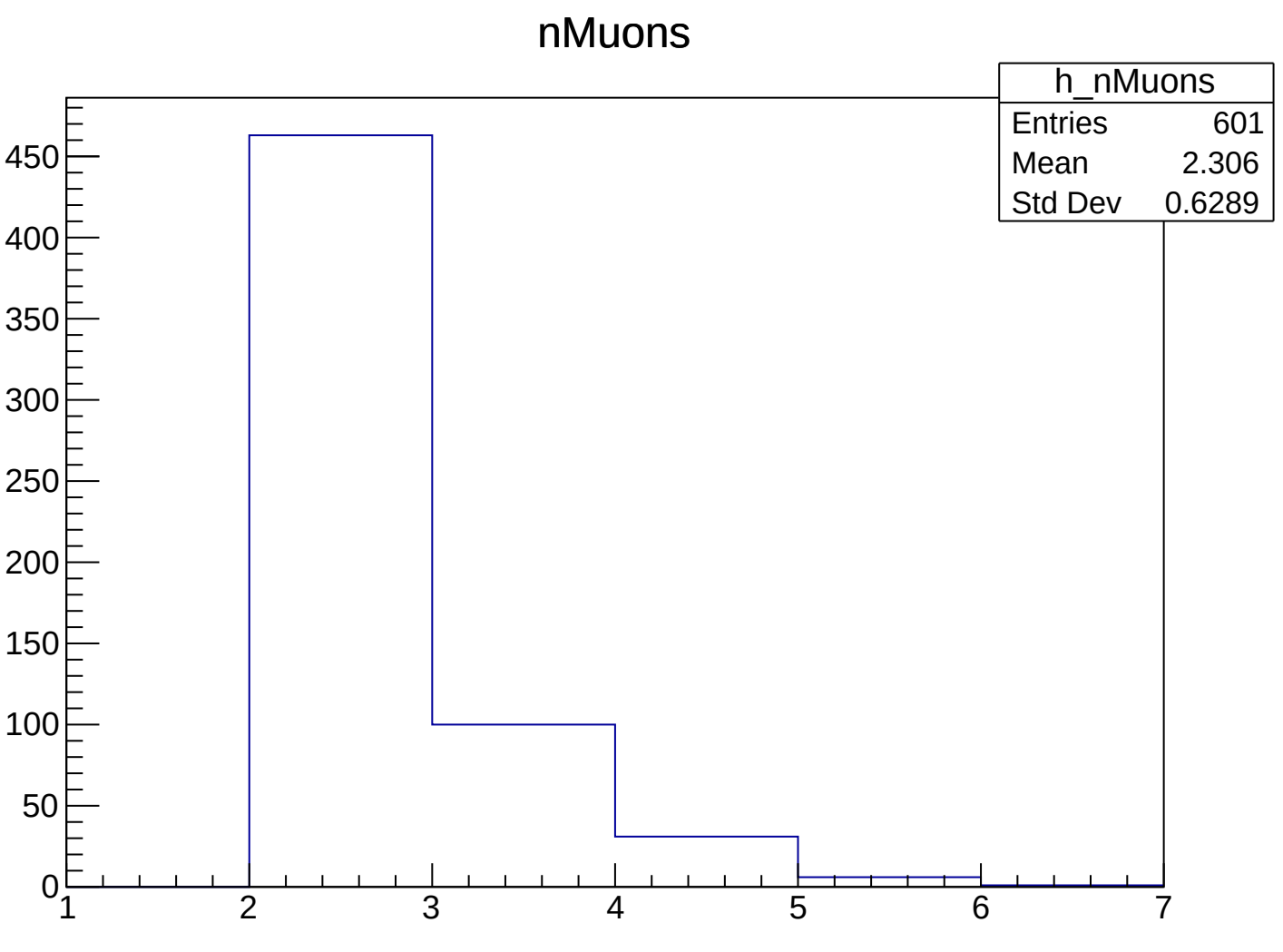

runNumber
h_runNumber
Entries 601
Mean 1
Std Dev 0
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3

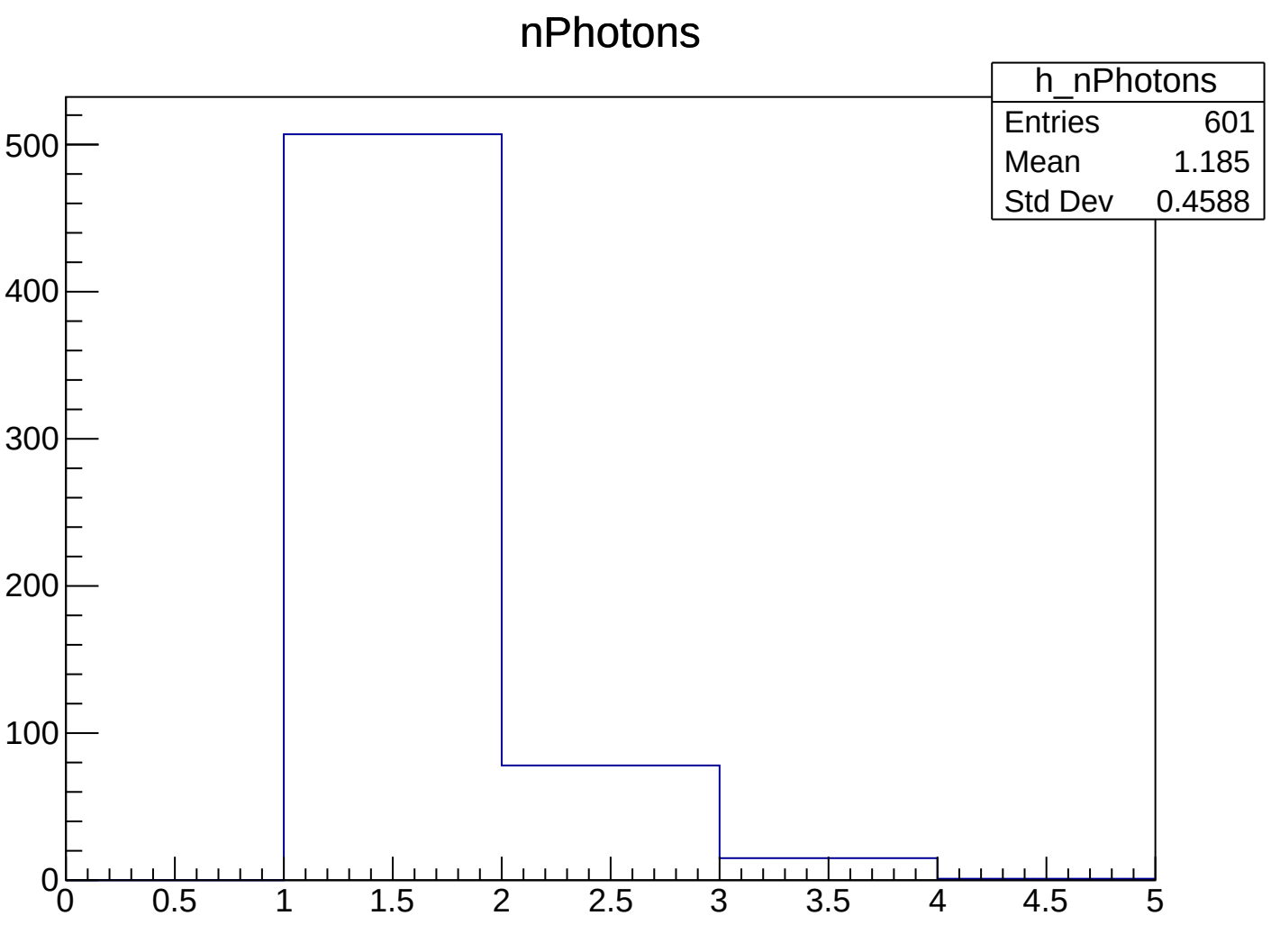

eventNumber
h_eventNumber
Entries 601
Mean 1.676e+06
Std Dev 1.017e+06
0
2
4
6
8
10
12
14
16
18
0
500
1000
1500
2000
2500
3000
3500
×10^3

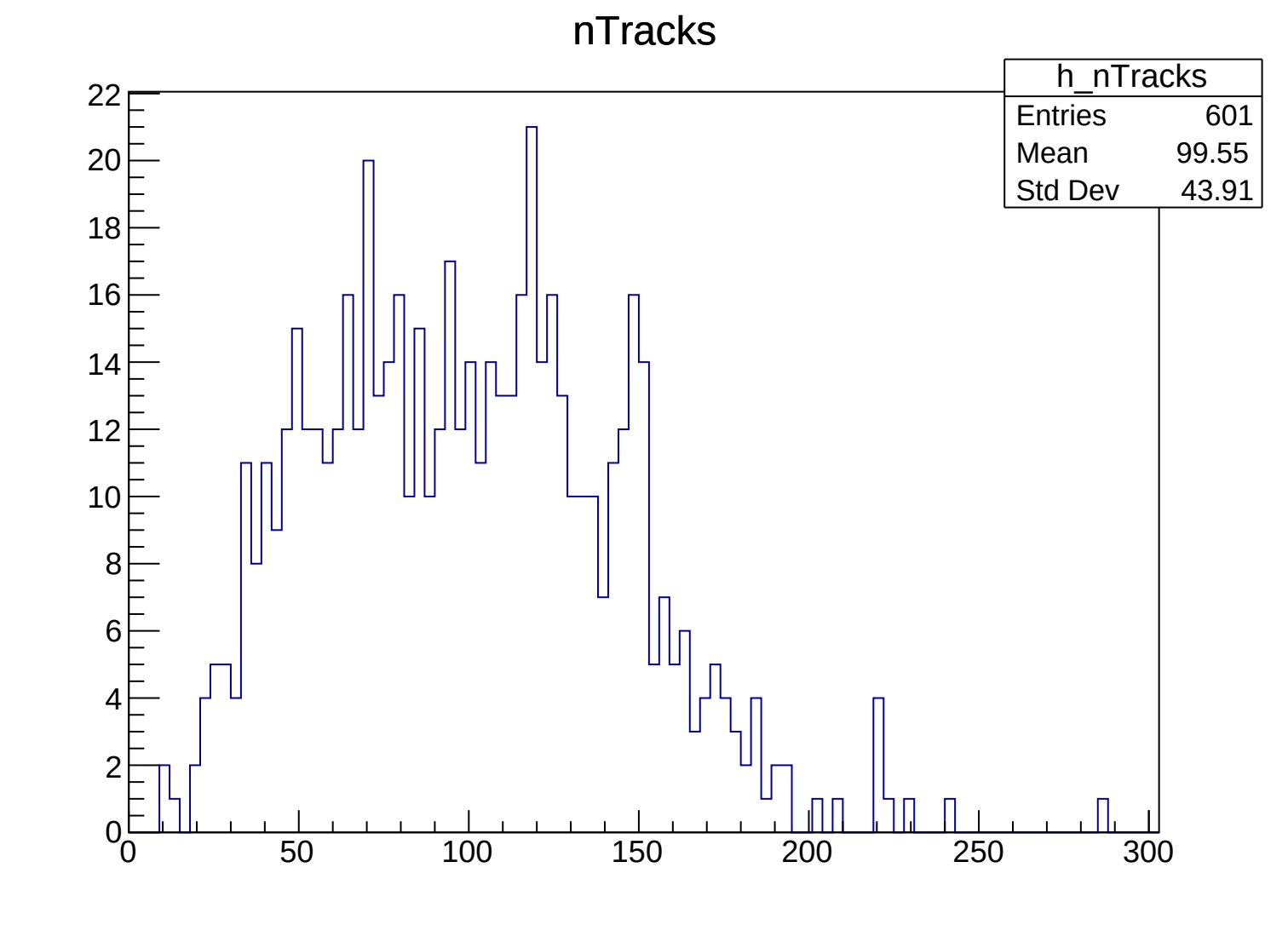

lumiSection
h_lumiSection
Entries 601
Mean 1.676e+04
Std Dev 1.017e+04
0
2
4
6
8
10
12
14
16
18
0
5000
10000
15000
20000
25000
30000
35000

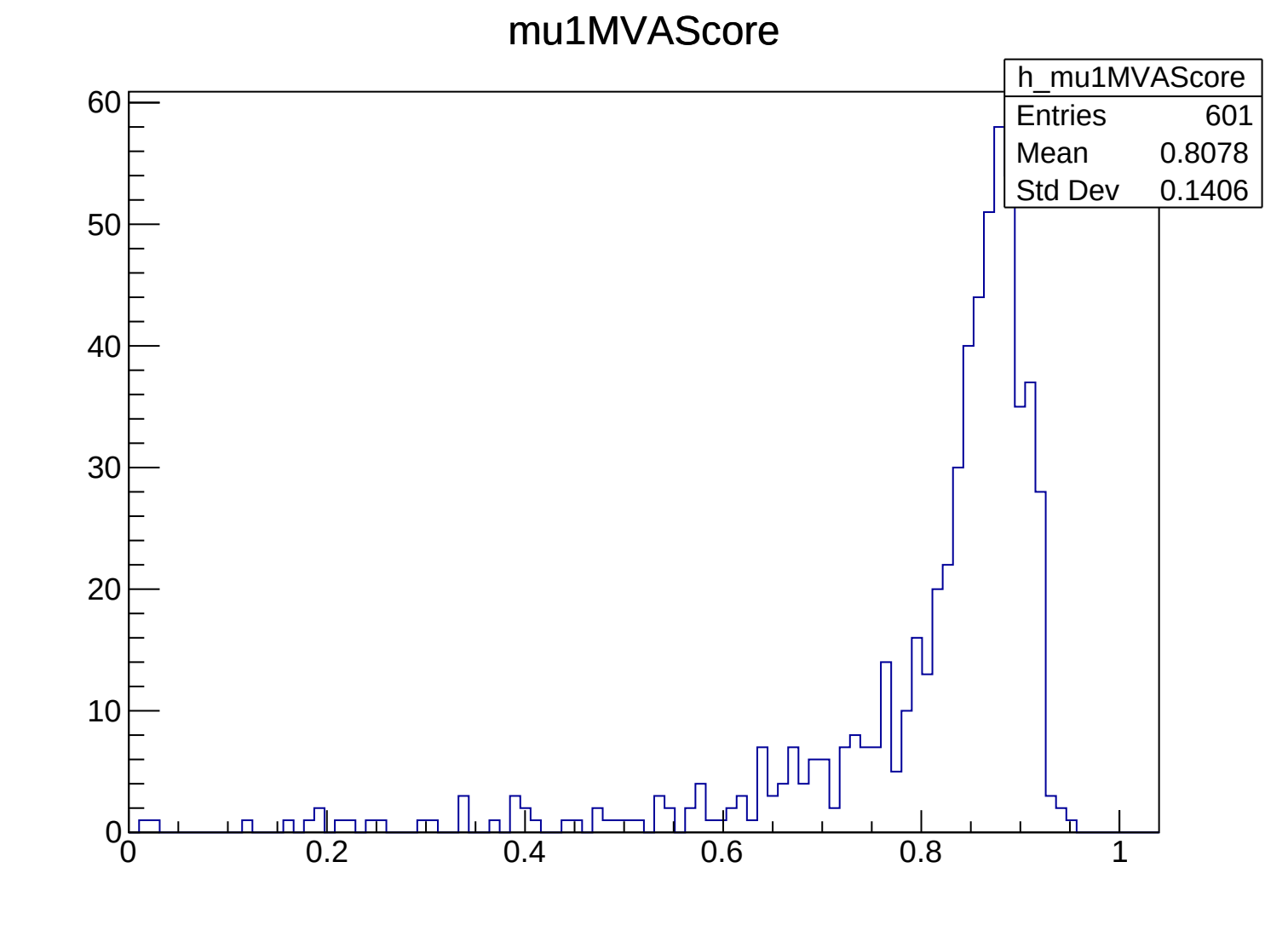

PUinteraction
h_PUinteraction
Entries 601
Mean 0
Std Dev 0
600
500
400
300
200
100
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

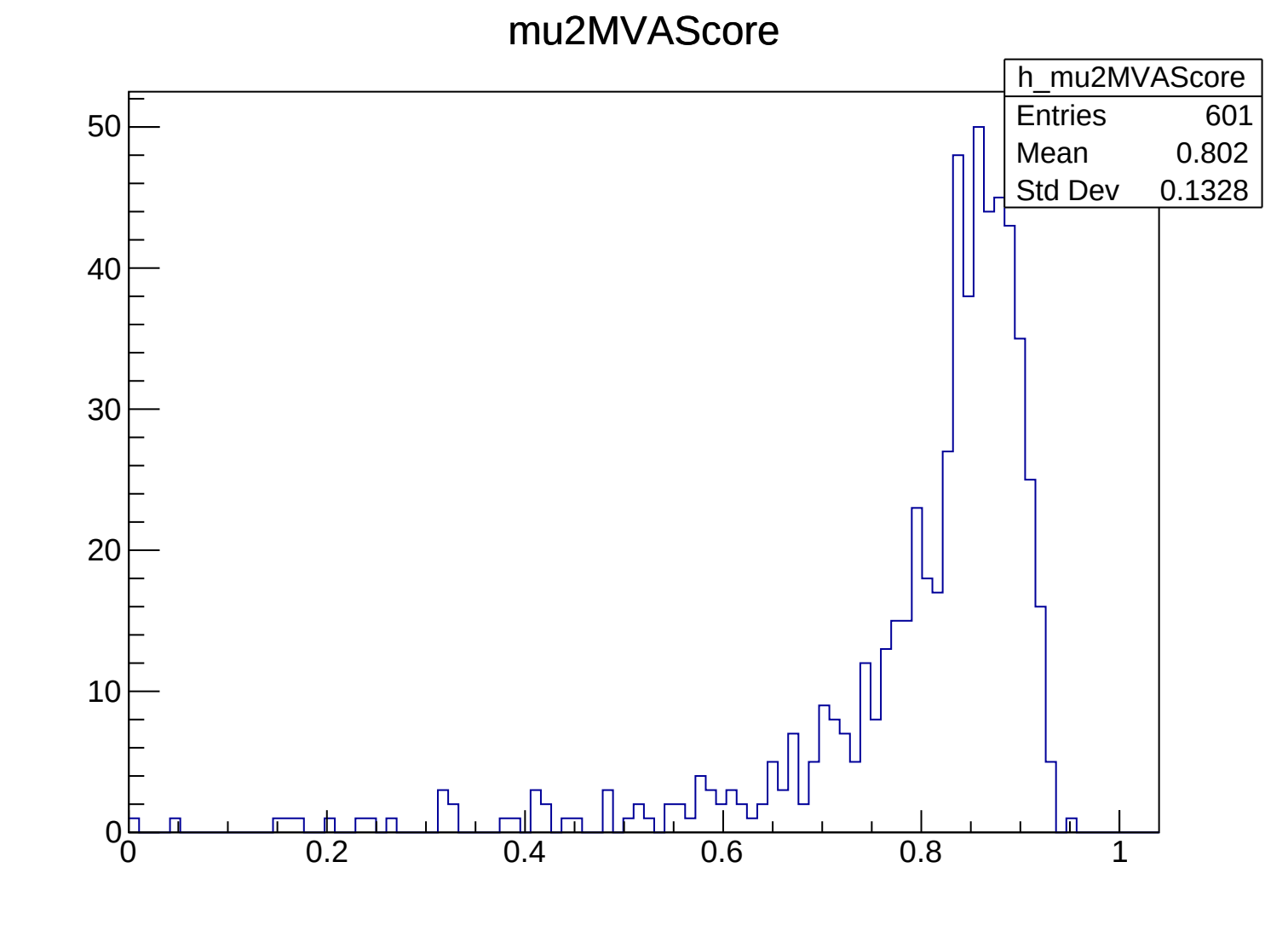

PUTrueinteraction
h_PUTrueinteraction
Entries 601
Mean 0
Std Dev 0
600
500
400
300
200
100
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

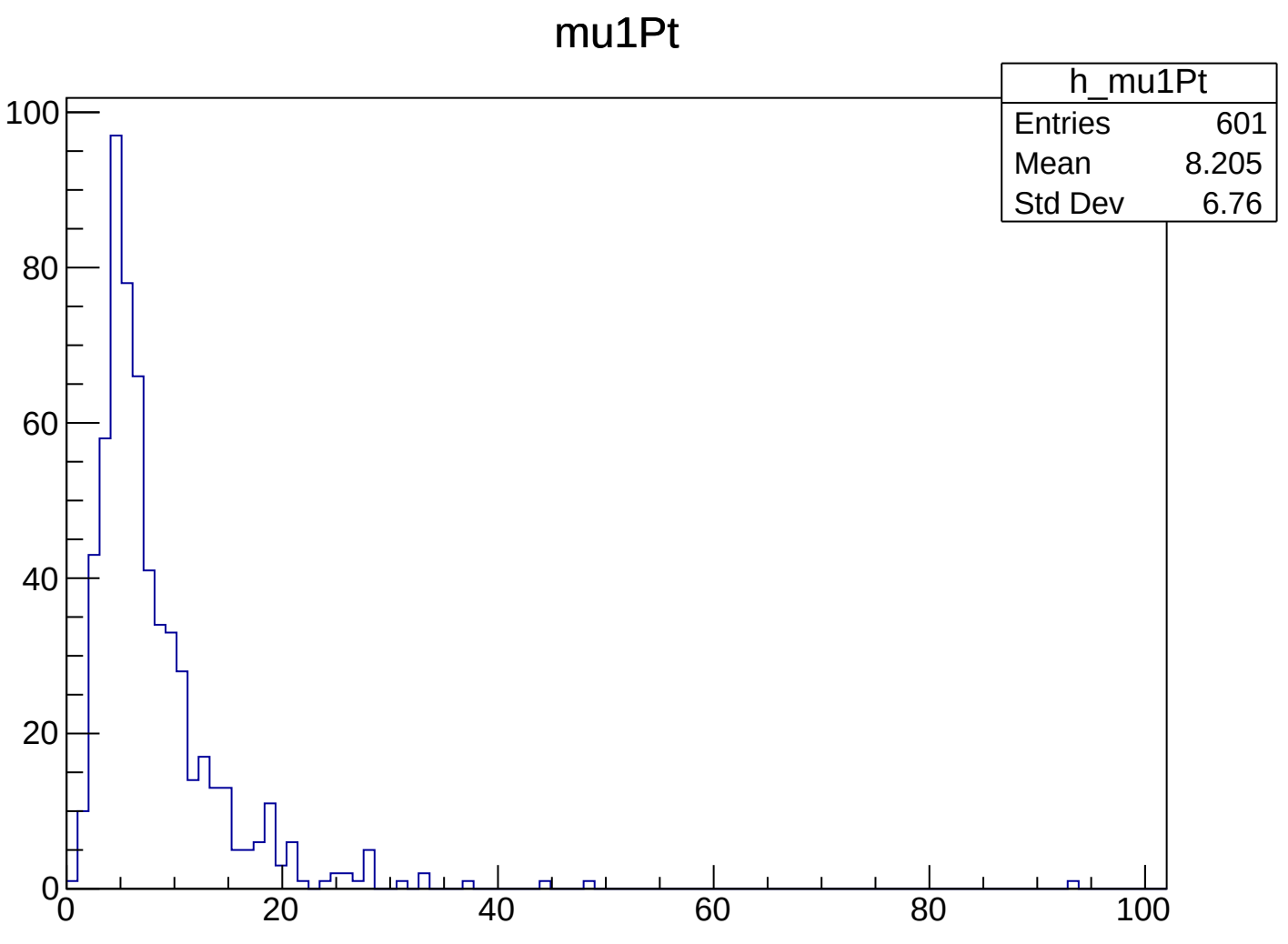

nMuons
h_nMuons
Entries 601
Mean 2.306
Std Dev 0.6289
450
400
350
300
250
200
150
100
50
0
1
2
3
4
5
6
7

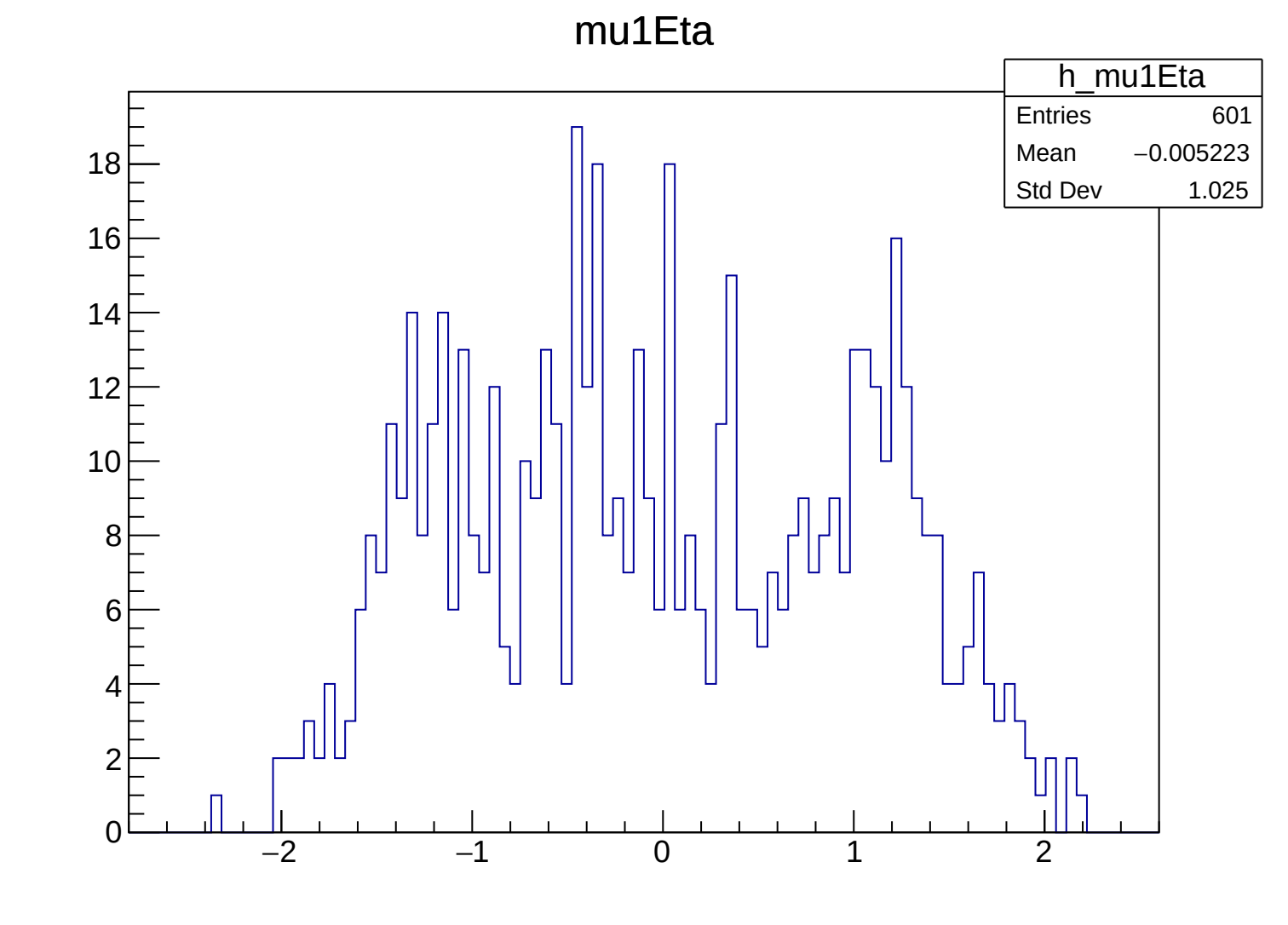

nPhotons
h_nPhotons
Entries 601
Mean 1.185
Std Dev 0.4588
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

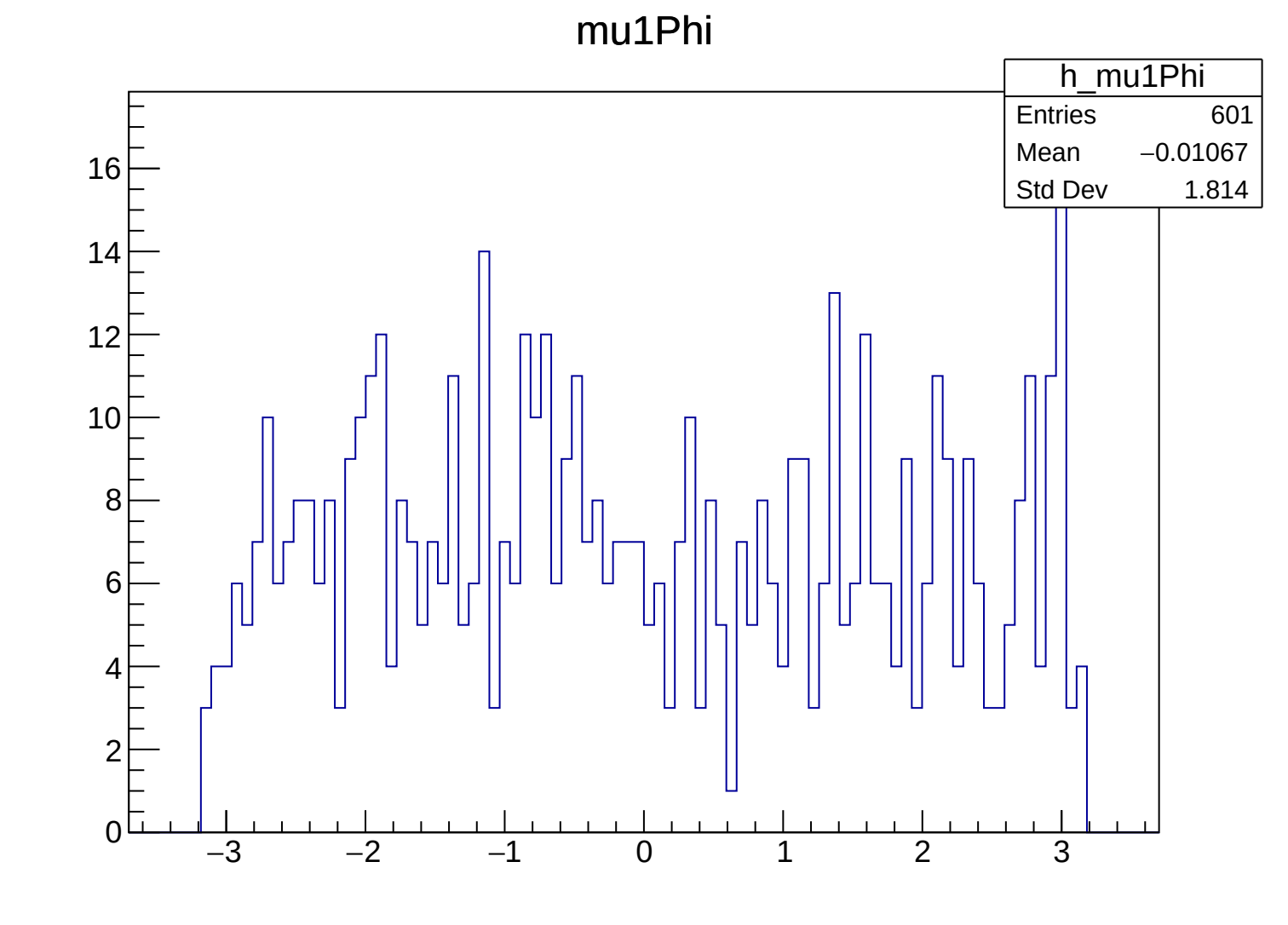

nTracks
h_nTracks
Entries 601
Mean 99.55
Std Dev 43.91
0
2
4
6
8
10
12
14
16
18
20
22
0
50
100
150
200
250
300

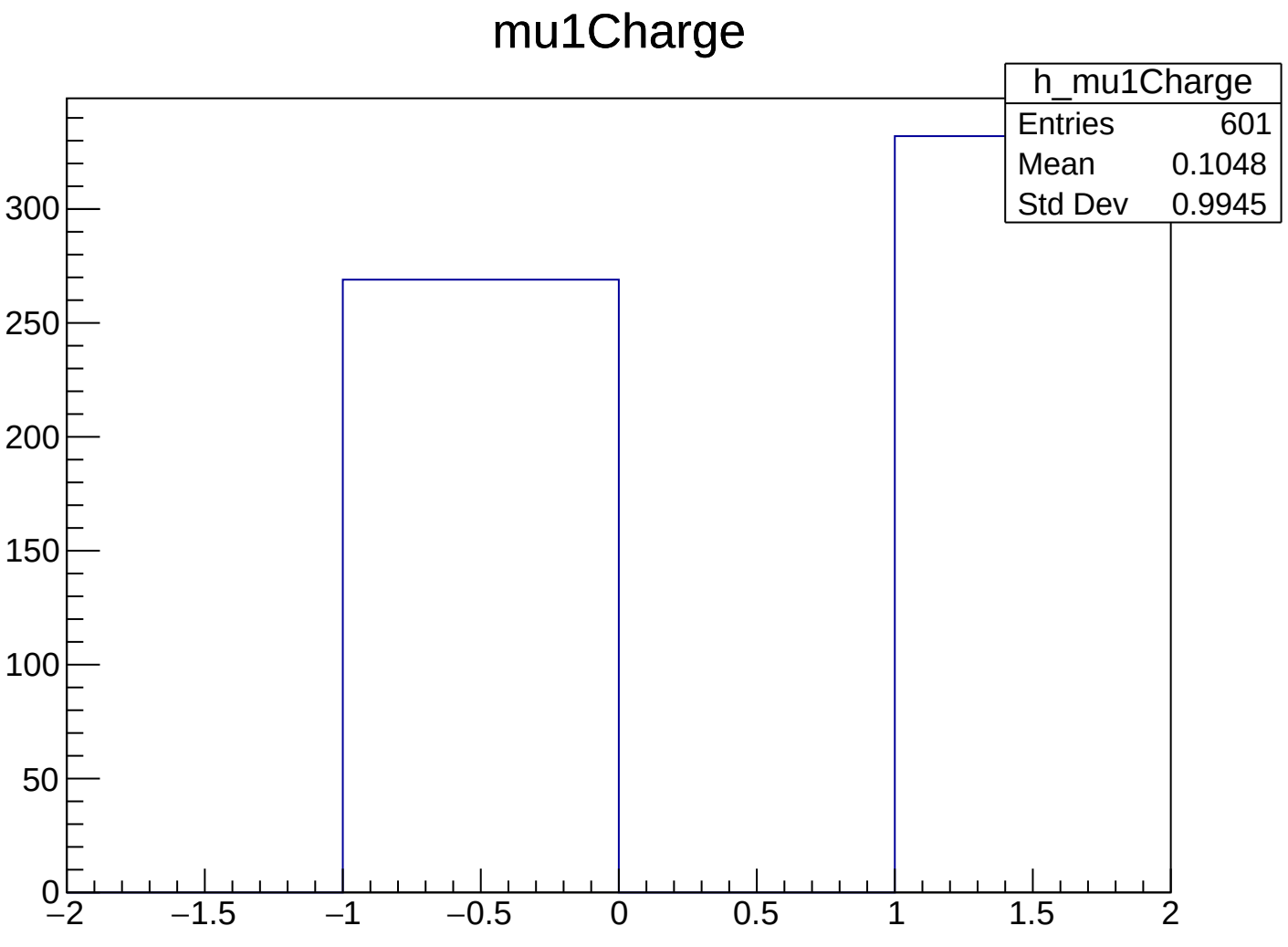

mu1MVAScore
h_mu1MVAScore
Entries 601
Mean 0.8078
Std Dev 0.1406
60
50
40
30
20
10
0
0
0.2
0.4
0.6
0.8
1

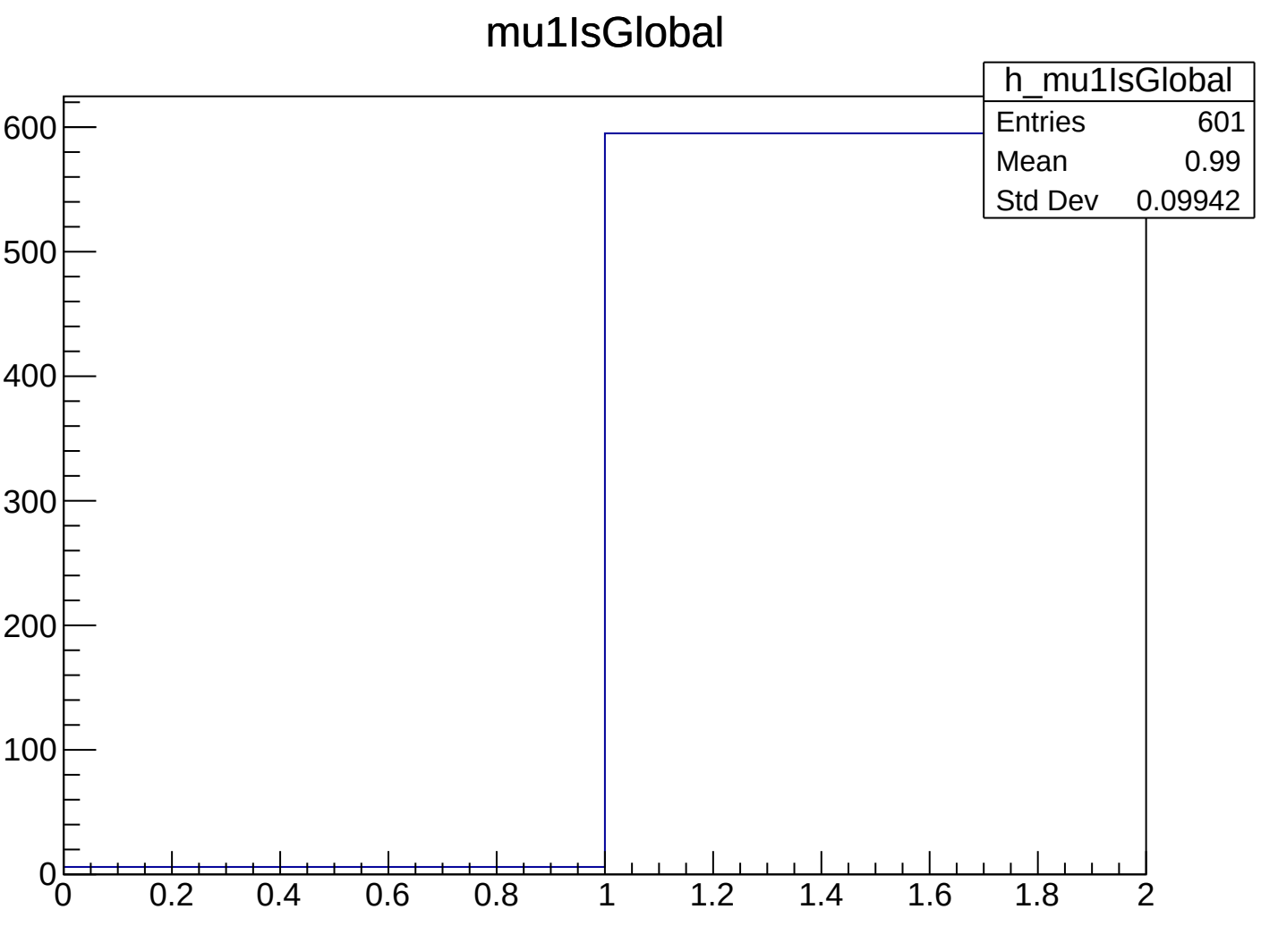

mu2MVAScore
h_mu2MVAScore
Entries 601
Mean 0.802
Std Dev 0.1328
50
40
30
20
10
0
0
0.2
0.4
0.6
0.8
1

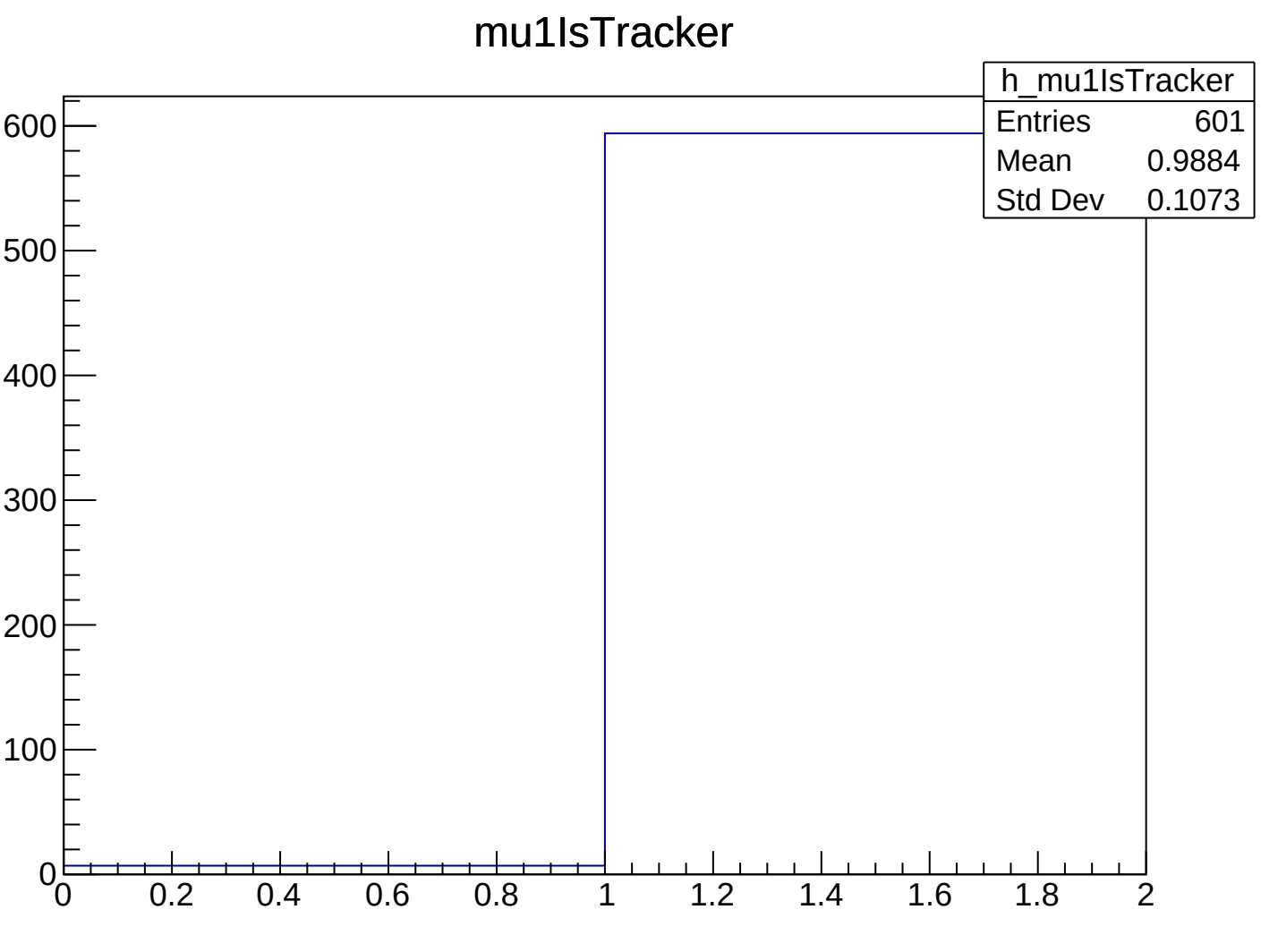

mu1Pt
h_mu1Pt
Entries 601
Mean 8.205
Std Dev 6.76
100
80
60
40
20
0
0
20
40
60
80
100

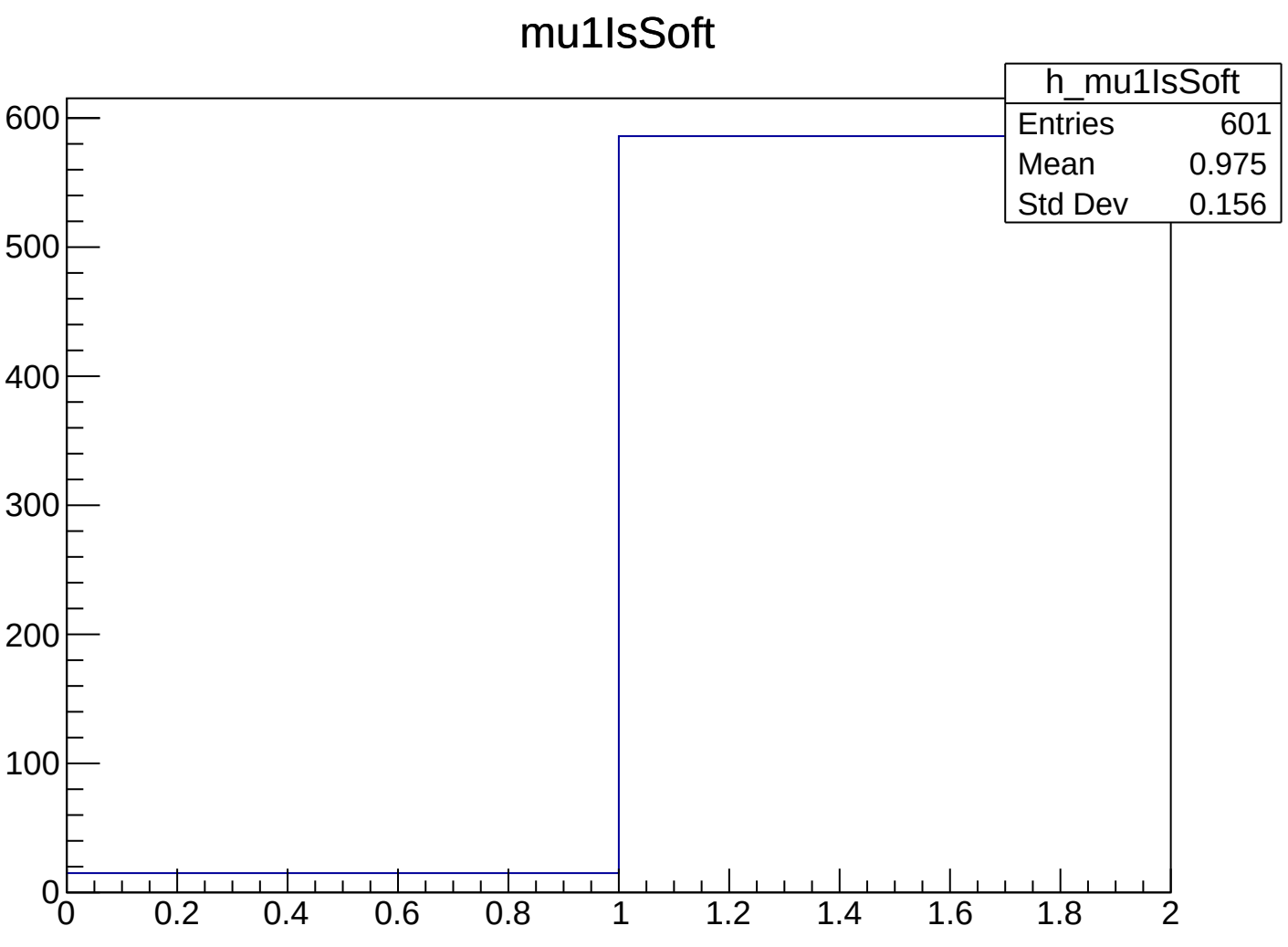

mu1Eta
h_mu1Eta
Entries 601
Mean −0.005223
Std Dev 1.025
18
16
14
12
10
8
6
4
2
0
−2
−1
0
1
2

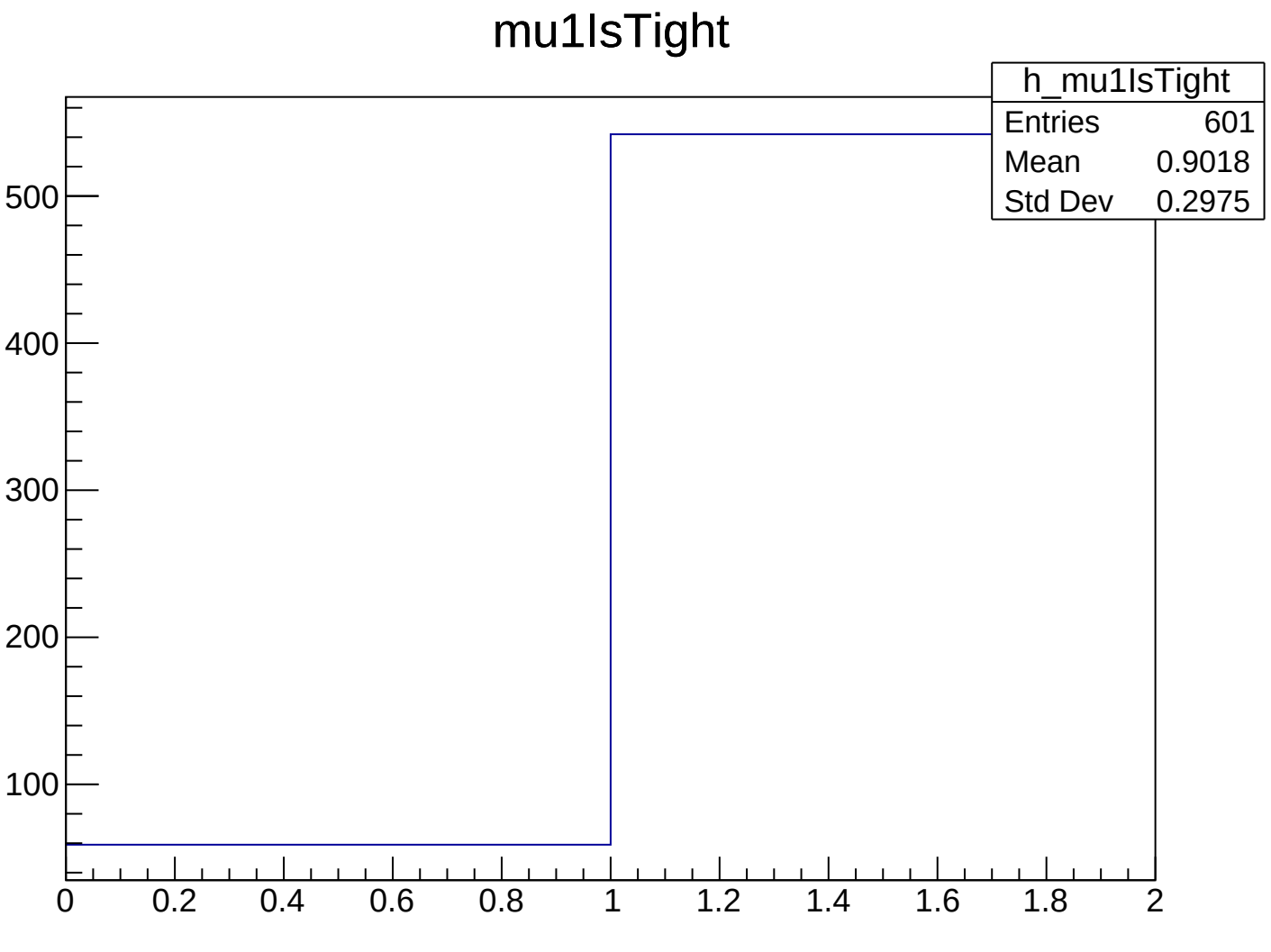

mu1Phi
h_mu1Phi
Entries 601
Mean −0.01067
Std Dev 1.814
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

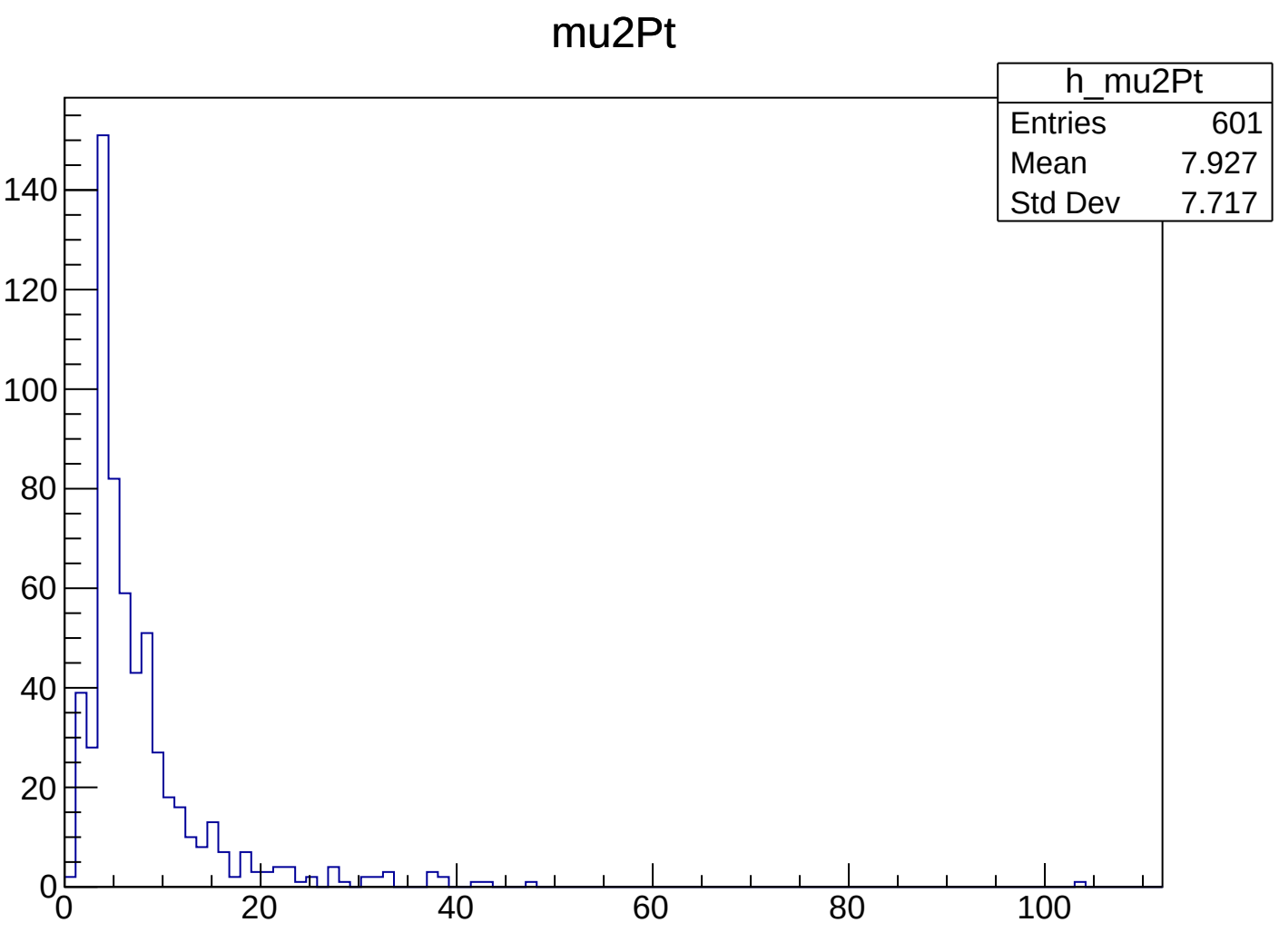

mu1Charge
h_mu1Charge
Entries 601
Mean 0.1048
Std Dev 0.9945
300
250
200
150
100
50
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

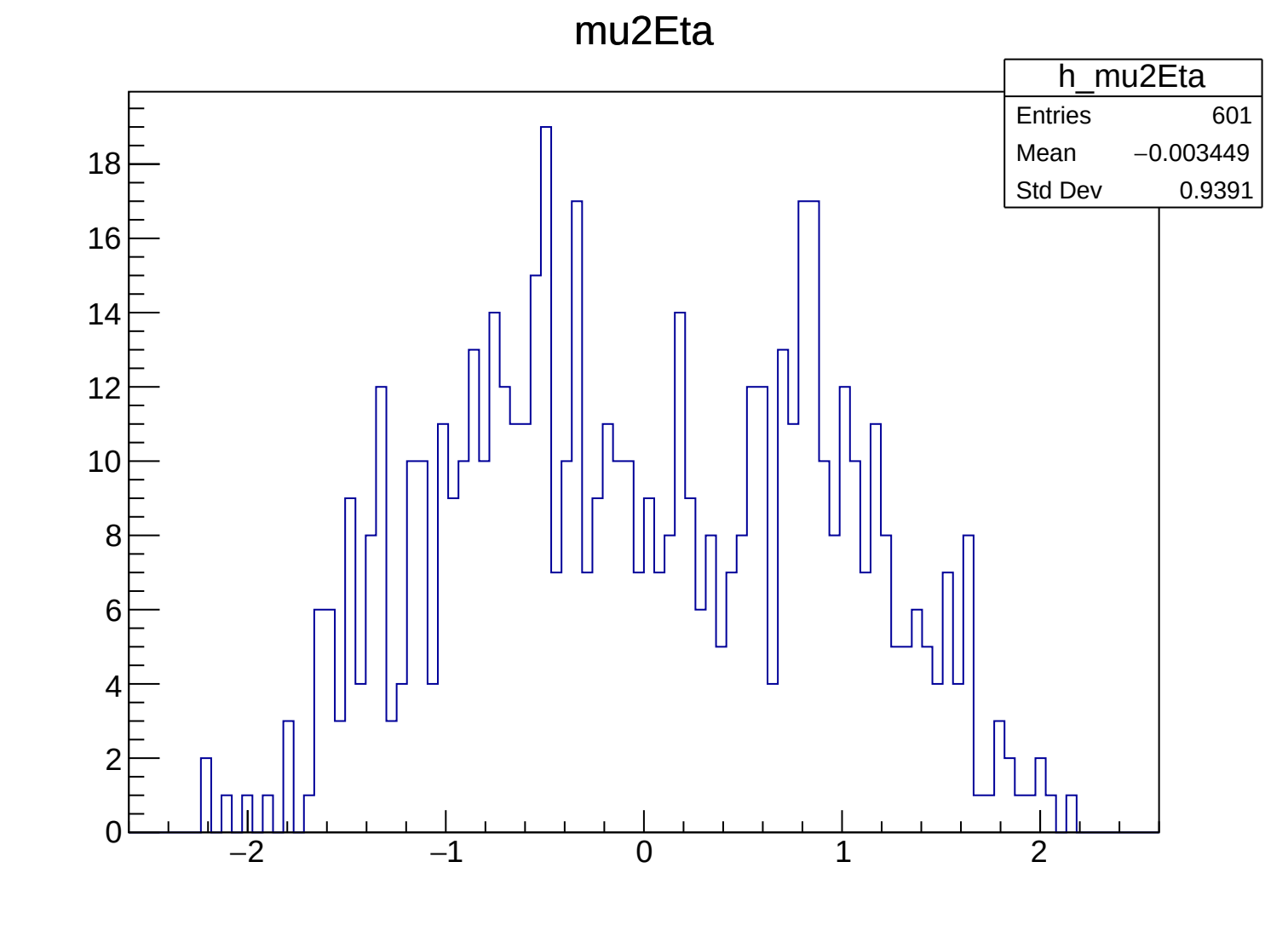

mu1IsGlobal
h_mu1IsGlobal
Entries 601
Mean 0.99
Std Dev 0.09942
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

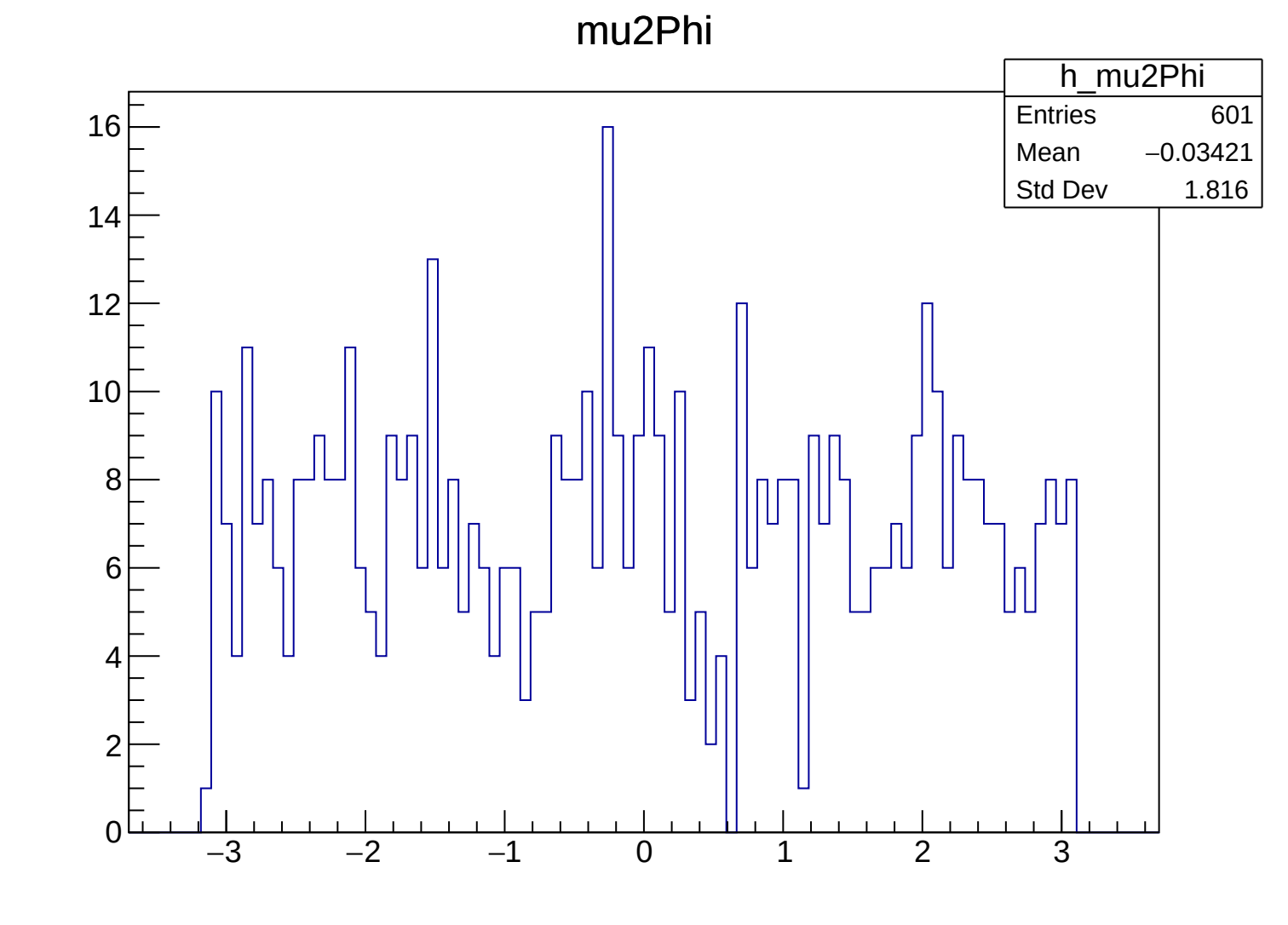

mu1IsTracker
h_mu1IsTracker
Entries 601
Mean 0.9884
Std Dev 0.1073
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

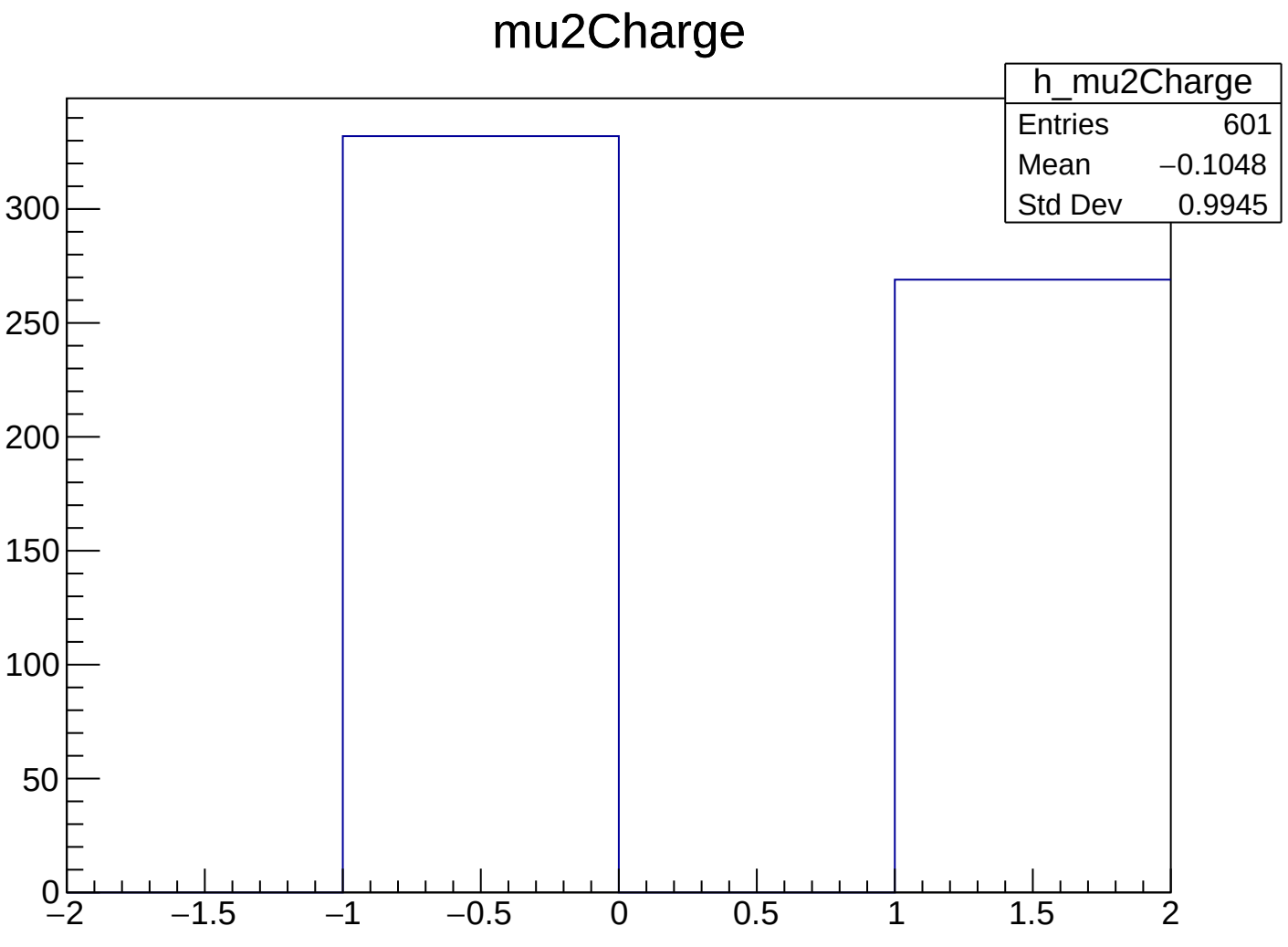

mu1IsSoft
h_mu1IsSoft
Entries 601
Mean 0.975
Std Dev 0.156
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

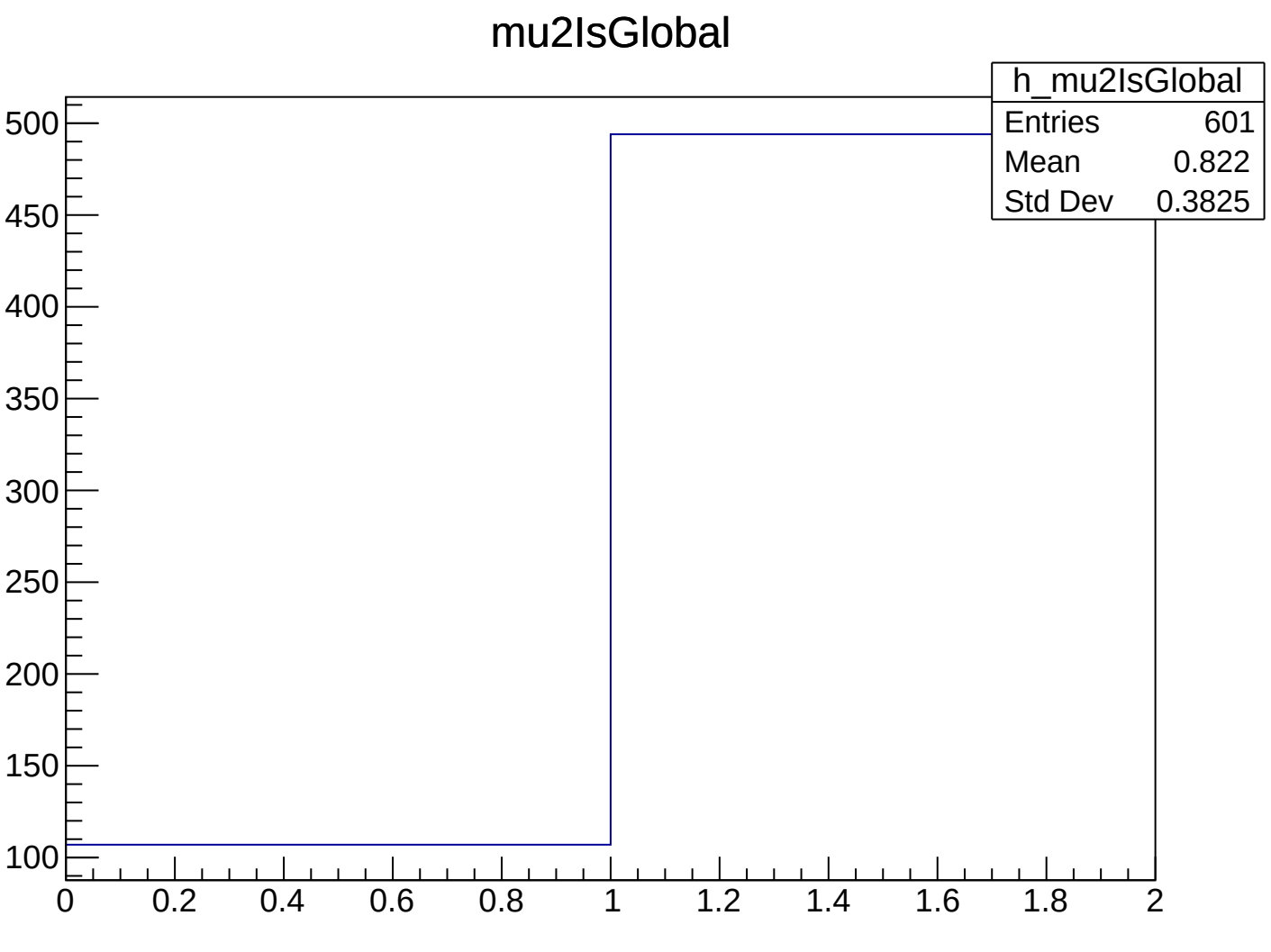

mu1IsTight
h_mu1IsTight
Entries 601
Mean 0.9018
Std Dev 0.2975
500
400
300
200
100
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

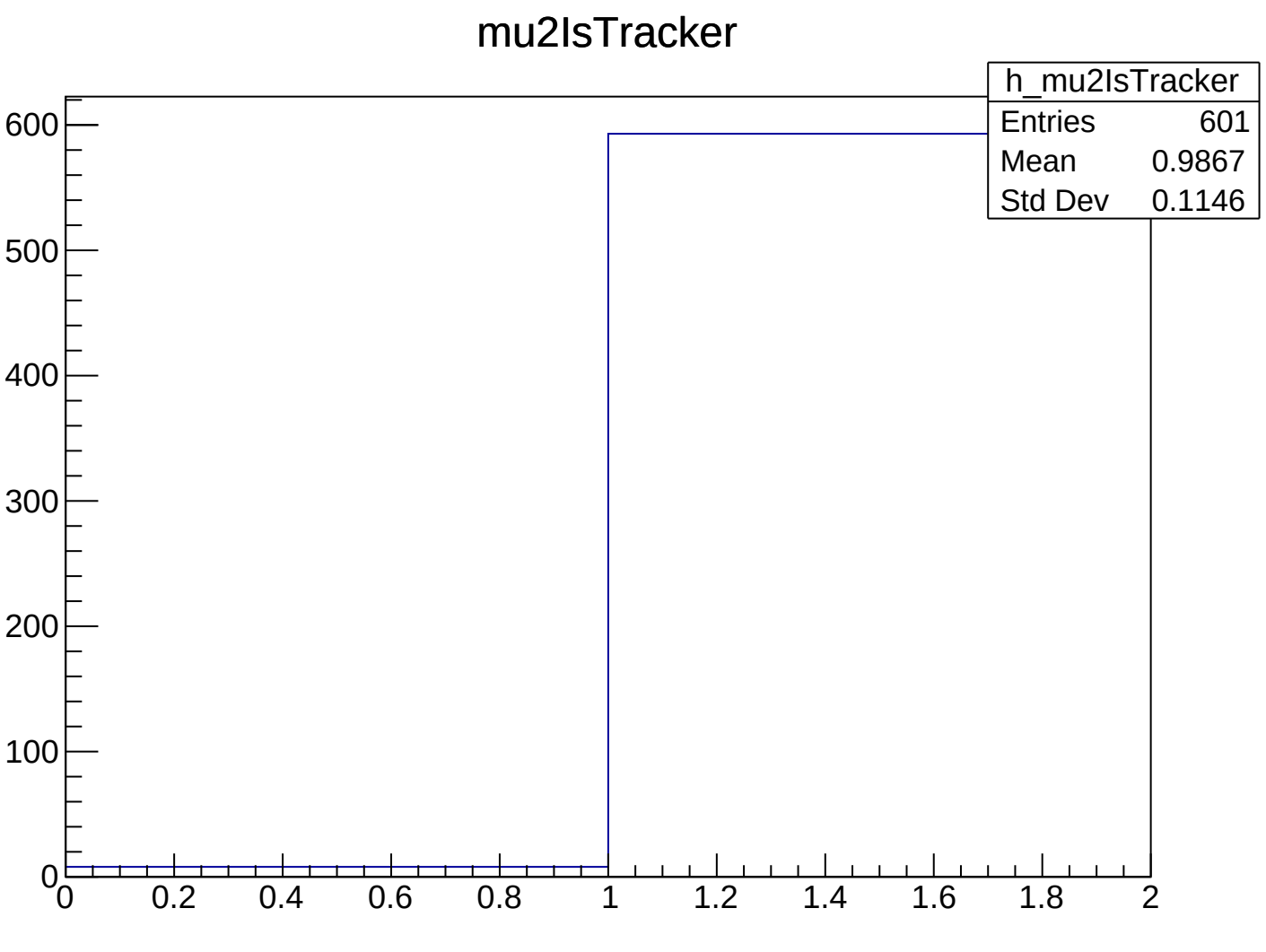

mu2Pt
h_mu2Pt
Entries 601
Mean 7.927
Std Dev 7.717
140
120
100
80
60
40
20
0
0
20
40
60
80
100

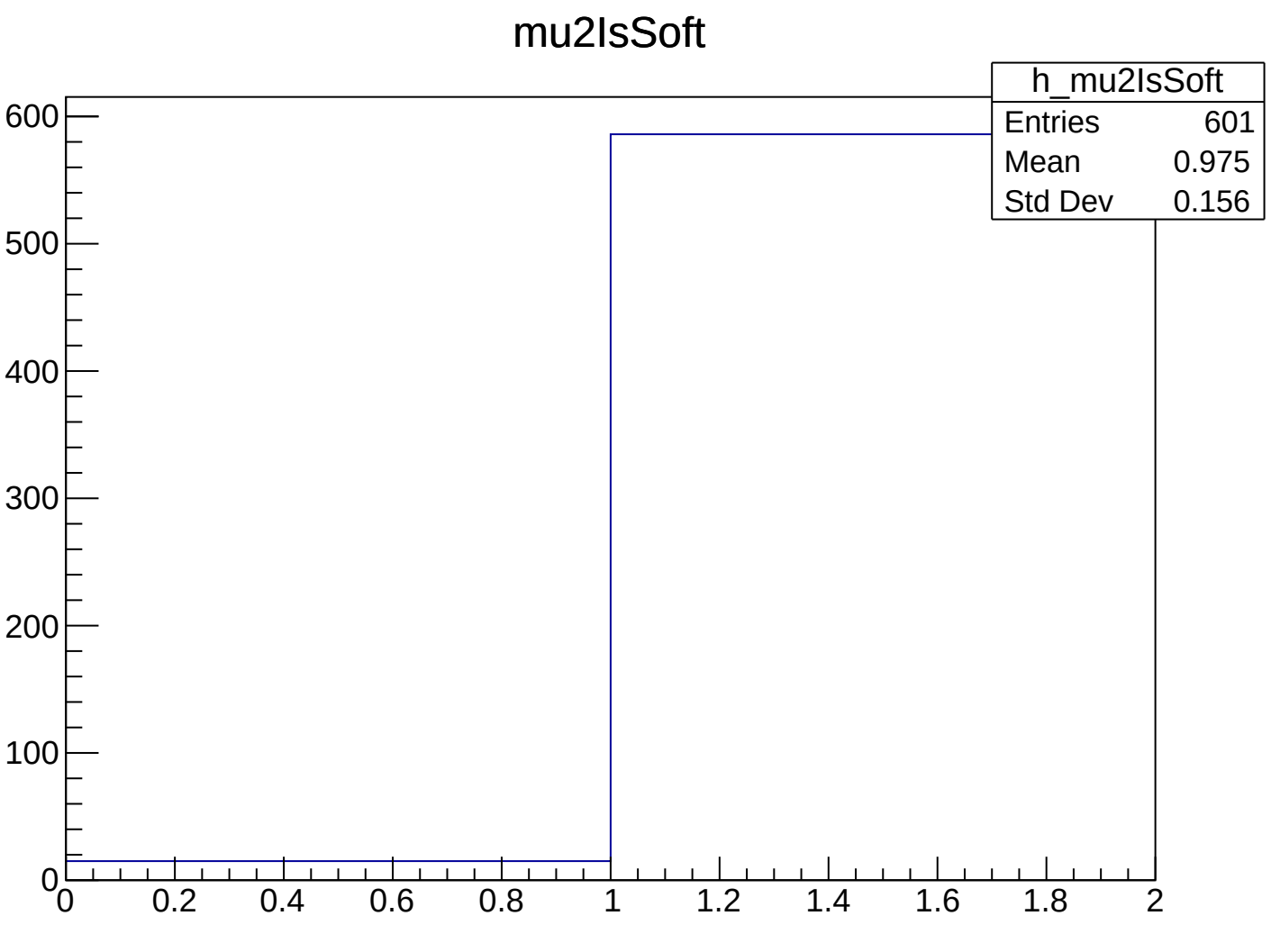

mu2Eta
h_mu2Eta
Entries 601
Mean −0.003449
Std Dev 0.9391
18
16
14
12
10
8
6
4
2
0
−2
−1
0
1
2

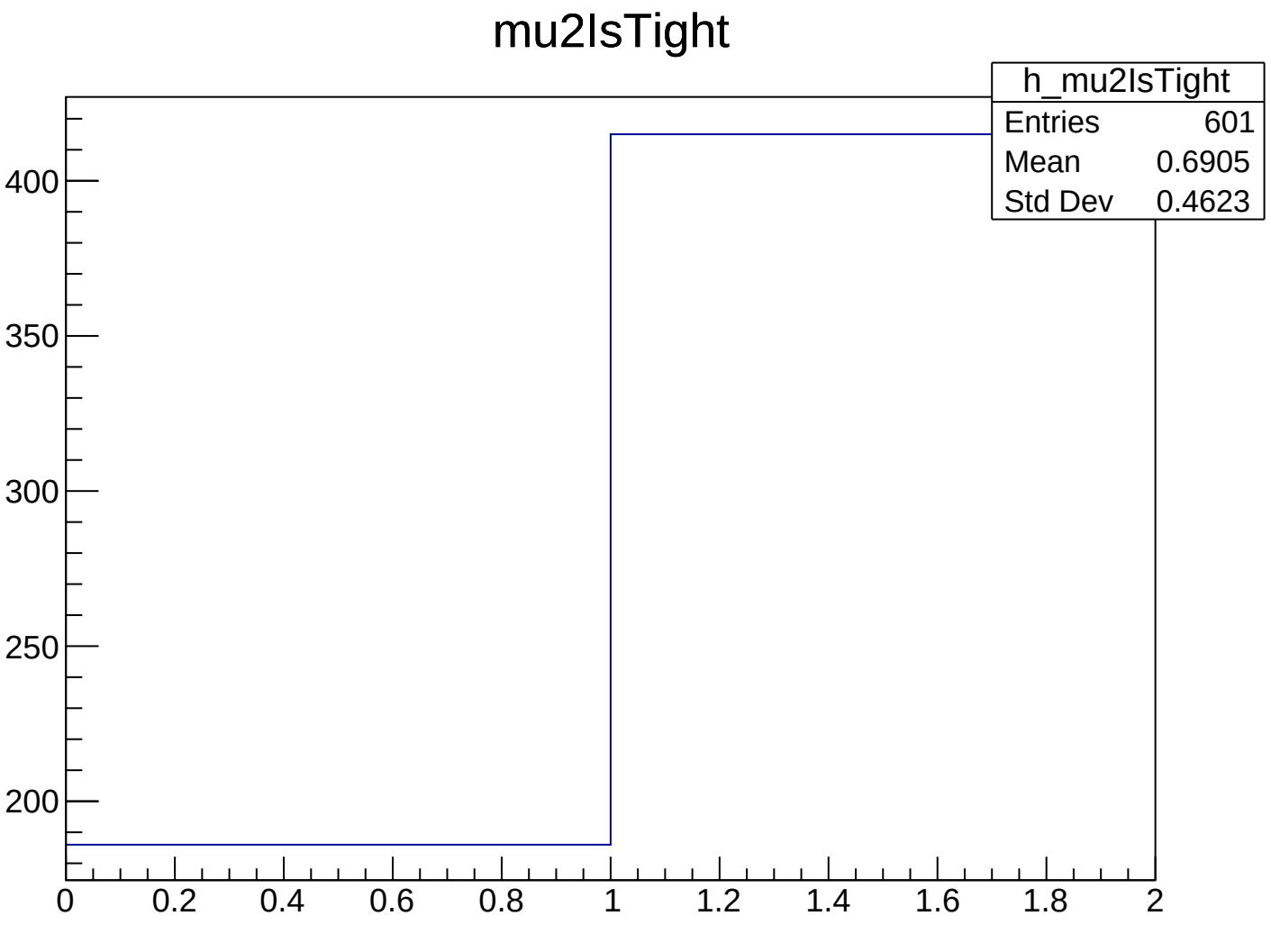

mu2Phi
h_mu2Phi
Entries 601
Mean −0.03421
Std Dev 1.816
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

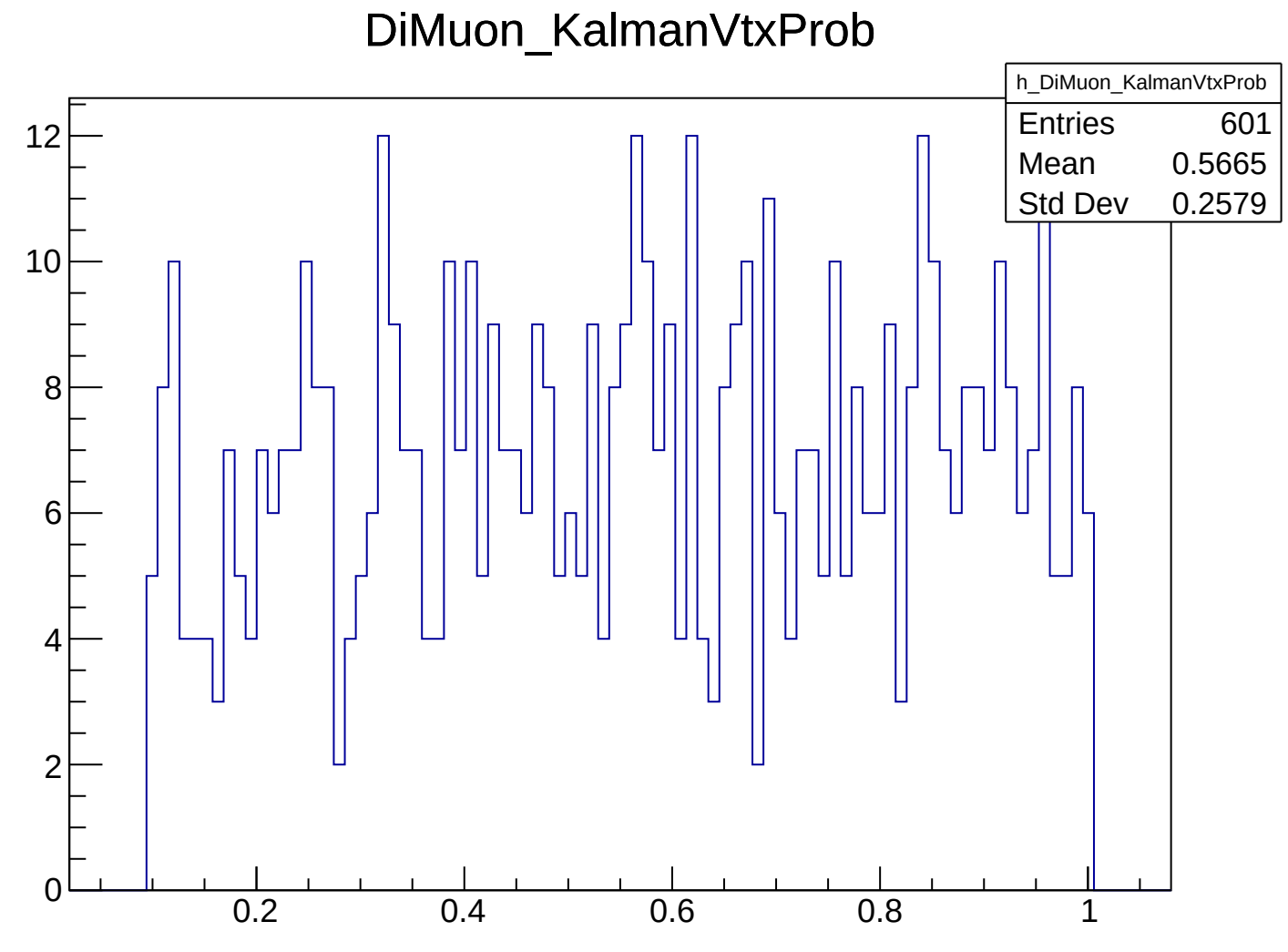

mu2Charge
h_mu2Charge
Entries 601
Mean −0.1048
Std Dev 0.9945
300
250
200
150
100
50
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

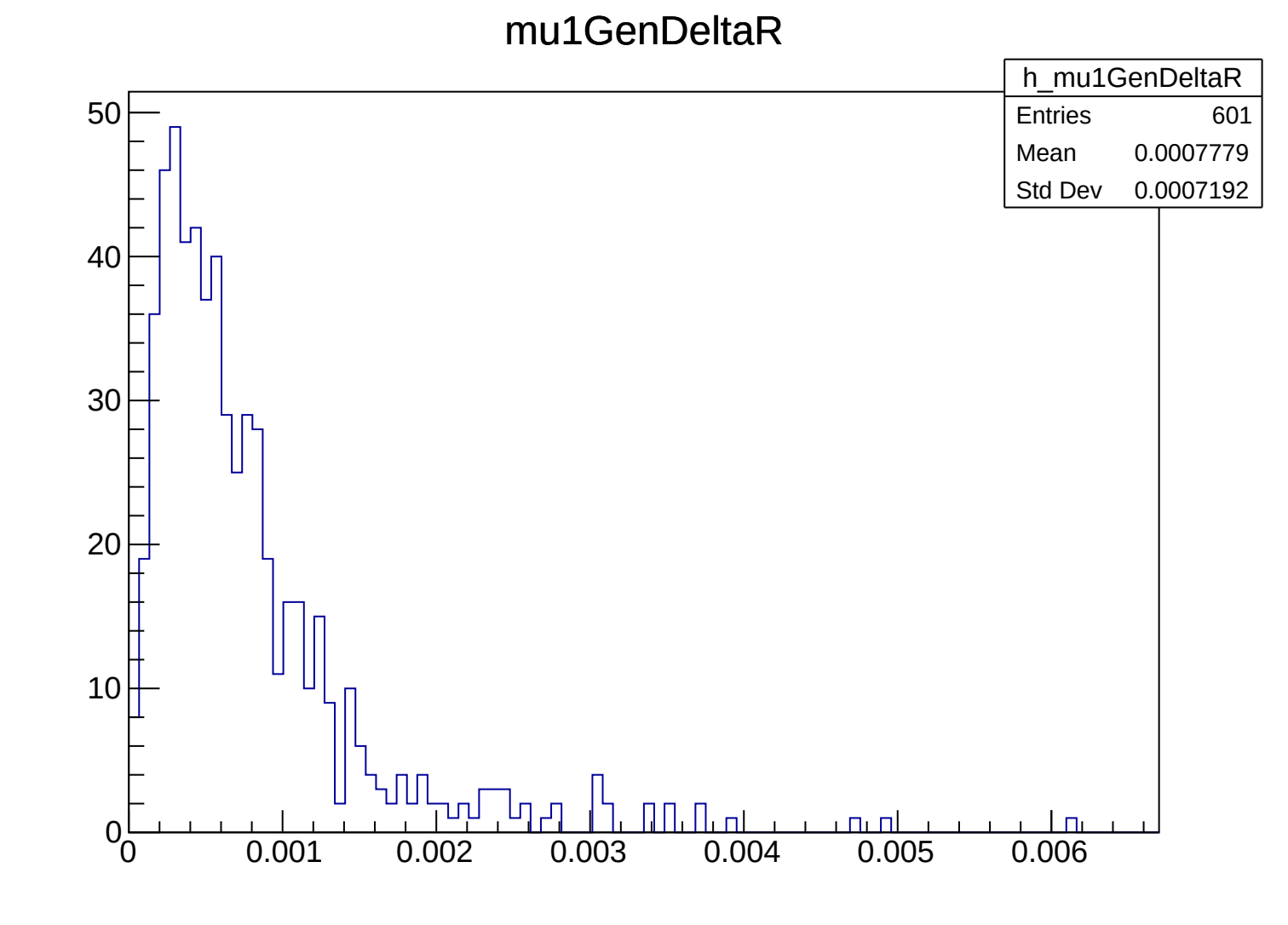

mu2IsGlobal
h_mu2IsGlobal
Entries 601
Mean 0.822
Std Dev 0.3825
500
450
400
350
300
250
200
150
100
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

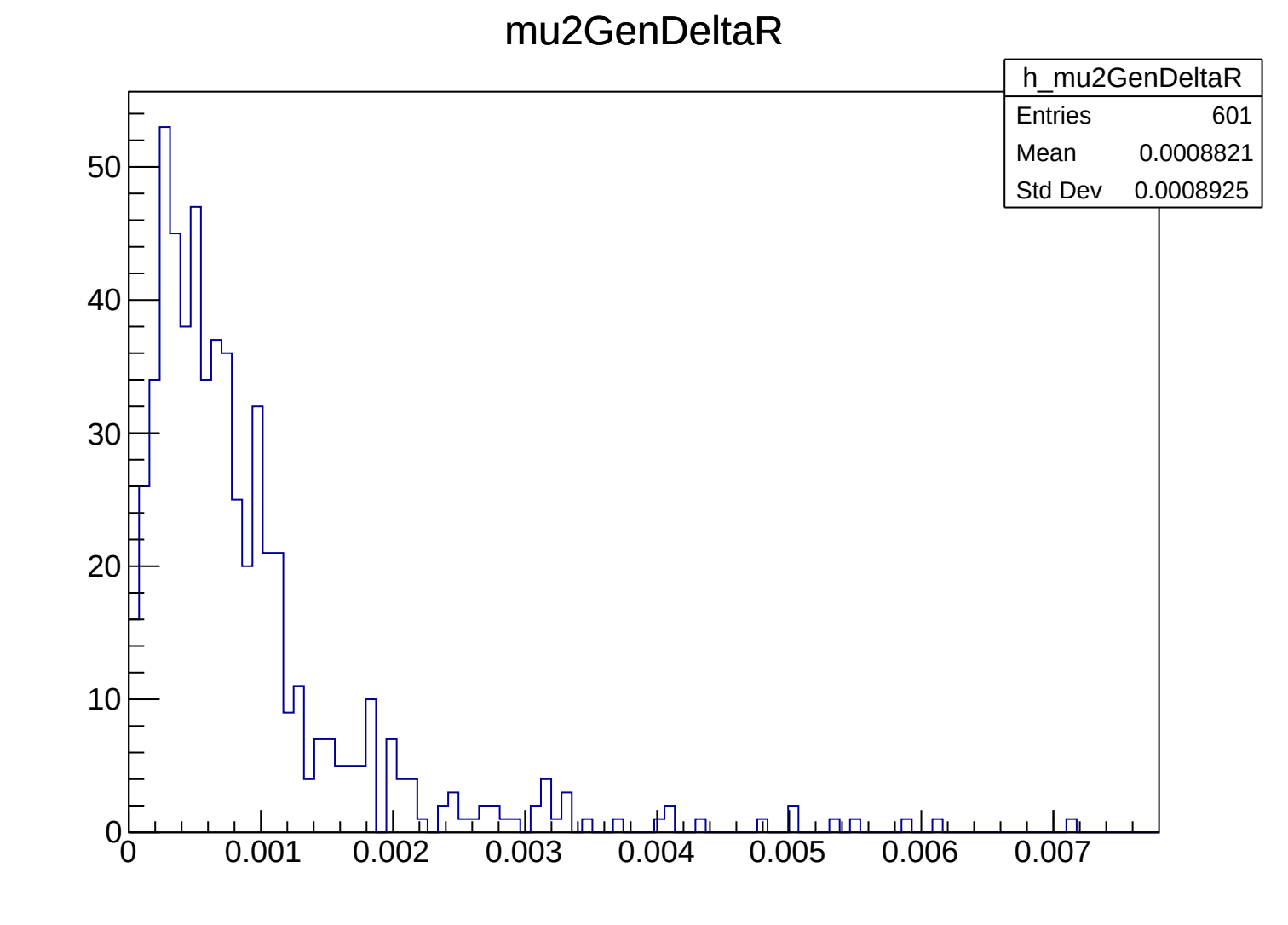

mu2IsTracker
h_mu2IsTracker
Entries 601
Mean 0.9867
Std Dev 0.1146
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

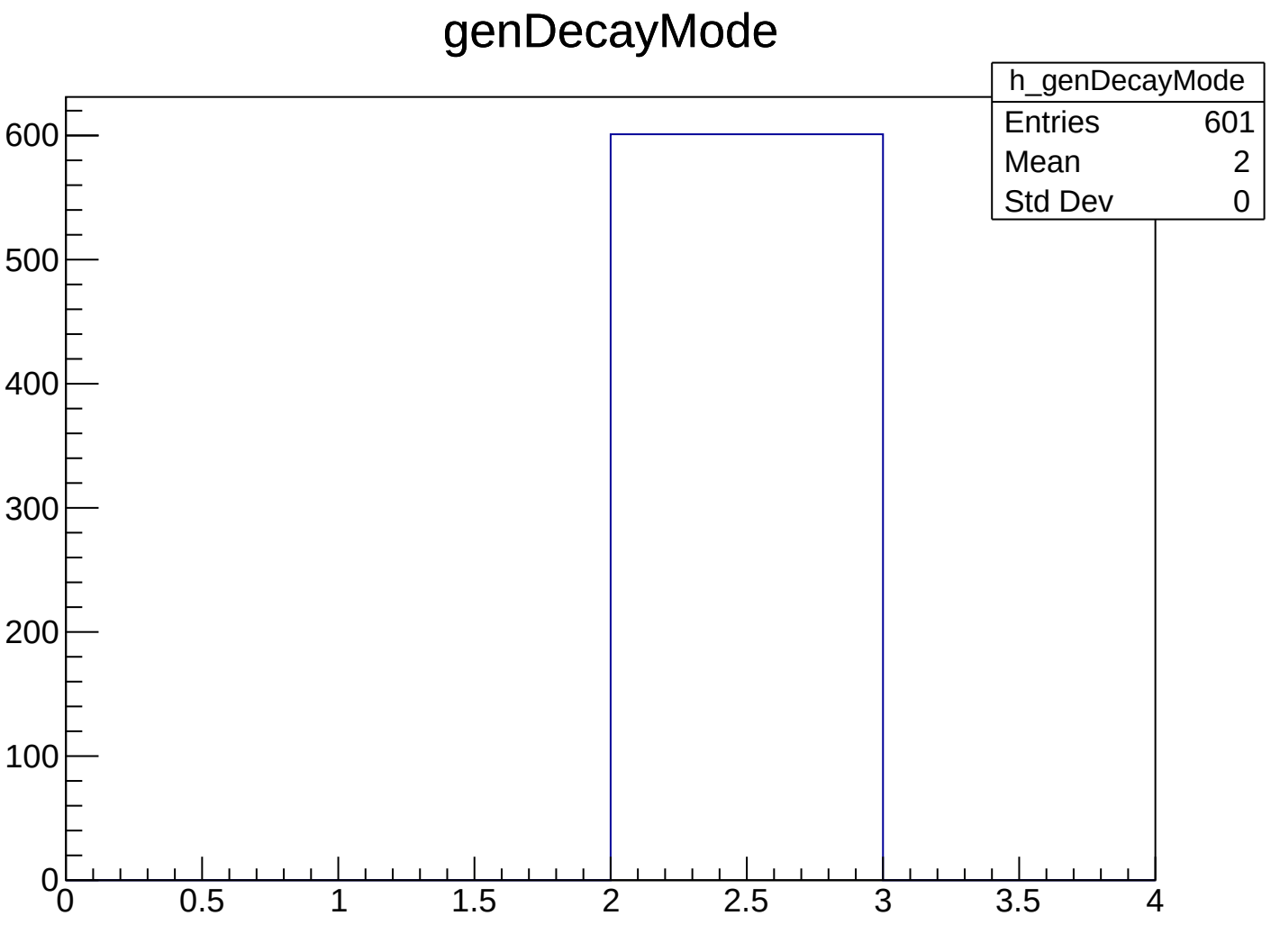

mu2IsSoft
h_mu2IsSoft
Entries 601
Mean 0.975
Std Dev 0.156
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

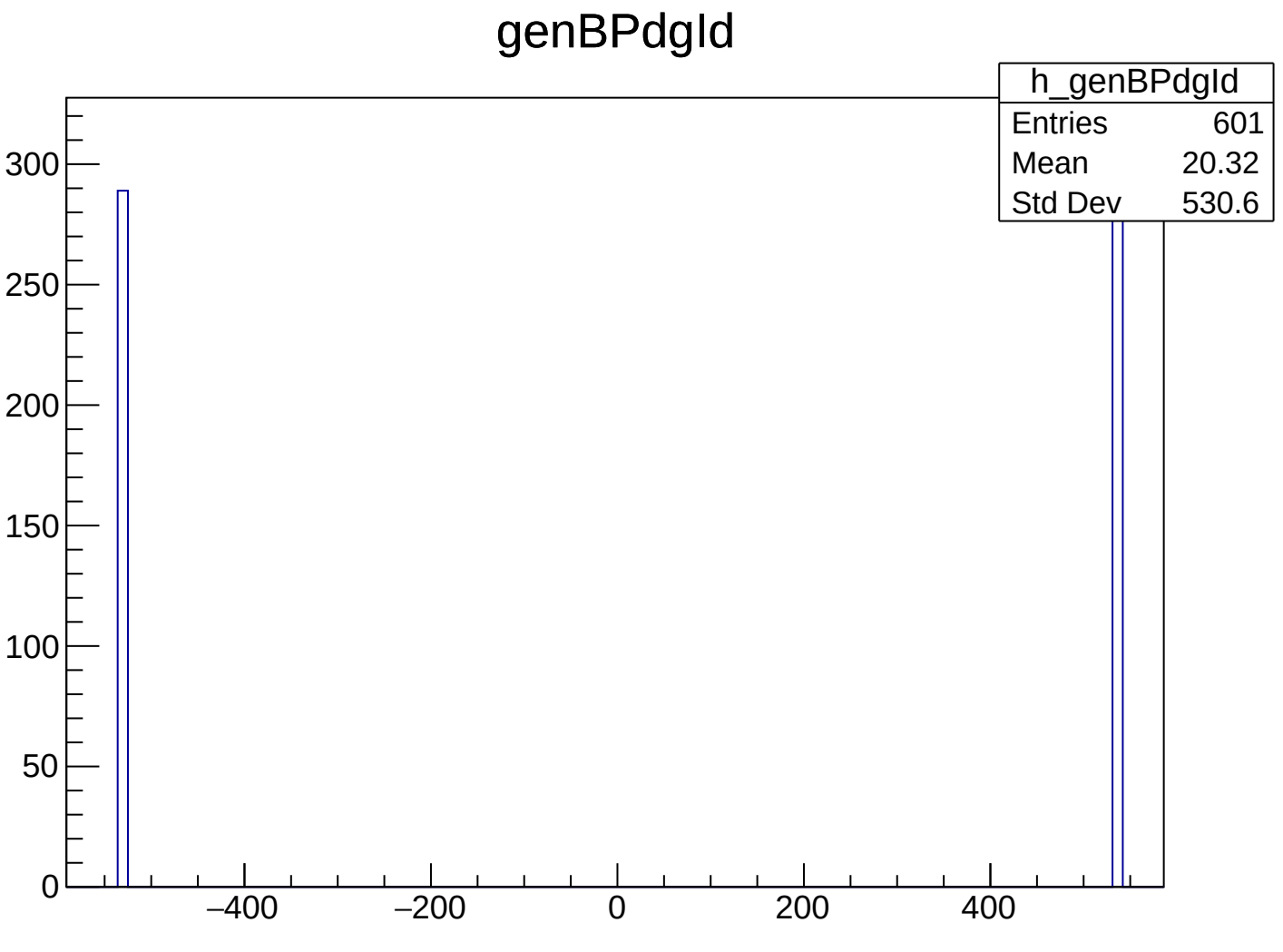

mu2IsTight
h_mu2IsTight
Entries 601
Mean 0.6905
Std Dev 0.4623
400
350
300
250
200
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

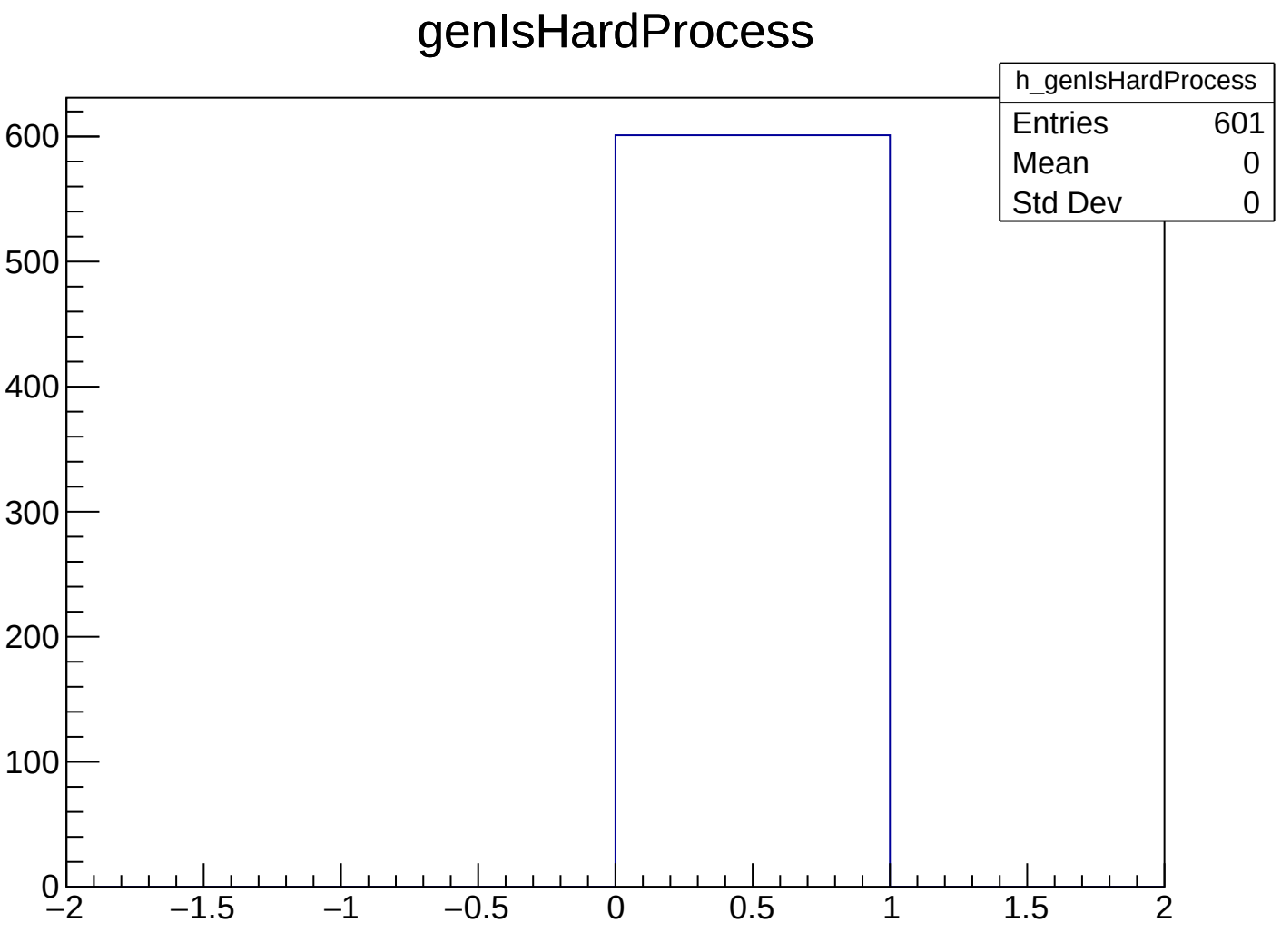

DiMuon_KalmanVtxProb
h_DiMuon_KalmanVtxProb
Entries 601
Mean 0.5665
Std Dev 0.2579
0
2
4
6
8
10
12
0.2
0.4
0.6
0.8
1

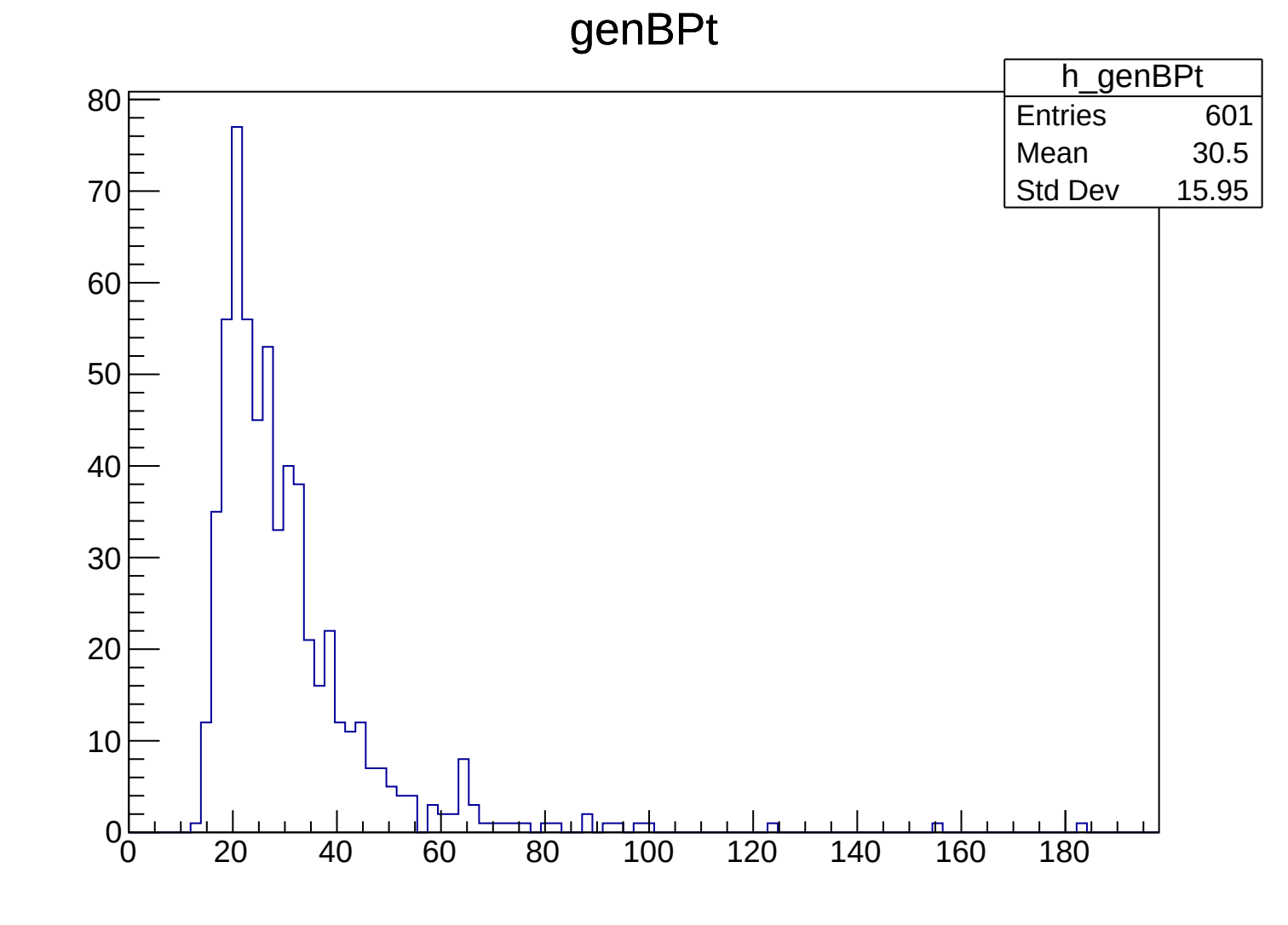

mu1GenDeltaR
h_mu1GenDeltaR
Entries 601
Mean 0.0007779
Std Dev 0.0007192
50
40
30
20
10
0
0
0.001
0.002
0.003
0.004
0.005
0.006

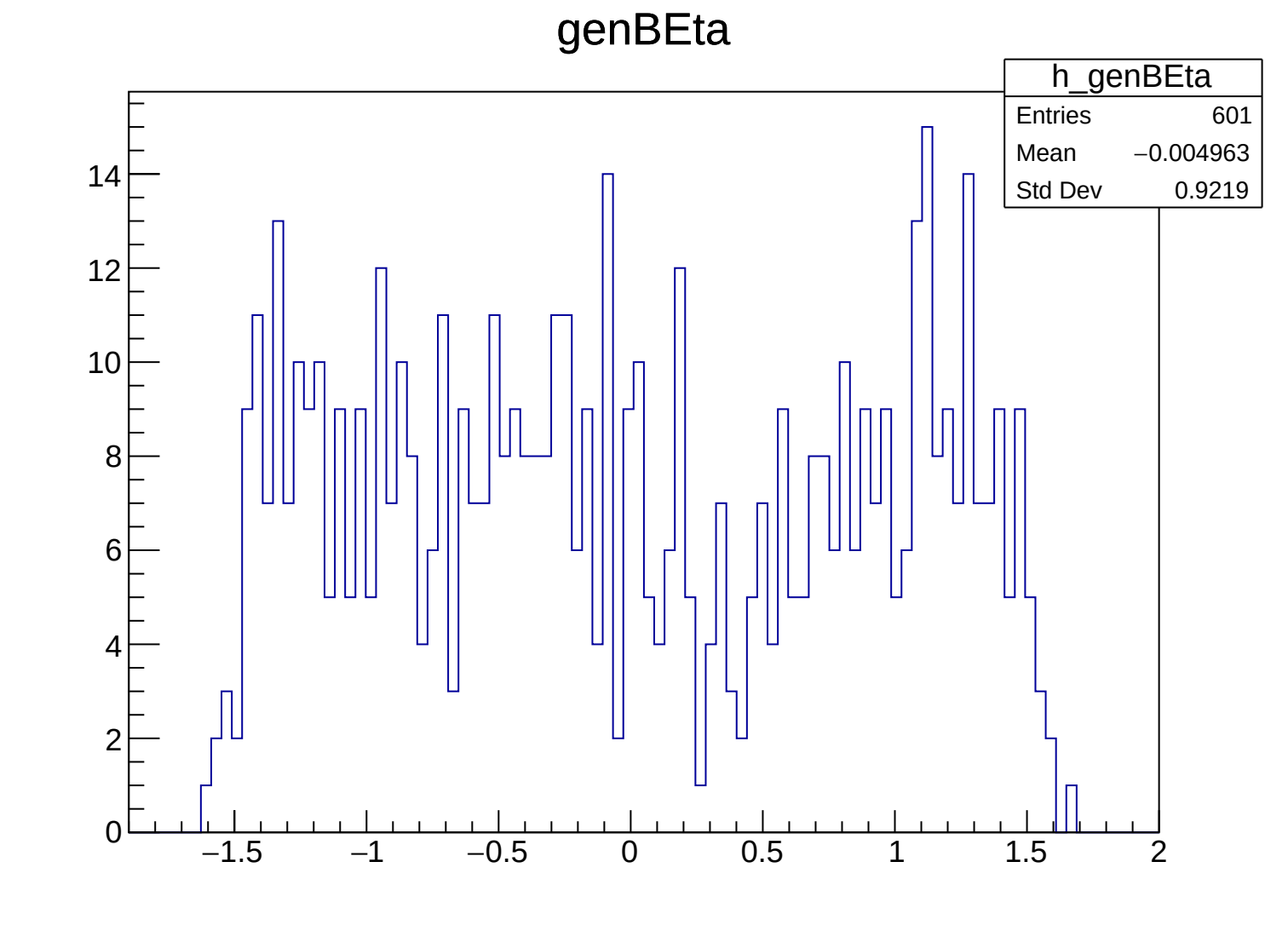

mu2GenDeltaR
h_mu2GenDeltaR
Entries 601
Mean 0.0008821
Std Dev 0.0008925
50
40
30
20
10
0
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007

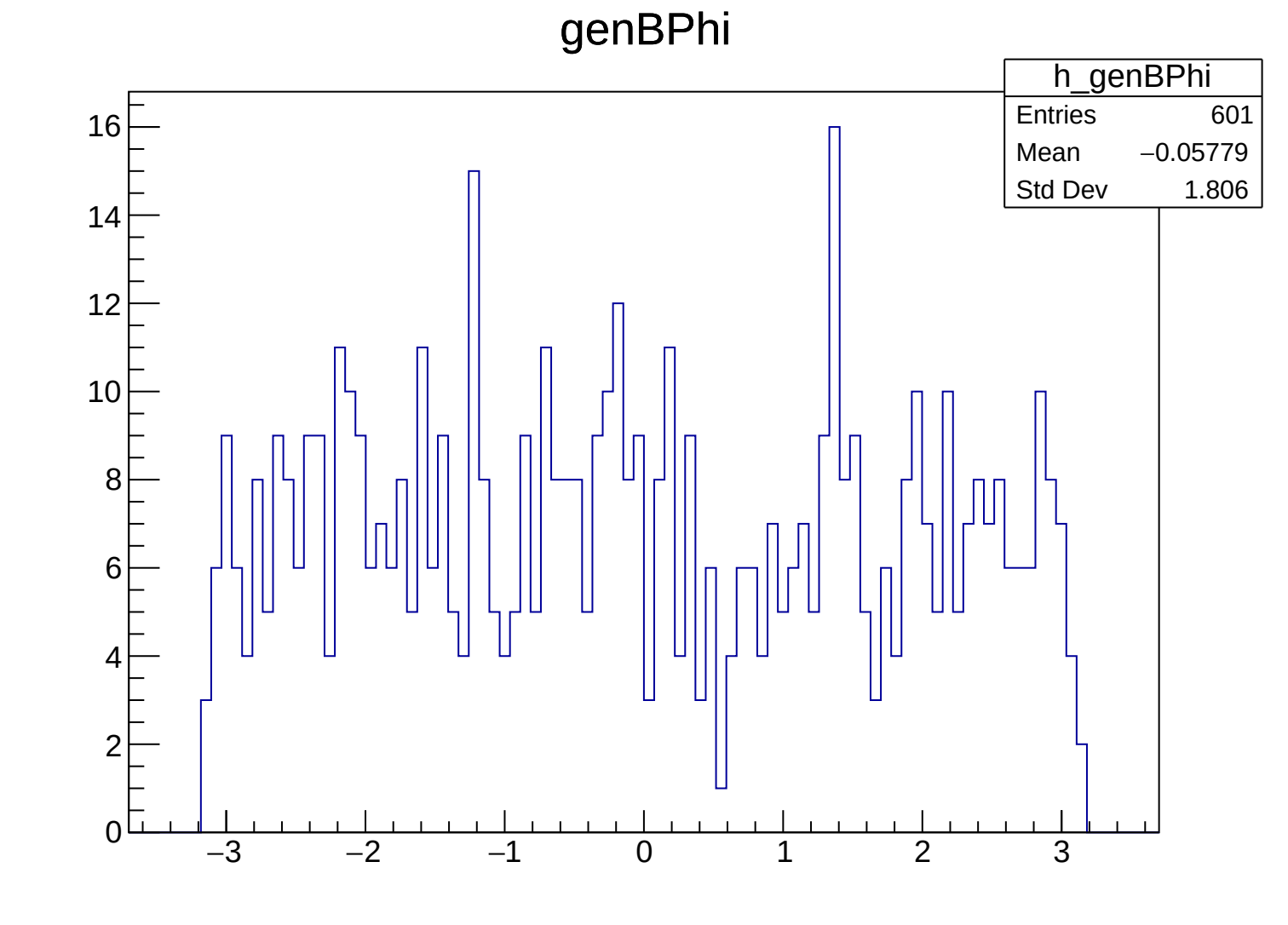

genDecayMode
h_genDecayMode
Entries 601
Mean 2
Std Dev 0
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
3.5
4

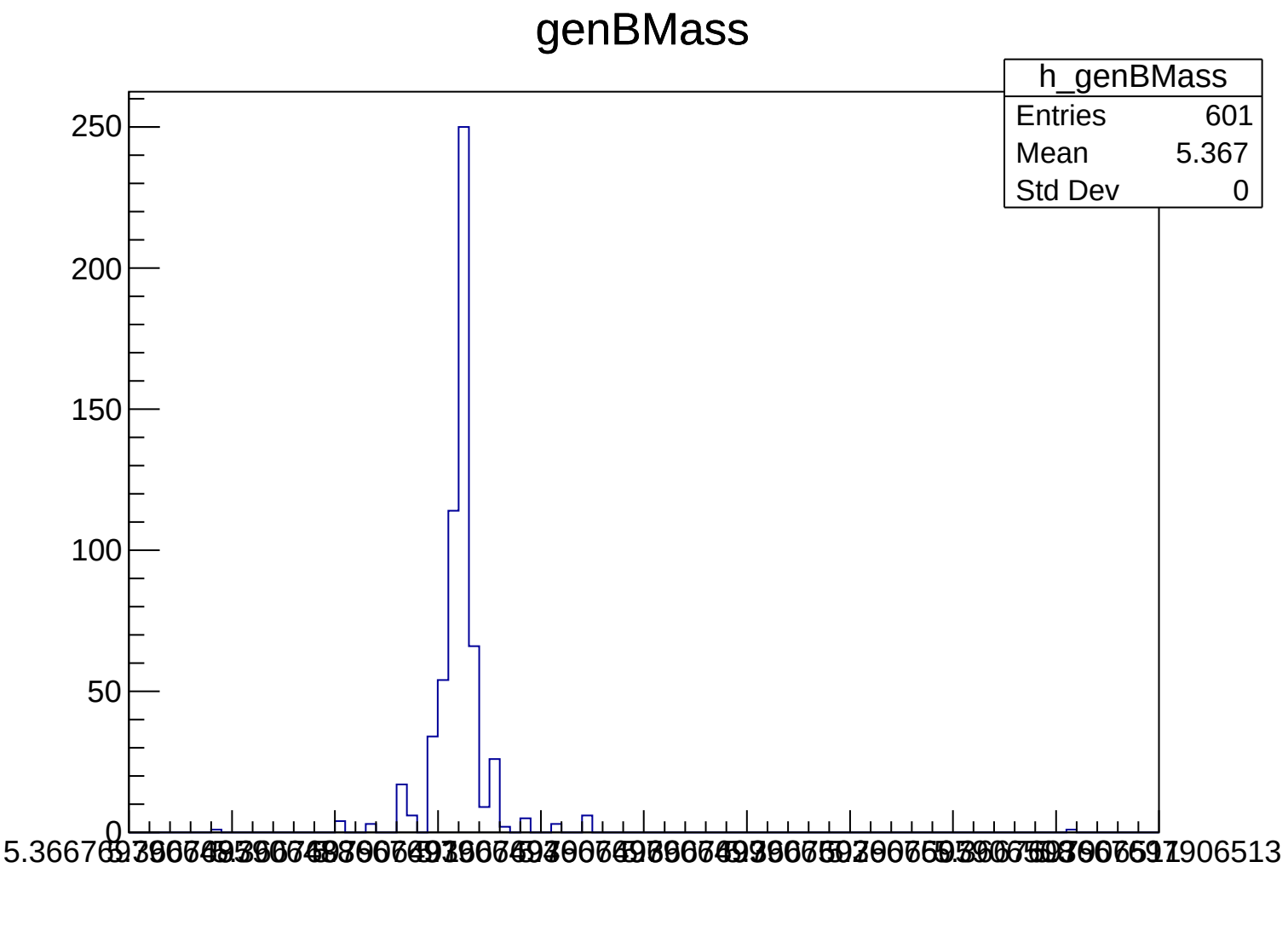

genBPdgId
h_genBPdgId
Entries 601
Mean 20.32
Std Dev 530.6
300
250
200
150
100
50
0
–400
–200
0
200
400

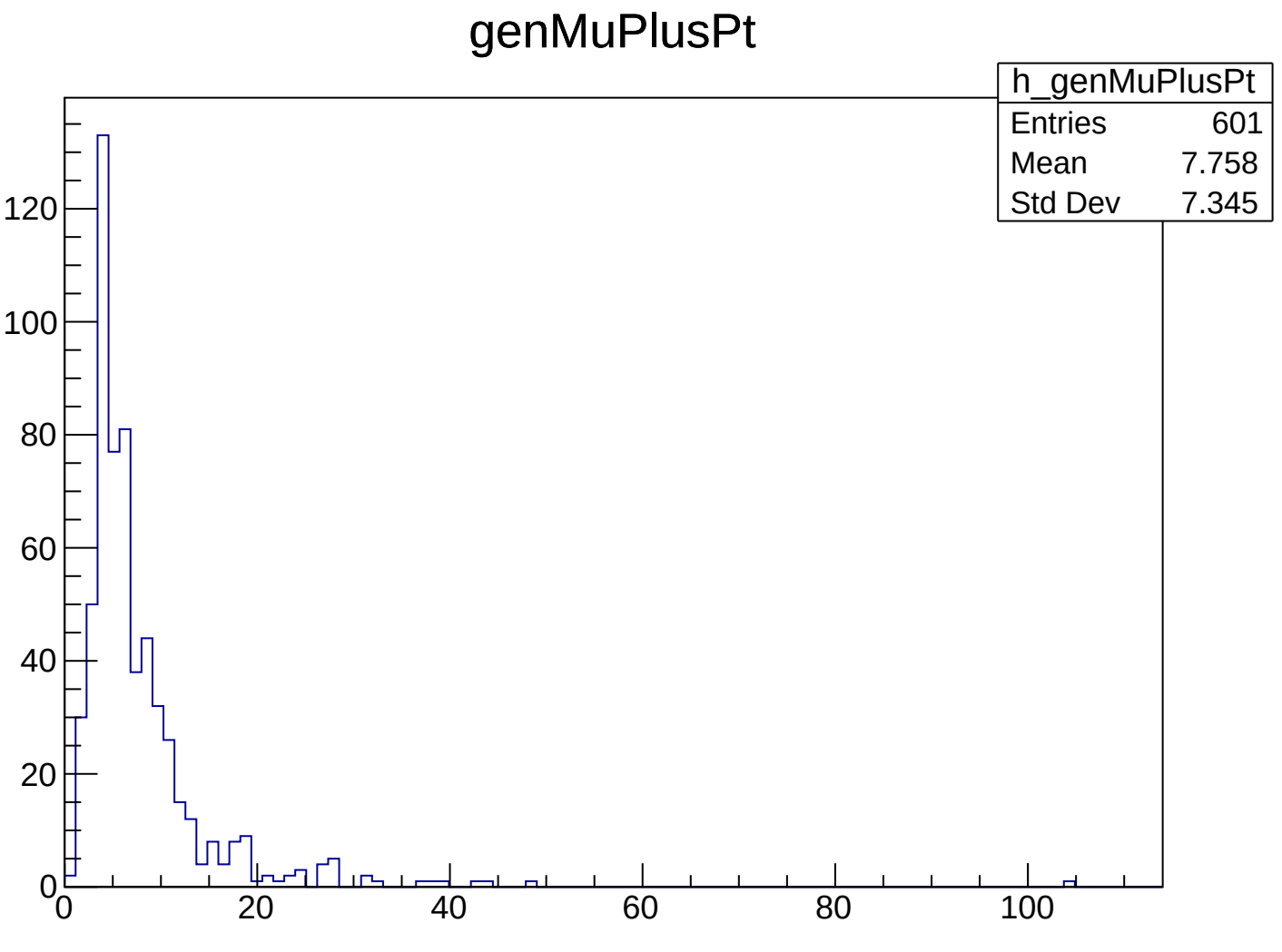

genIsHardProcess
h_genIsHardProcess
Entries 601
Mean 0
Std Dev 0
600
500
400
300
200
100
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

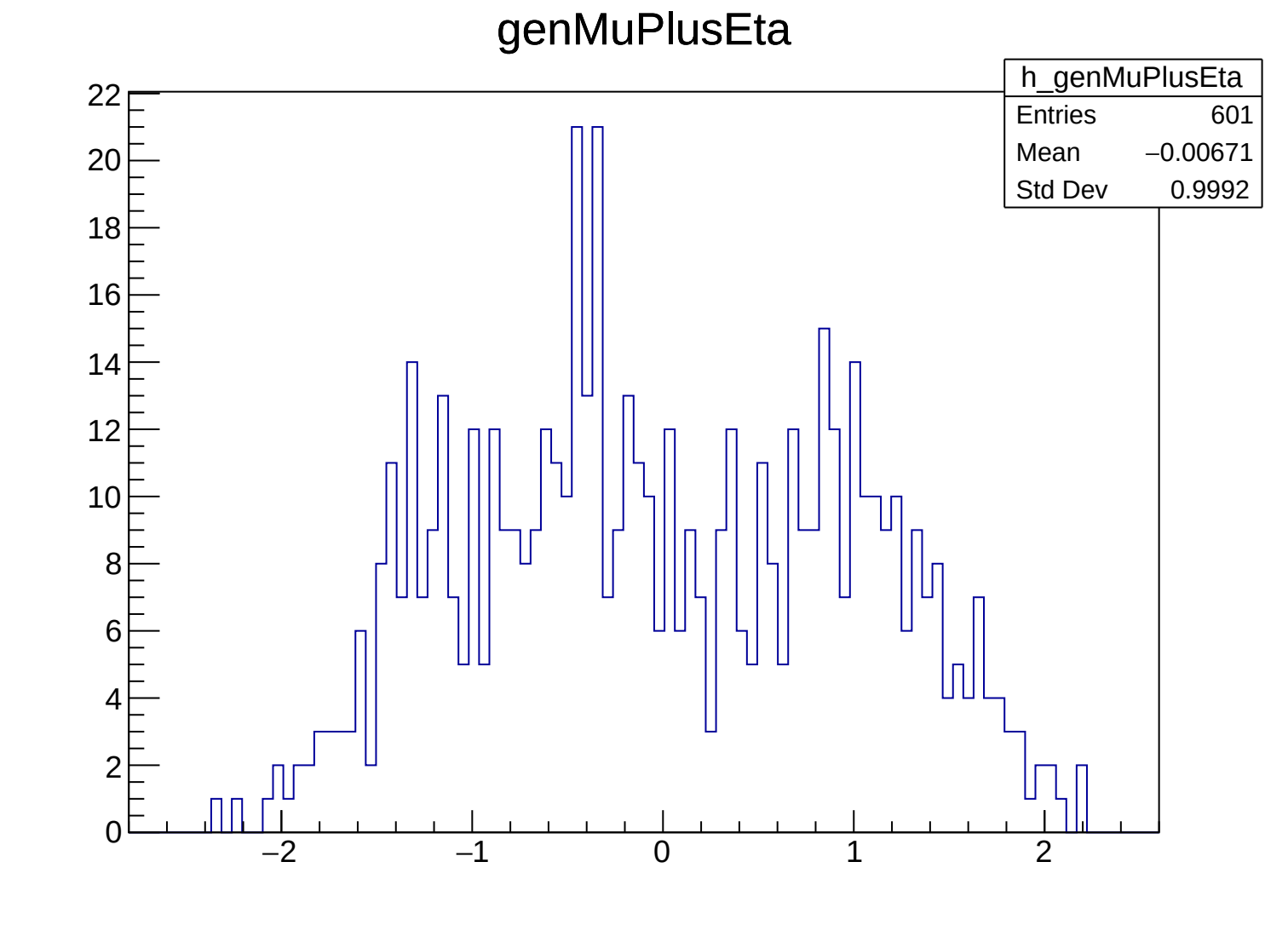

genBPt
h_genBPt
Entries 601
Mean 30.5
Std Dev 15.95
0
10
20
30
40
50
60
70
80
0
20
40
60
80
100
120
140
160
180

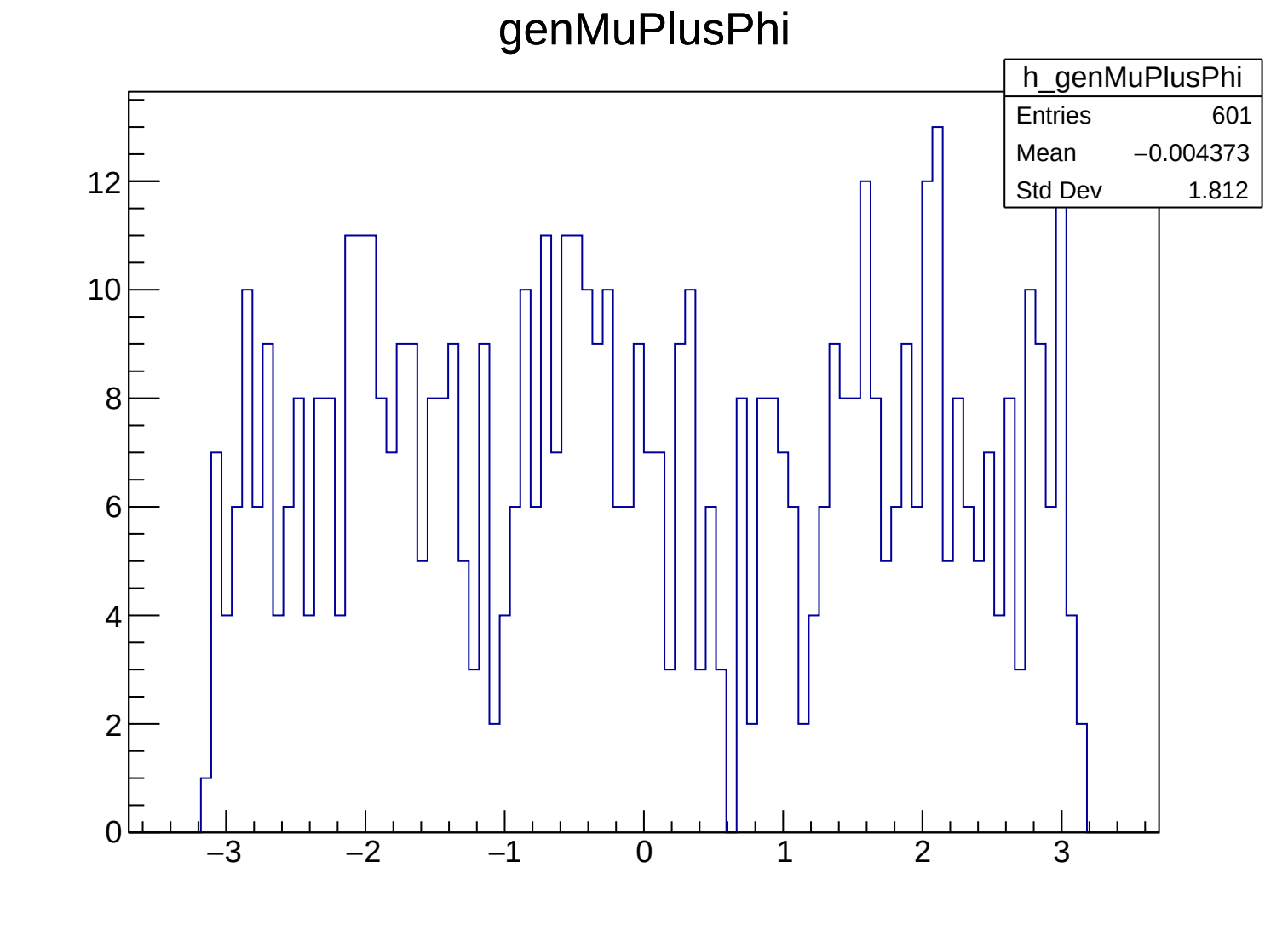

genBEta
h_genBEta
Entries 601
Mean −0.004963
Std Dev 0.9219
0
2
4
6
8
10
12
14
−1.5
−1
−0.5
0
0.5
1
1.5
2

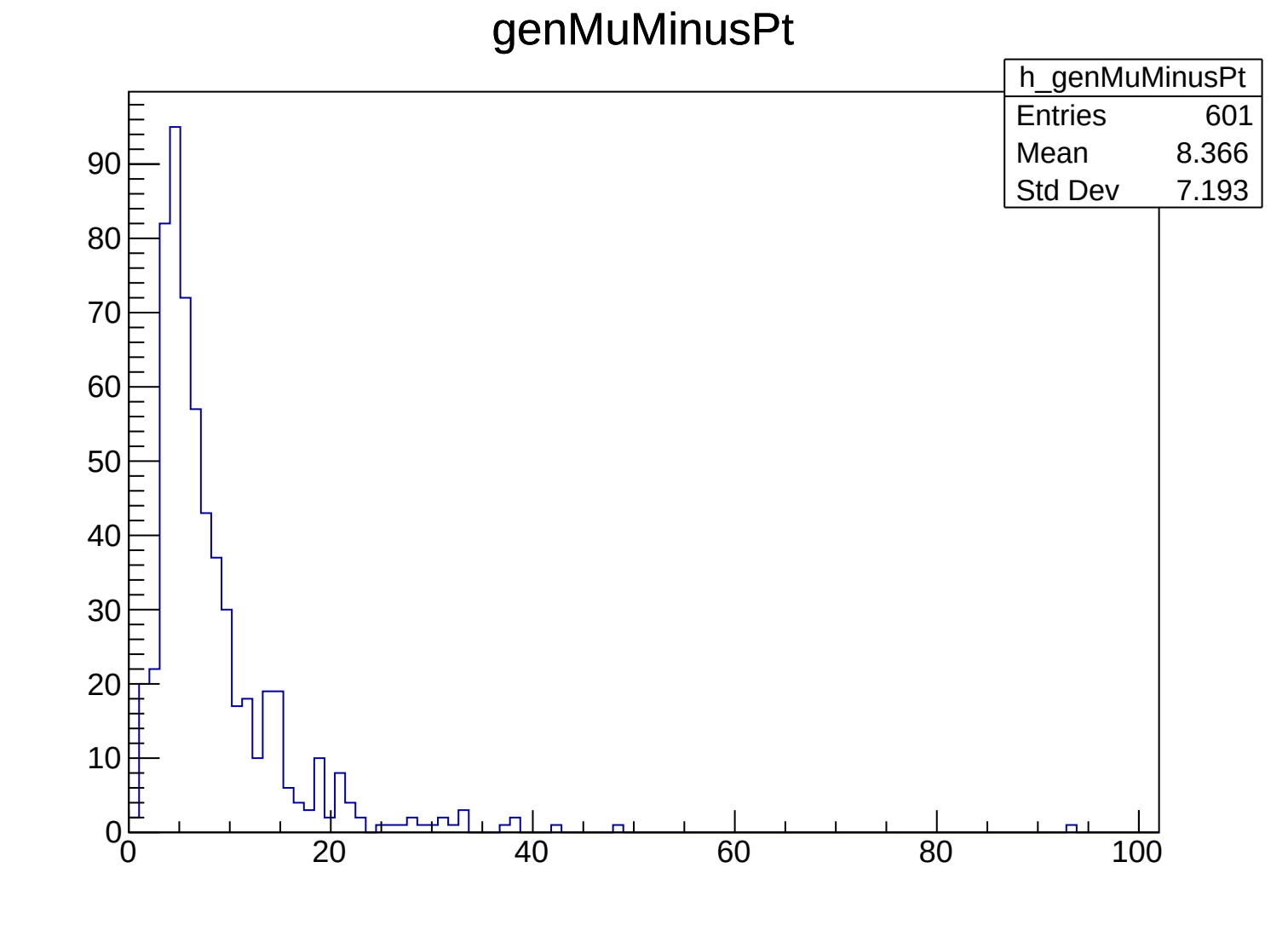

genBPhi
h_genBPhi
Entries 601
Mean −0.05779
Std Dev 1.806
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

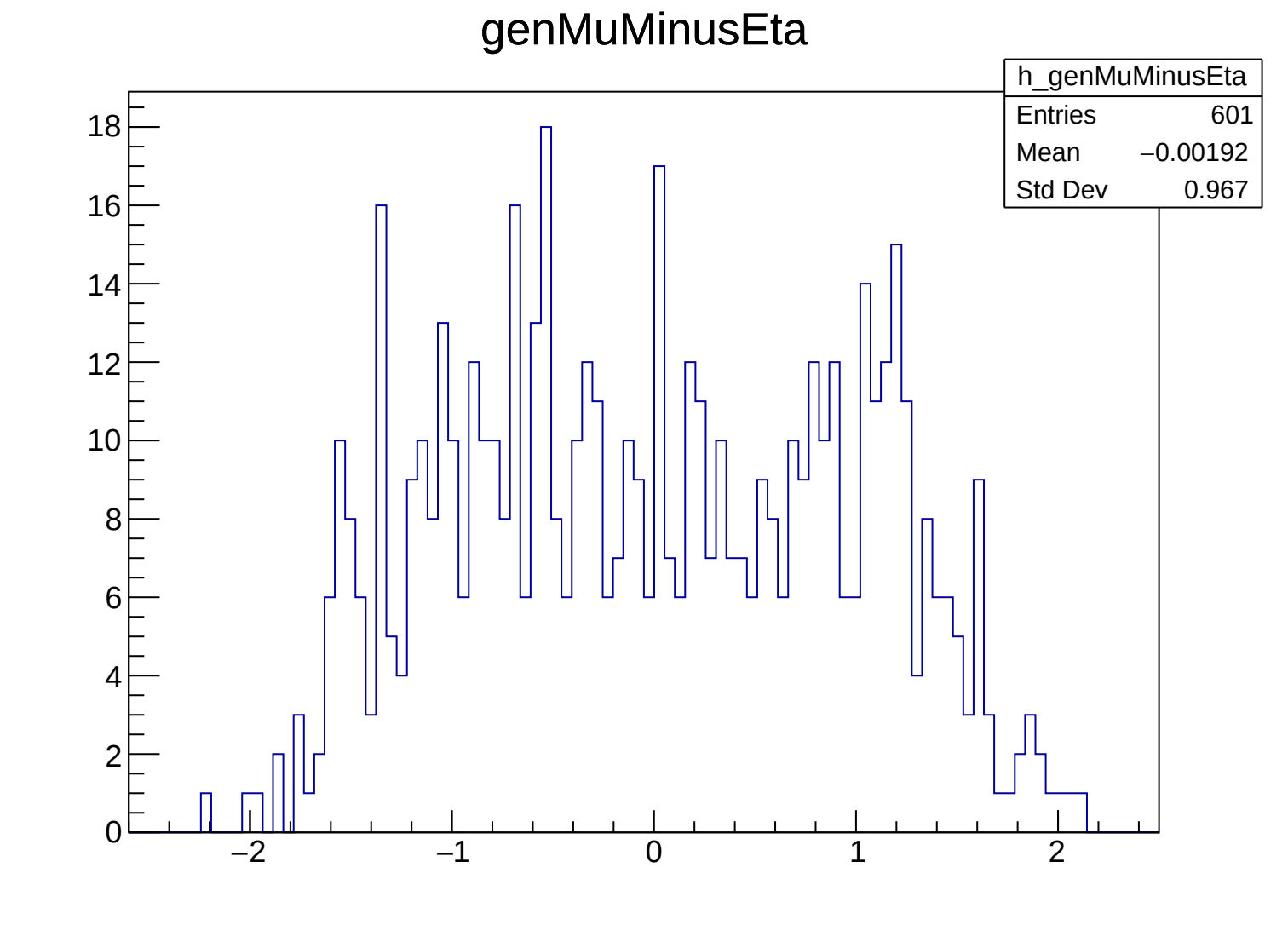

genBMass
h_genBMass
Entries 601
Mean 5.367
Std Dev 0
250
200
150
100
50
0
5.366765
5.367906513

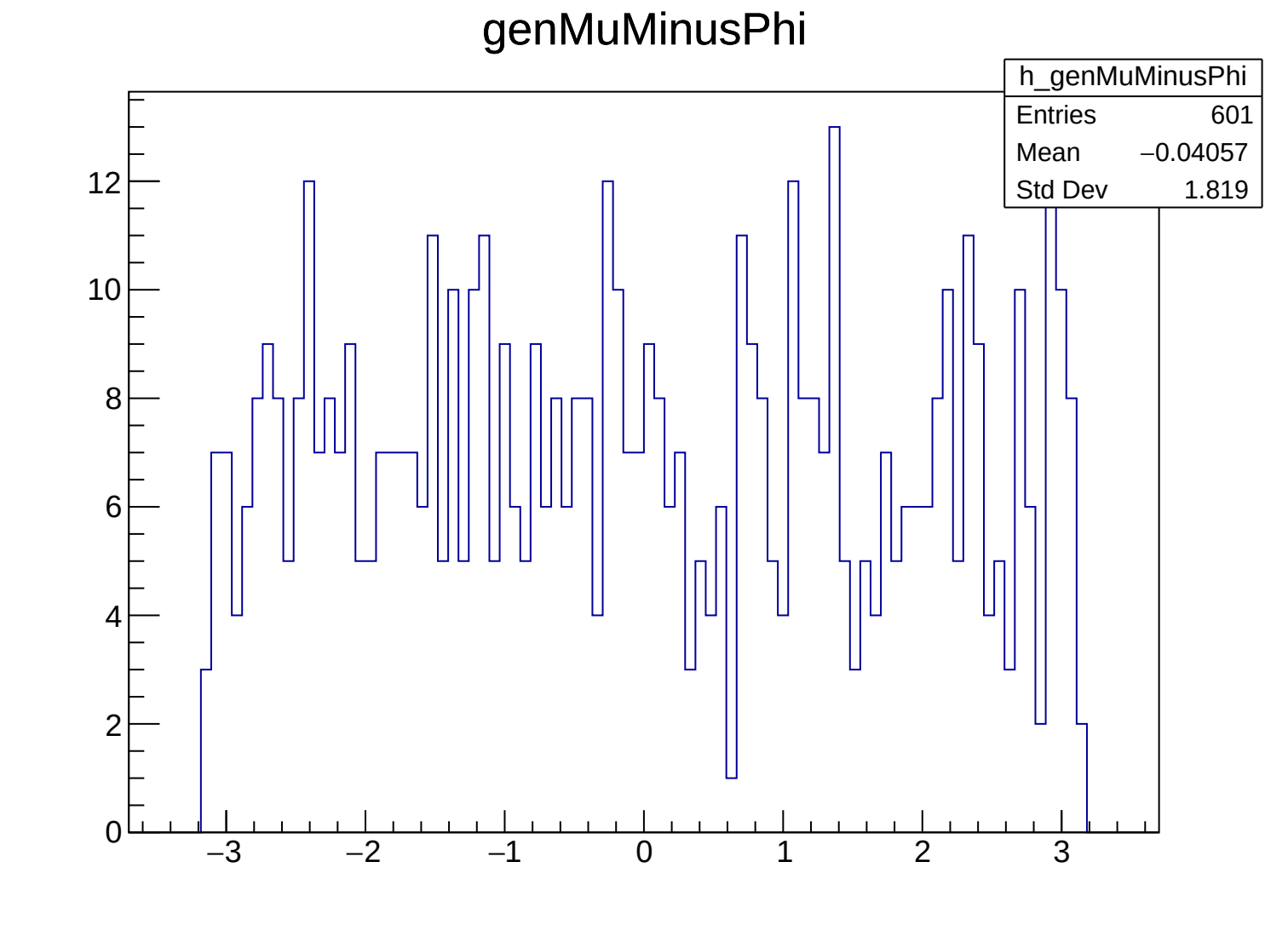

genMuPlusPt
h_genMuPlusPt
Entries 601
Mean 7.758
Std Dev 7.345
120
100
80
60
40
20
0
0
20
40
60
80
100

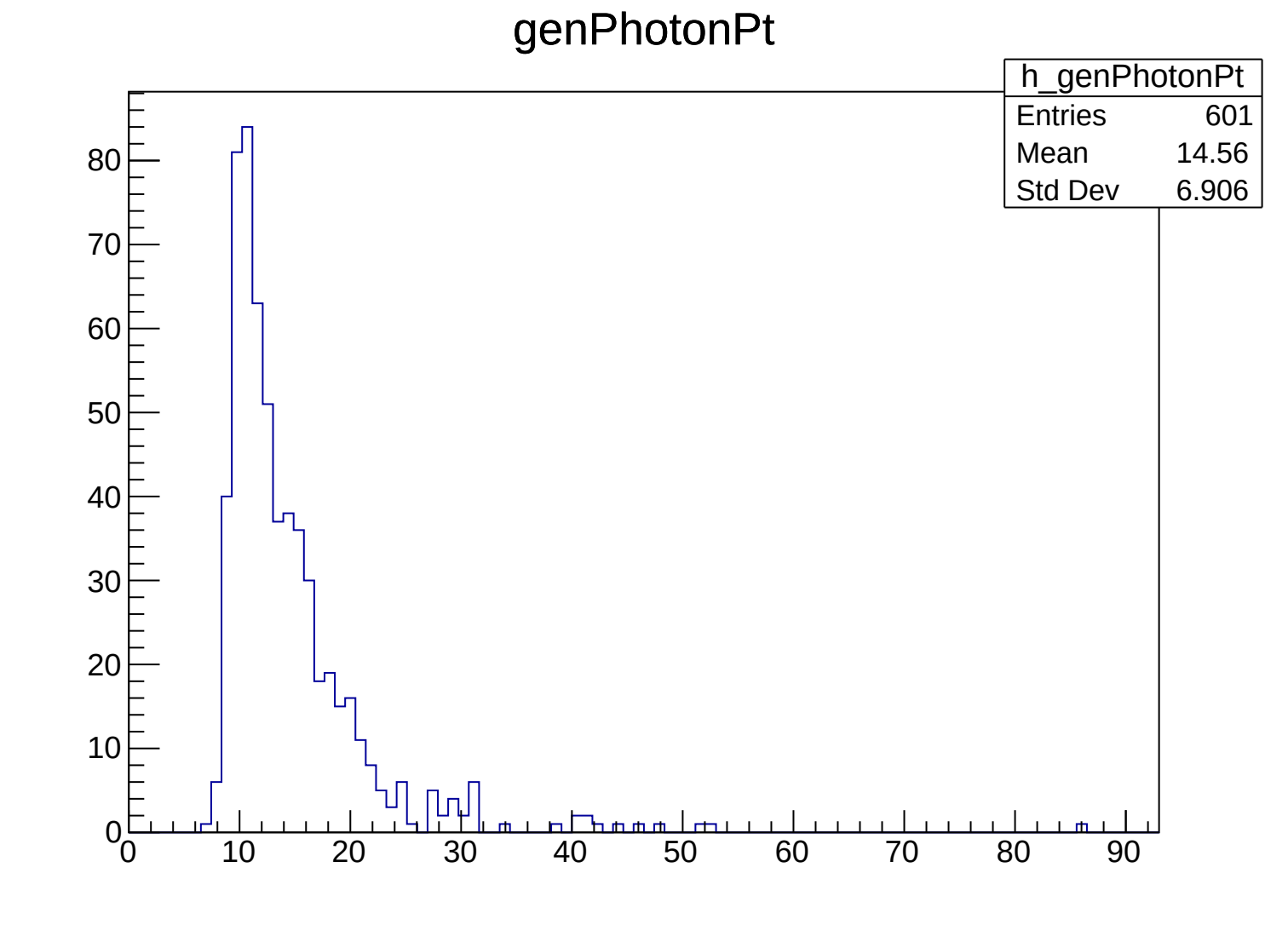

genMuPlusEta
h_genMuPlusEta
Entries 601
Mean −0.00671
Std Dev 0.9992
22
20
18
16
14
12
10
8
6
4
2
0
−2
−1
0
1
2

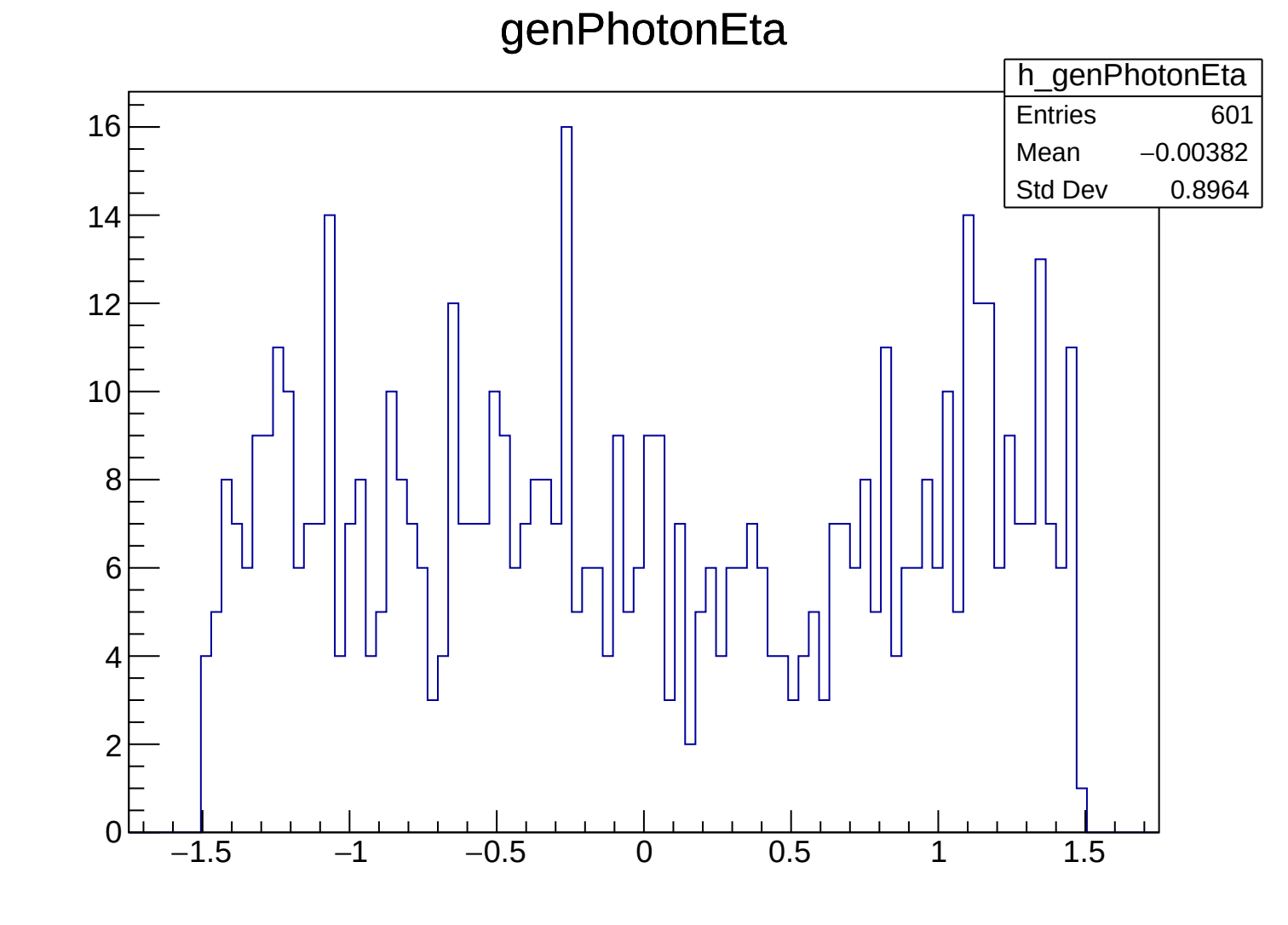

genMuPlusPhi
h_genMuPlusPhi
Entries 601
Mean −0.004373
Std Dev 1.812
0
2
4
6
8
10
12
−3
−2
−1
0
1
2
3

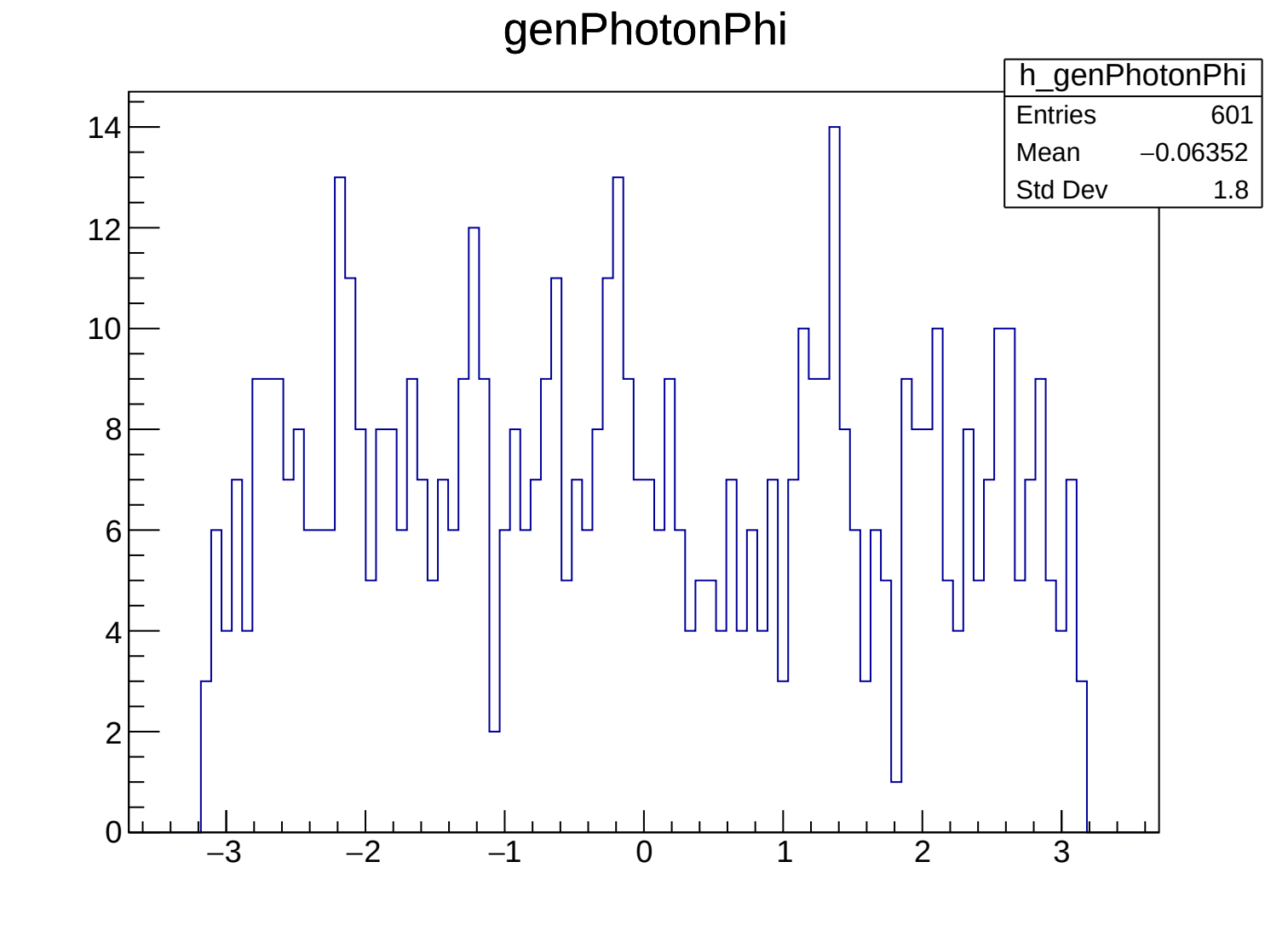

genMuMinusPt
h_genMuMinusPt
Entries 601
Mean 8.366
Std Dev 7.193
0
10
20
30
40
50
60
70
80
90
0
20
40
60
80
100

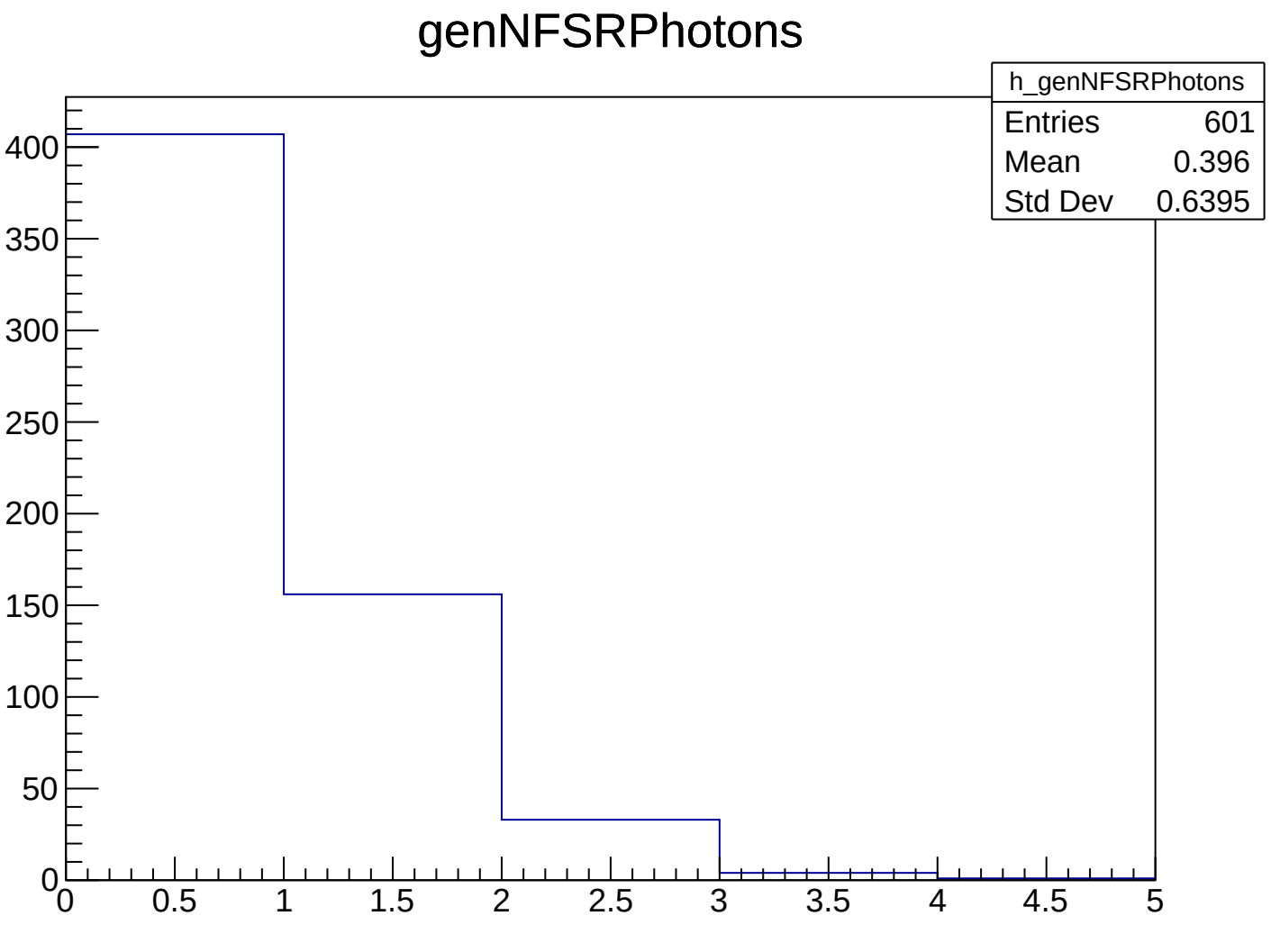

genMuMinusEta
h_genMuMinusEta
Entries 601
Mean −0.00192
Std Dev 0.967
18
16
14
12
10
8
6
4
2
0
−2
−1
0
1
2

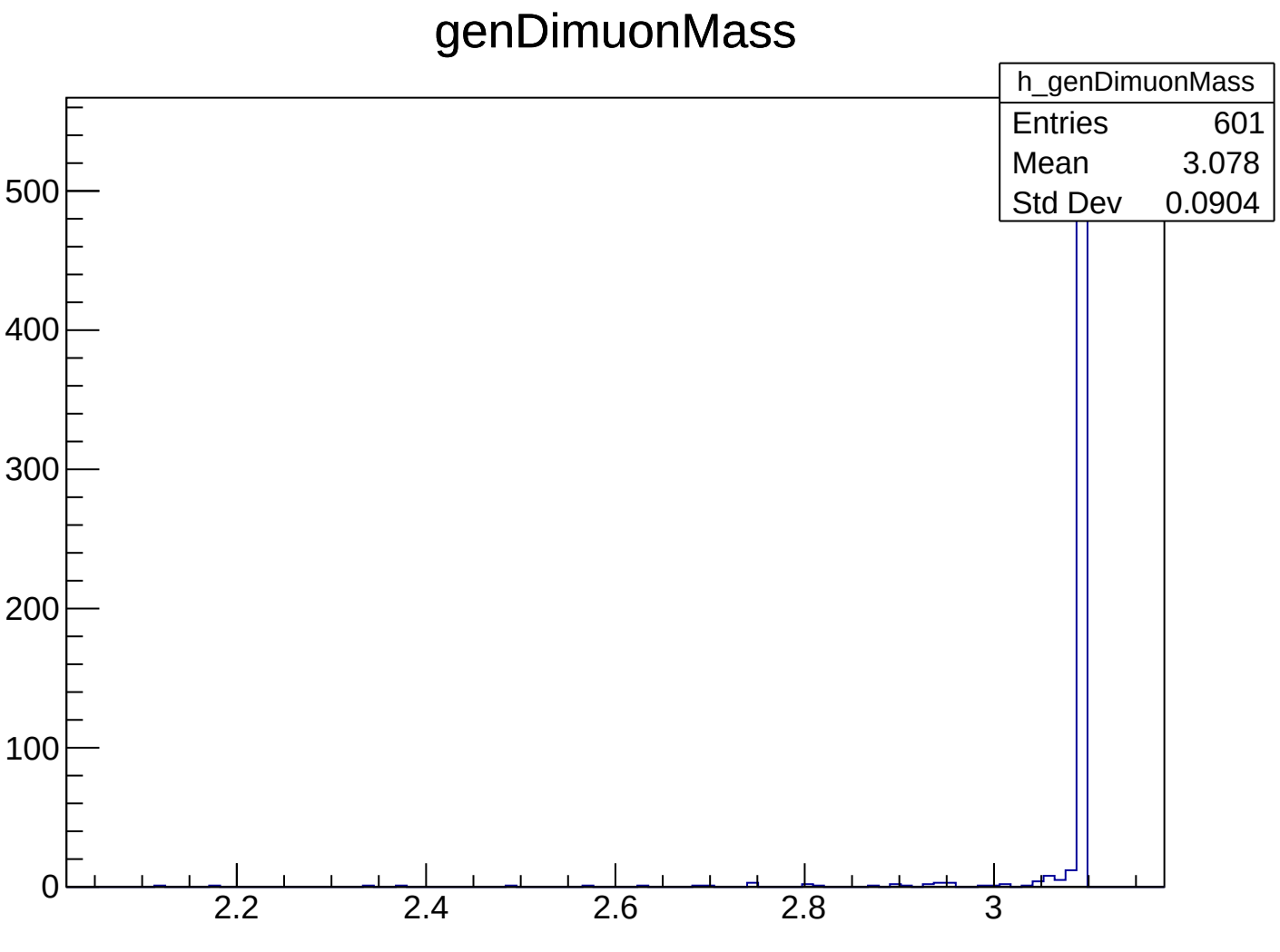

genMuMinusPhi
h_genMuMinusPhi
Entries 601
Mean −0.04057
Std Dev 1.819
0
2
4
6
8
10
12
−3
−2
−1
0
1
2
3

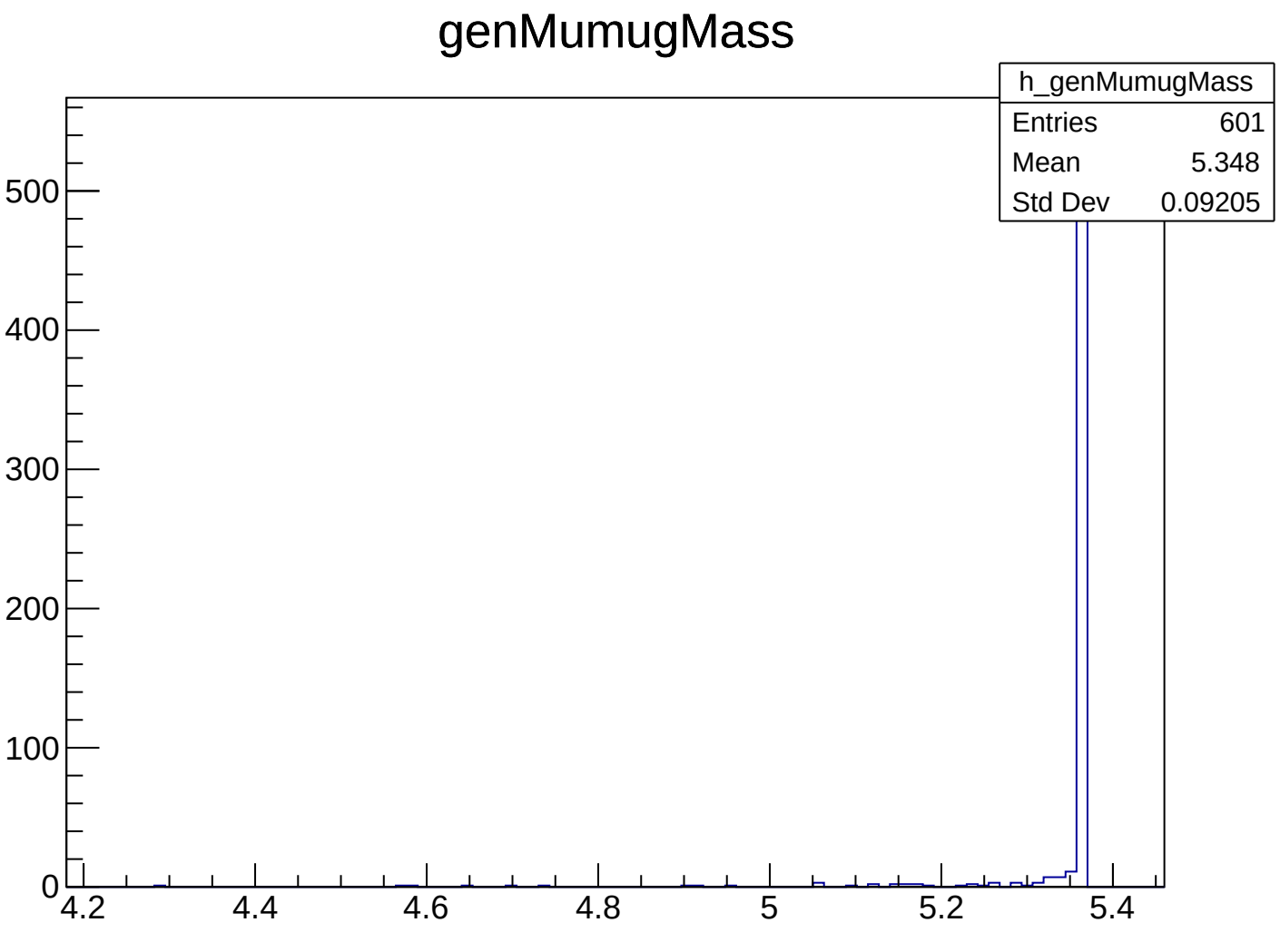

genPhotonPt
h_genPhotonPt
Entries 601
Mean 14.56
Std Dev 6.906
0
10
20
30
40
50
60
70
80
90

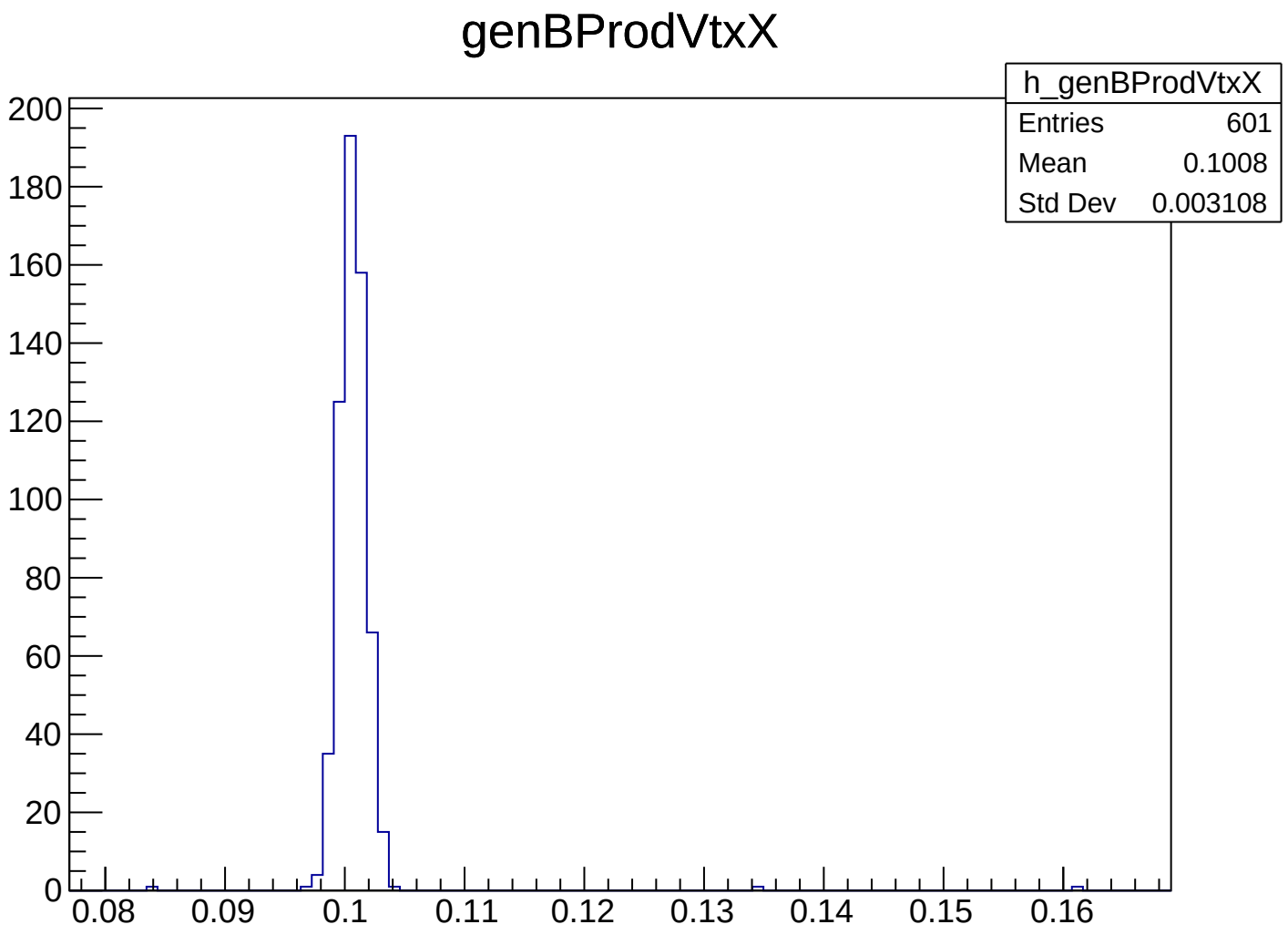

genPhotonEta
h_genPhotonEta
Entries 601
Mean −0.00382
Std Dev 0.8964
16
14
12
10
8
6
4
2
0
−1.5
−1
−0.5
0
0.5
1
1.5

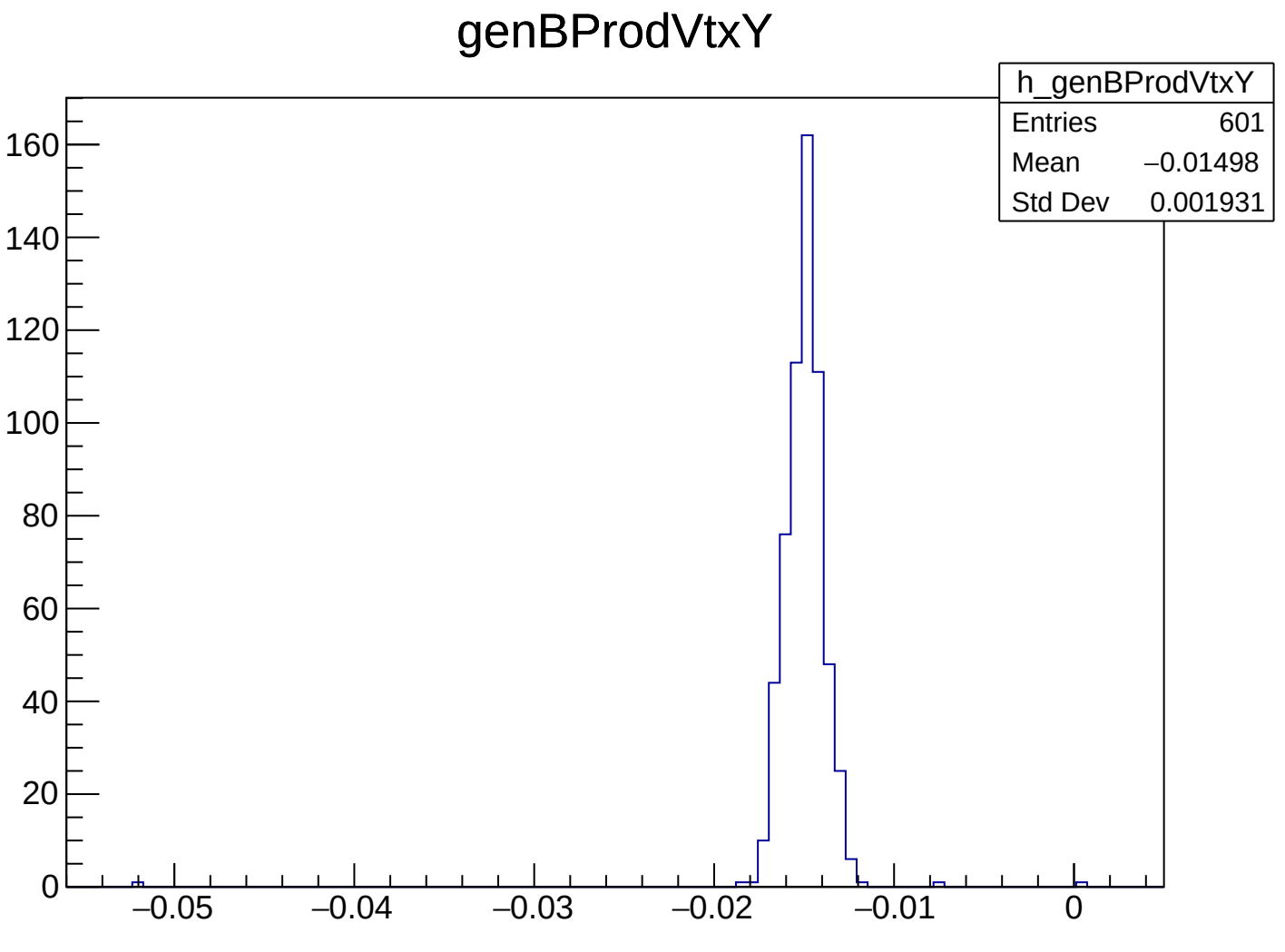

genPhotonPhi
h_genPhotonPhi
Entries 601
Mean −0.06352
Std Dev 1.8
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

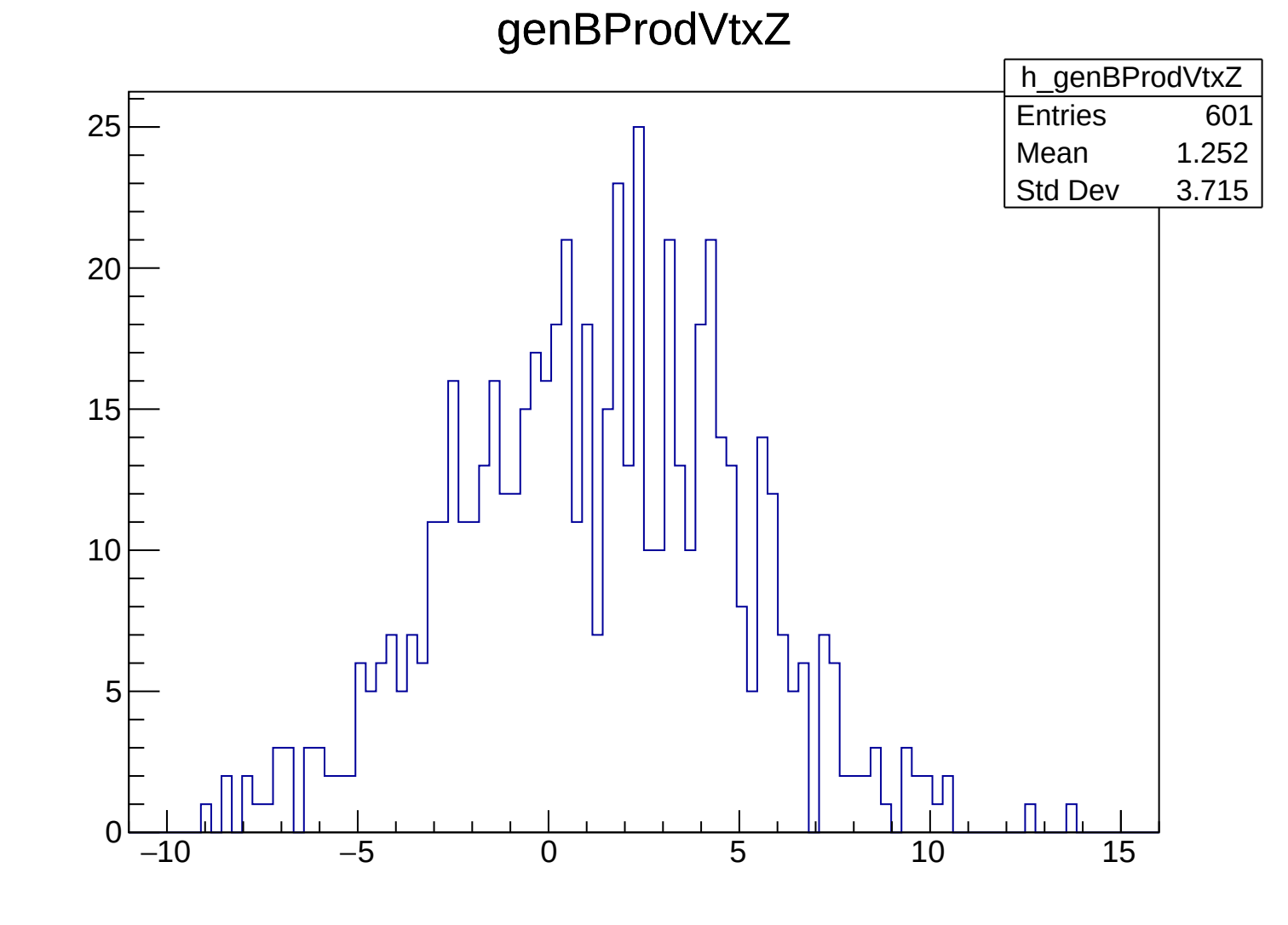

genNFSRPhotons
h_genNFSRPhotons
Entries 601
Mean 0.396
Std Dev 0.6395
400
350
300
250
200
150
100
50
0
0
0.5
1
1.5
2
2.5
3
3.5
4
4.5
5

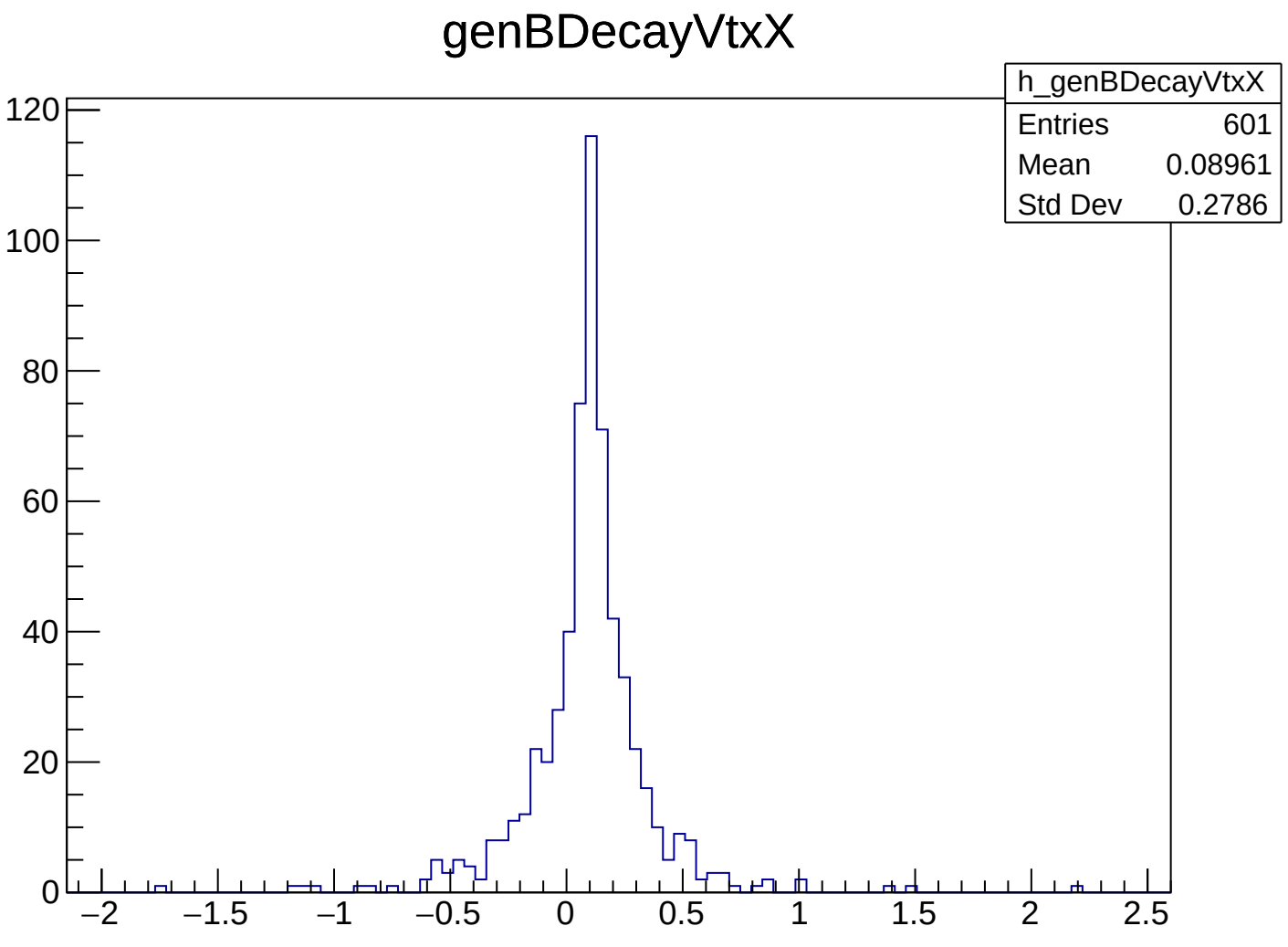

genDimuonMass
h_genDimuonMass
Entries 601
Mean 3.078
Std Dev 0.0904
500
400
300
200
100
0
2.2
2.4
2.6
2.8
3

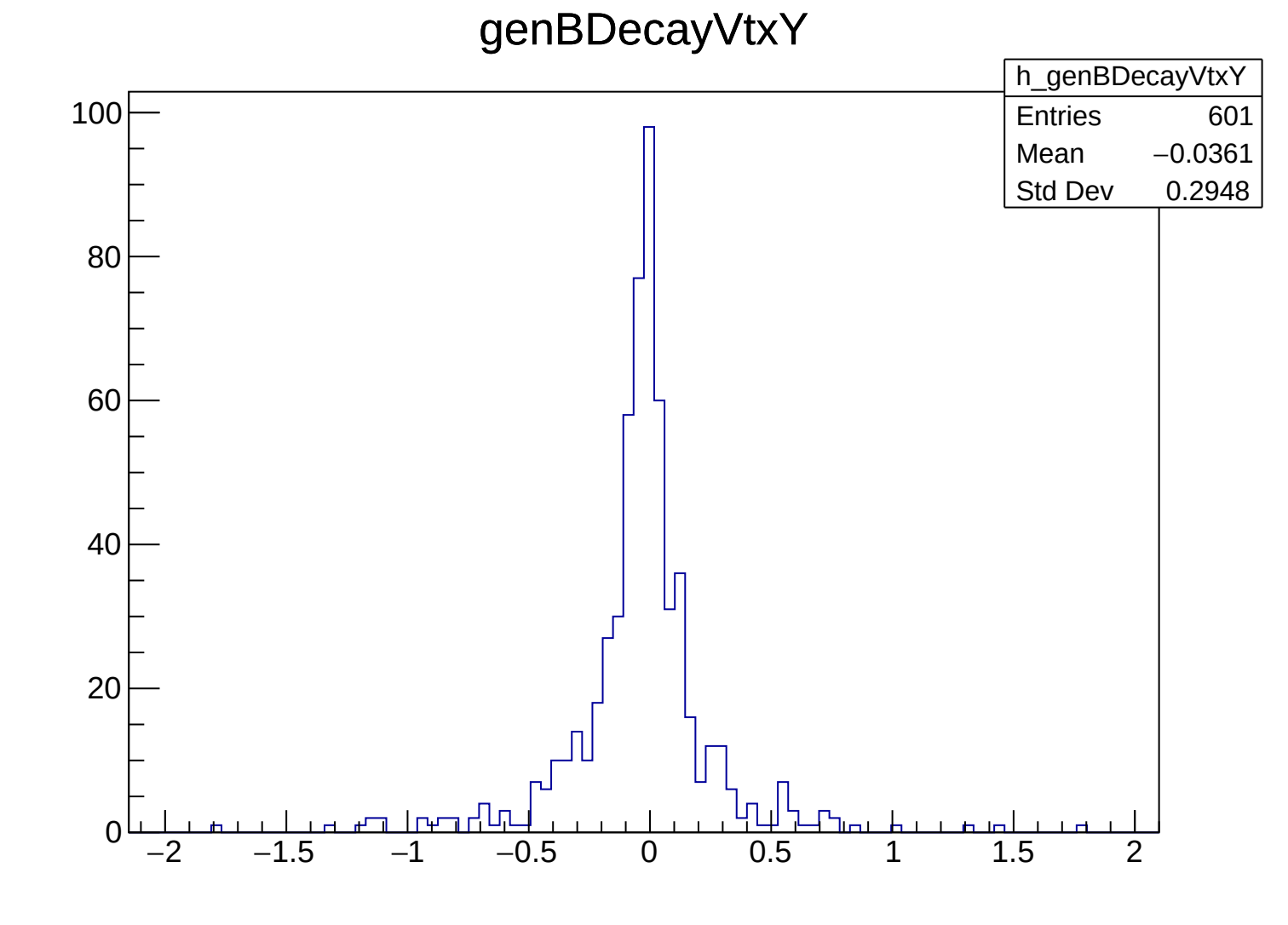

genMumugMass
h_genMumugMass
Entries 601
Mean 5.348
Std Dev 0.09205
500
400
300
200
100
0
4.2
4.4
4.6
4.8
5
5.2
5.4

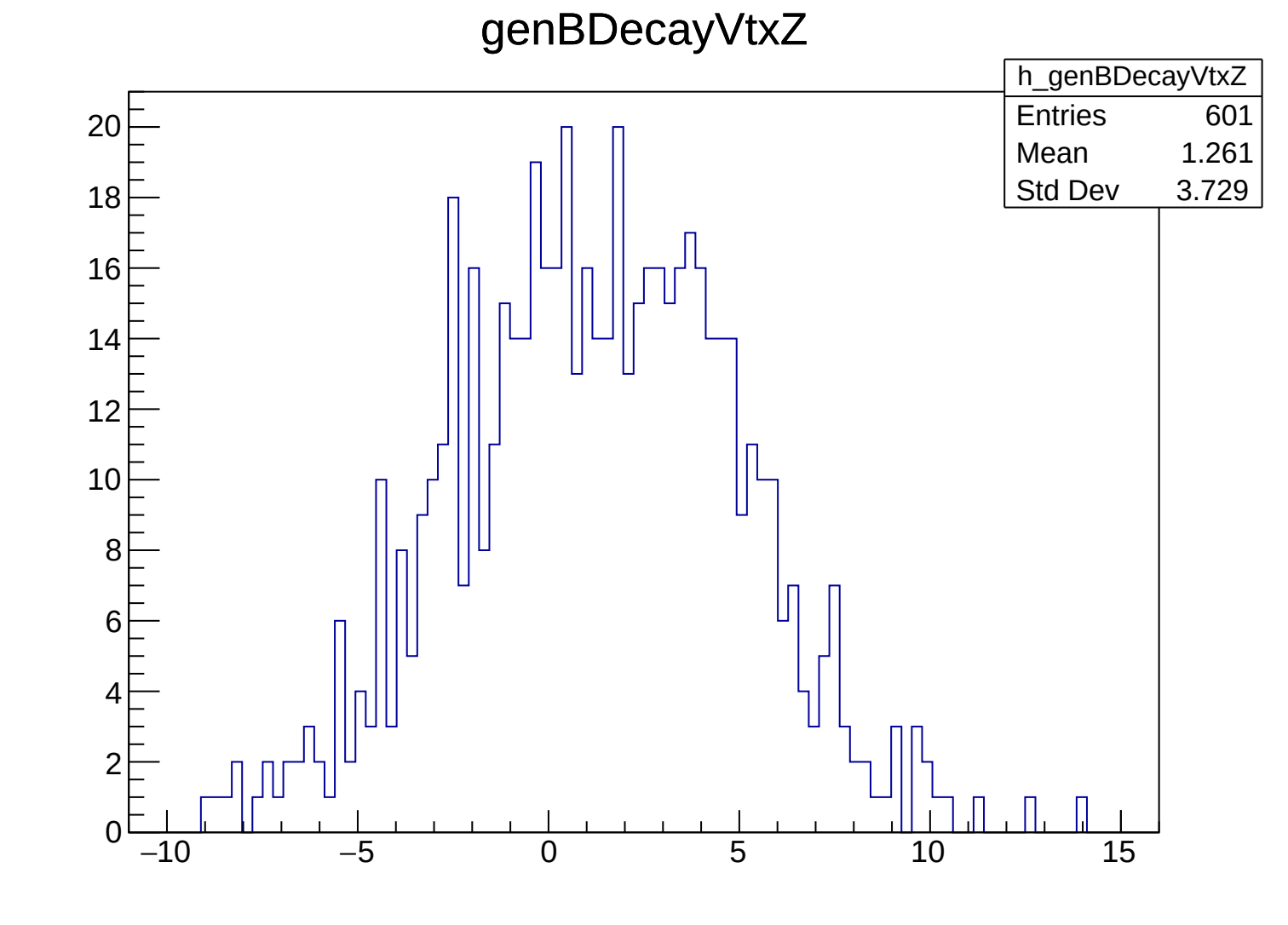

genBProdVtxX
h_genBProdVtxX
Entries 601
Mean 0.1008
Std Dev 0.003108
200
180
160
140
120
100
80
60
40
20
0
0.08
0.09
0.1
0.11
0.12
0.13
0.14
0.15
0.16

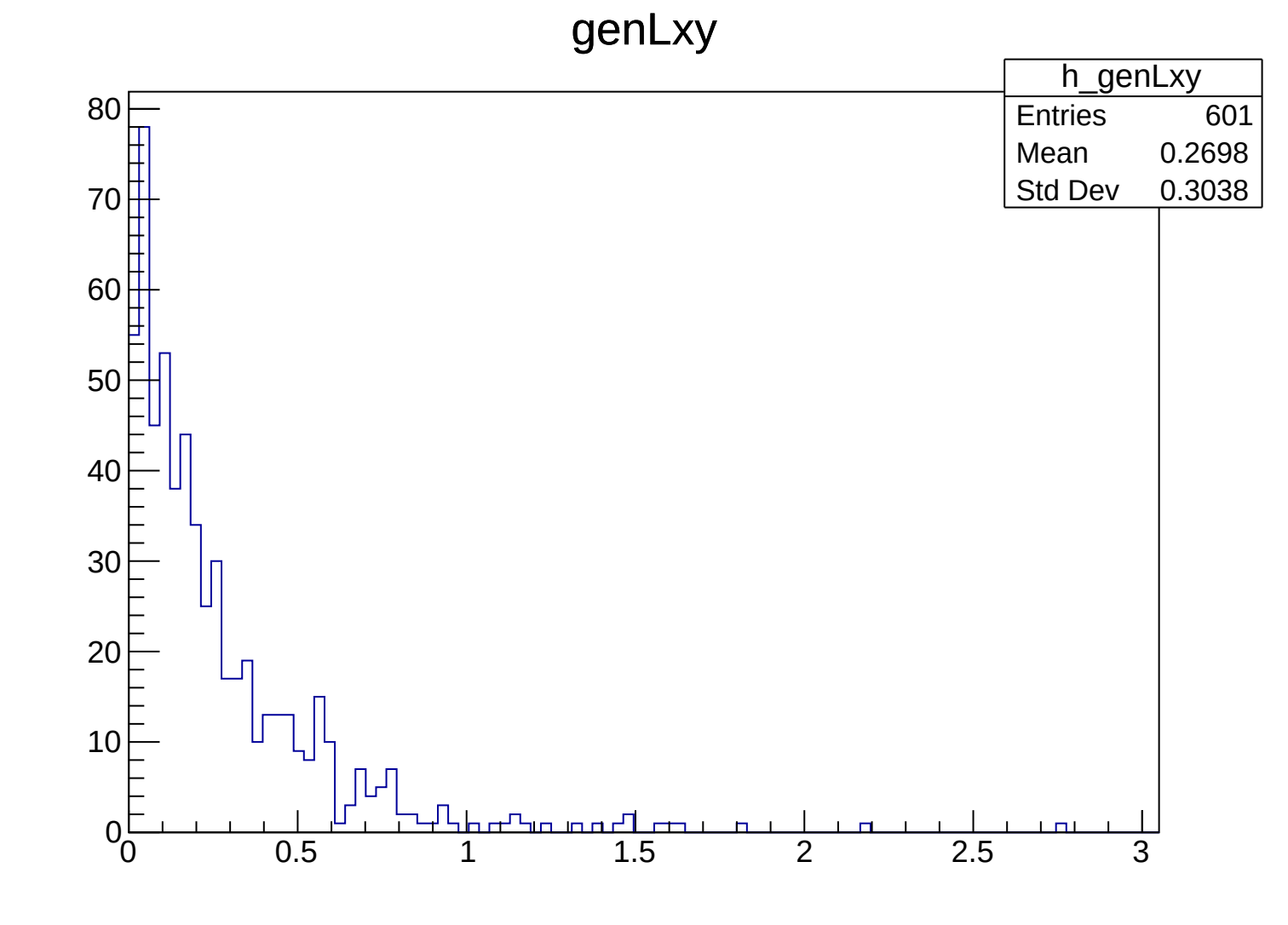

genBProdVtxY
h_genBProdVtxY
Entries 601
Mean −0.01498
Std Dev 0.001931
160
140
120
100
80
60
40
20
0
−0.05
−0.04
−0.03
−0.02
−0.01
0

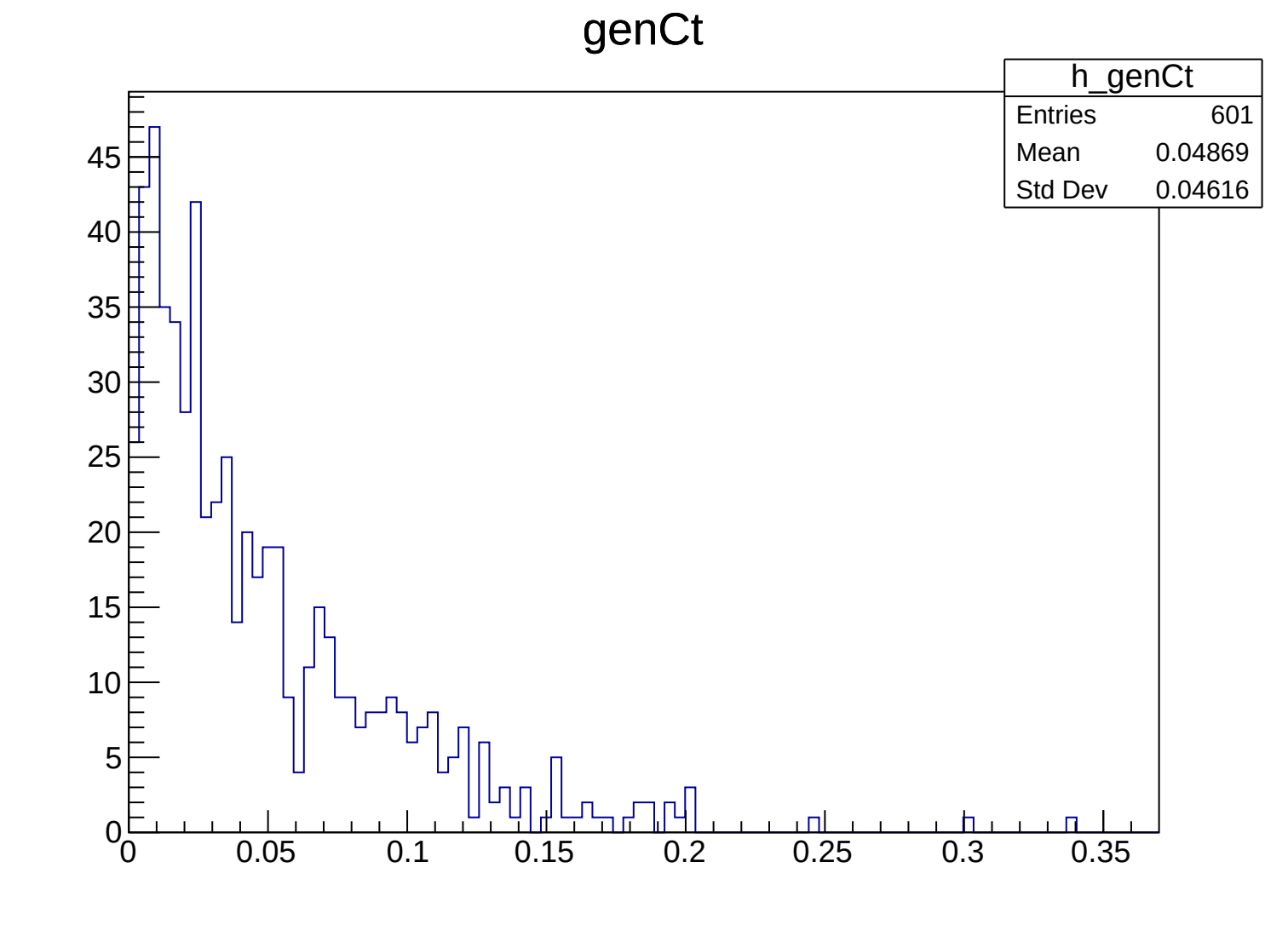

genBProdVtxZ
h_genBProdVtxZ
Entries 601
Mean 1.252
Std Dev 3.715
25
20
15
10
5
0
−10
−5
0
5
10
15

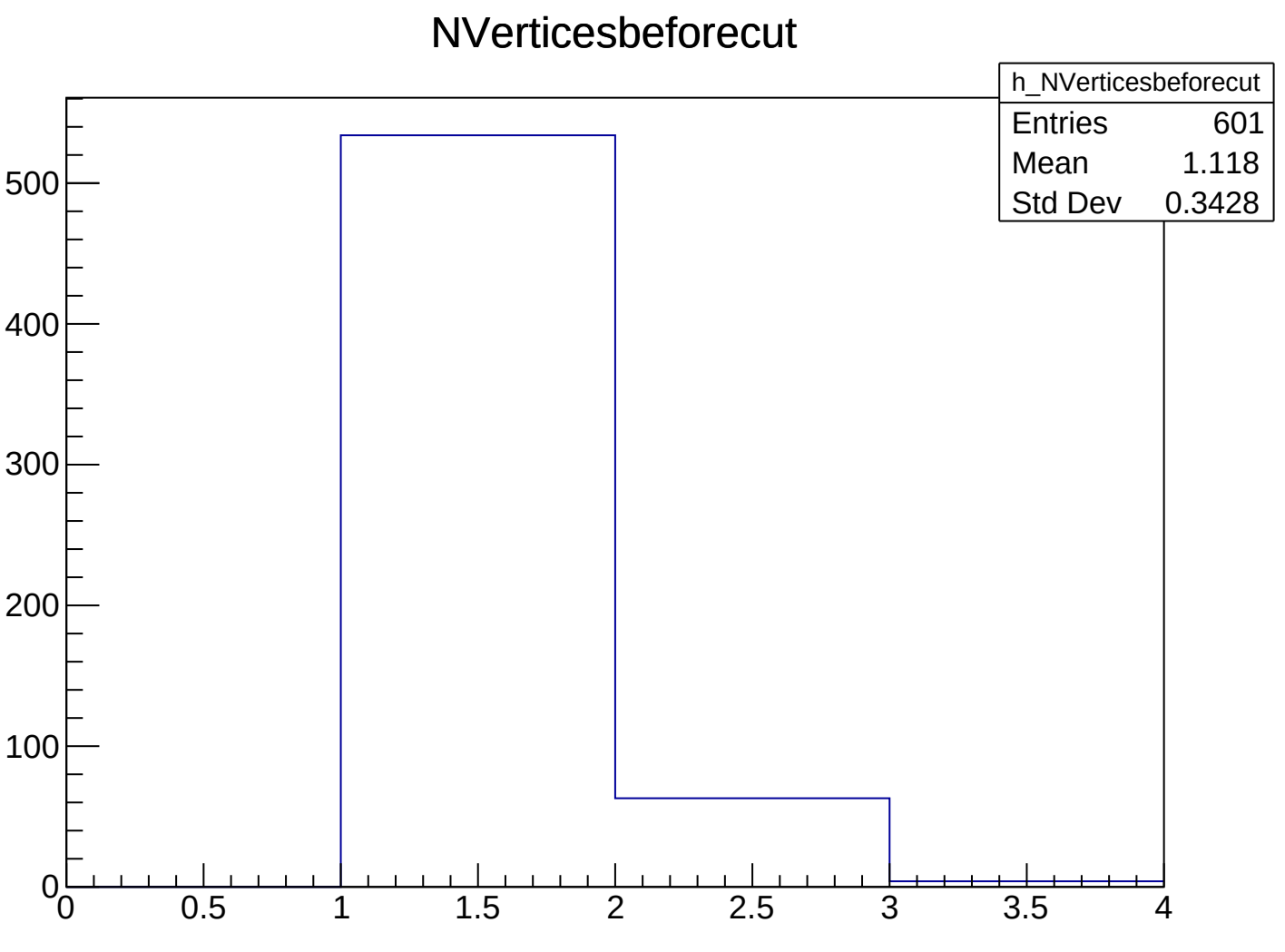

genBDecayVtxX
h_genBDecayVtxX
Entries 601
Mean 0.08961
Std Dev 0.2786
120
100
80
60
40
20
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
2.5

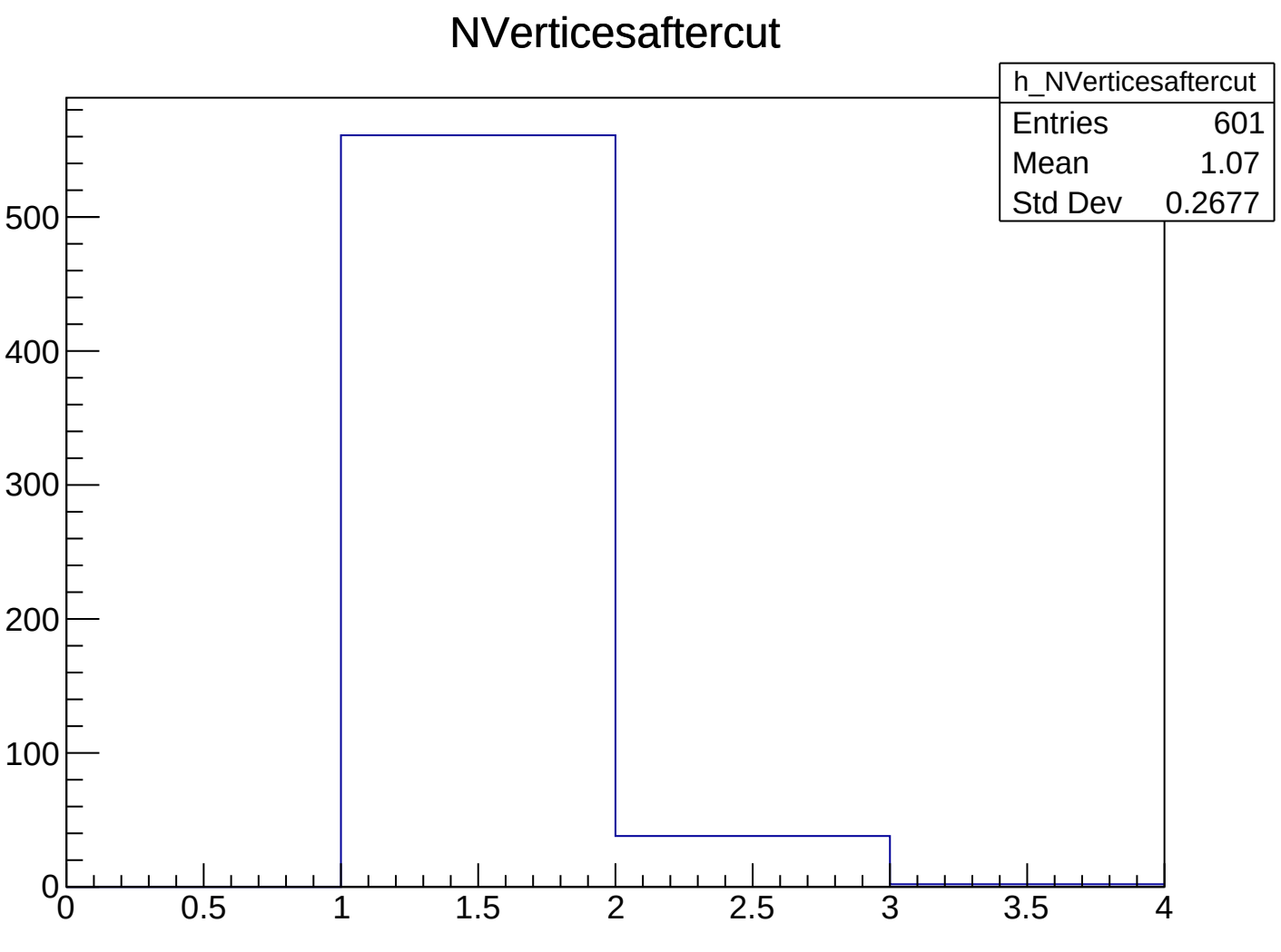

genBDecayVtxY
h_genBDecayVtxY
Entries 601
Mean −0.0361
Std Dev 0.2948
100
80
60
40
20
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

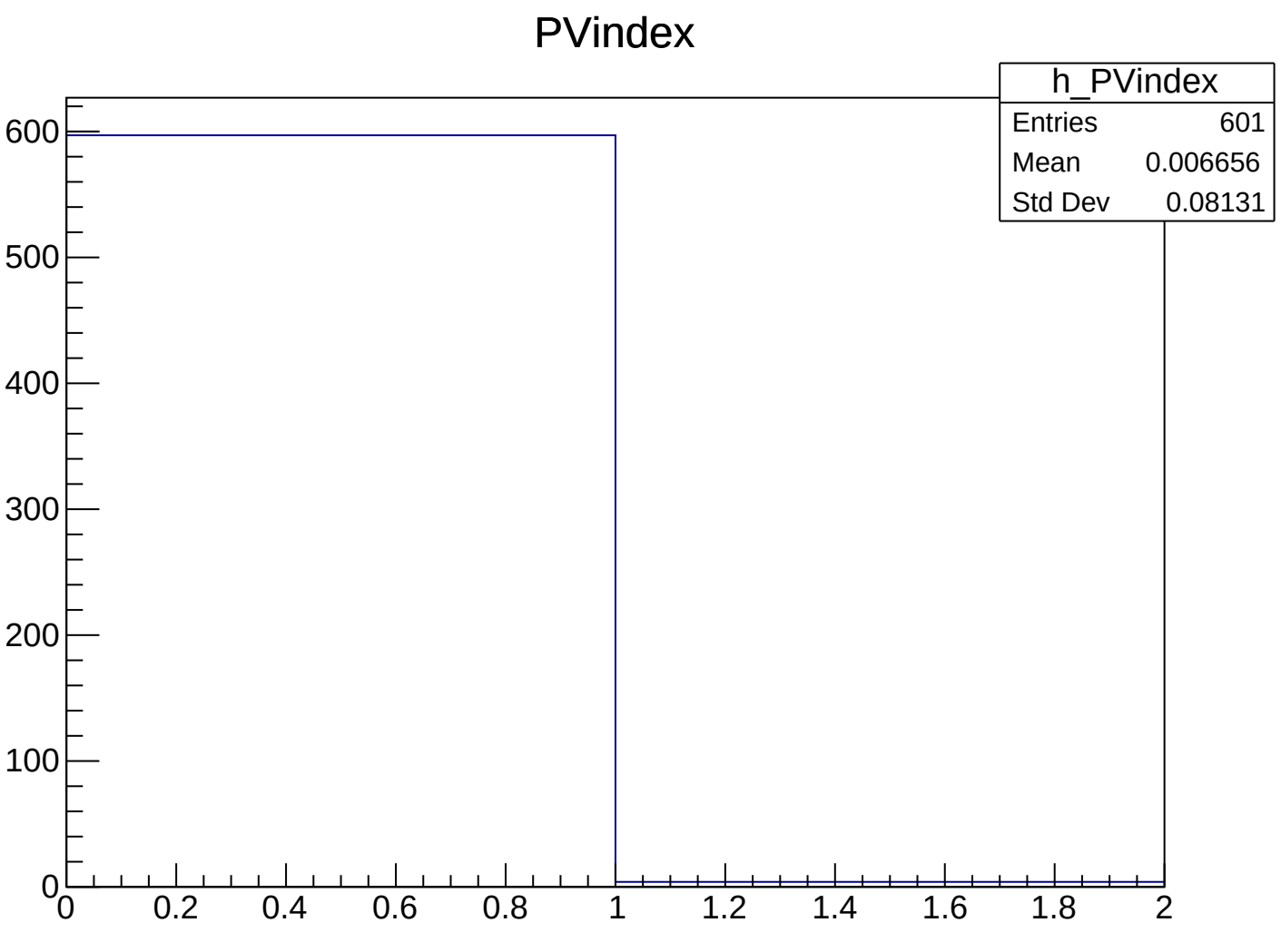

genBDecayVtxZ
h_genBDecayVtxZ
Entries 601
Mean 1.261
Std Dev 3.729
20
18
16
14
12
10
8
6
4
2
0
−10
−5
0
5
10
15

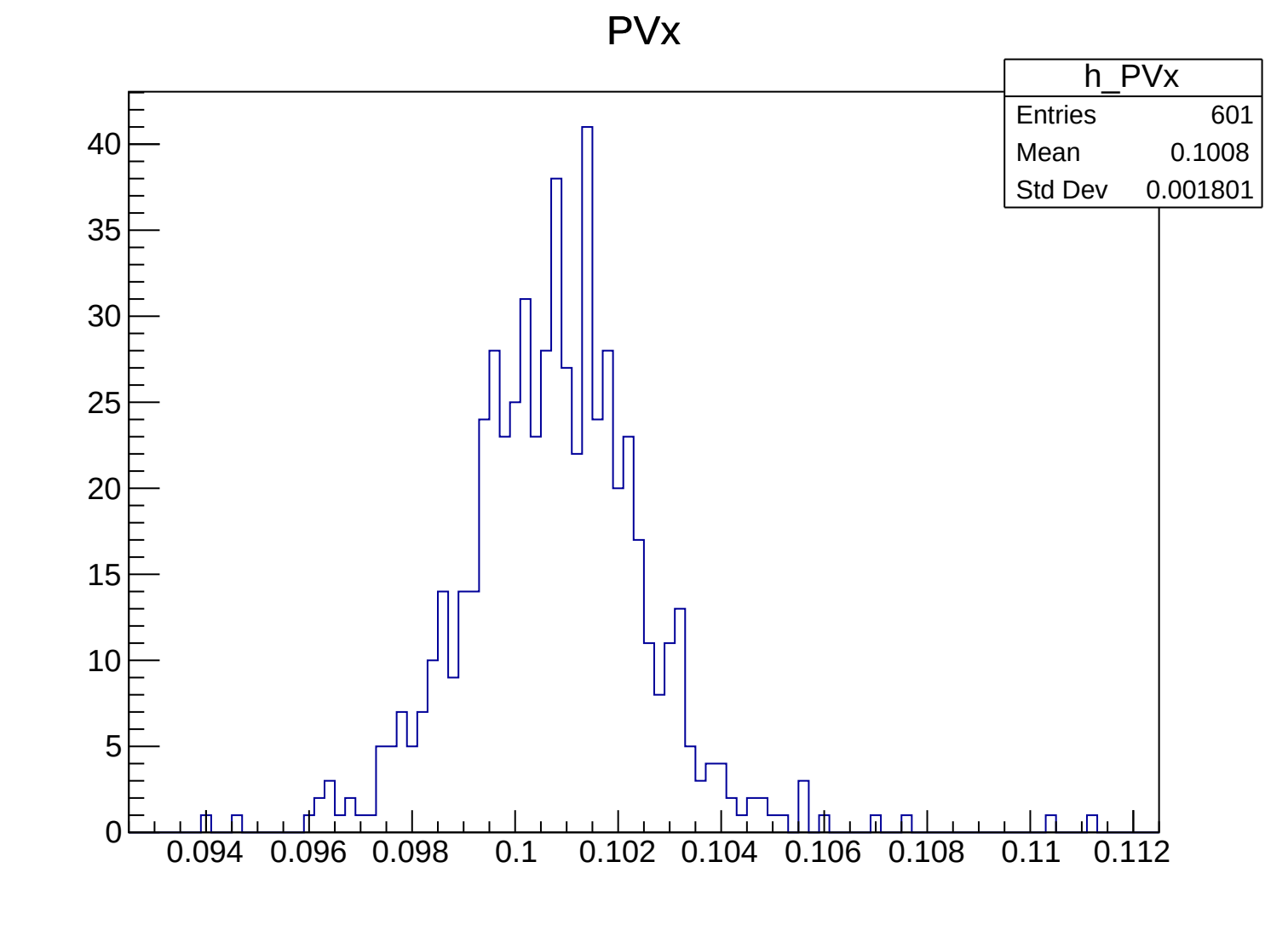

genLxy
h_genLxy
Entries 601
Mean 0.2698
Std Dev 0.3038
80
70
60
50
40
30
20
10
0
0
0.5
1
1.5
2
2.5
3

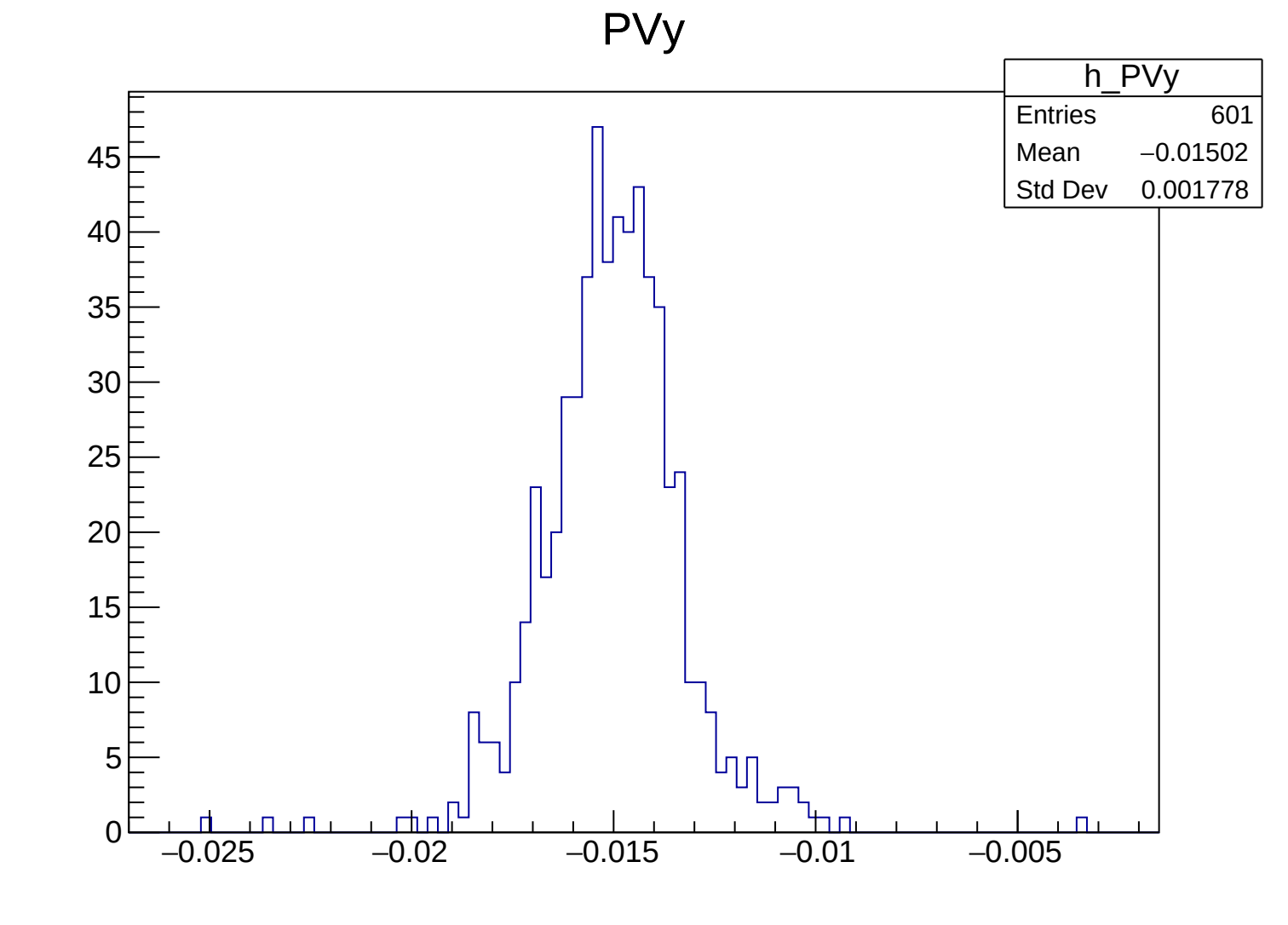

genCt
h_genCt
Entries 601
Mean 0.04869
Std Dev 0.04616
45
40
35
30
25
20
15
10
5
0
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35

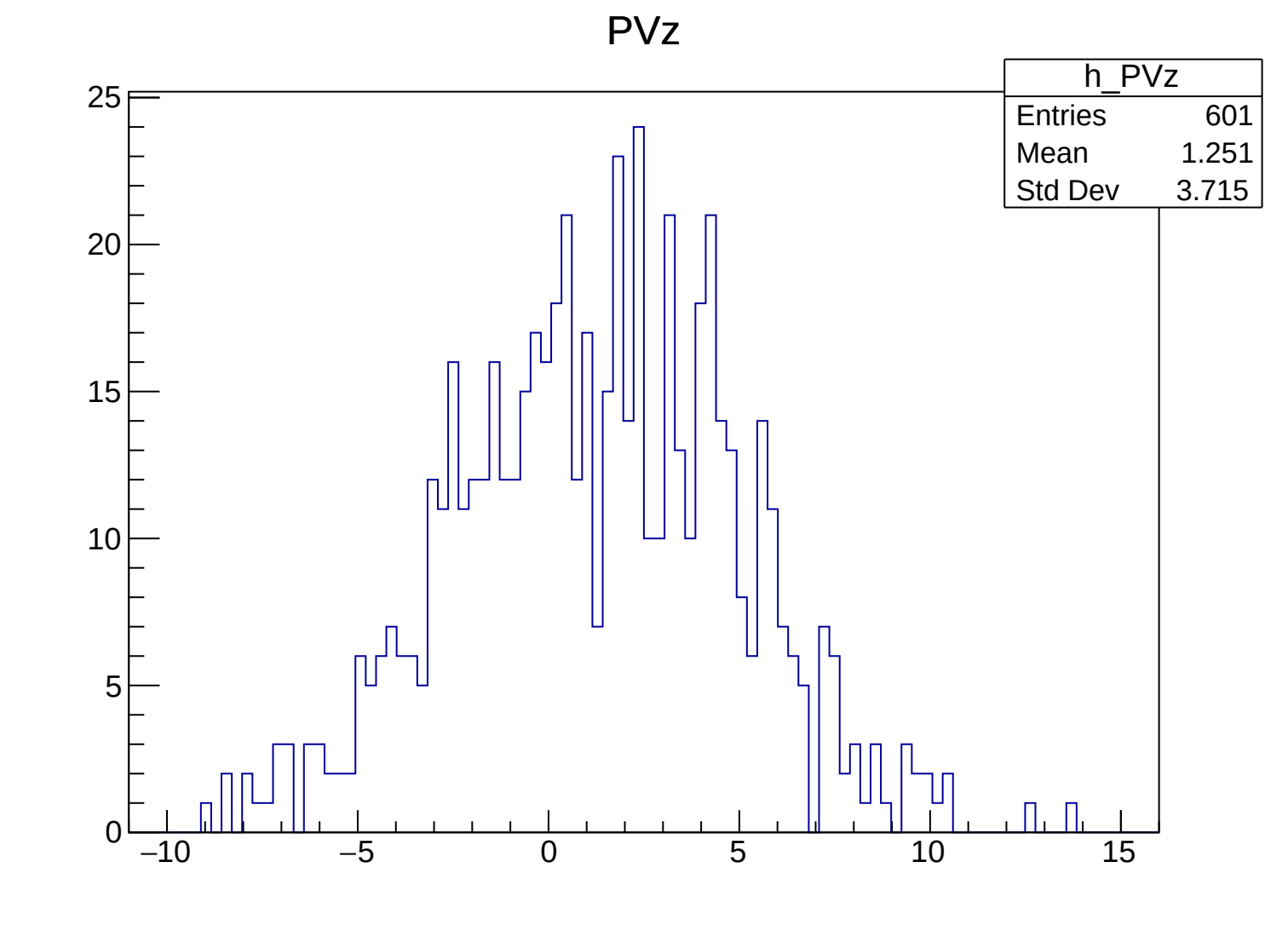

NVerticesbeforecut
h_NVerticesbeforecut
Entries 601
Mean 1.118
Std Dev 0.3428
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
3.5
4

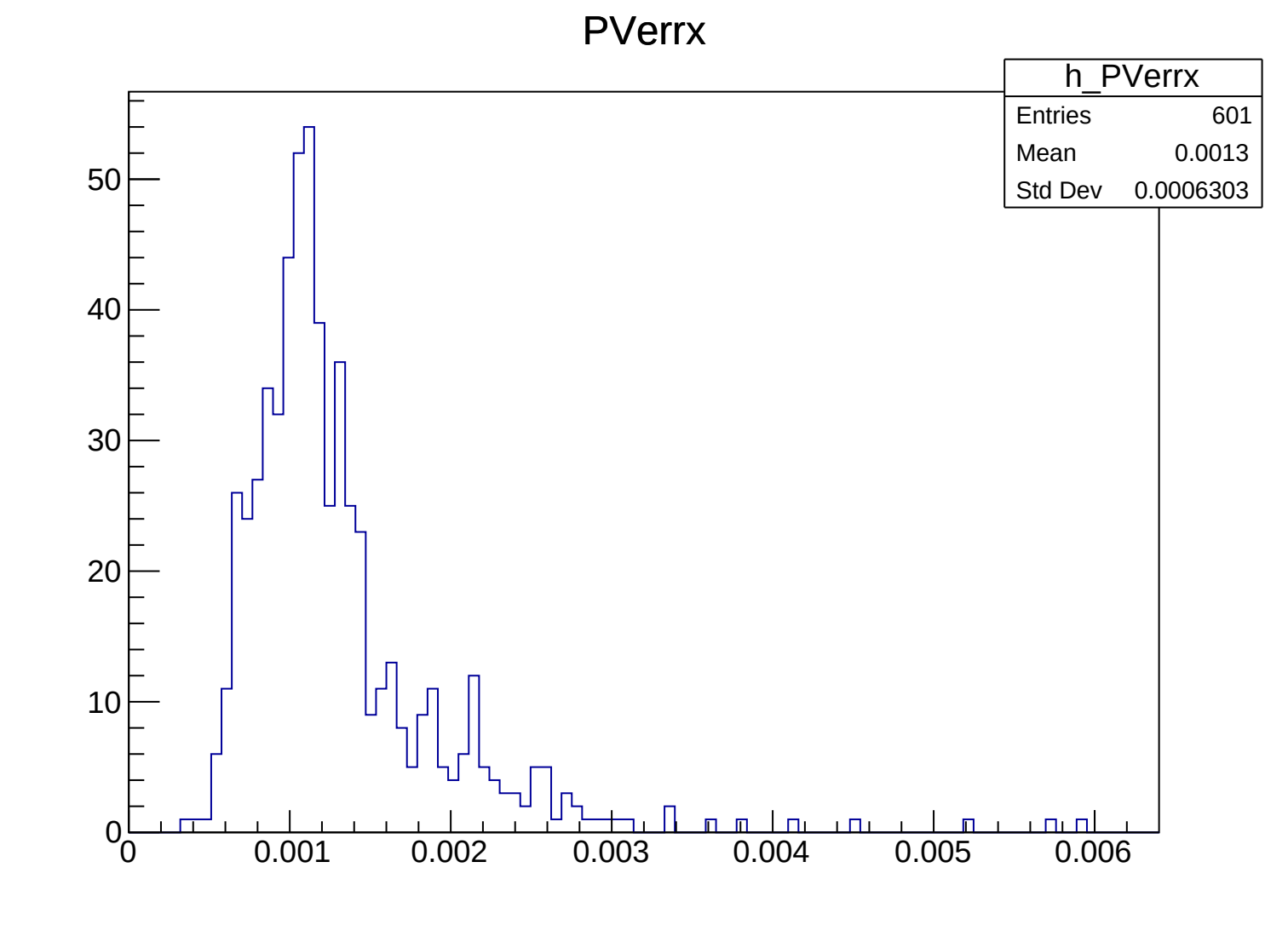

NVerticesaftercut
h_NVerticesaftercut
Entries 601
Mean 1.07
Std Dev 0.2677
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3
3.5
4

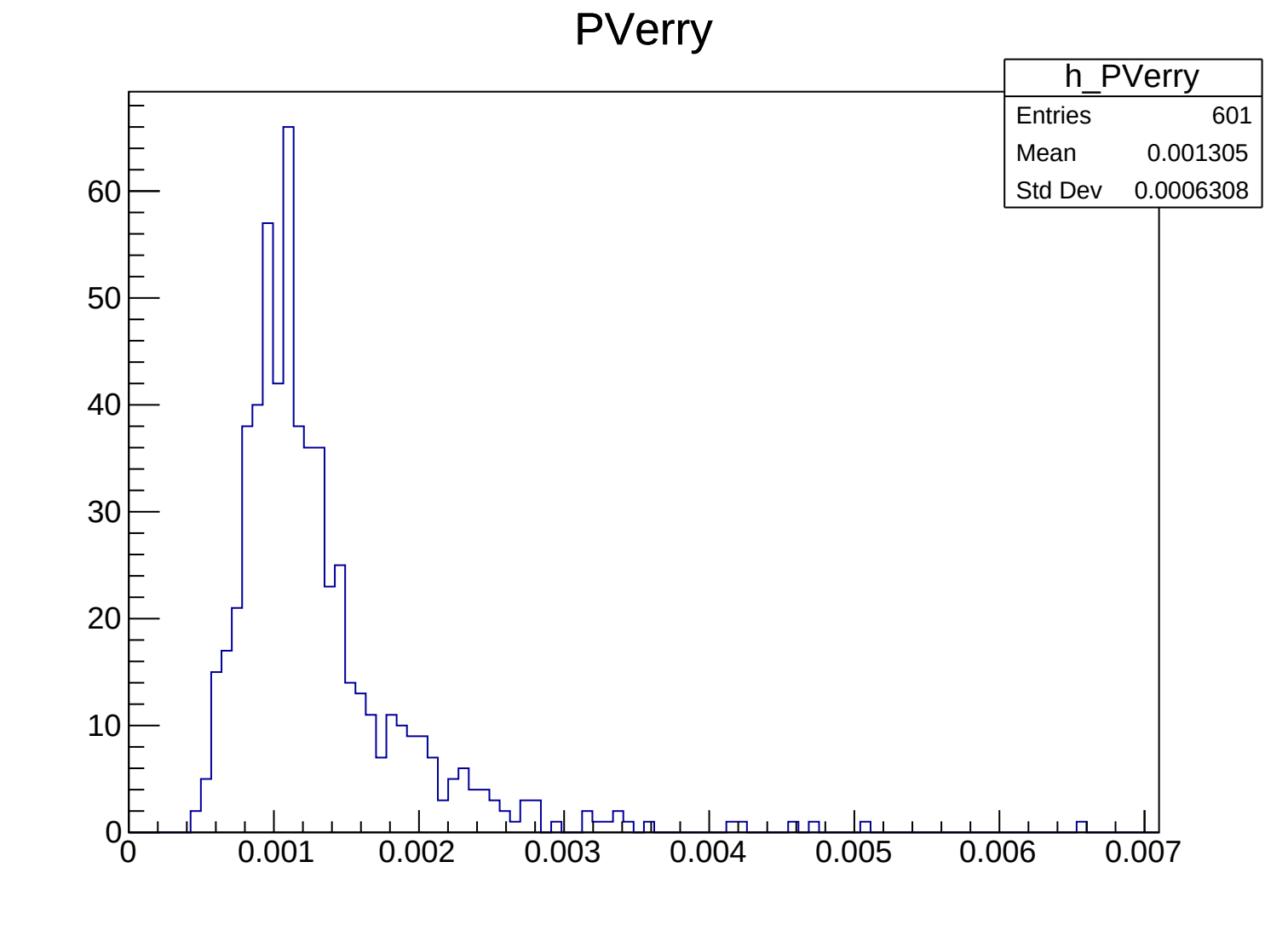

PVindex
h_PVindex
Entries 601
Mean 0.006656
Std Dev 0.08131
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

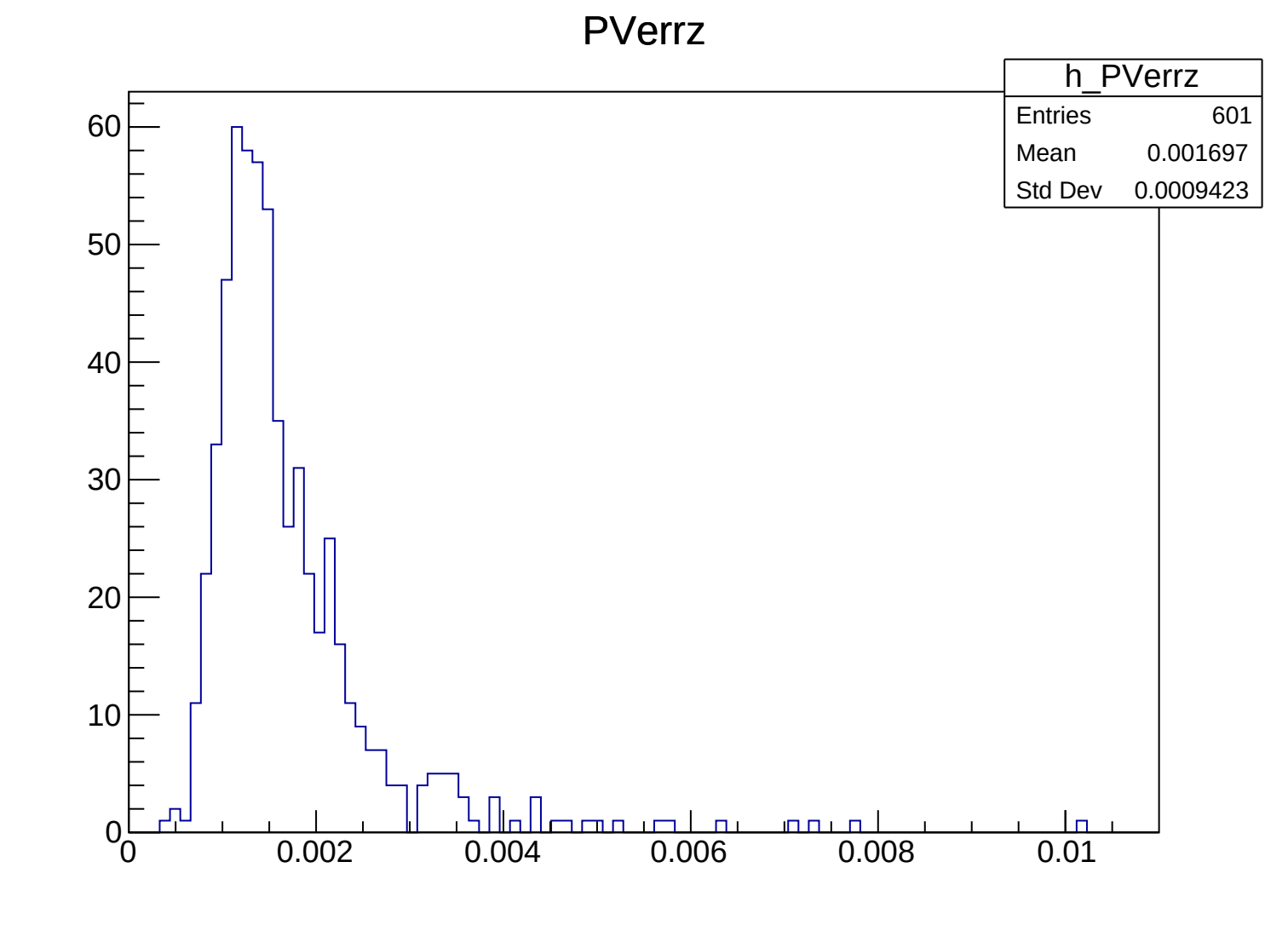

PVx
h_PVx
Entries 601
Mean 0.1008
Std Dev 0.001801
40
35
30
25
20
15
10
5
0
0.094
0.096
0.098
0.1
0.102
0.104
0.106
0.108
0.11
0.112

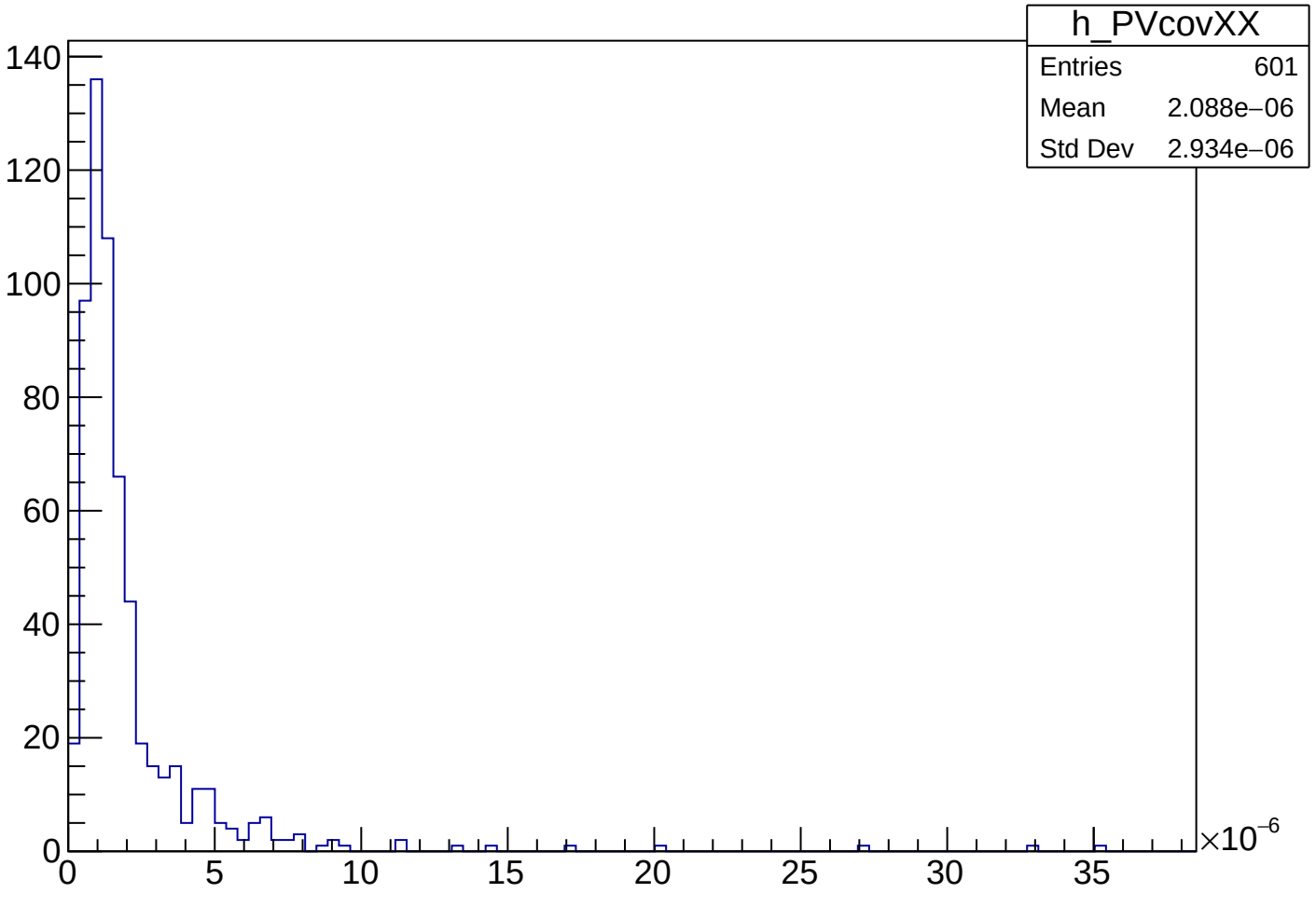

PVy
h_PVy
Entries 601
Mean −0.01502
Std Dev 0.001778
45
40
35
30
25
20
15
10
5
0
−0.025
−0.02
−0.015
−0.01
−0.005

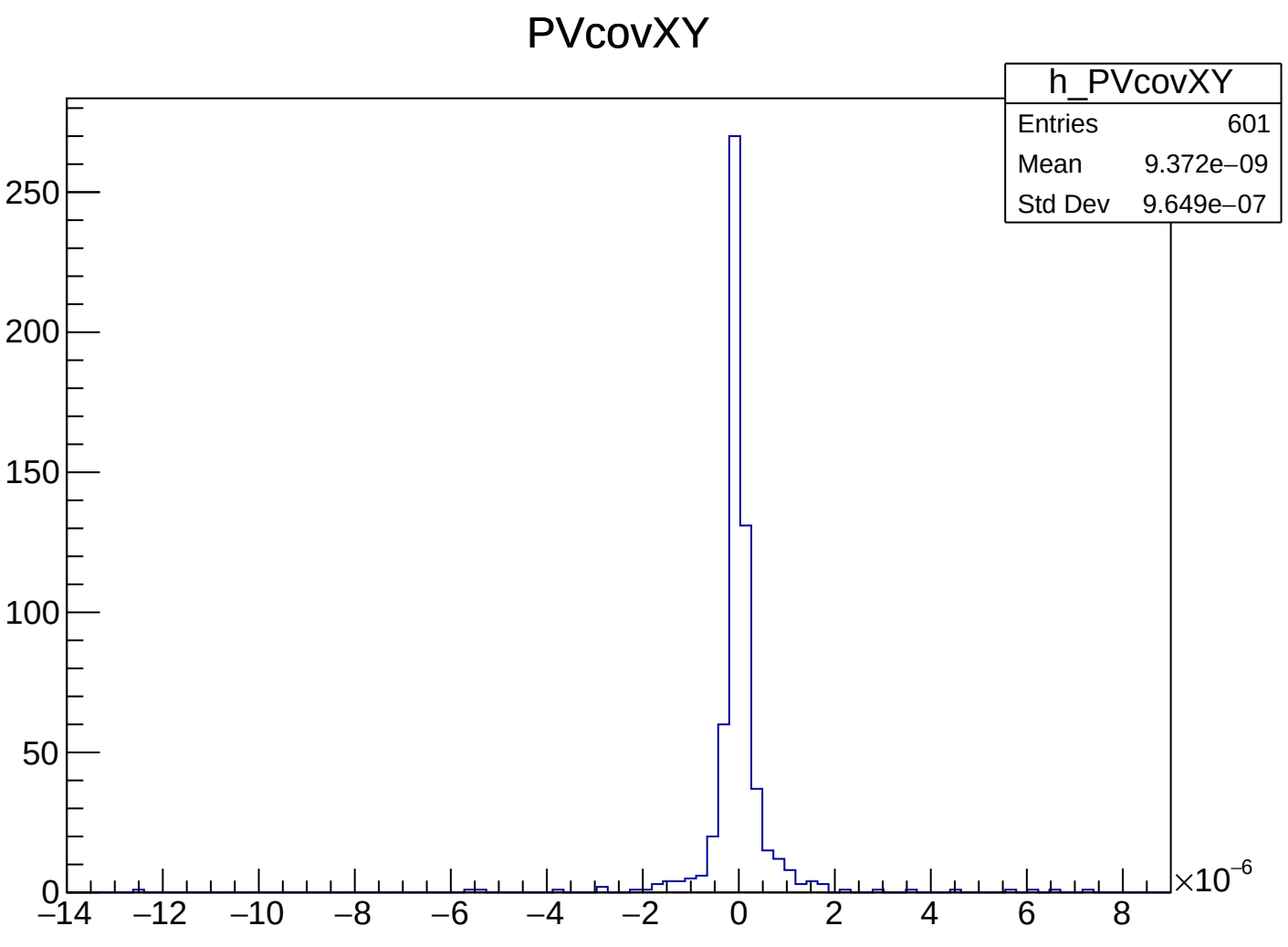

PVz
h_PVz
Entries 601
Mean 1.251
Std Dev 3.715
25
20
15
10
5
0
−10
−5
0
5
10
15

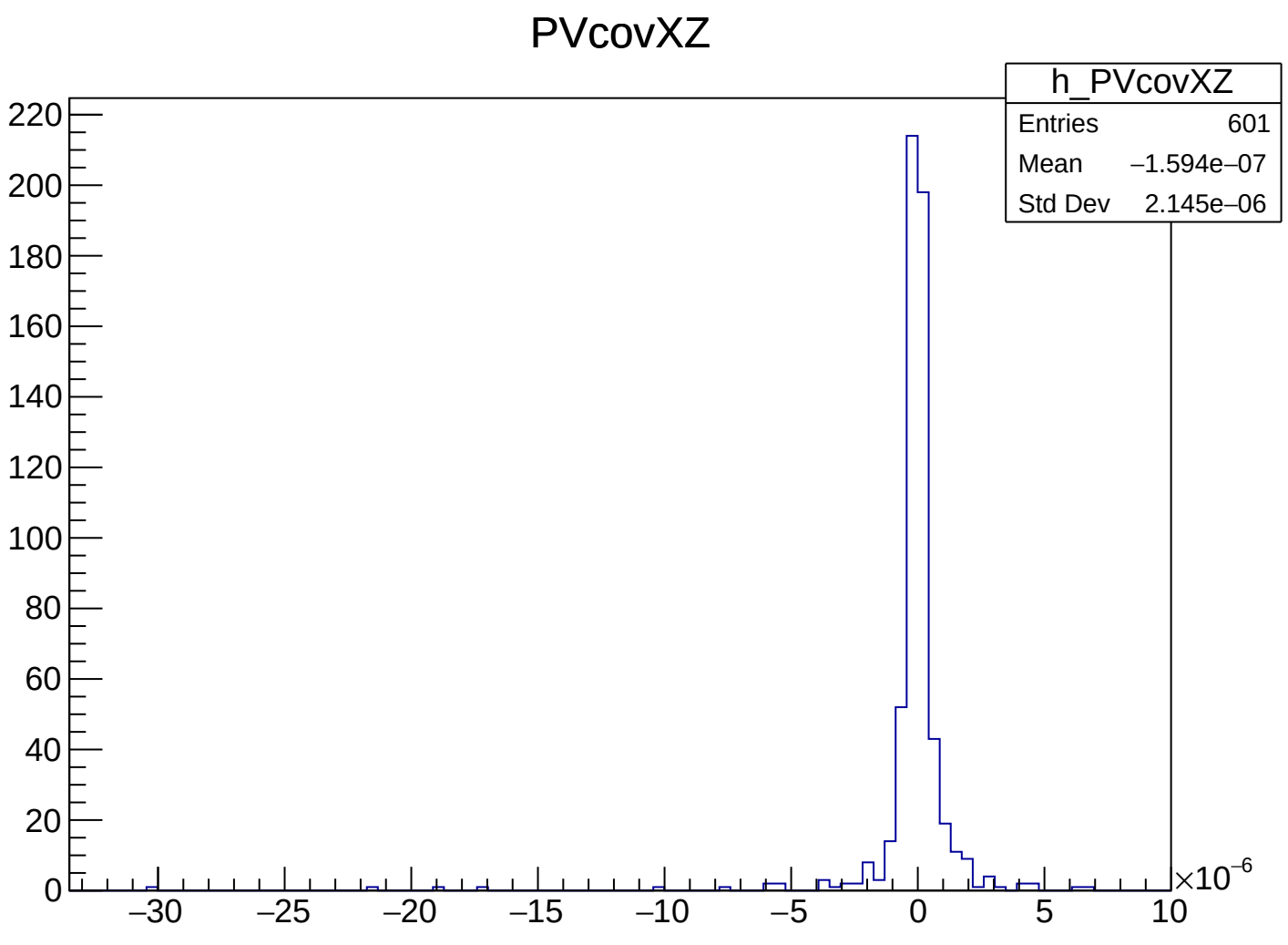

PVerrx
h_PVerrx
Entries 601
Mean 0.0013
Std Dev 0.0006303
50
40
30
20
10
0
0
0.001
0.002
0.003
0.004
0.005
0.006

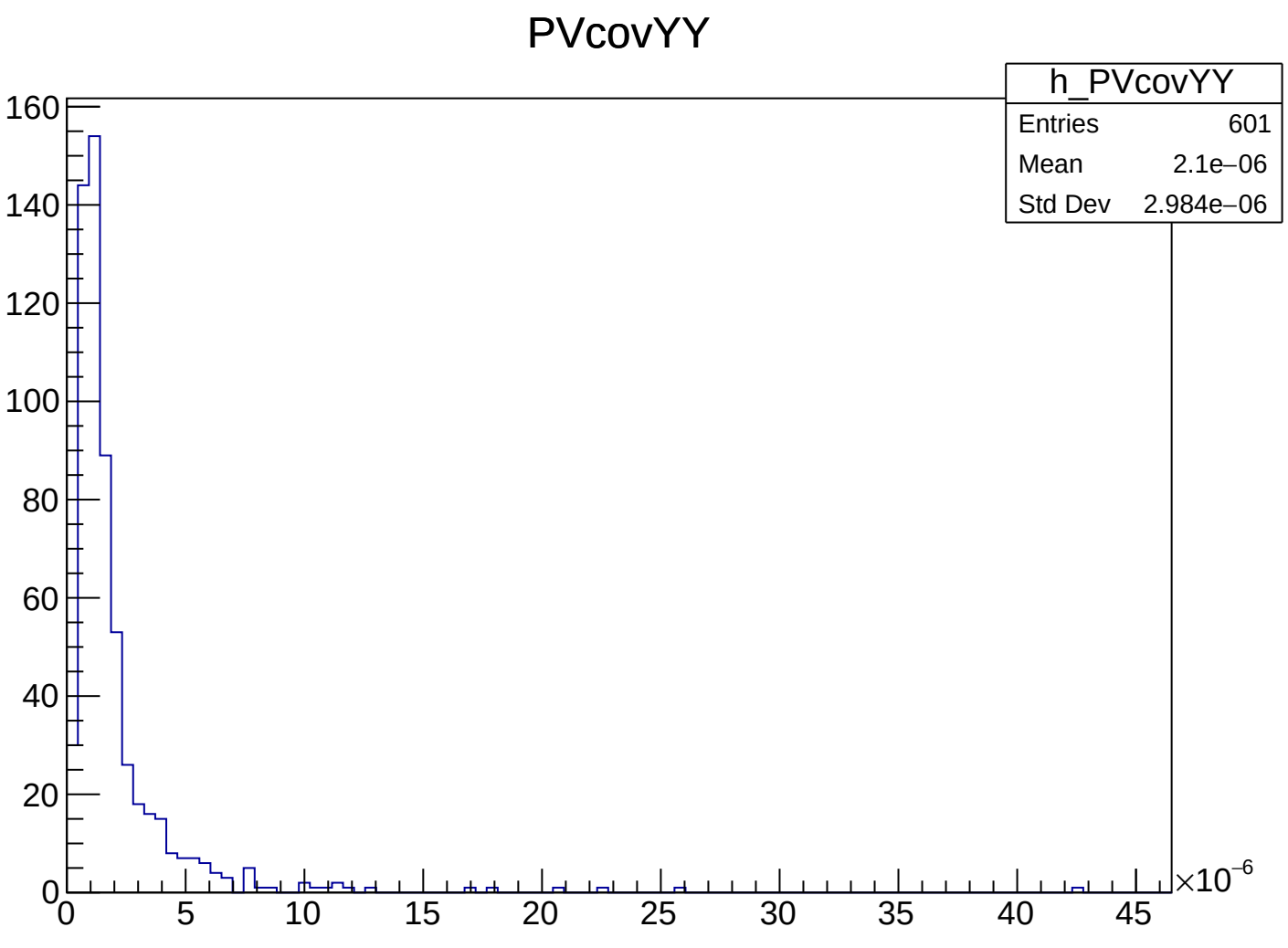

PVerry
h_PVerry
Entries 601
Mean 0.001305
Std Dev 0.0006308
60
50
40
30
20
10
0
0
0.001
0.002
0.003
0.004
0.005
0.006
0.007

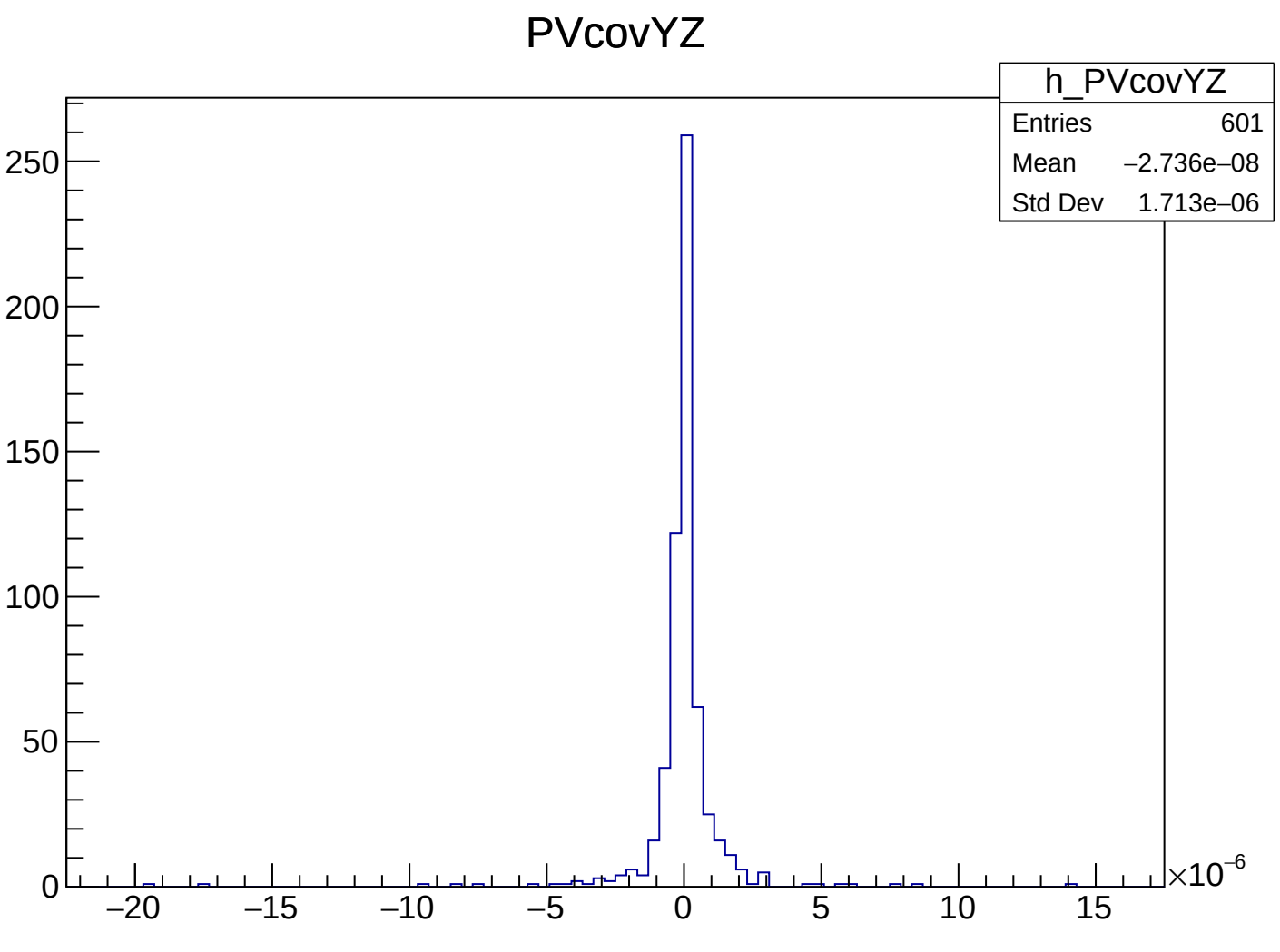

PVerrz
h_PVerrz
Entries 601
Mean 0.001697
Std Dev 0.0009423
60
50
40
30
20
10
0
0
0.002
0.004
0.006
0.008
0.01

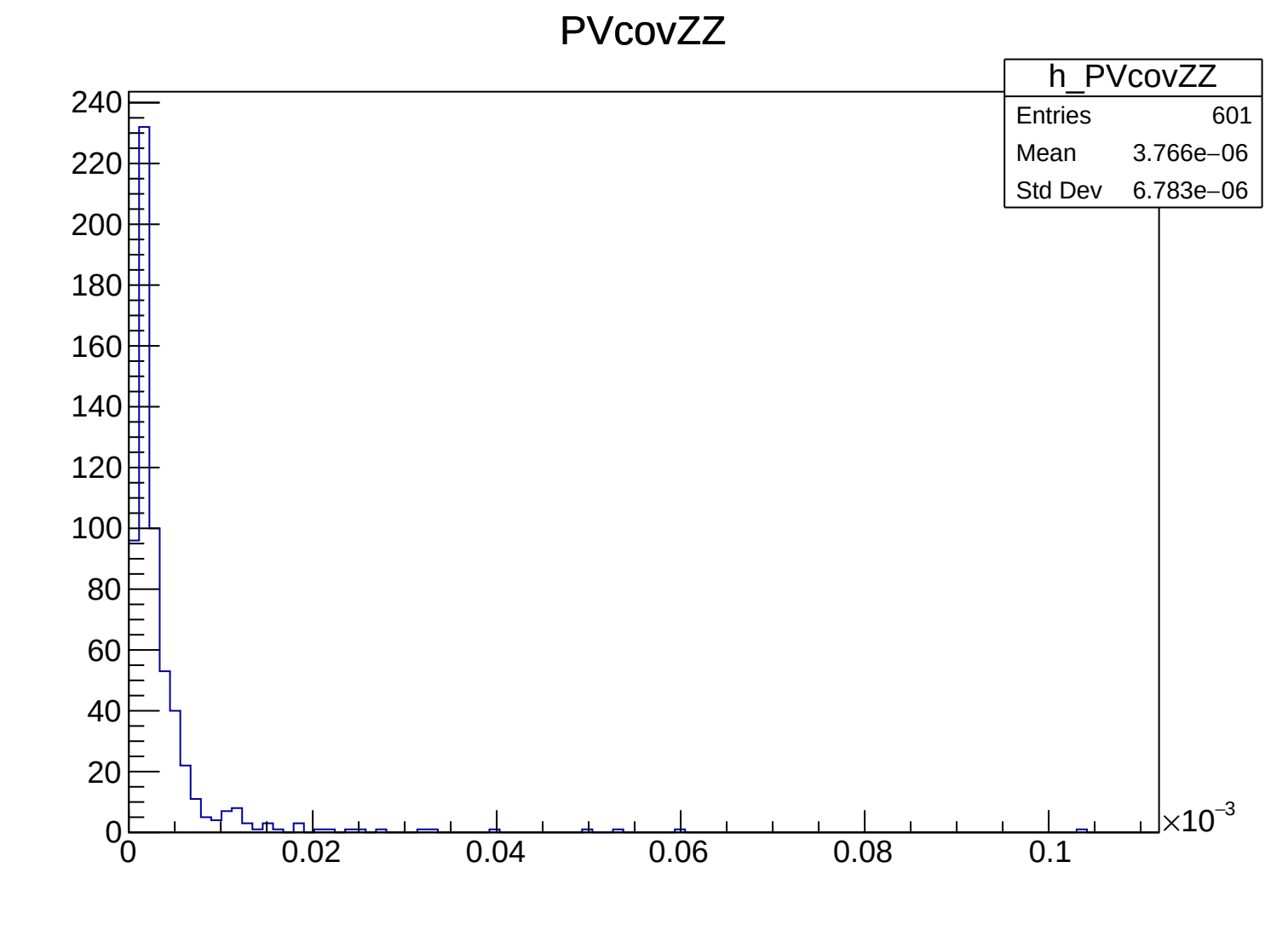

PVcovXX
h_PVcovXX
Entries 601
Mean 2.088e−06
Std Dev 2.934e−06
140
120
100
80
60
40
20
0
0
5
10
15
20
25
30
35
×10−6

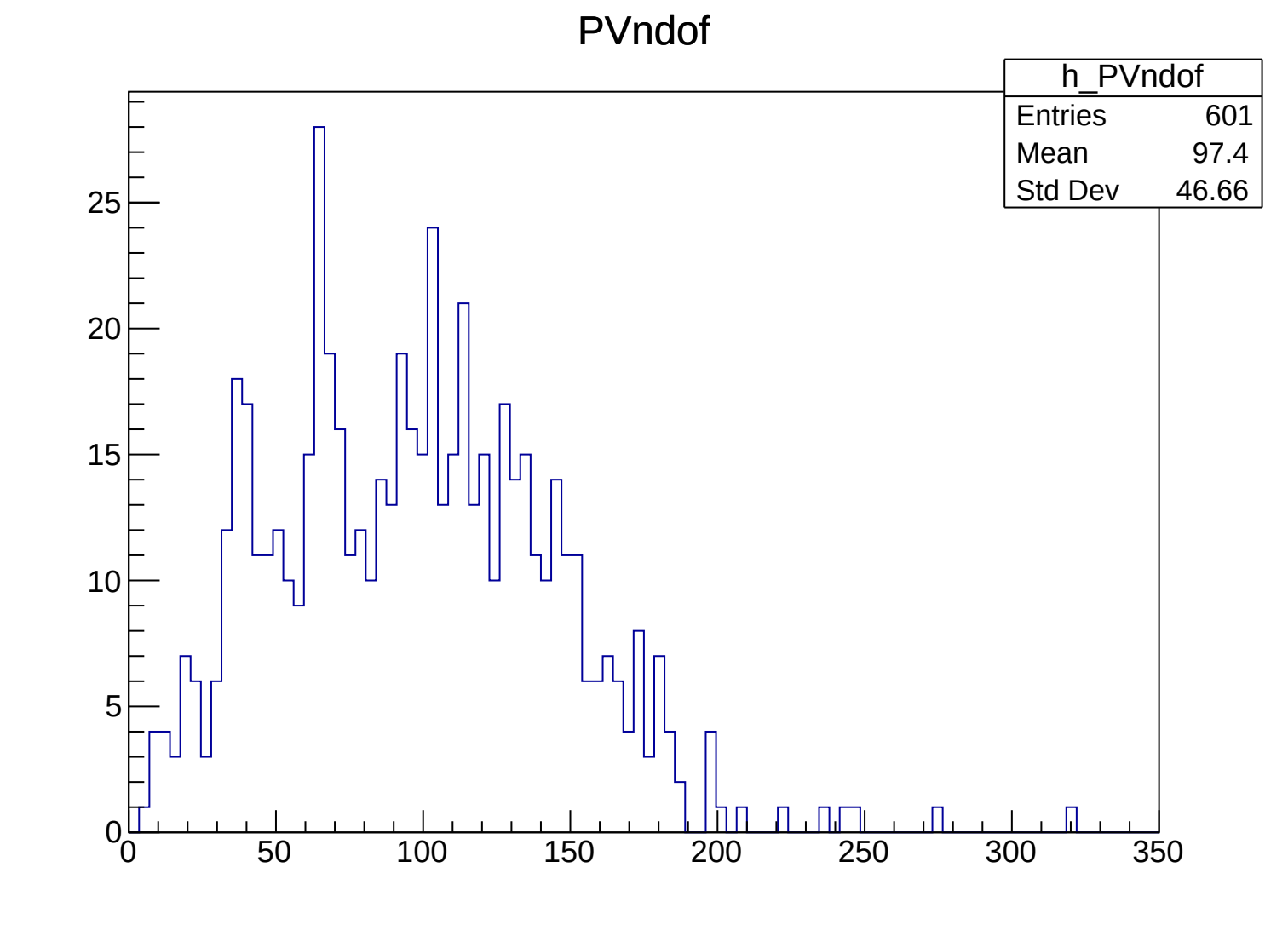

PVcovXY
h_PVcovXY
Entries 601
Mean 9.372e−09
Std Dev 9.649e−07
250
200
150
100
50
0
−14
−12
−10
−8
−6
−4
−2
0
2
4
6
8
×10^−6

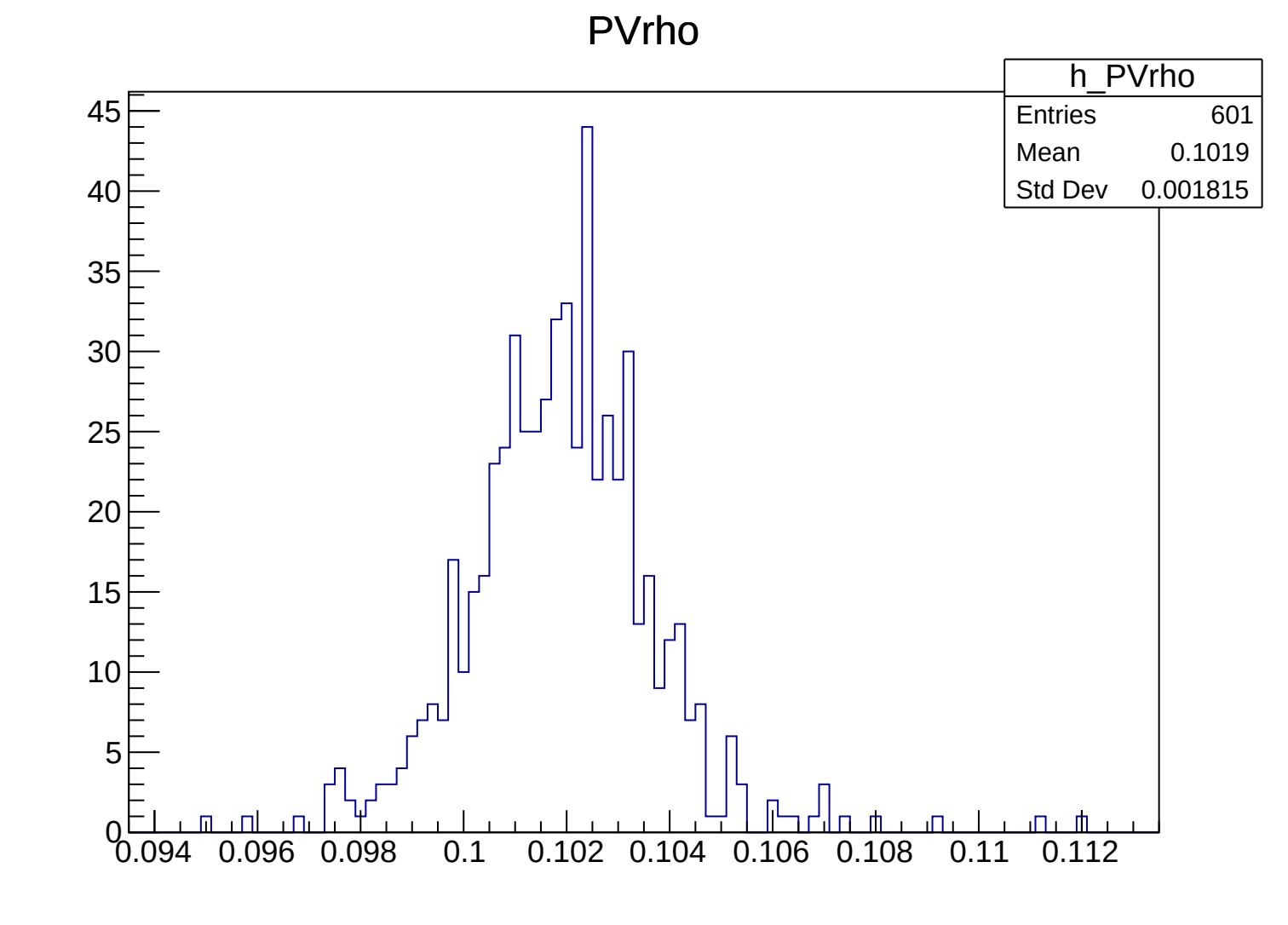

PVcovXZ
h_PVcovXZ
Entries 601
Mean −1.594e−07
Std Dev 2.145e−06
220
200
180
160
140
120
100
80
60
40
20
0
−30
−25
−20
−15
−10
−5
0
5
10
×10^−6

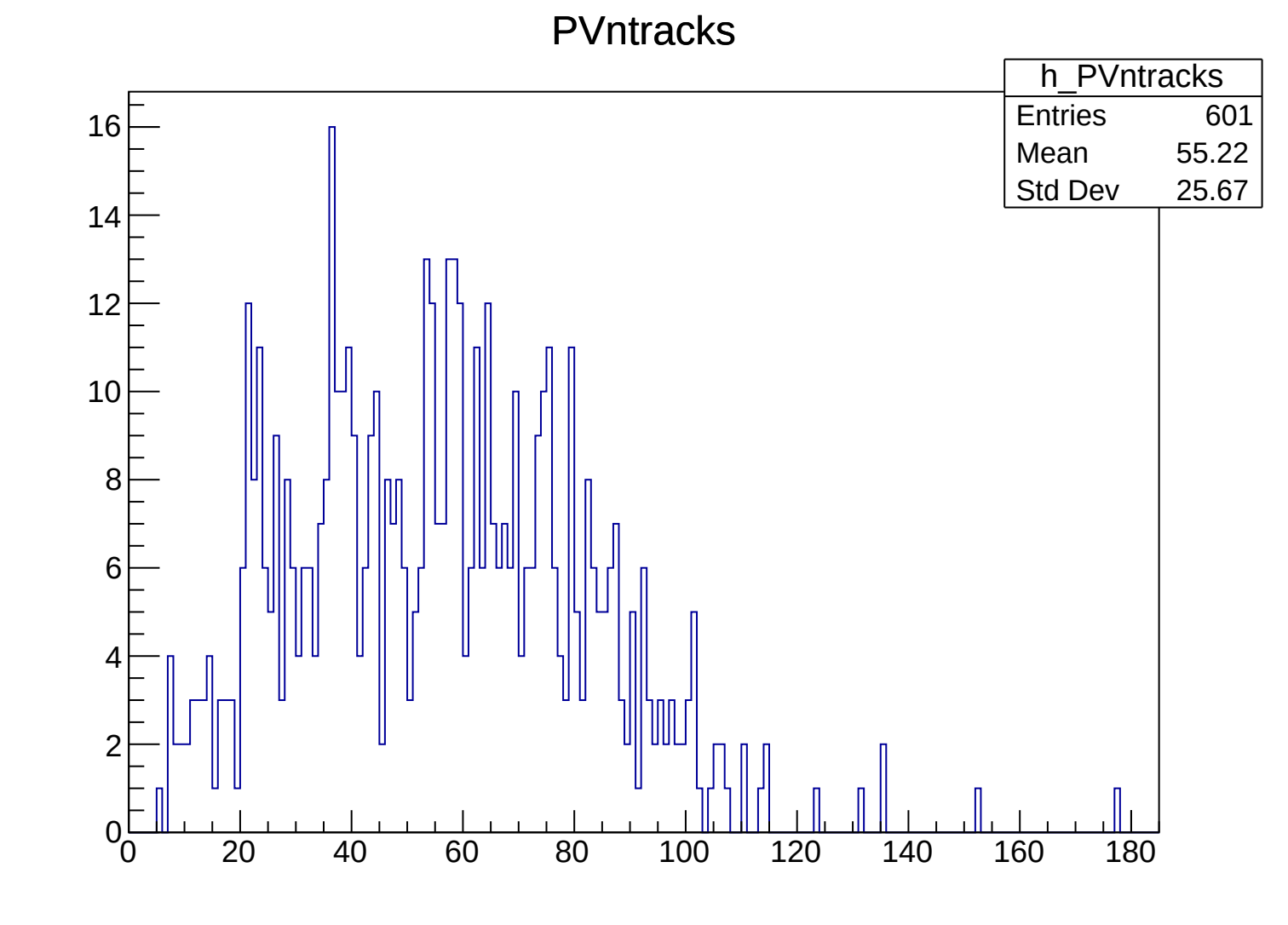

PVcovYY
h_PVcovYY
Entries 601
Mean 2.1e−06
Std Dev 2.984e−06
160
140
120
100
80
60
40
20
0
0
5
10
15
20
25
30
35
40
45
×10⁻⁶

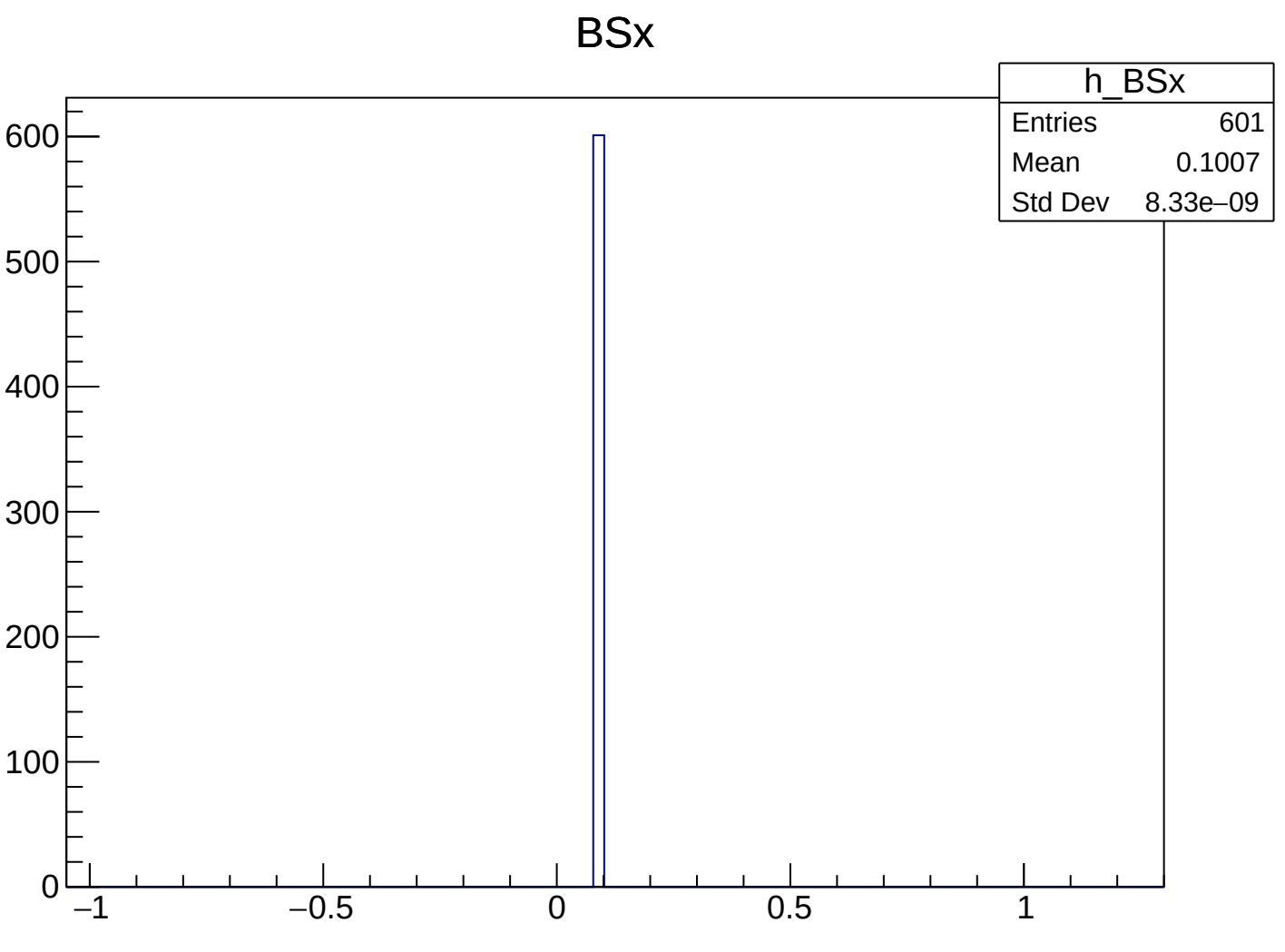

PVcovYZ
h_PVcovYZ
Entries 601
Mean −2.736e−08
Std Dev 1.713e−06
250
200
150
100
50
0
−20
−15
−10
−5
0
5
10
15
×10^−6

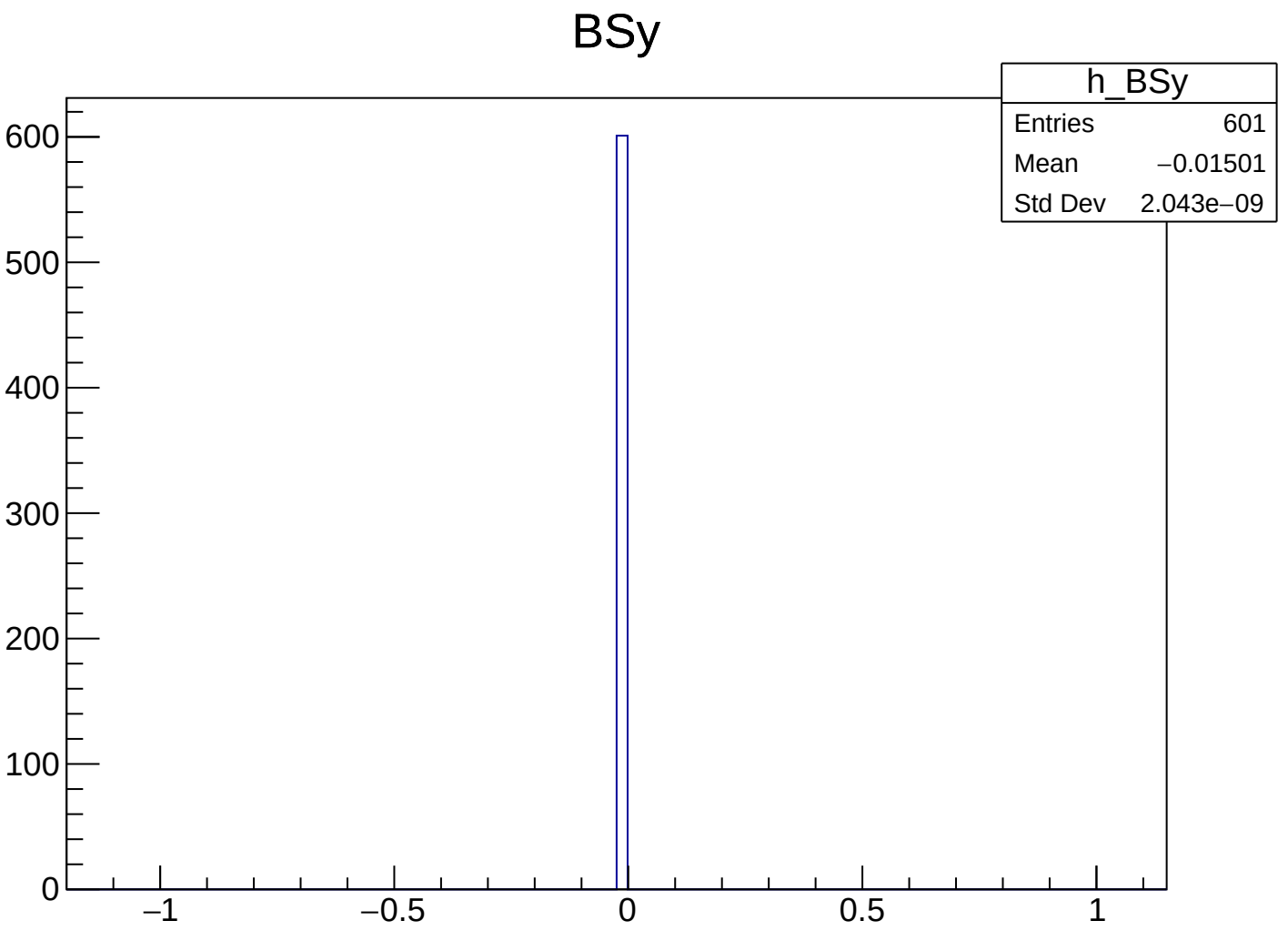

PVcovZZ
h_PVcovZZ
Entries 601
Mean 3.766e−06
Std Dev 6.783e−06
240
220
200
180
160
140
120
100
80
60
40
20
0
0
0.02
0.04
0.06
0.08
0.1
×10⁻³

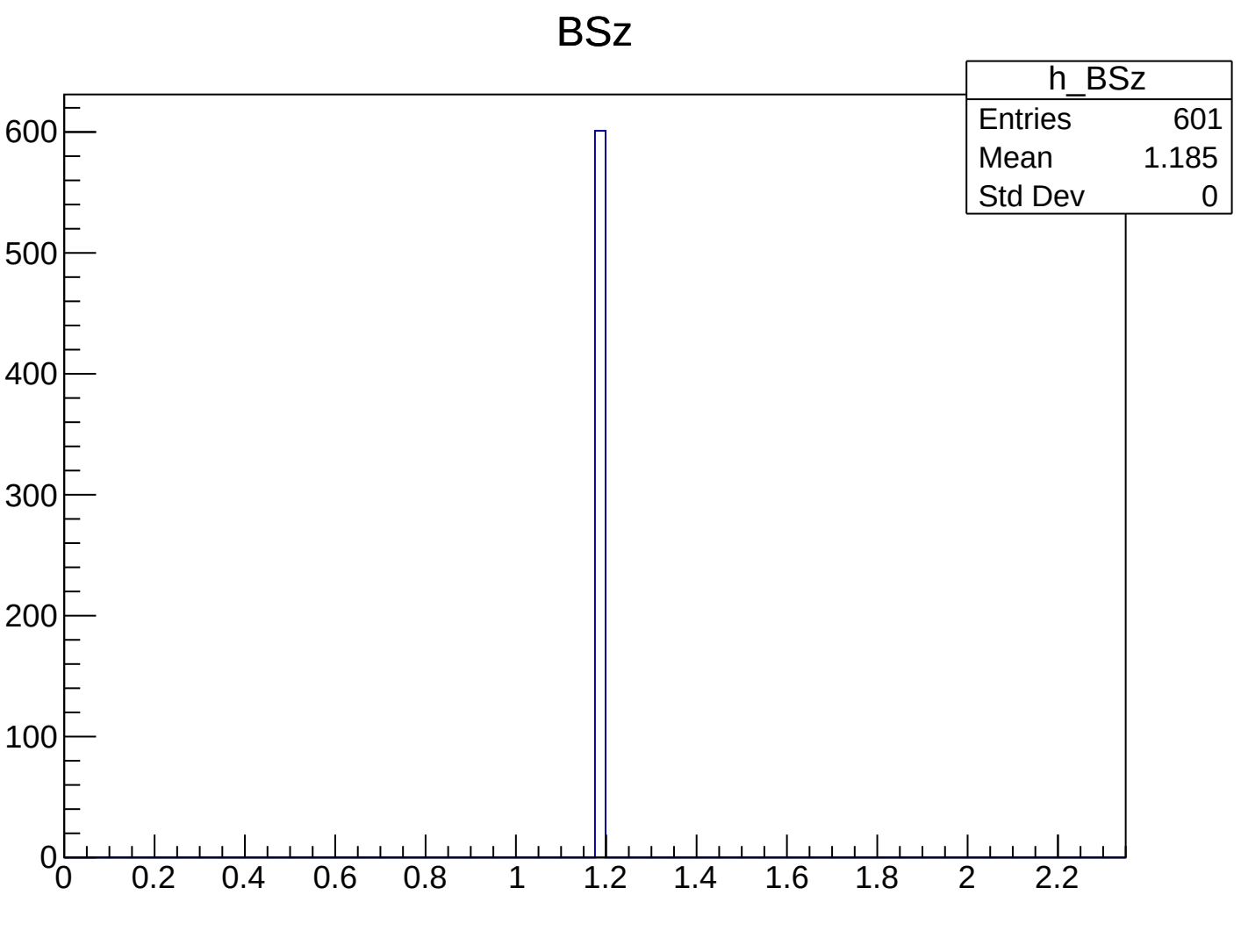

PVndof
h_PVndof
Entries 601
Mean 97.4
Std Dev 46.66
0
5
10
15
20
25
0
50
100
150
200
250
300
350

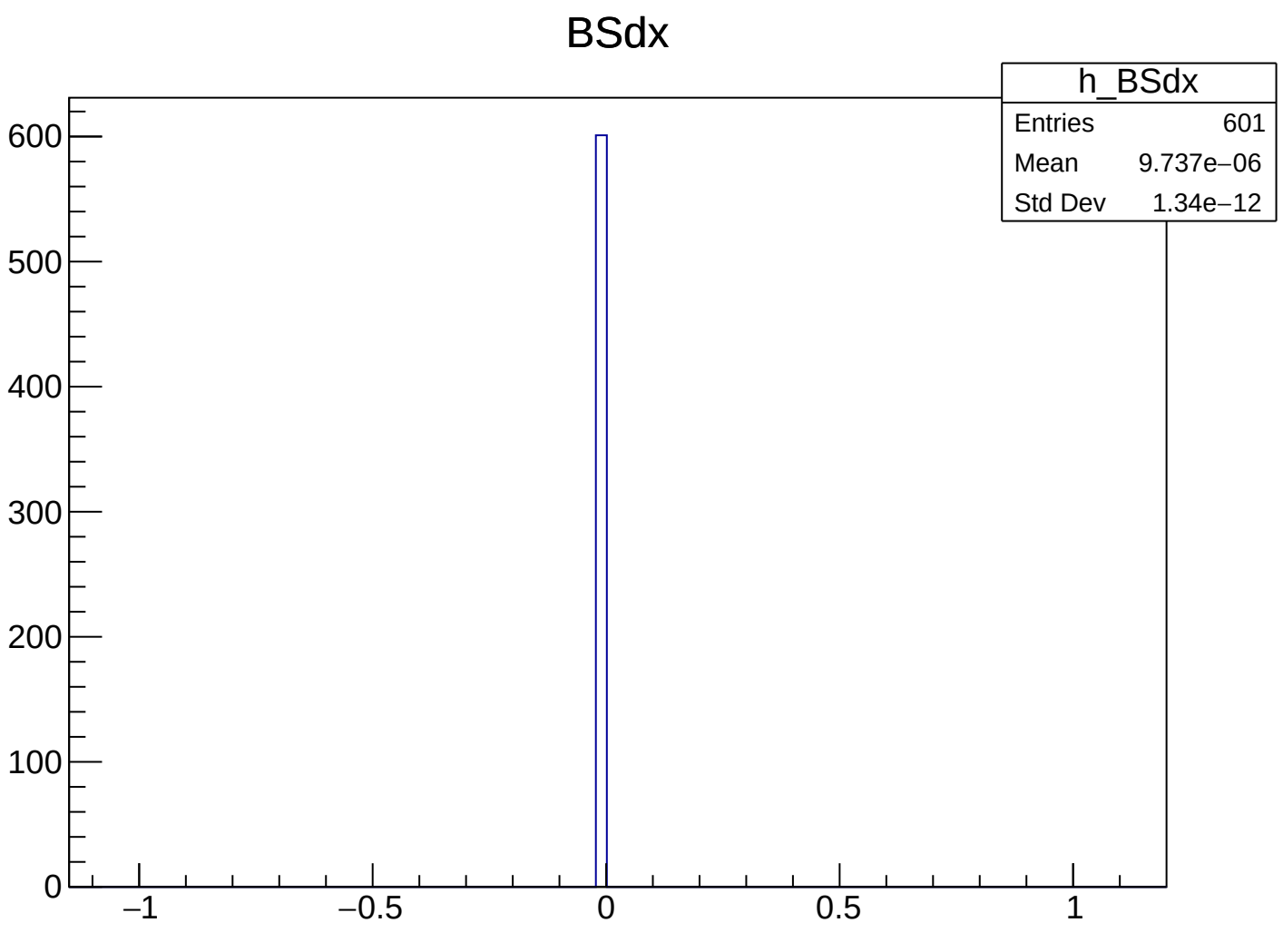

PVrho
h_PVrho
Entries 601
Mean 0.1019
Std Dev 0.001815
45
40
35
30
25
20
15
10
5
0
0.094
0.096
0.098
0.1
0.102
0.104
0.106
0.108
0.11
0.112

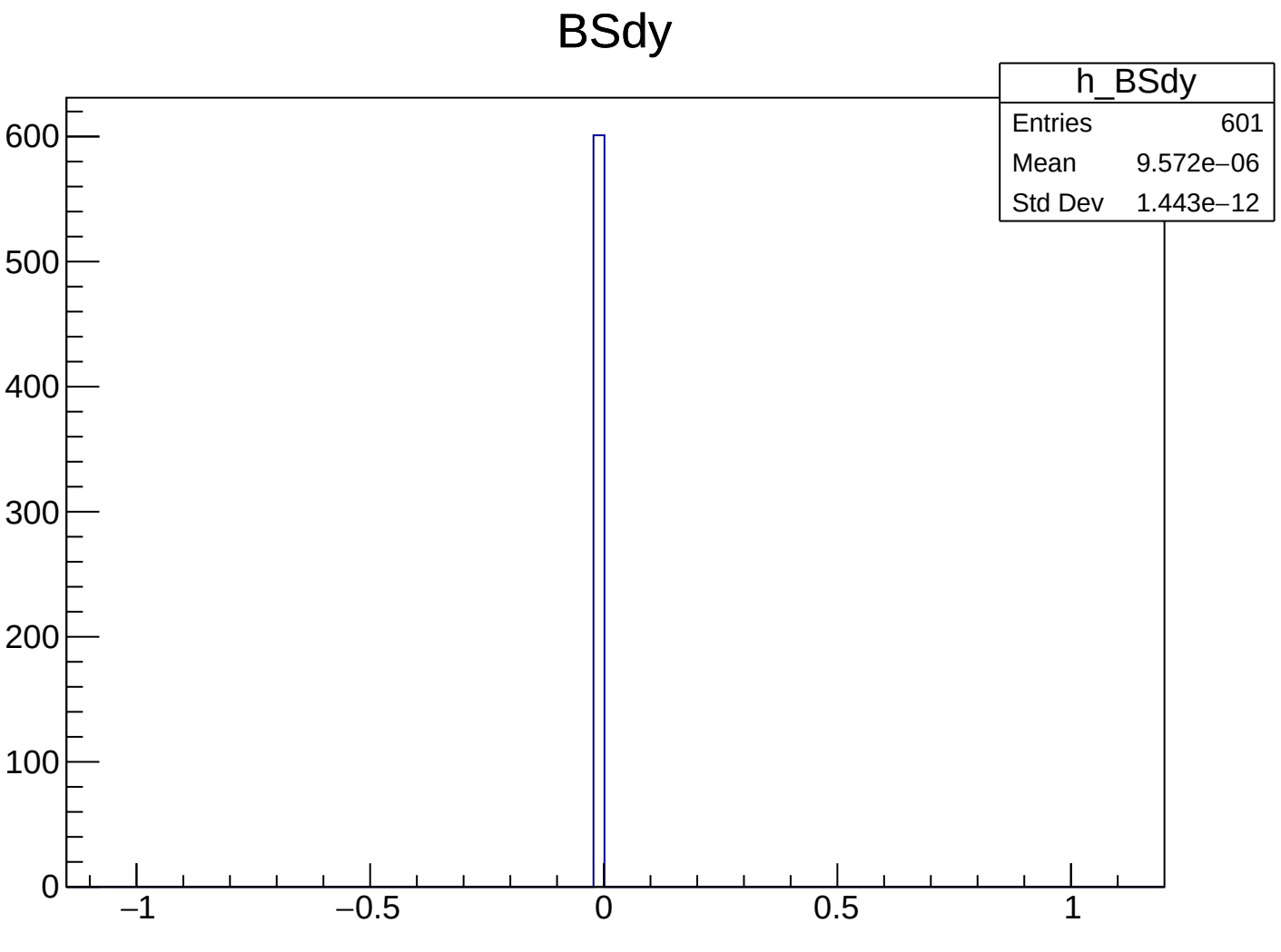

PVntracks
h_PVntracks
Entries 601
Mean 55.22
Std Dev 25.67
0
2
4
6
8
10
12
14
16
0
20
40
60
80
100
120
140
160
180

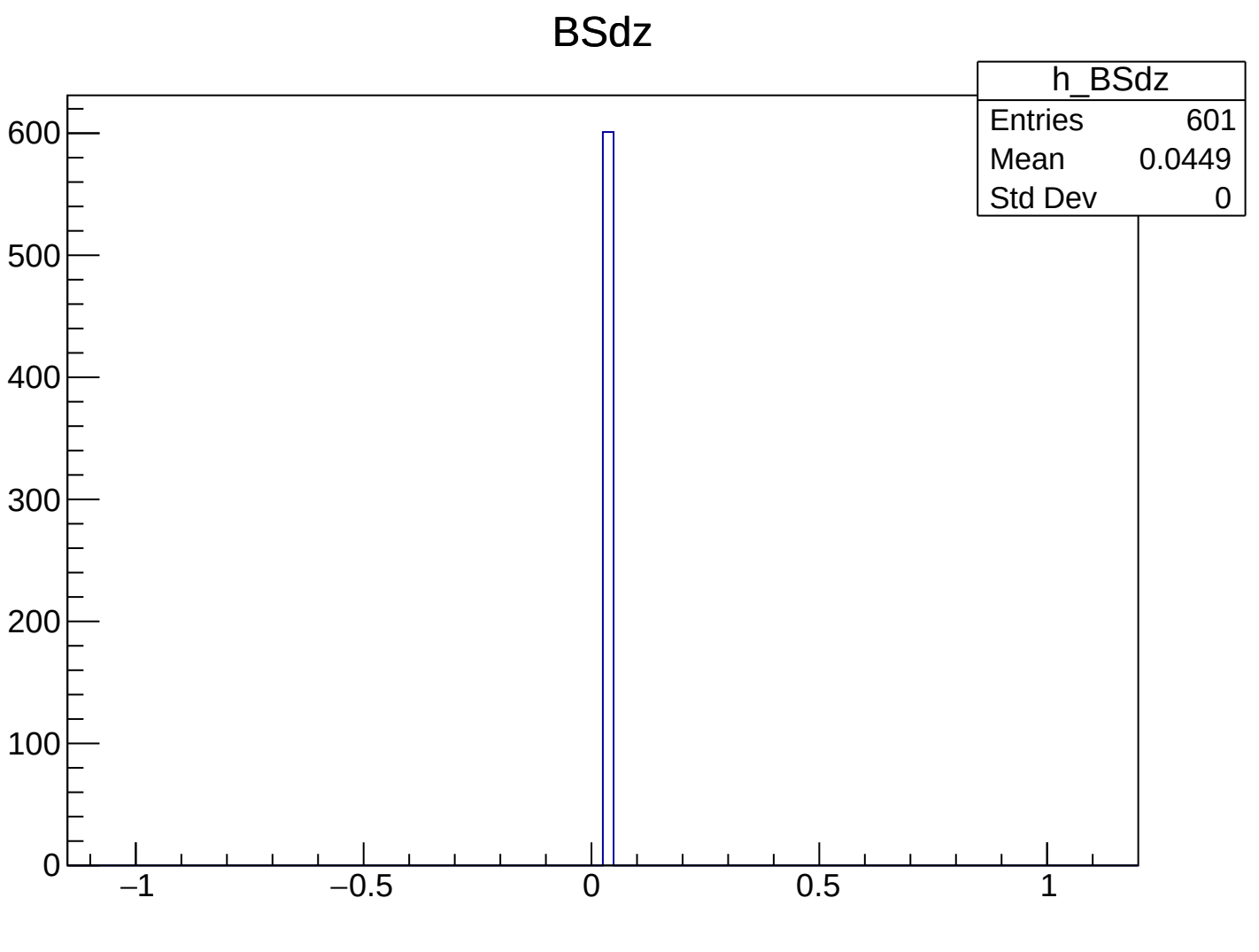

BSx
h_BSx
Entries 601
Mean 0.1007
Std Dev 8.33e−09
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

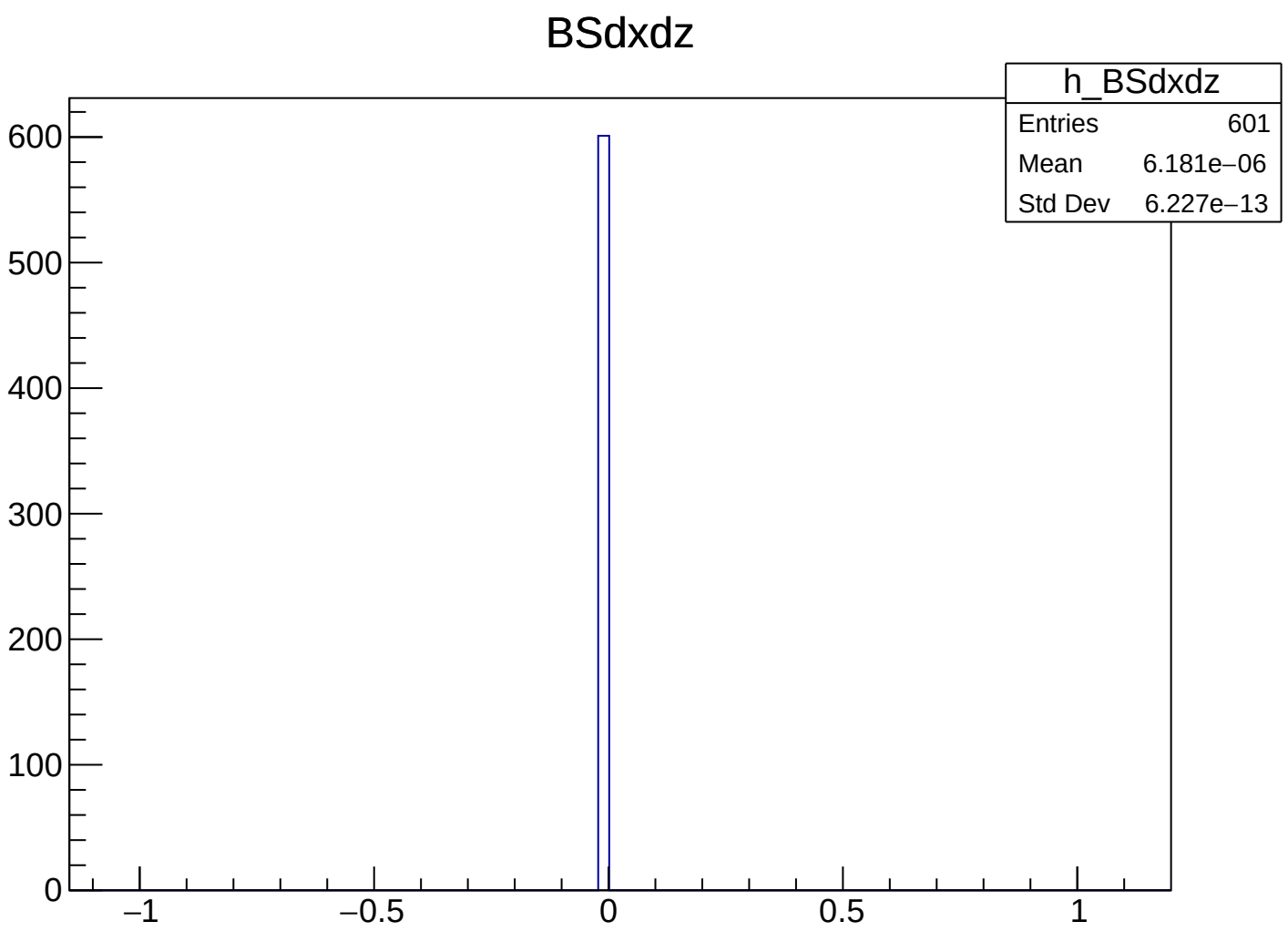

BSy
h_BSy
Entries 601
Mean −0.01501
Std Dev 2.043e−09
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

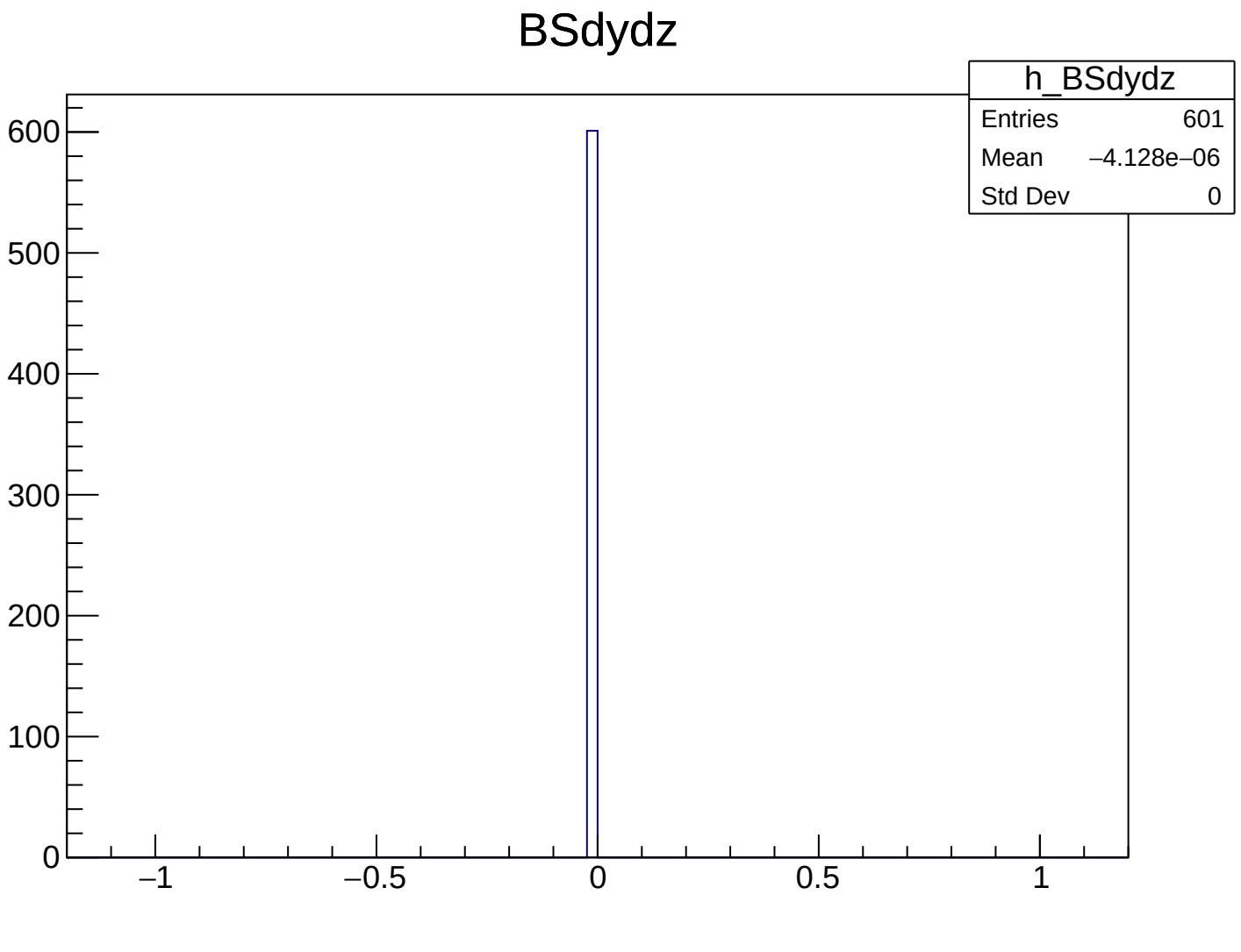

BSz
h_BSz
Entries 601
Mean 1.185
Std Dev 0
600
500
400
300
200
100
0
0 0.2 0.4 0.6 0.8 1 1.2 1.4 1.6 1.8 2 2.2

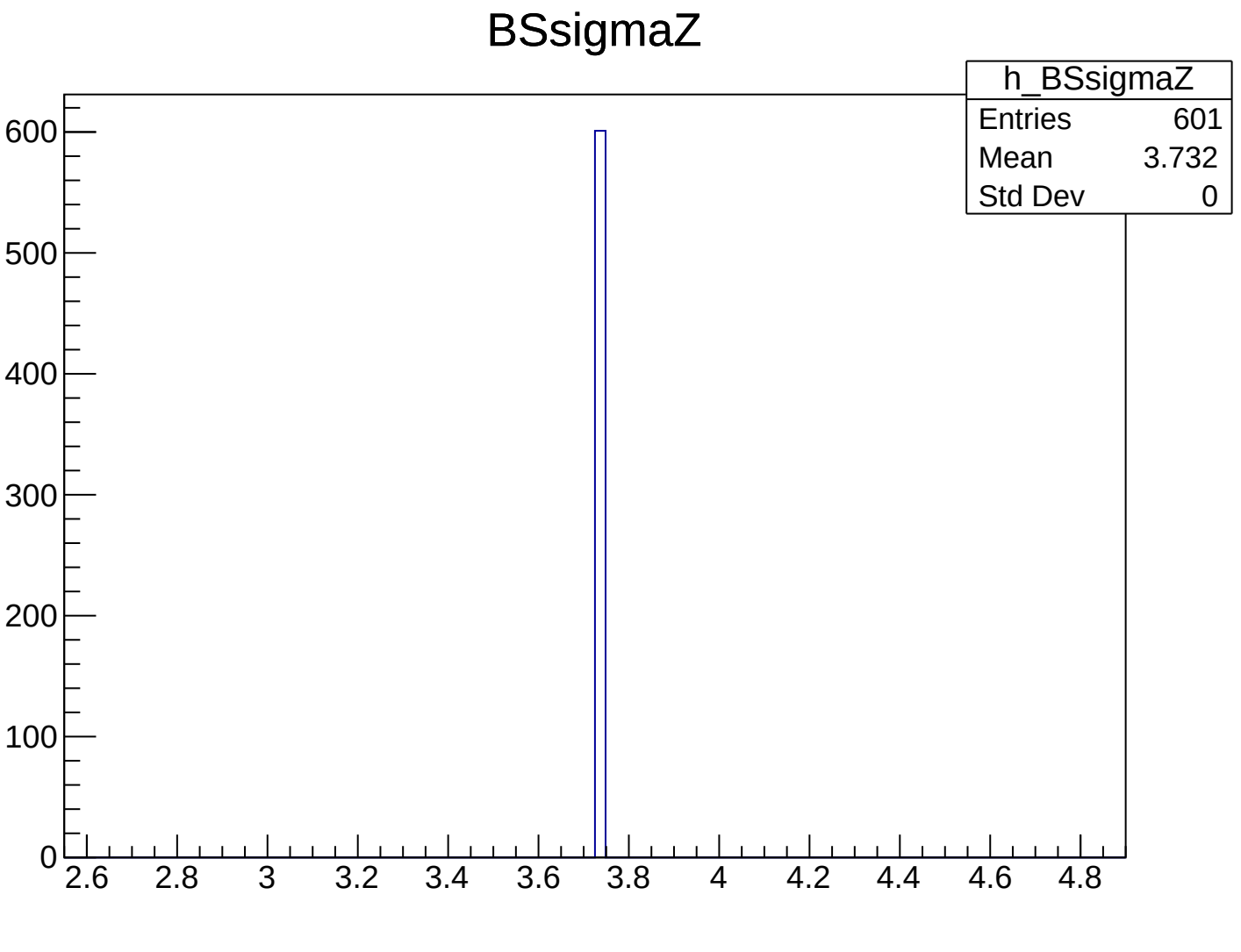

BSdx
h_BSdx
Entries 601
Mean 9.737e−06
Std Dev 1.34e−12
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

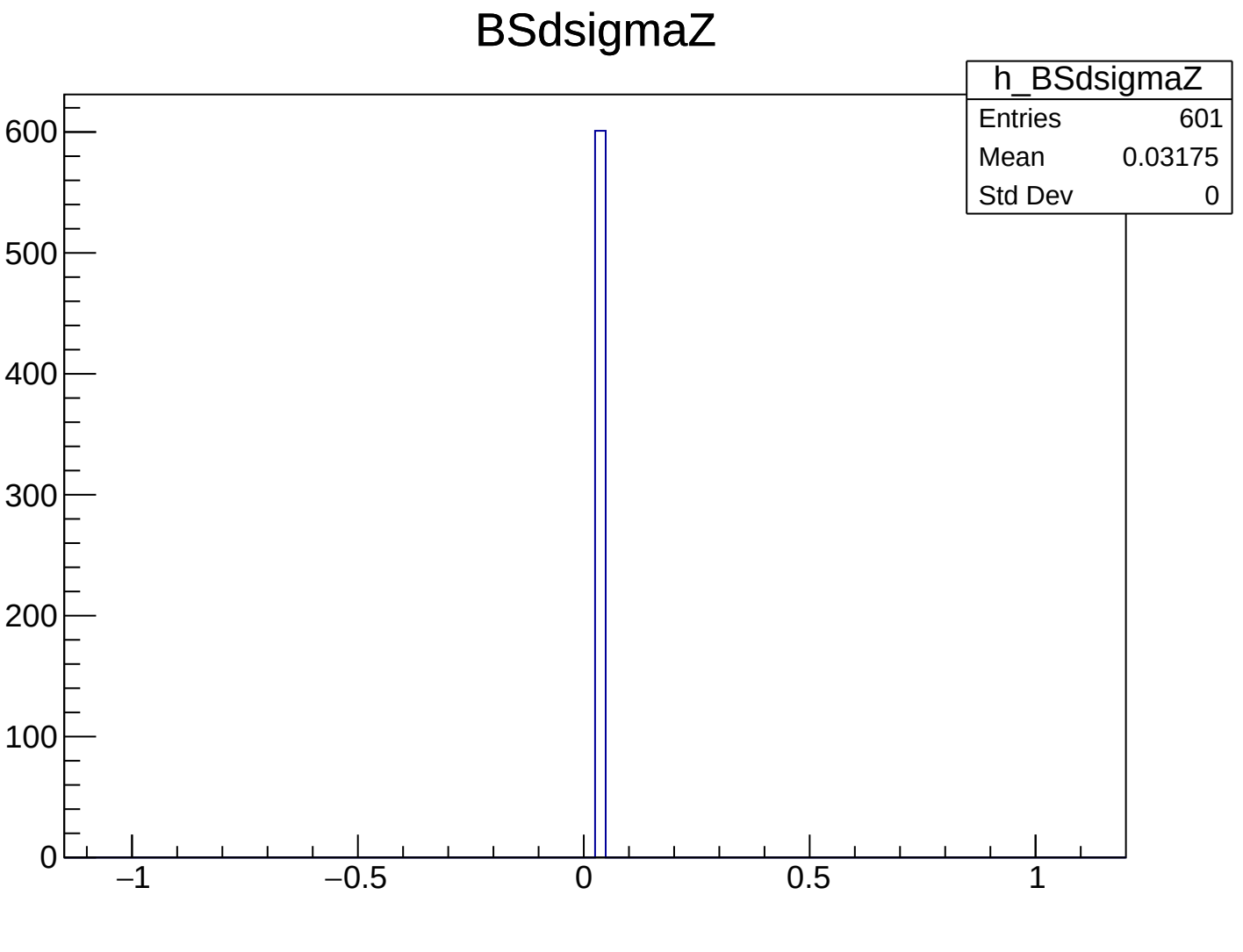

BSdy
h_BSdy
Entries 601
Mean 9.572e−06
Std Dev 1.443e−12
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

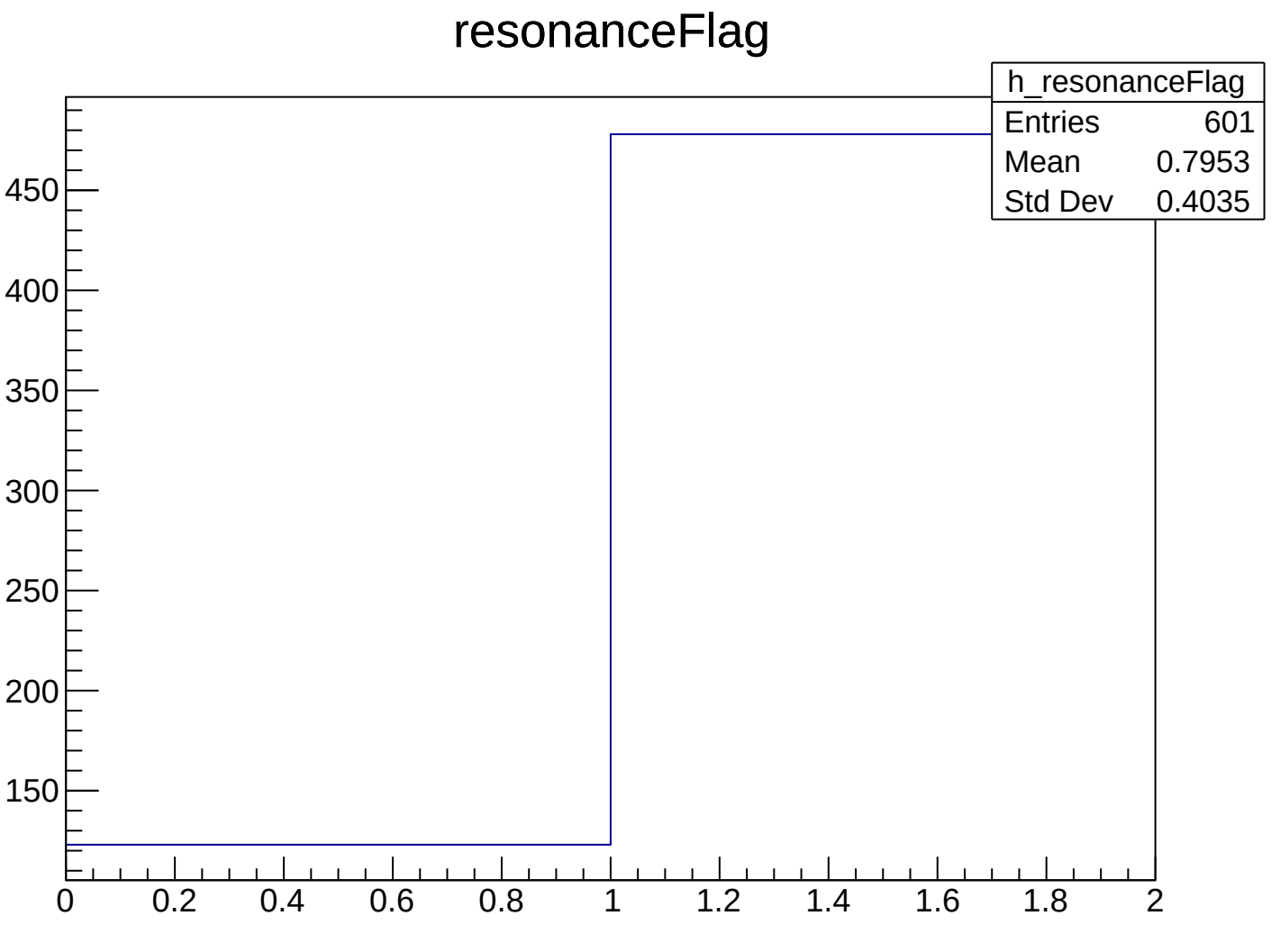

BSdz
h_BSdz
Entries 601
Mean 0.0449
Std Dev 0
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

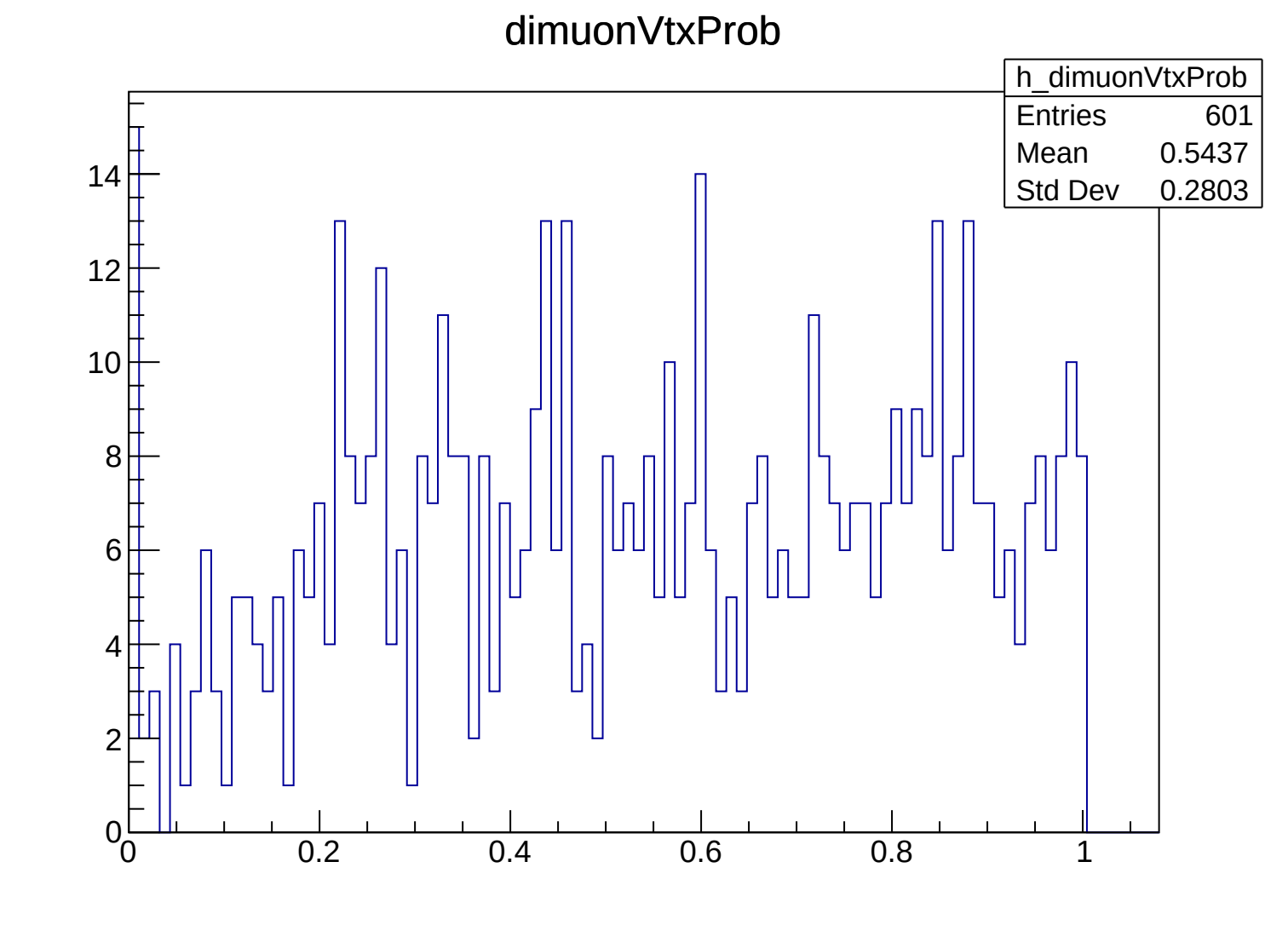

BSdxdz
h_BSdxdz
Entries 601
Mean 6.181e−06
Std Dev 6.227e−13
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

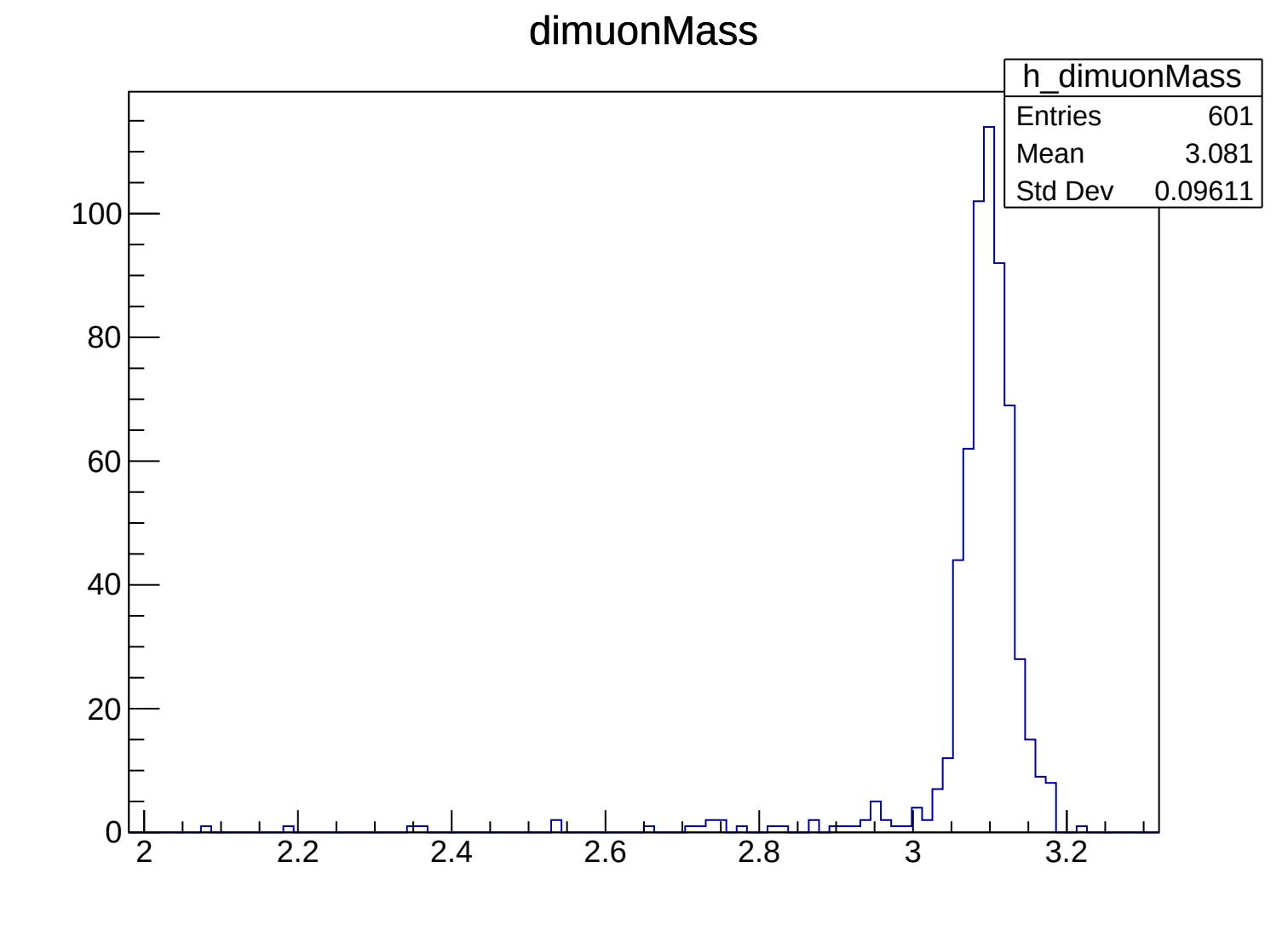

BSdydz
h_BSdydz
Entries 601
Mean –4.128e–06
Std Dev 0
600
500
400
300
200
100
0
–1
–0.5
0
0.5
1

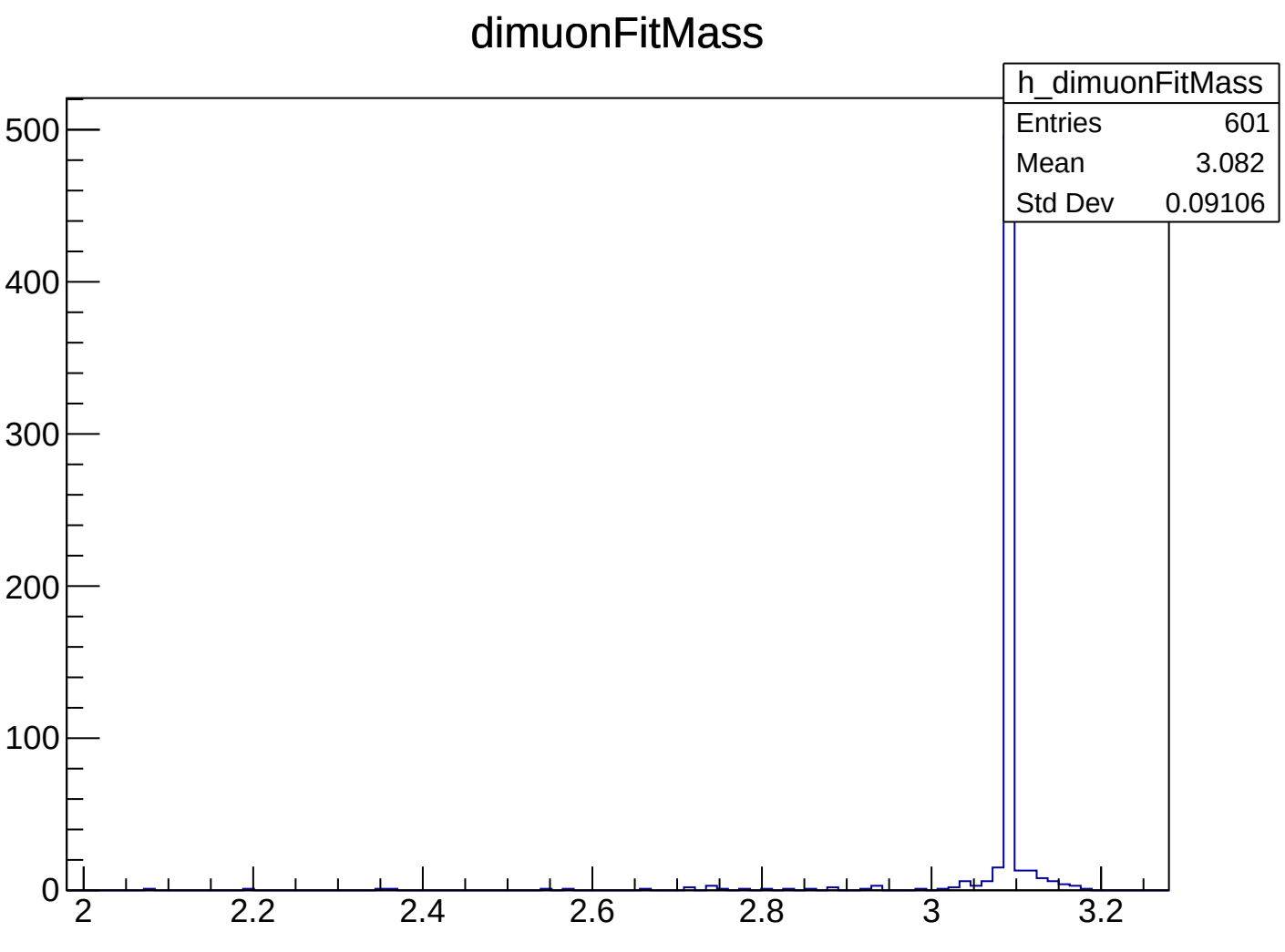

BSsigmaZ
h_BSsigmaZ
Entries 601
Mean 3.732
Std Dev 0
600
500
400
300
200
100
0
2.6
2.8
3
3.2
3.4
3.6
3.8
4
4.2
4.4
4.6
4.8

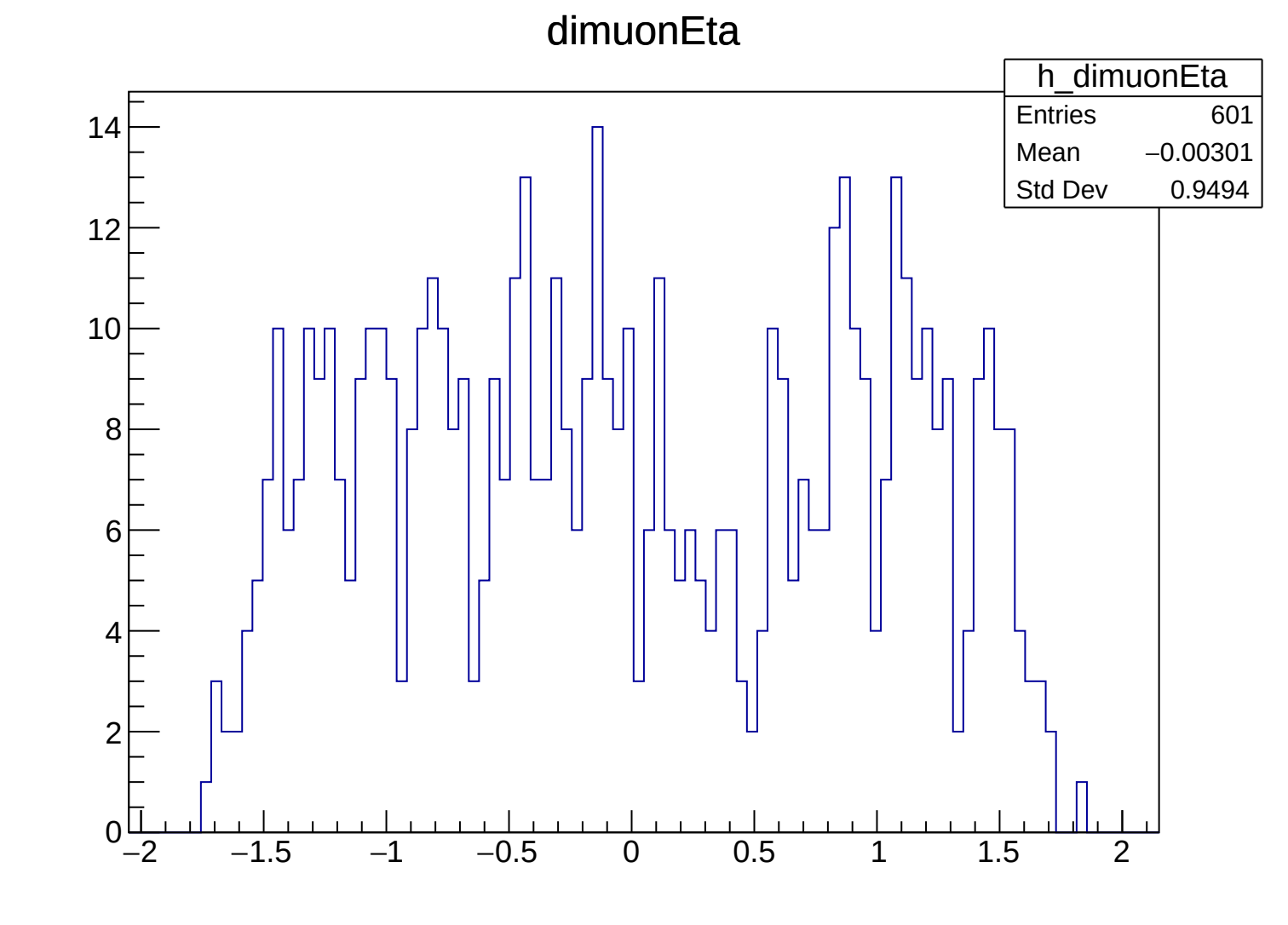

BSdsigmaZ
h_BSdsigmaZ
Entries 601
Mean 0.03175
Std Dev 0
600
500
400
300
200
100
0
−1
−0.5
0
0.5
1

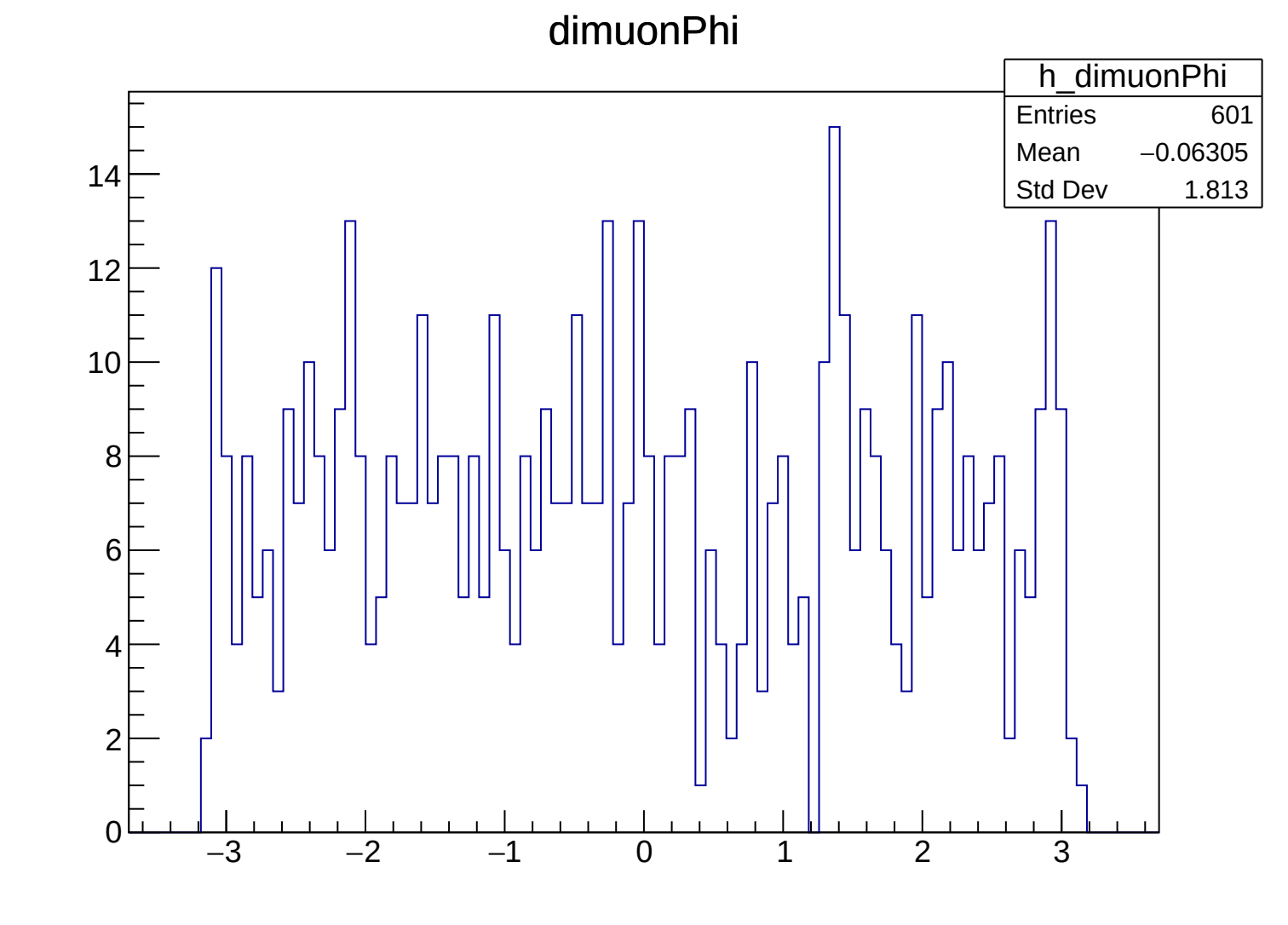

resonanceFlag
h_resonanceFlag
Entries 601
Mean 0.7953
Std Dev 0.4035
450
400
350
300
250
200
150
0
0.2
0.4
0.6
0.8
1
1.2
1.4
1.6
1.8
2

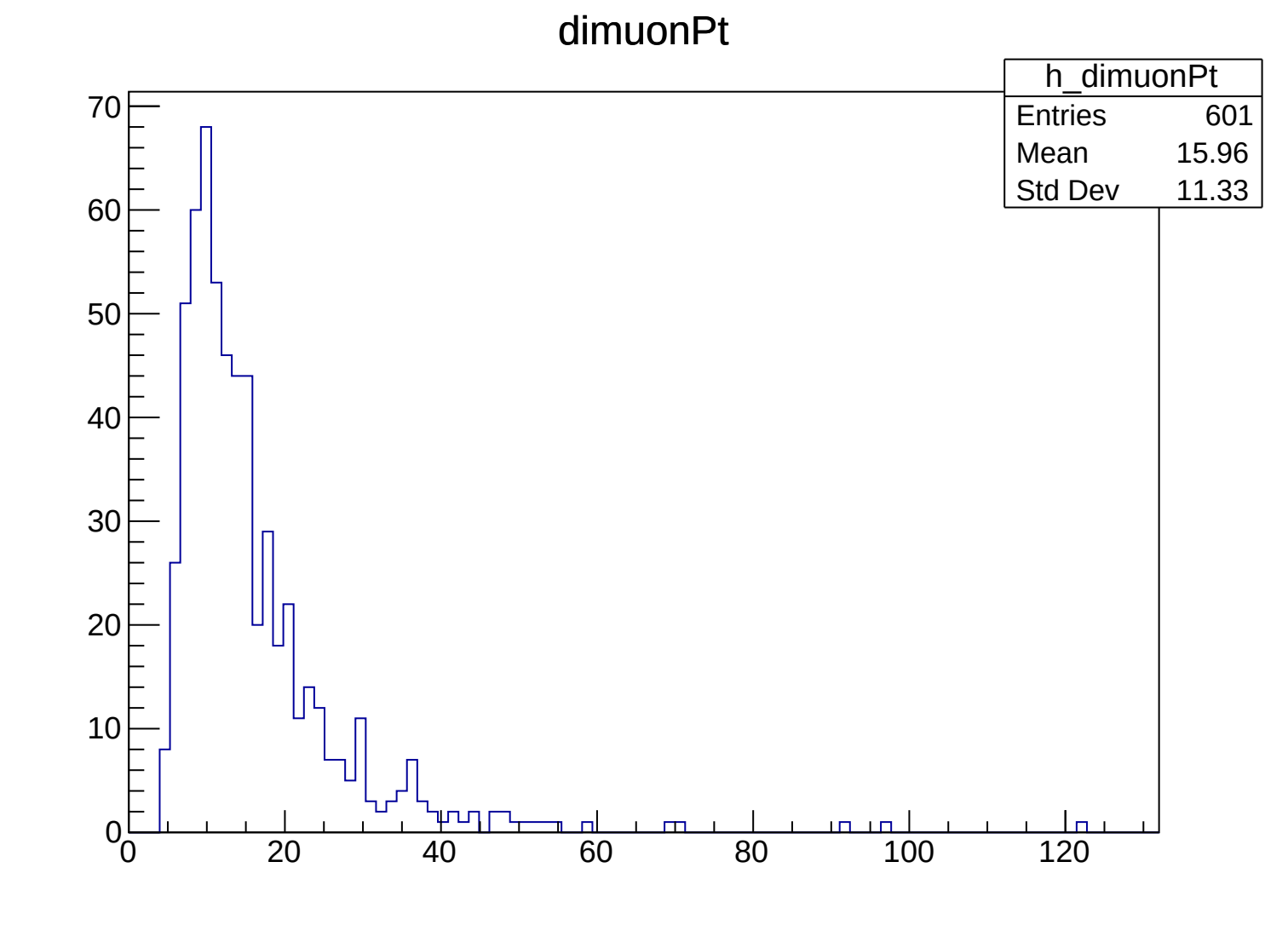

dimuonVtxProb
h_dimuonVtxProb
Entries 601
Mean 0.5437
Std Dev 0.2803
0
2
4
6
8
10
12
14
0
0.2
0.4
0.6
0.8
1

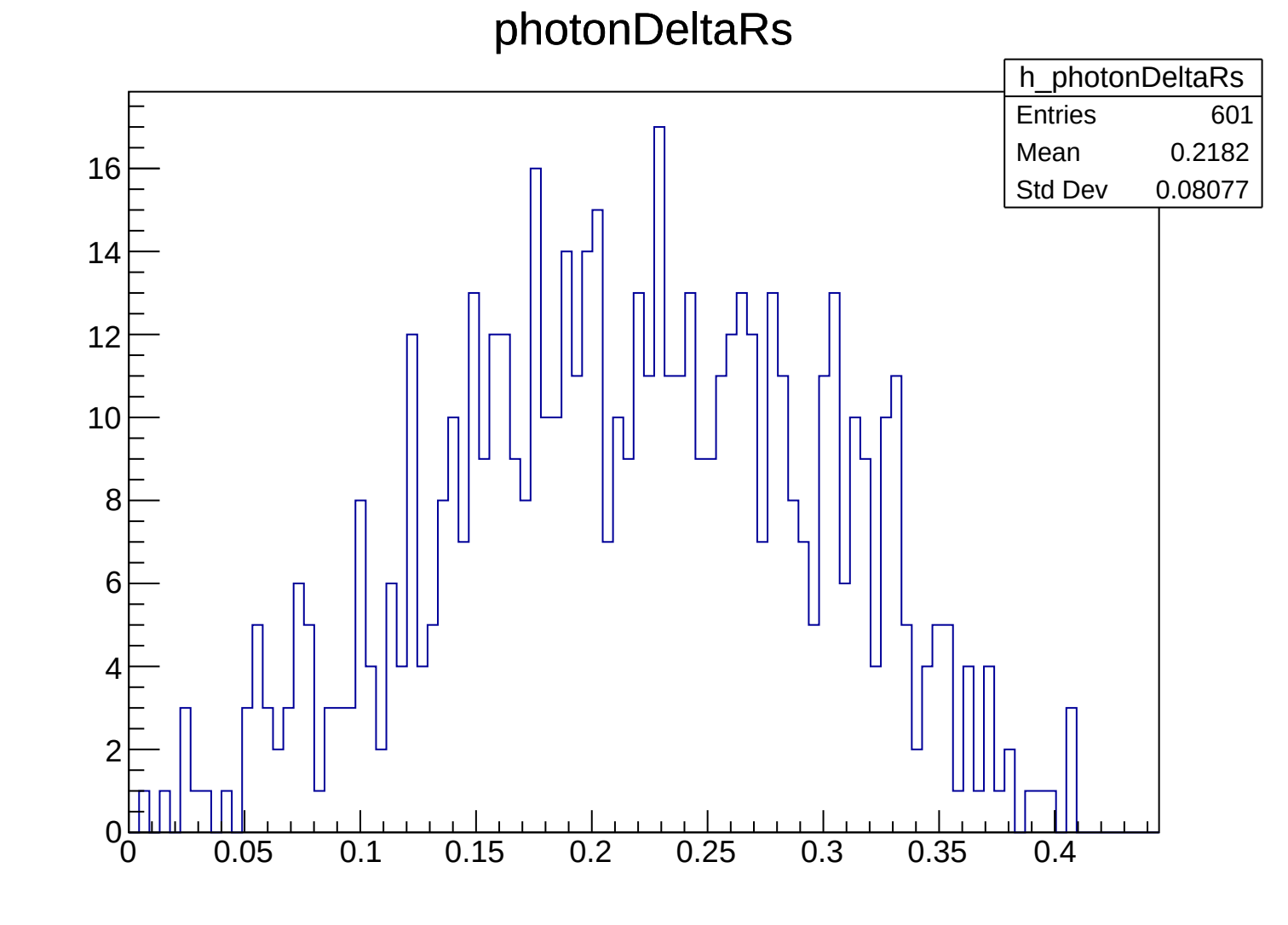

dimuonMass
h_dimuonMass
Entries 601
Mean 3.081
Std Dev 0.09611
100
80
60
40
20
0
2
2.2
2.4
2.6
2.8
3
3.2

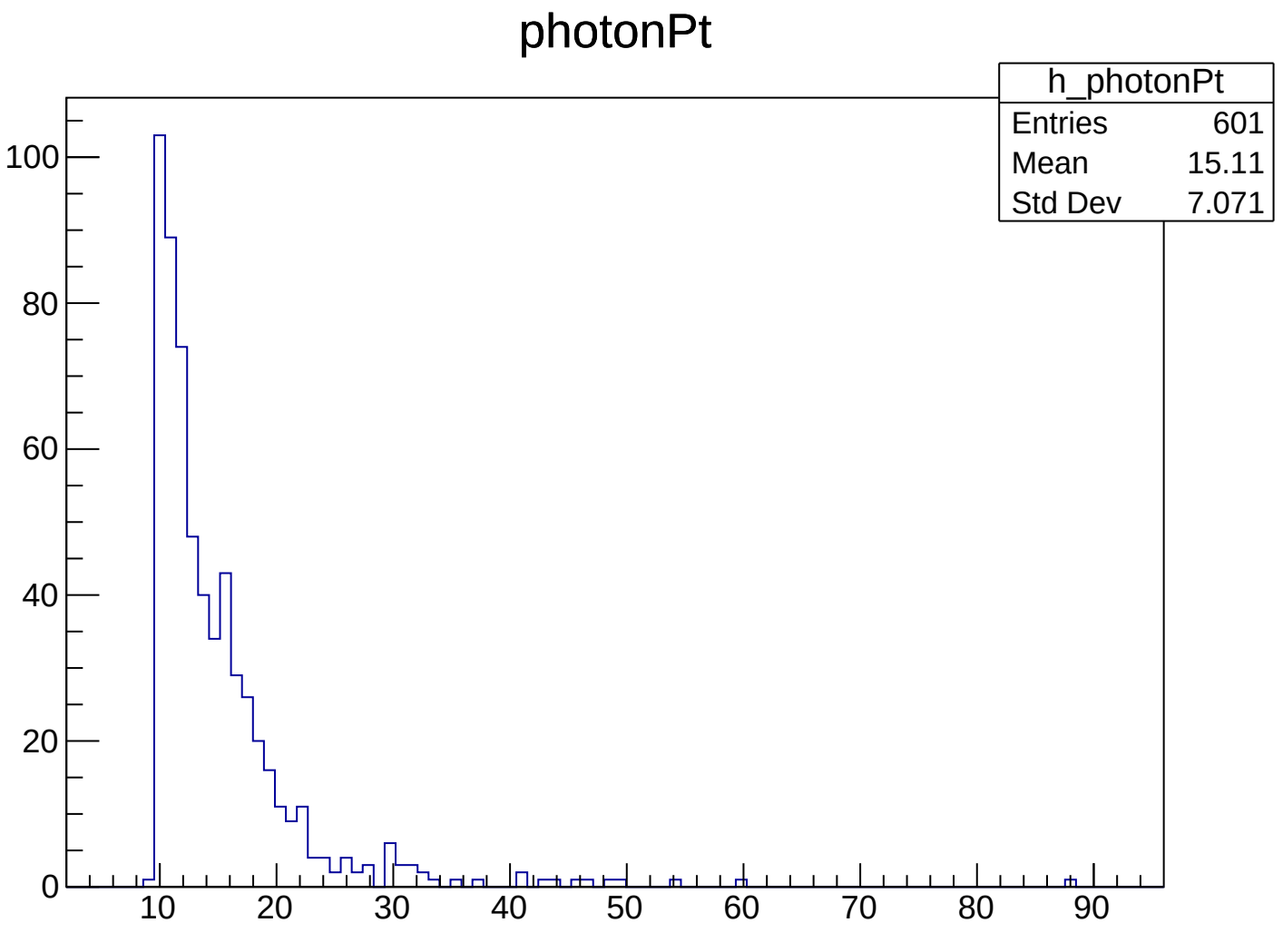

dimuonFitMass
h_dimuonFitMass
Entries 601
Mean 3.082
Std Dev 0.09106
500
400
300
200
100
0
2
2.2
2.4
2.6
2.8
3
3.2

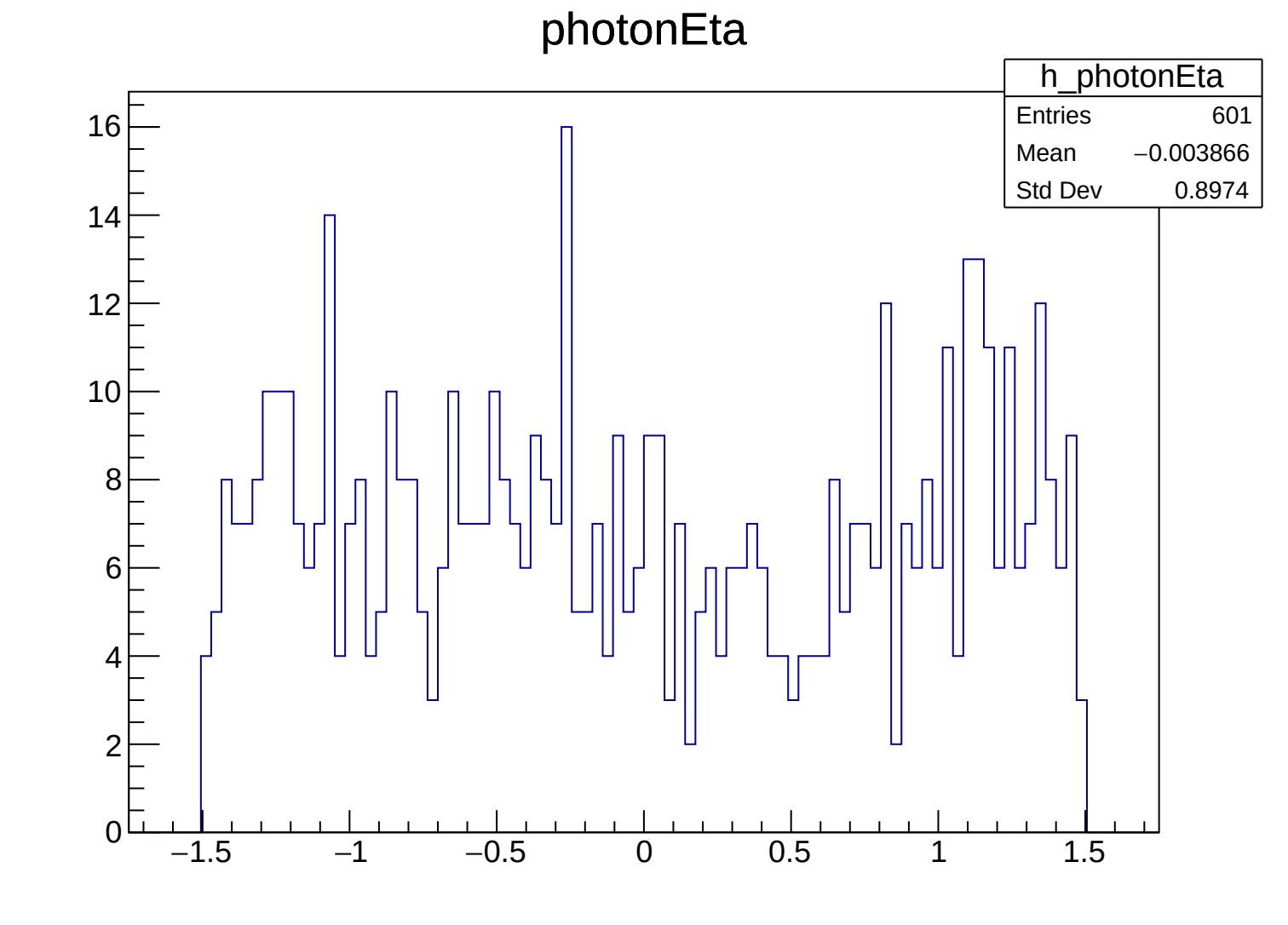

dimuonEta
h_dimuonEta
Entries 601
Mean −0.00301
Std Dev 0.9494
14
12
10
8
6
4
2
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

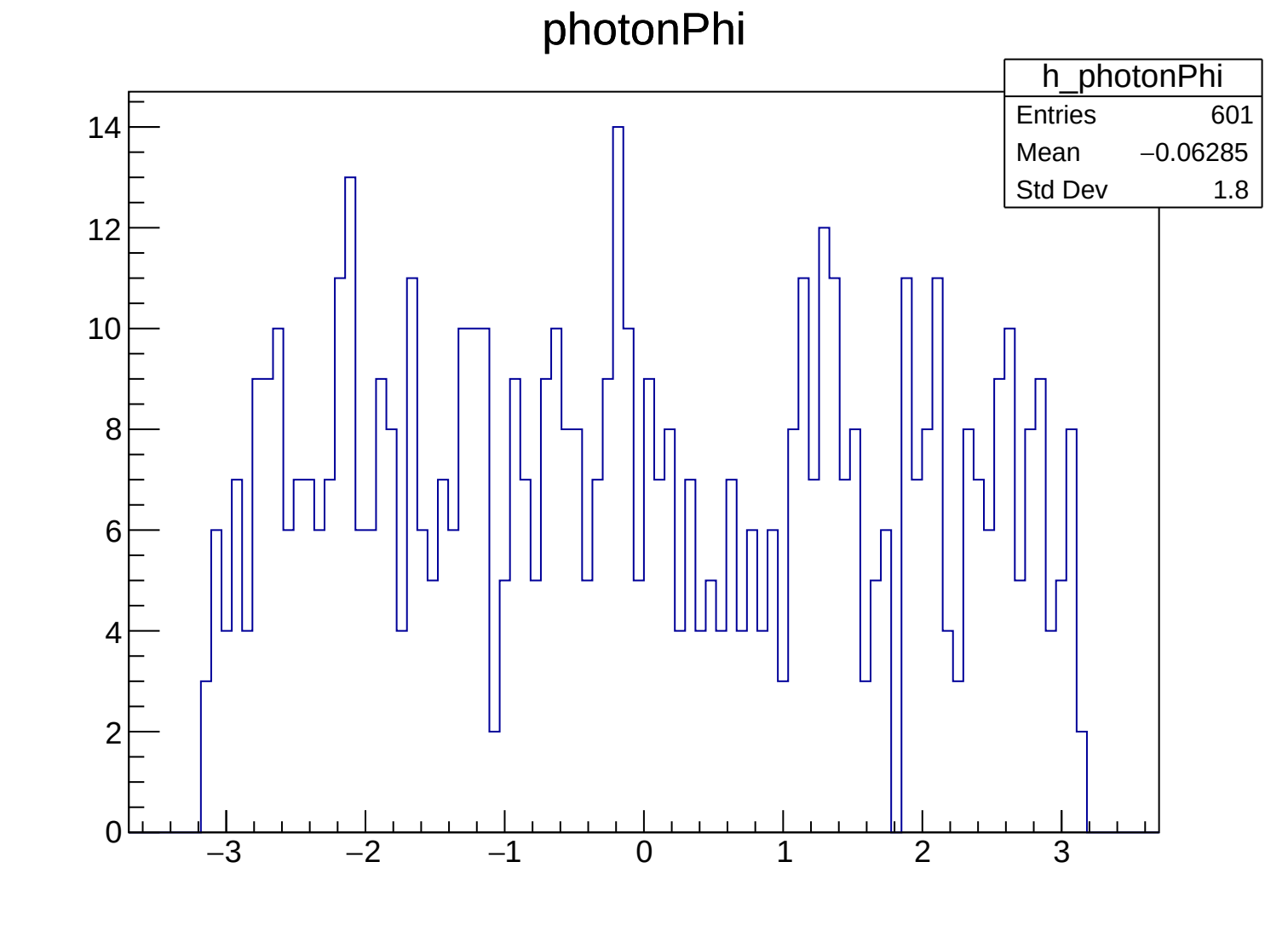

dimuonPhi
h_dimuonPhi
Entries 601
Mean −0.06305
Std Dev 1.813
0
2
4
6
8
10
12
14
−3
−2
−1
0
1
2
3

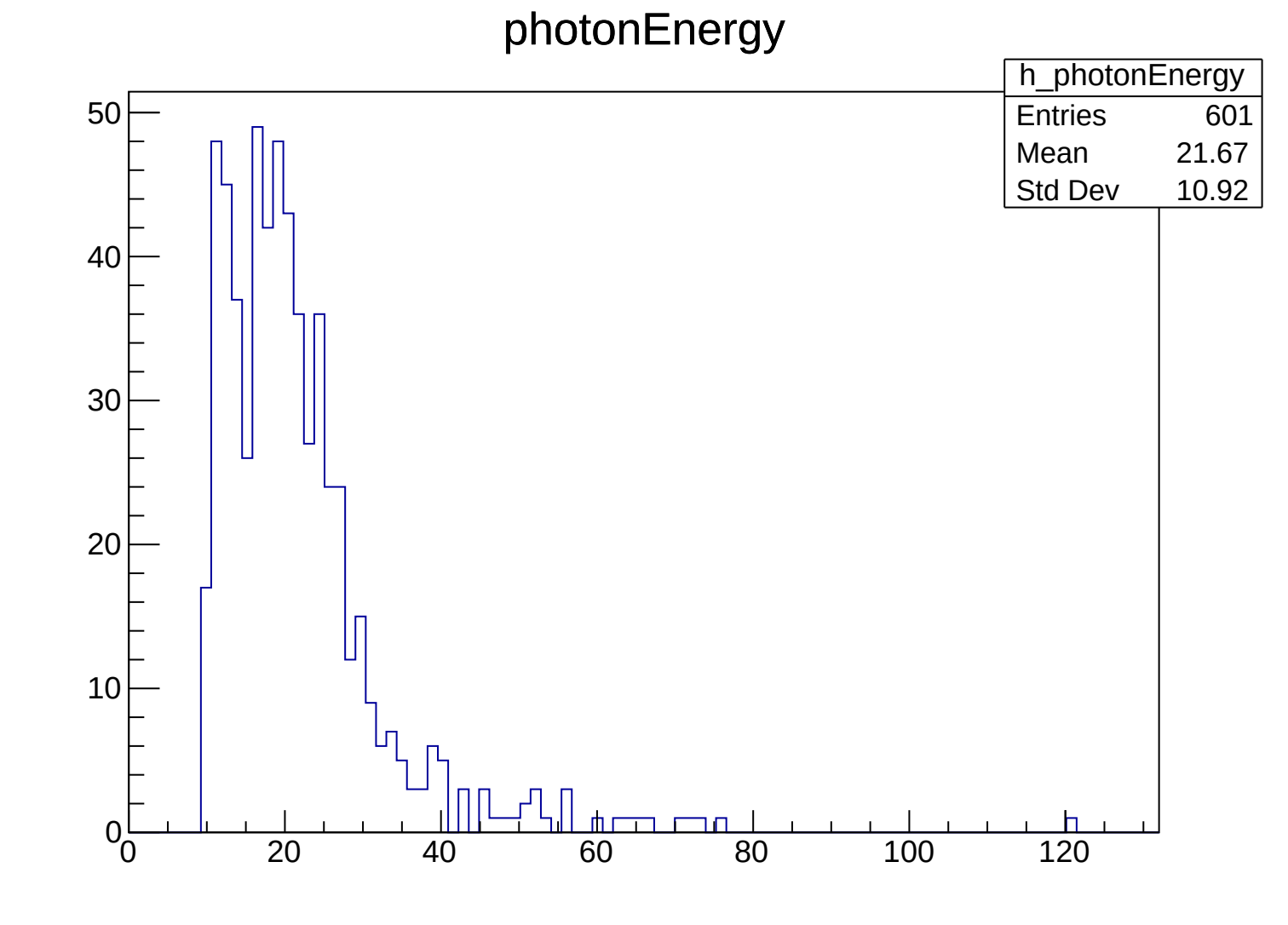

dimuonPt
h_dimuonPt
Entries 601
Mean 15.96
Std Dev 11.33
70
60
50
40
30
20
10
0
0
20
40
60
80
100
120

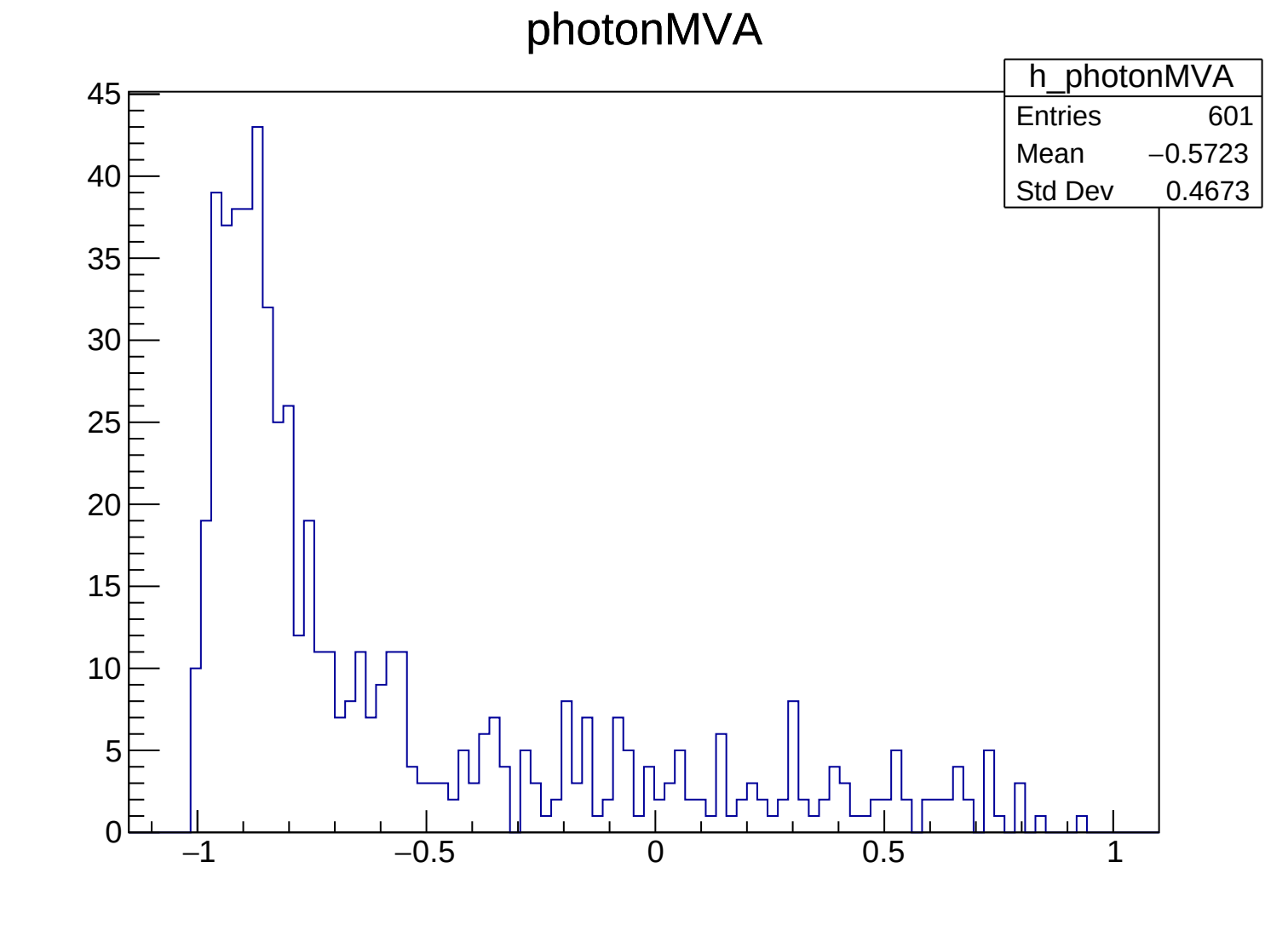

photonDeltaRs
h_photonDeltaRs
Entries 601
Mean 0.2182
Std Dev 0.08077
0
2
4
6
8
10
12
14
16
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35
0.4

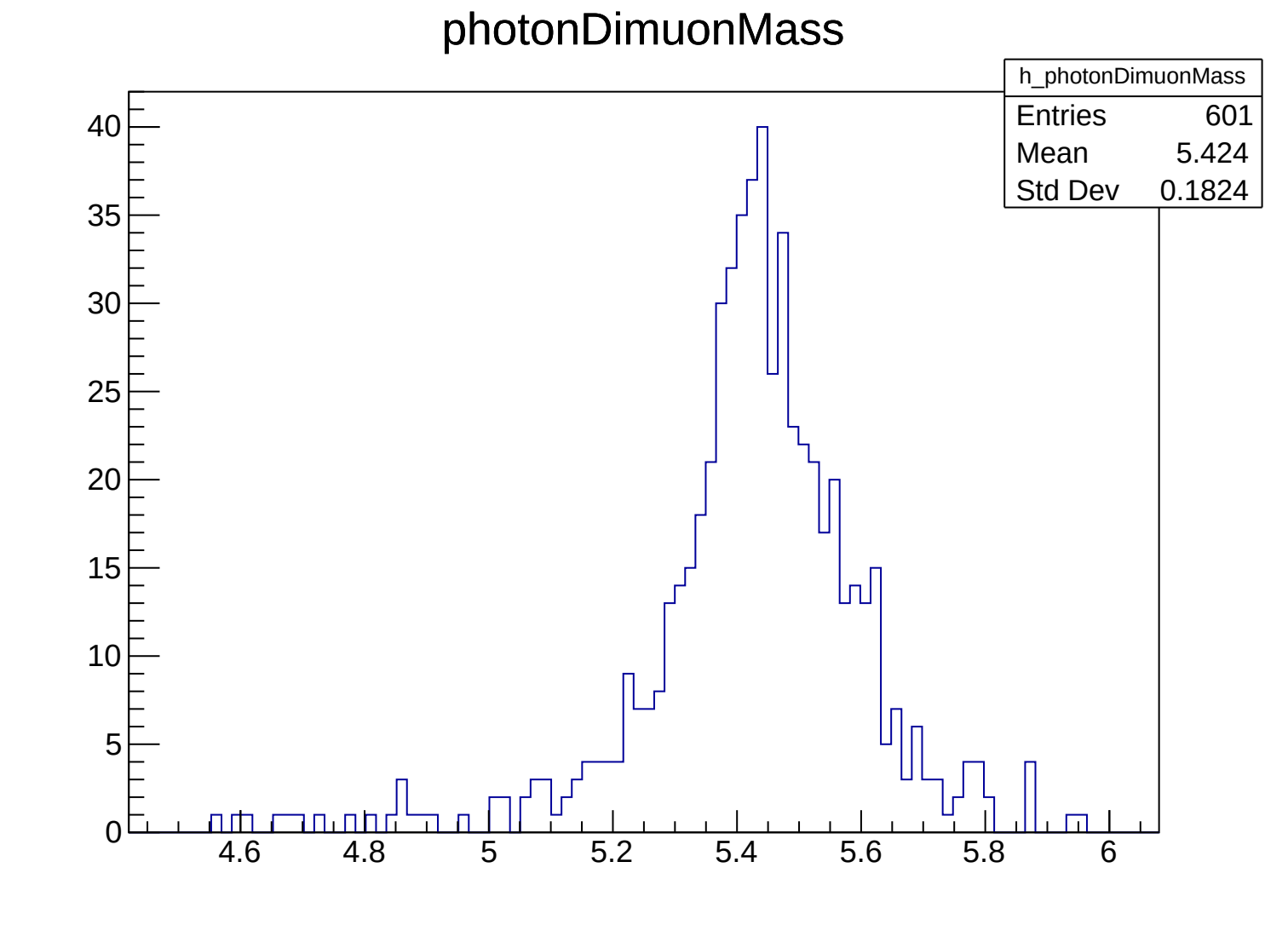

photonPt
h_photonPt
Entries 601
Mean 15.11
Std Dev 7.071
100
80
60
40
20
0
10
20
30
40
50
60
70
80
90

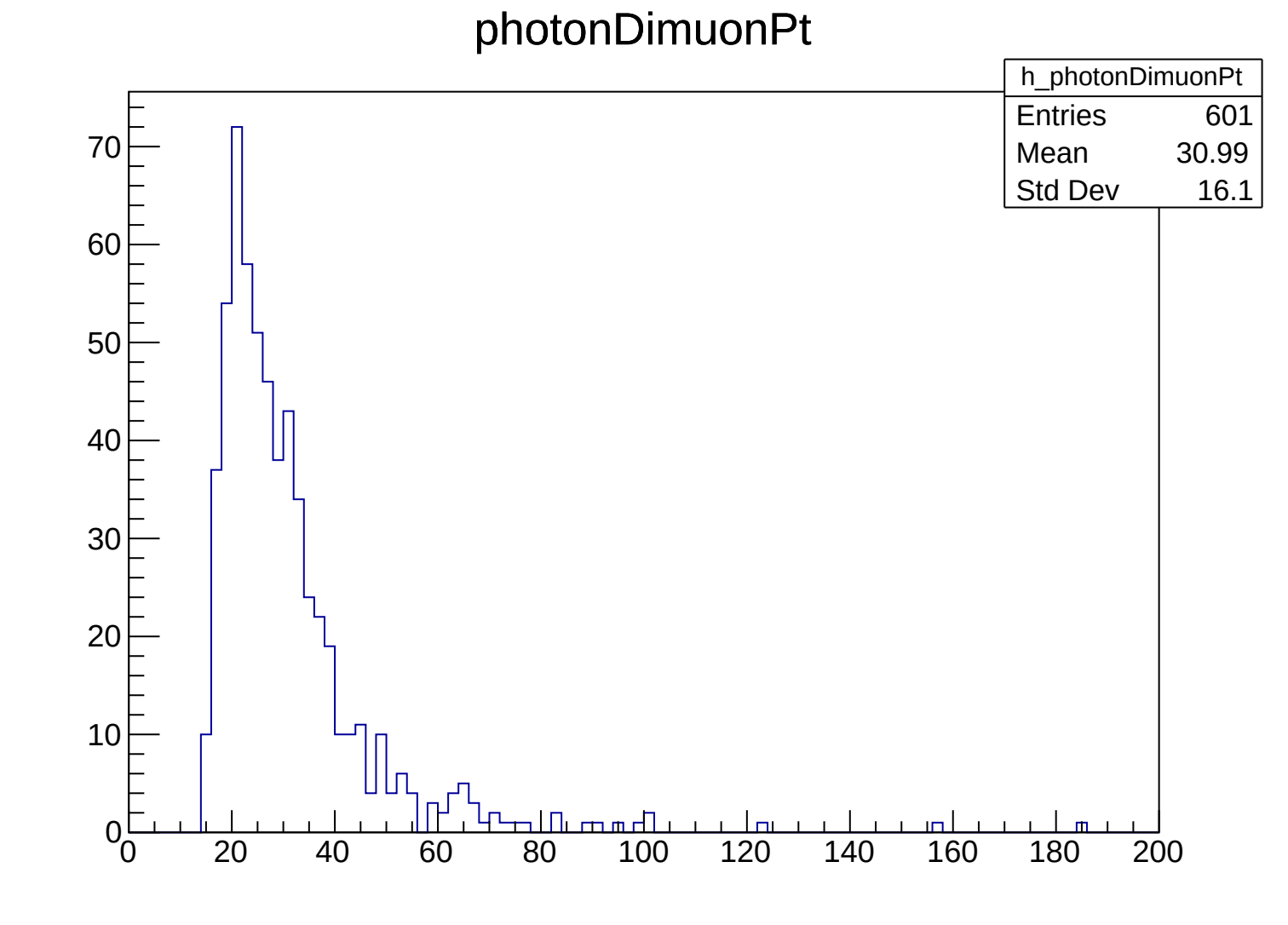

photonEta
h_photonEta
Entries 601
Mean −0.003866
Std Dev 0.8974
16
14
12
10
8
6
4
2
0
−1.5
−1
−0.5
0
0.5
1
1.5

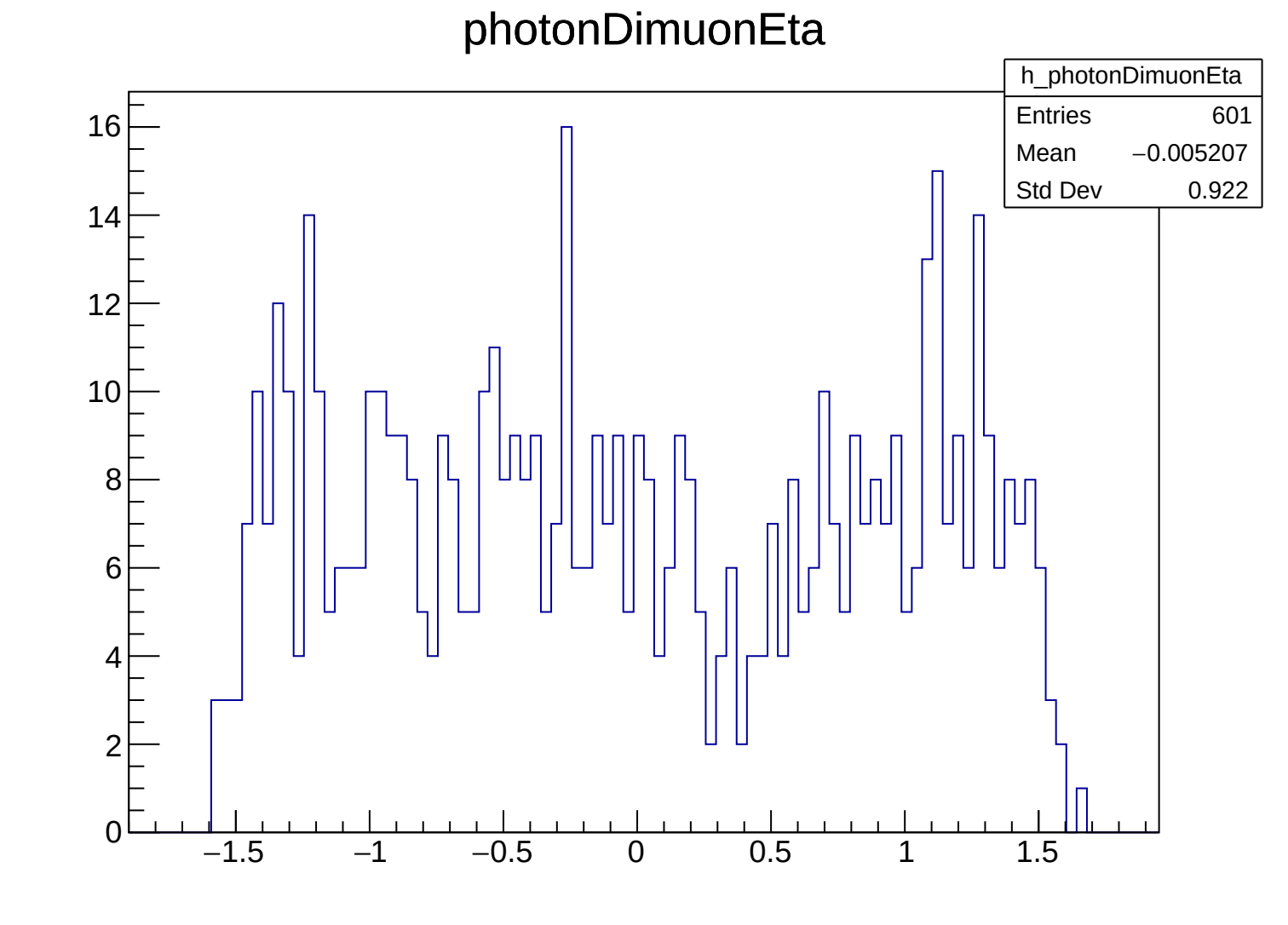

photonPhi
h_photonPhi
Entries 601
Mean −0.06285
Std Dev 1.8
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

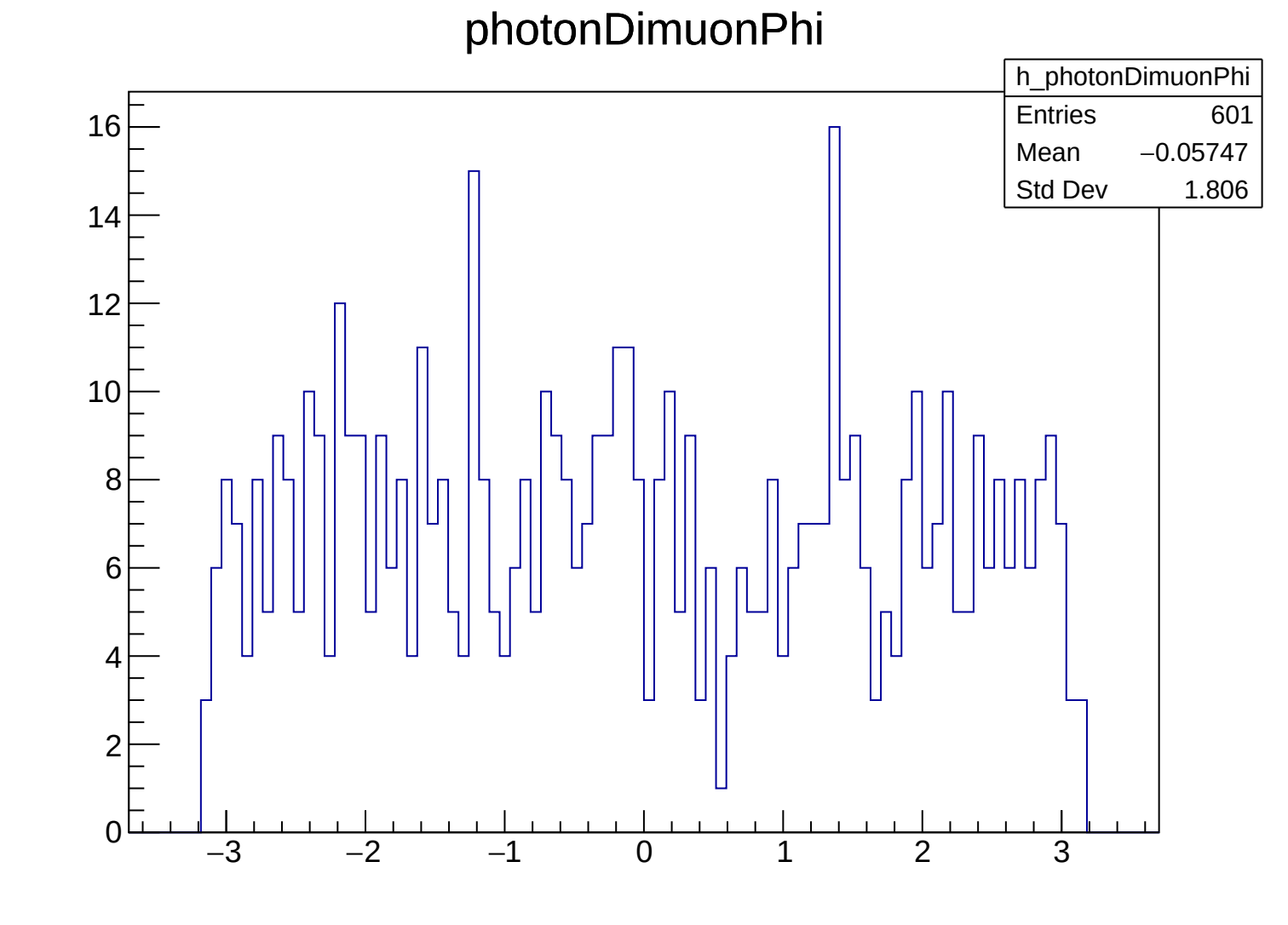

photonEnergy
h_photonEnergy
Entries 601
Mean 21.67
Std Dev 10.92
50
40
30
20
10
0
0
20
40
60
80
100
120

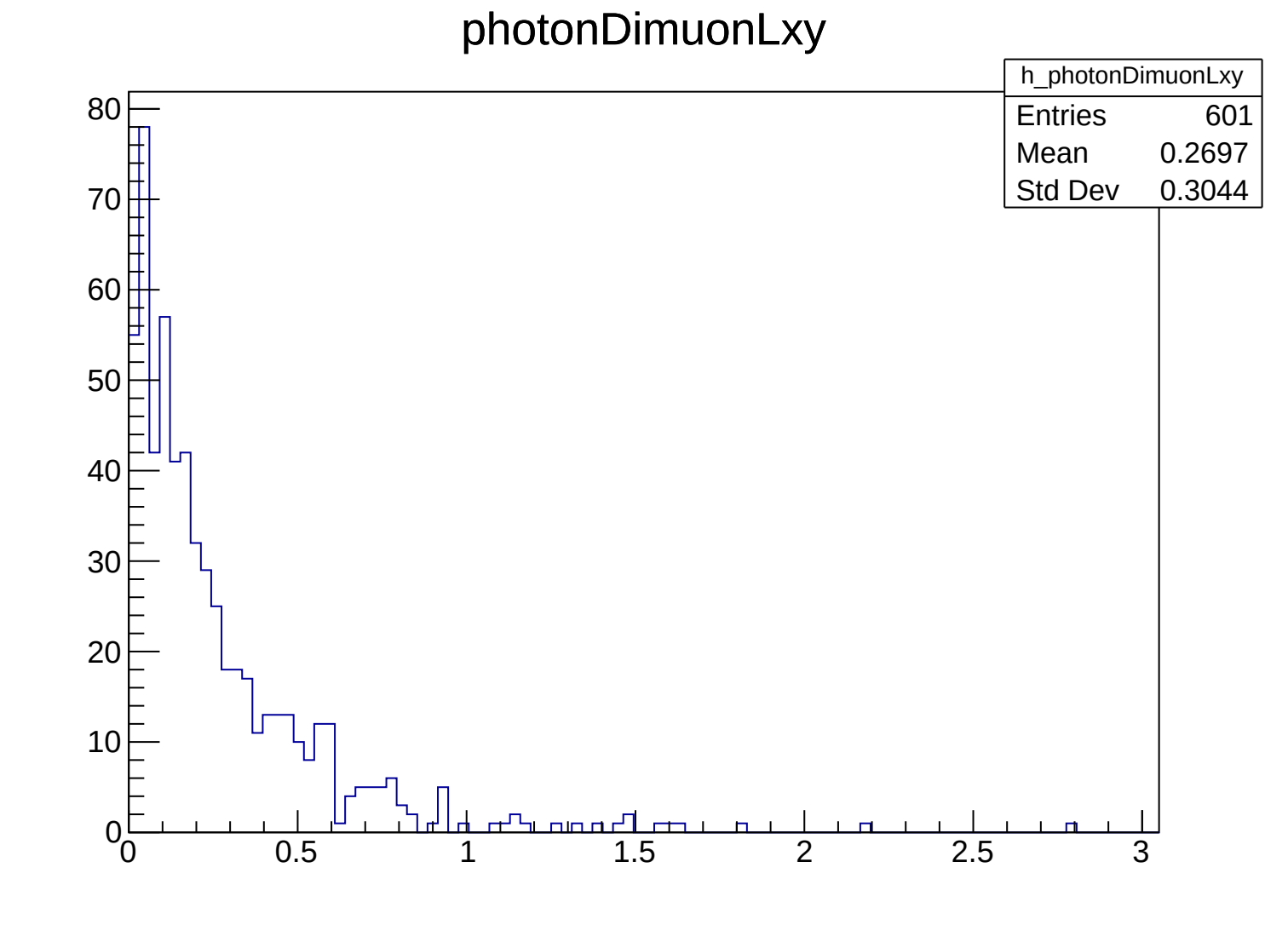

photonMVA
h_photonMVA
Entries 601
Mean −0.5723
Std Dev 0.4673
45
40
35
30
25
20
15
10
5
0
−1
−0.5
0
0.5
1

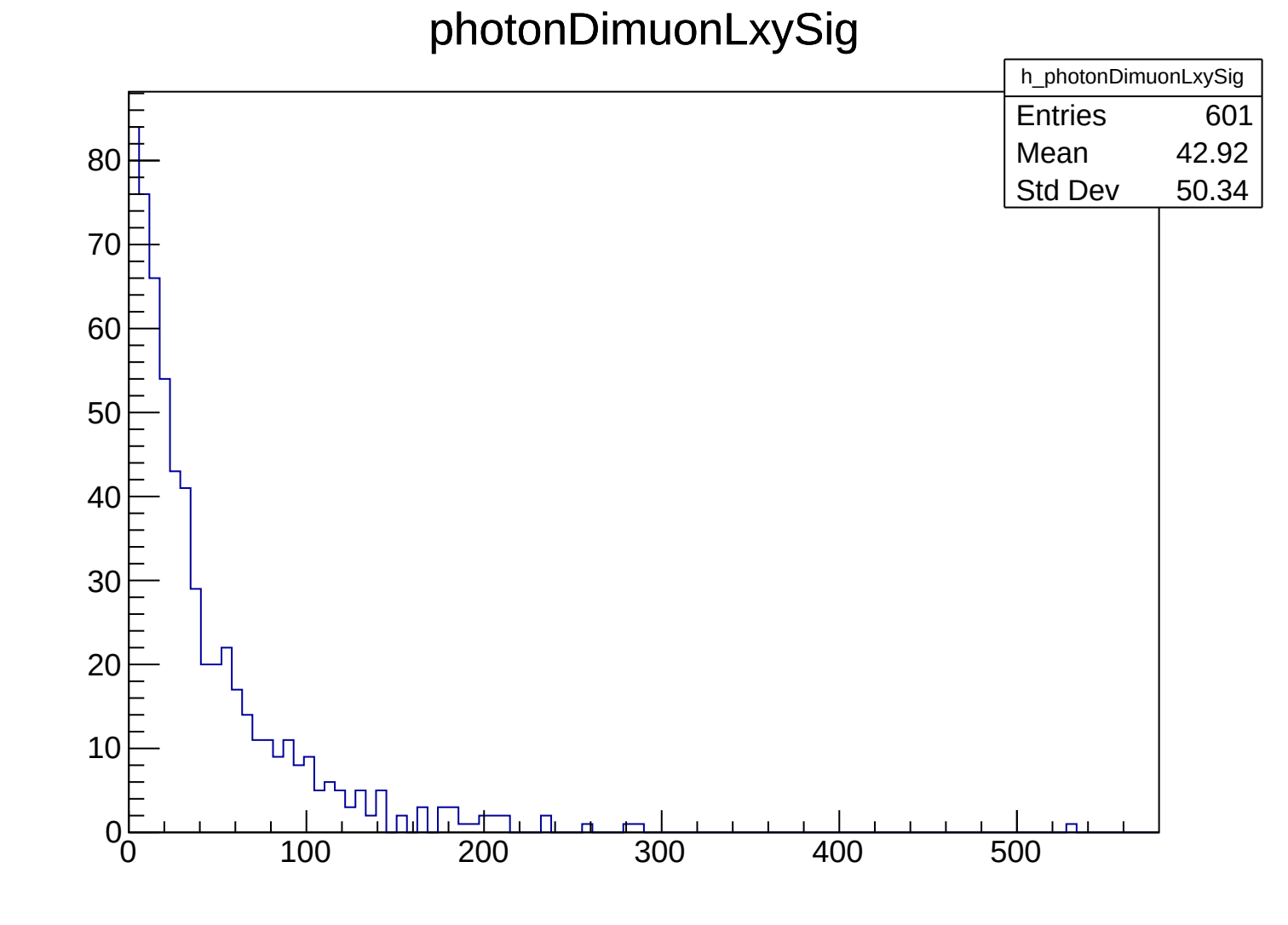

photonDimuonMass
h_photonDimuonMass
Entries 601
Mean 5.424
Std Dev 0.1824
40
35
30
25
20
15
10
5
0
4.6
4.8
5
5.2
5.4
5.6
5.8
6

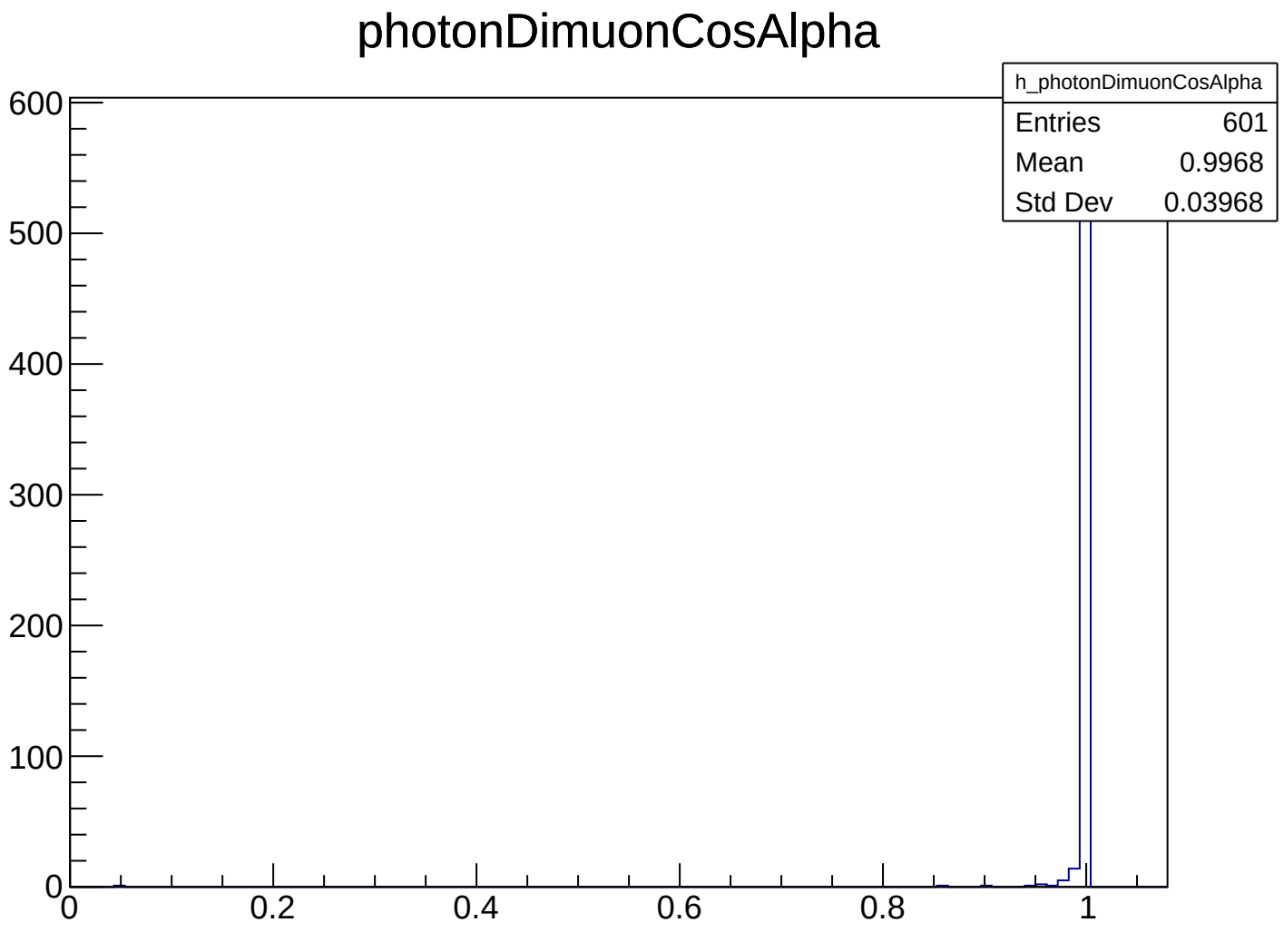

photonDimuonPt
h_photonDimuonPt
Entries 601
Mean 30.99
Std Dev 16.1
0
10
20
30
40
50
60
70
0
20
40
60
80
100
120
140
160
180
200

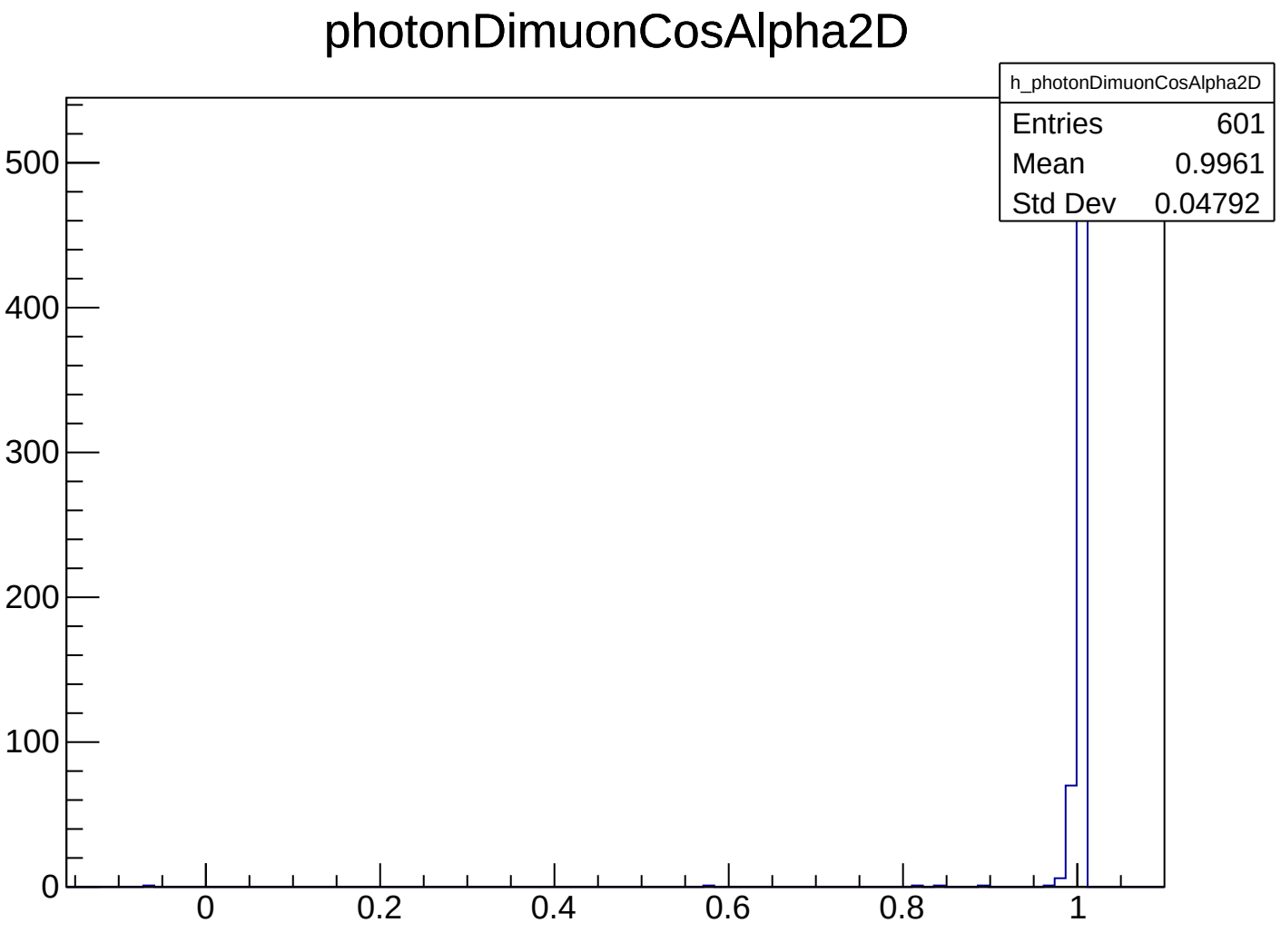

photonDimuonEta
h_photonDimuonEta
Entries 601
Mean −0.005207
Std Dev 0.922
16
14
12
10
8
6
4
2
0
−1.5
−1
−0.5
0
0.5
1
1.5

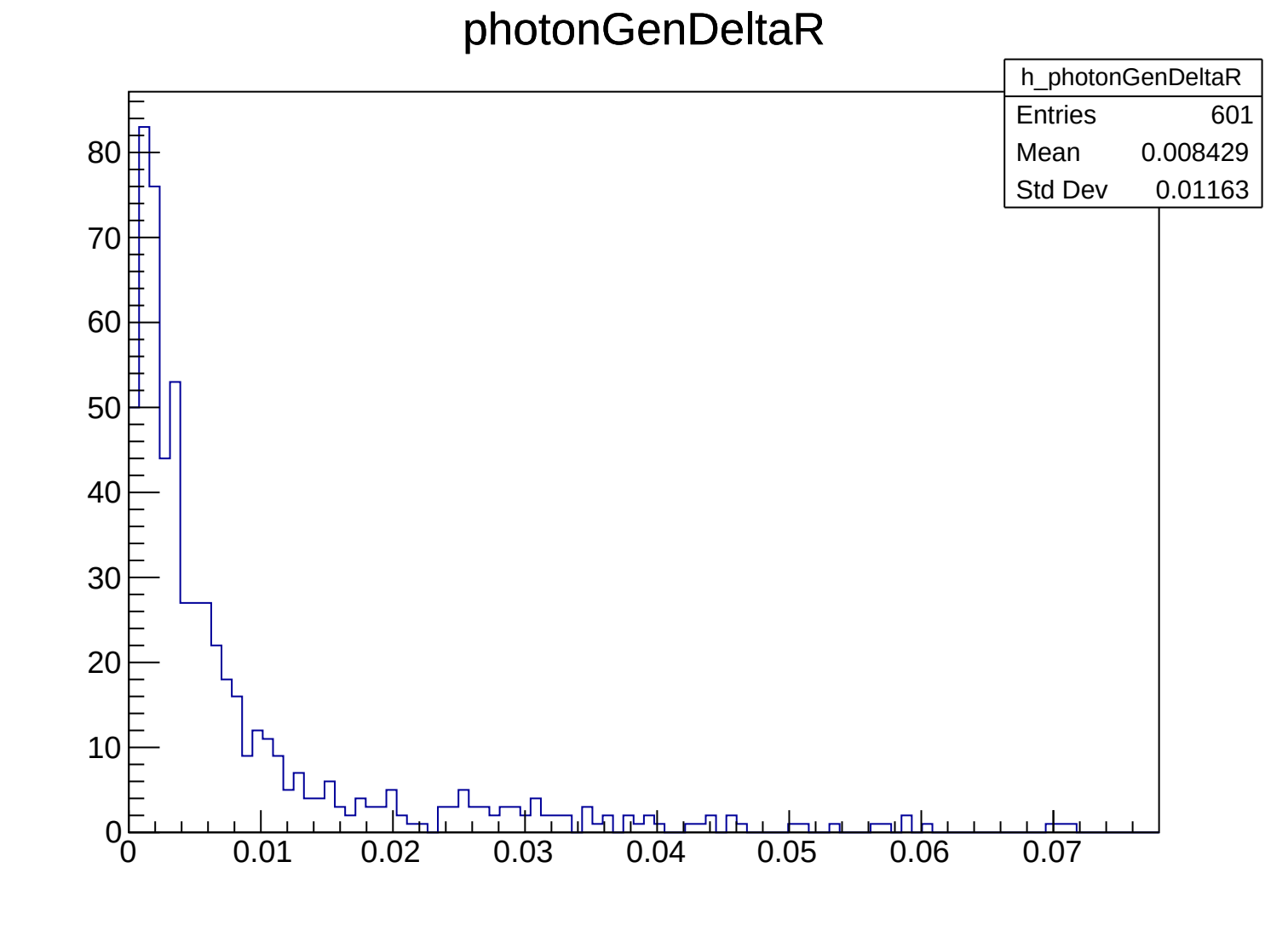

photonDimuonPhi
h_photonDimuonPhi
Entries 601
Mean −0.05747
Std Dev 1.806
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3

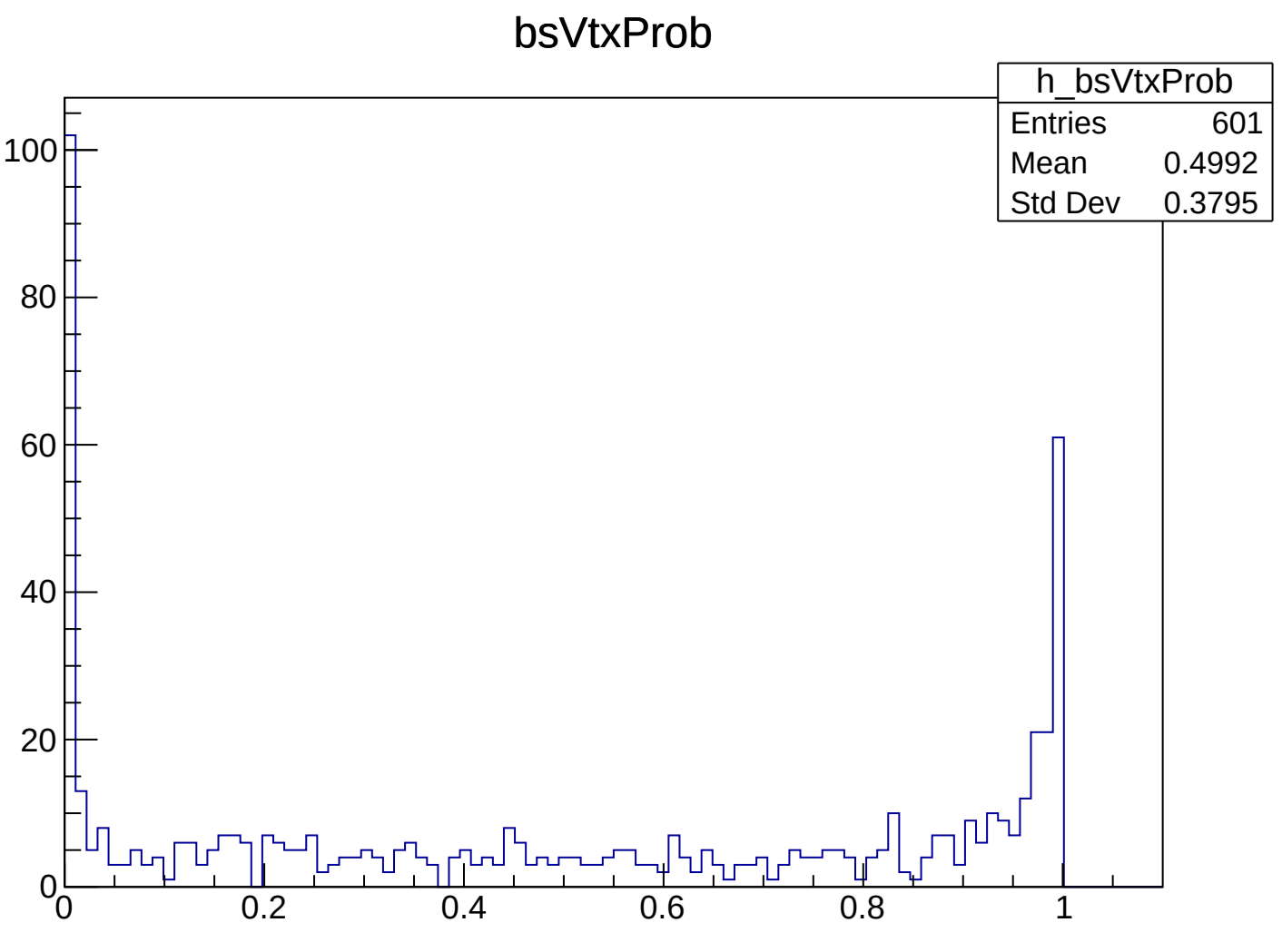

photonDimuonLxy
h_photonDimuonLxy
Entries 601
Mean 0.2697
Std Dev 0.3044
80
70
60
50
40
30
20
10
0
0
0.5
1
1.5
2
2.5
3

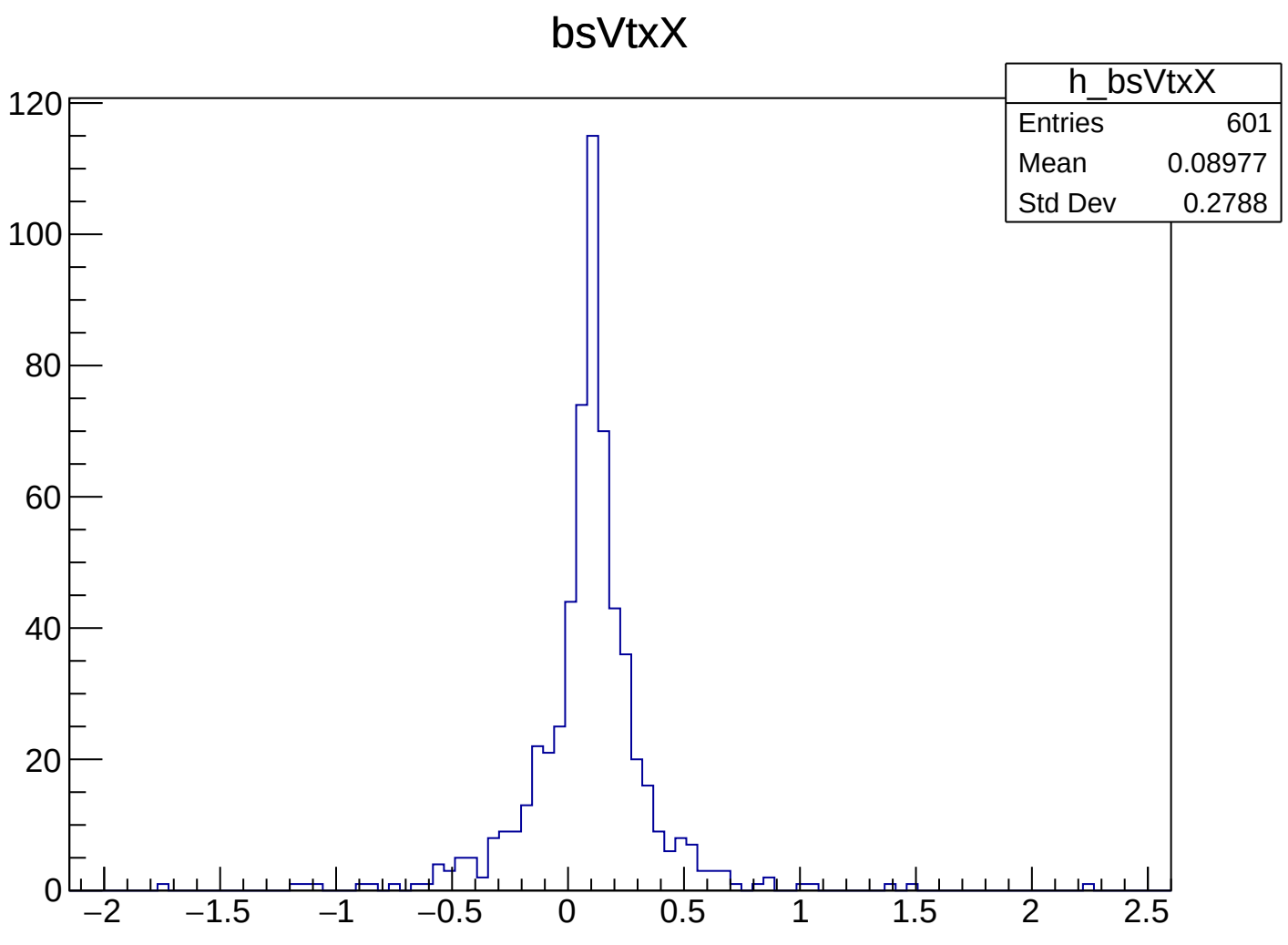

photonDimuonLxySig
h_photonDimuonLxySig
Entries 601
Mean 42.92
Std Dev 50.34
0
10
20
30
40
50
60
70
80
0
100
200
300
400
500

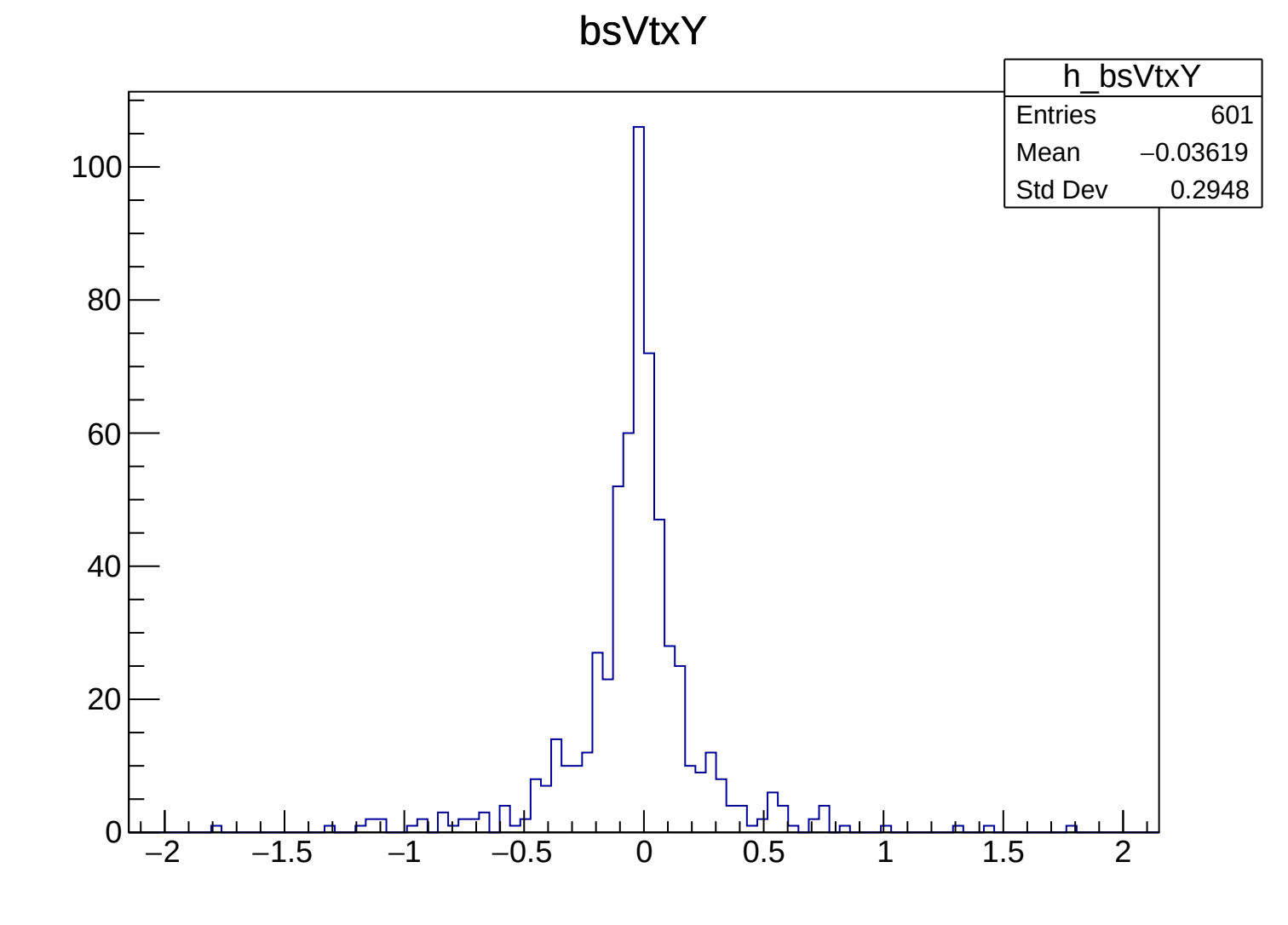

photonDimuonCosAlpha
h_photonDimuonCosAlpha
Entries 601
Mean 0.9968
Std Dev 0.03968
600
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1

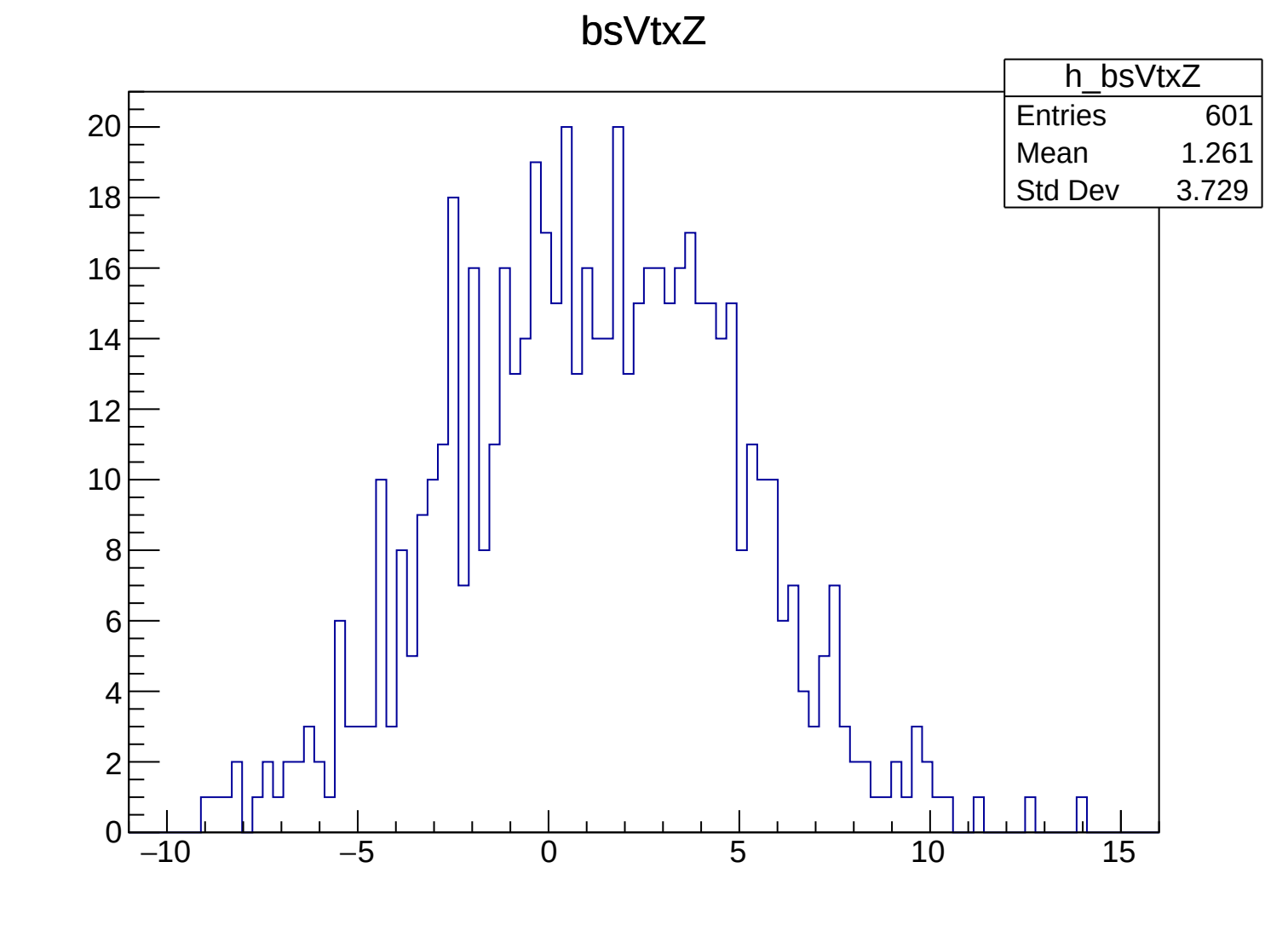

photonDimuonCosAlpha2D
h_photonDimuonCosAlpha2D
Entries 601
Mean 0.9961
Std Dev 0.04792
500
400
300
200
100
0
0
0.2
0.4
0.6
0.8
1

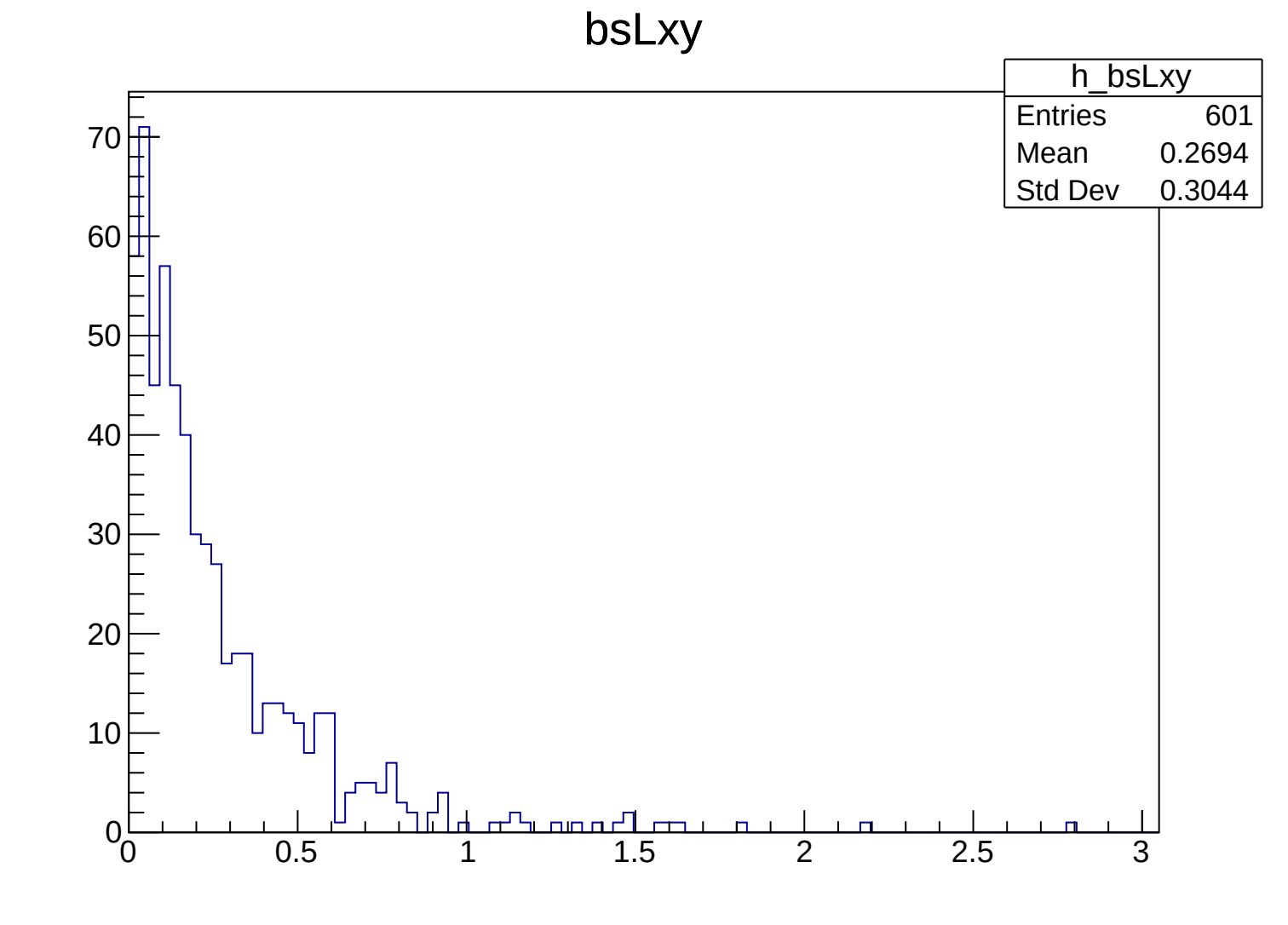

photonGenDeltaR
h_photonGenDeltaR
Entries 601
Mean 0.008429
Std Dev 0.01163
80
70
60
50
40
30
20
10
0
0
0.01
0.02
0.03
0.04
0.05
0.06
0.07

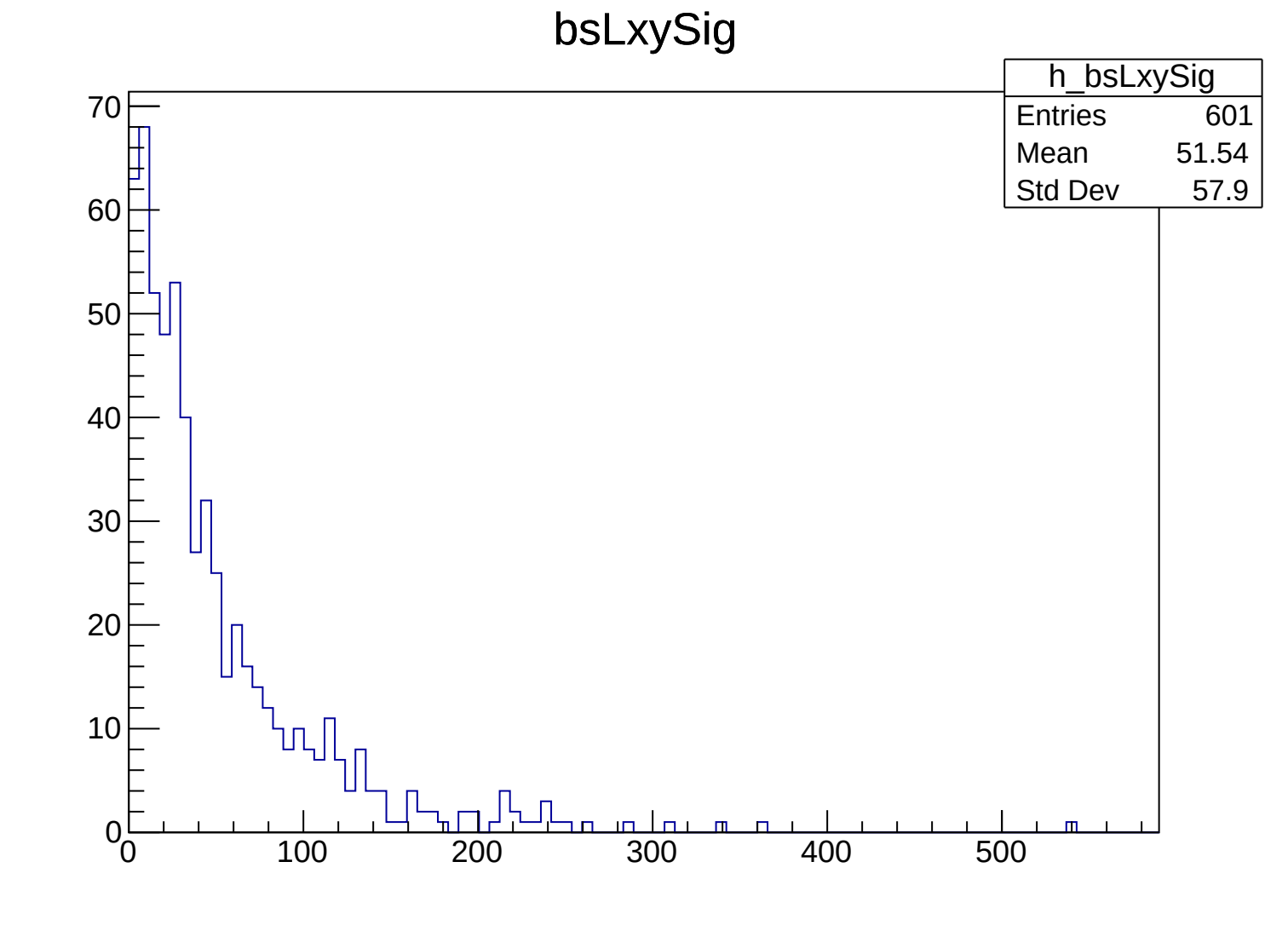

bsVtxProb
h_bsVtxProb
Entries 601
Mean 0.4992
Std Dev 0.3795
100
80
60
40
20
0
0
0.2
0.4
0.6
0.8
1

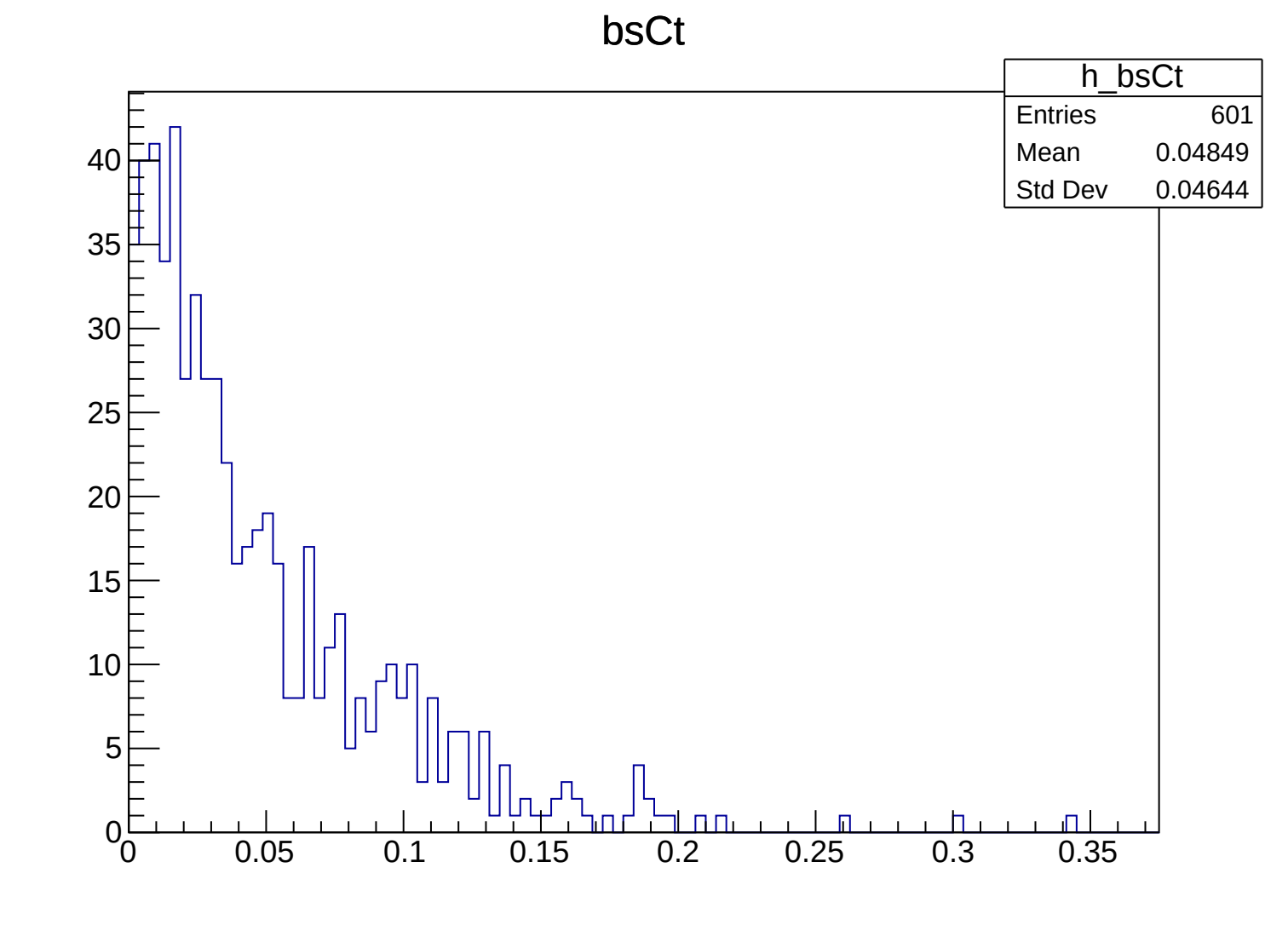

bsVtxX
h_bsVtxX
Entries 601
Mean 0.08977
Std Dev 0.2788
120
100
80
60
40
20
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2
2.5

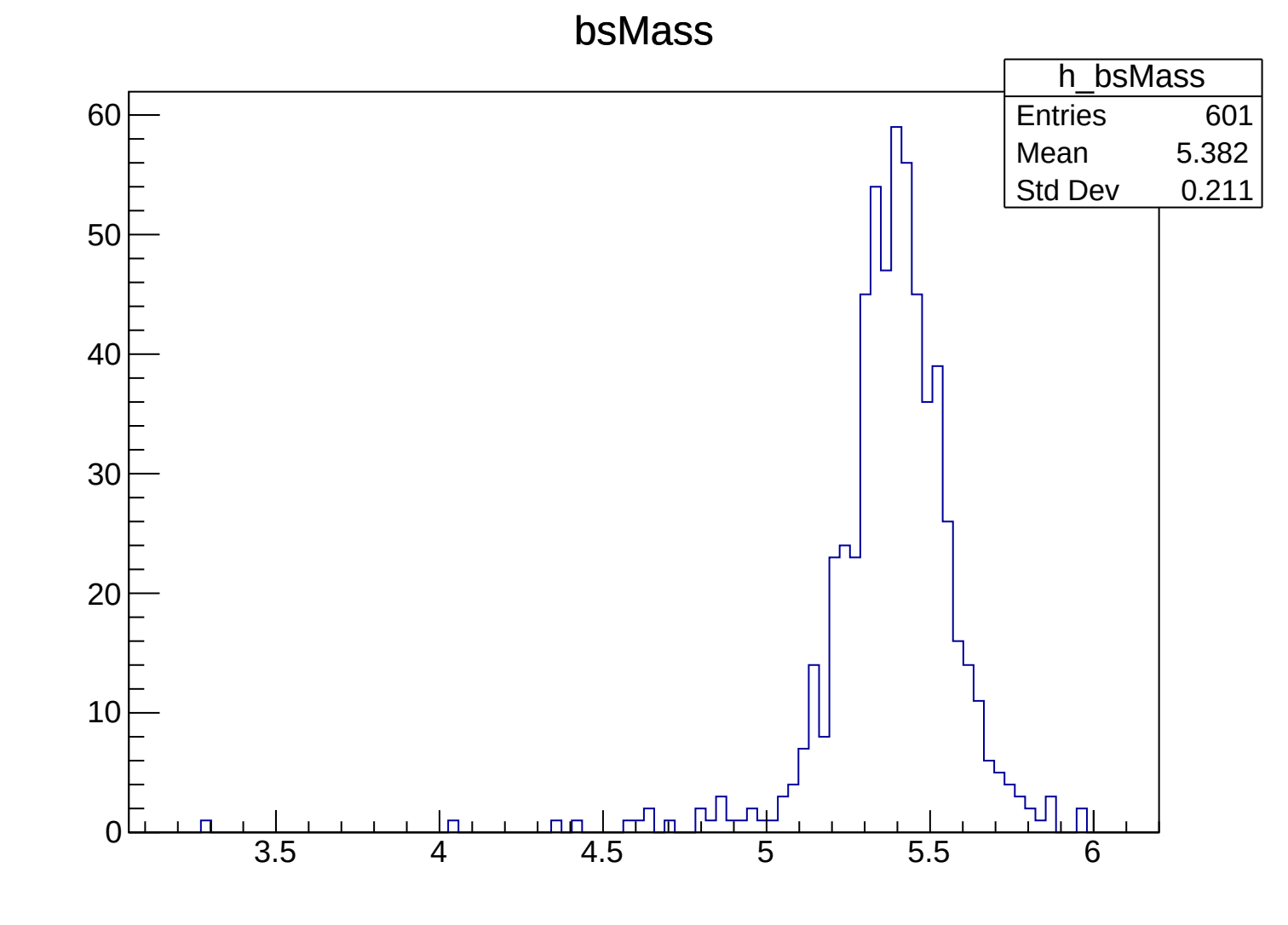

bsVtxY
h_bsVtxY
Entries 601
Mean −0.03619
Std Dev 0.2948
100
80
60
40
20
0
−2
−1.5
−1
−0.5
0
0.5
1
1.5
2

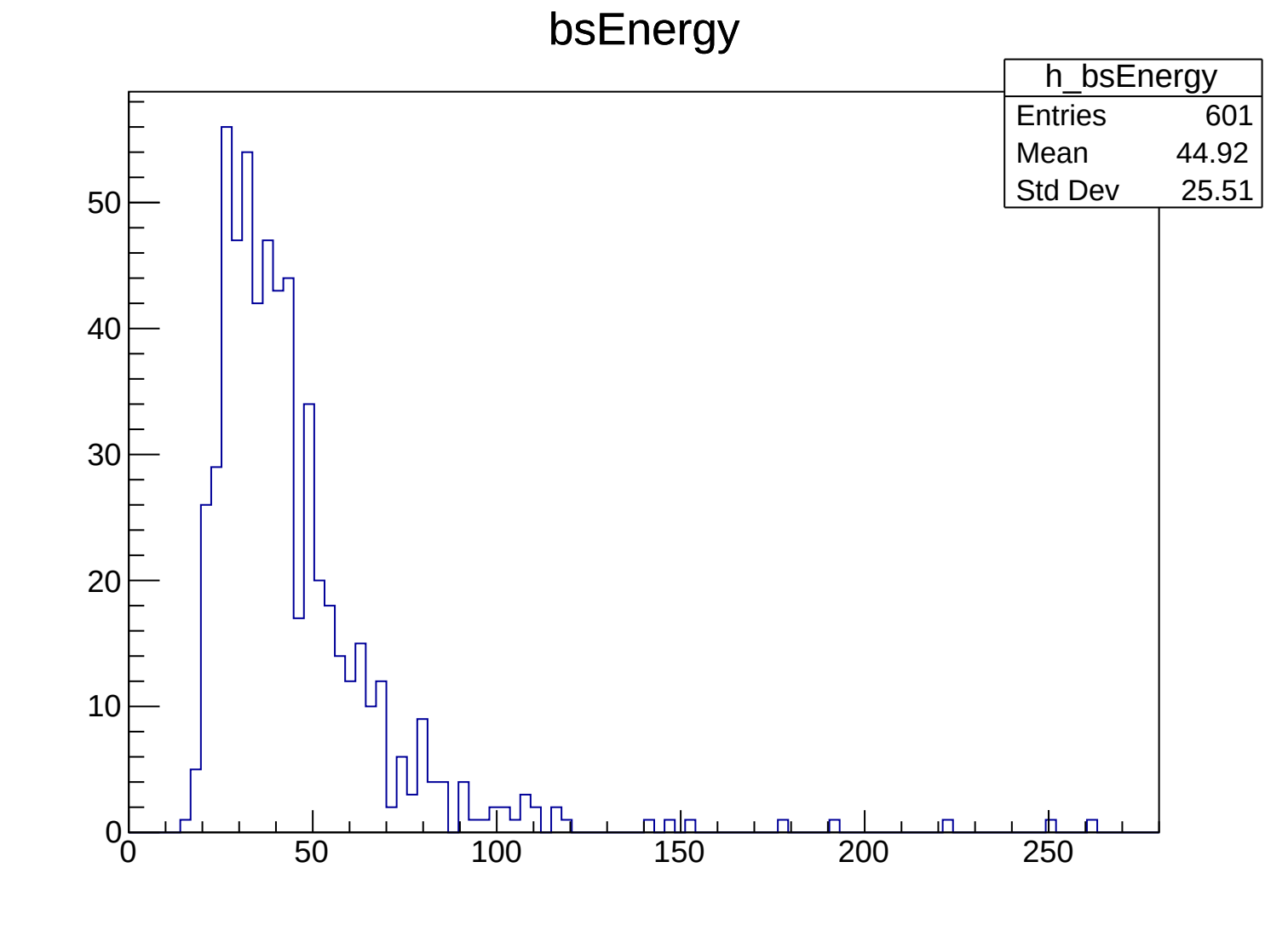

bsVtxZ
h_bsVtxZ
Entries 601
Mean 1.261
Std Dev 3.729
20
18
16
14
12
10
8
6
4
2
0
−10
−5
0
5
10
15

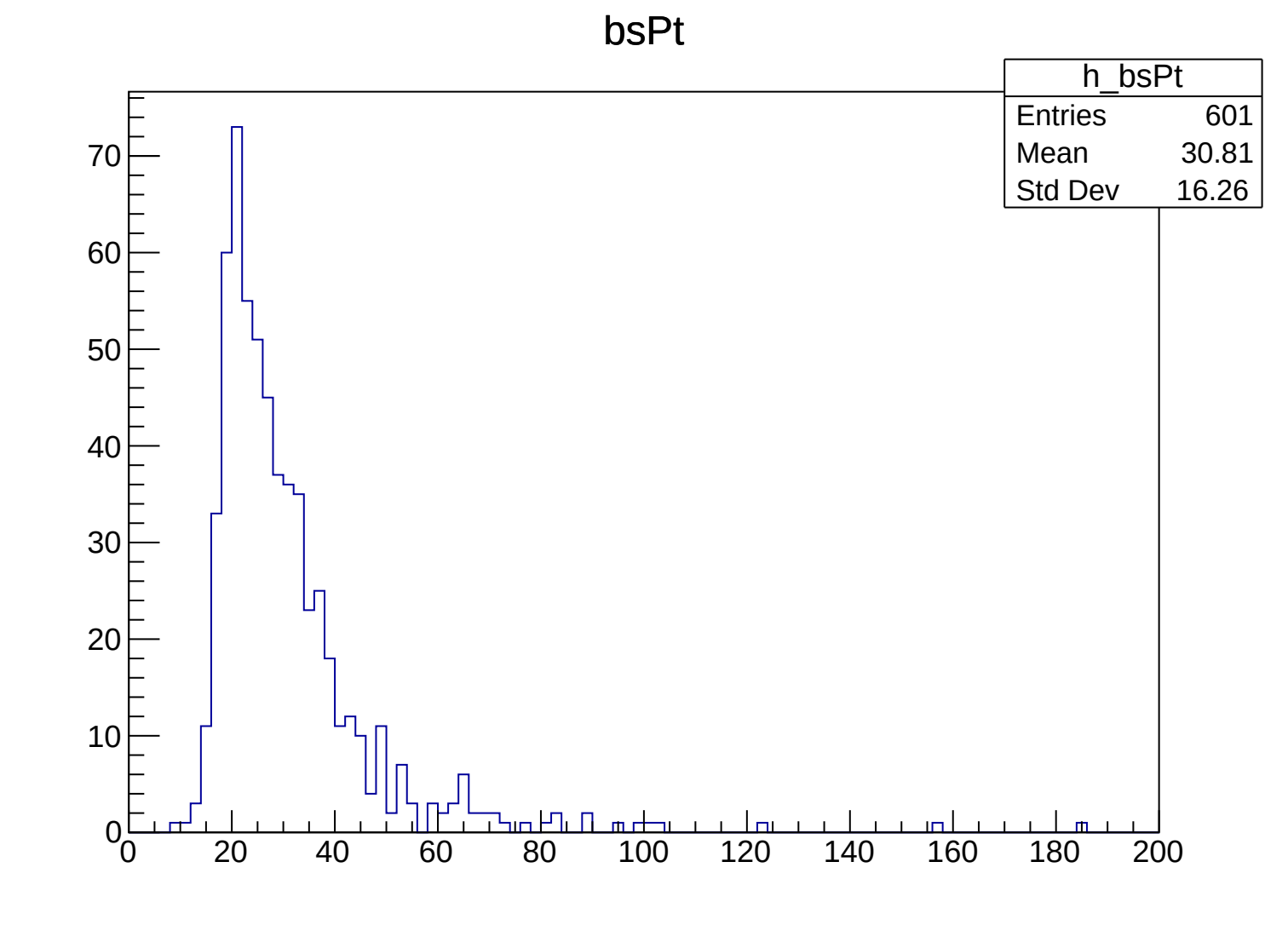

bsLxy
h_bsLxy
Entries 601
Mean 0.2694
Std Dev 0.3044
70
60
50
40
30
20
10
0
0
0.5
1
1.5
2
2.5
3

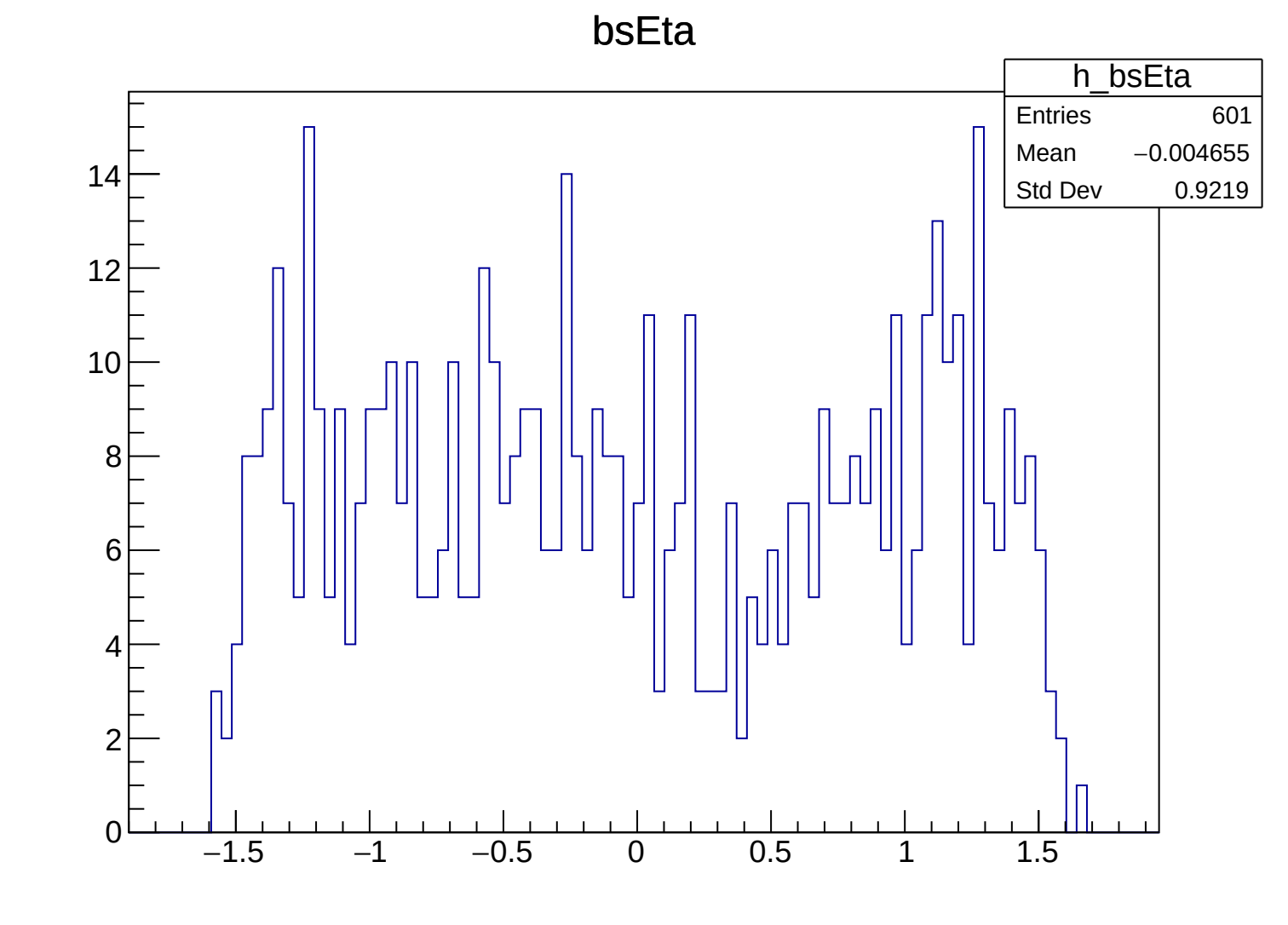

bsLxySig
h_bsLxySig
Entries 601
Mean 51.54
Std Dev 57.9
70
60
50
40
30
20
10
0
0
100
200
300
400
500

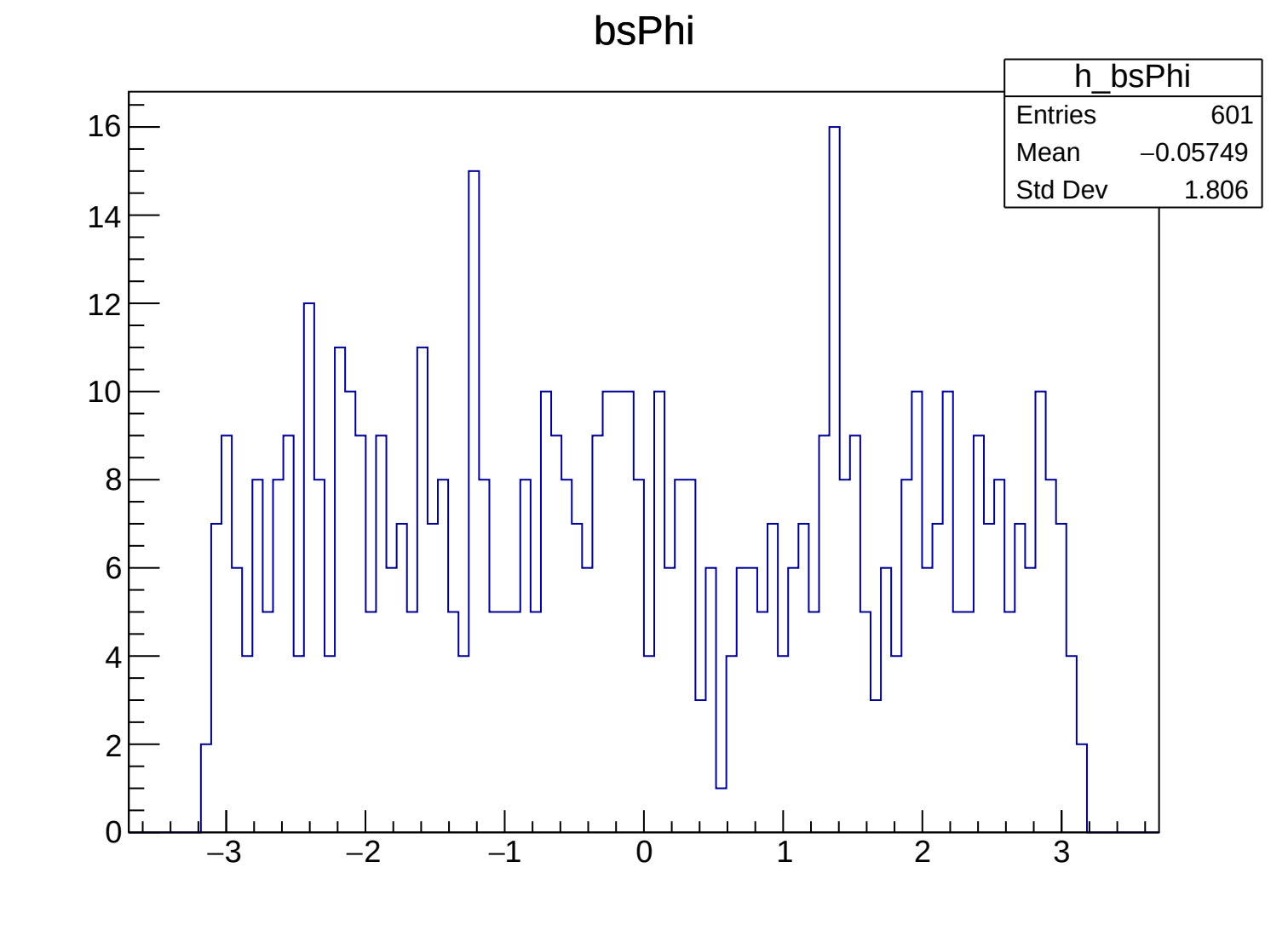

bsCt
h_bsCt
Entries 601
Mean 0.04849
Std Dev 0.04644
0
5
10
15
20
25
30
35
40
0
0.05
0.1
0.15
0.2
0.25
0.3
0.35

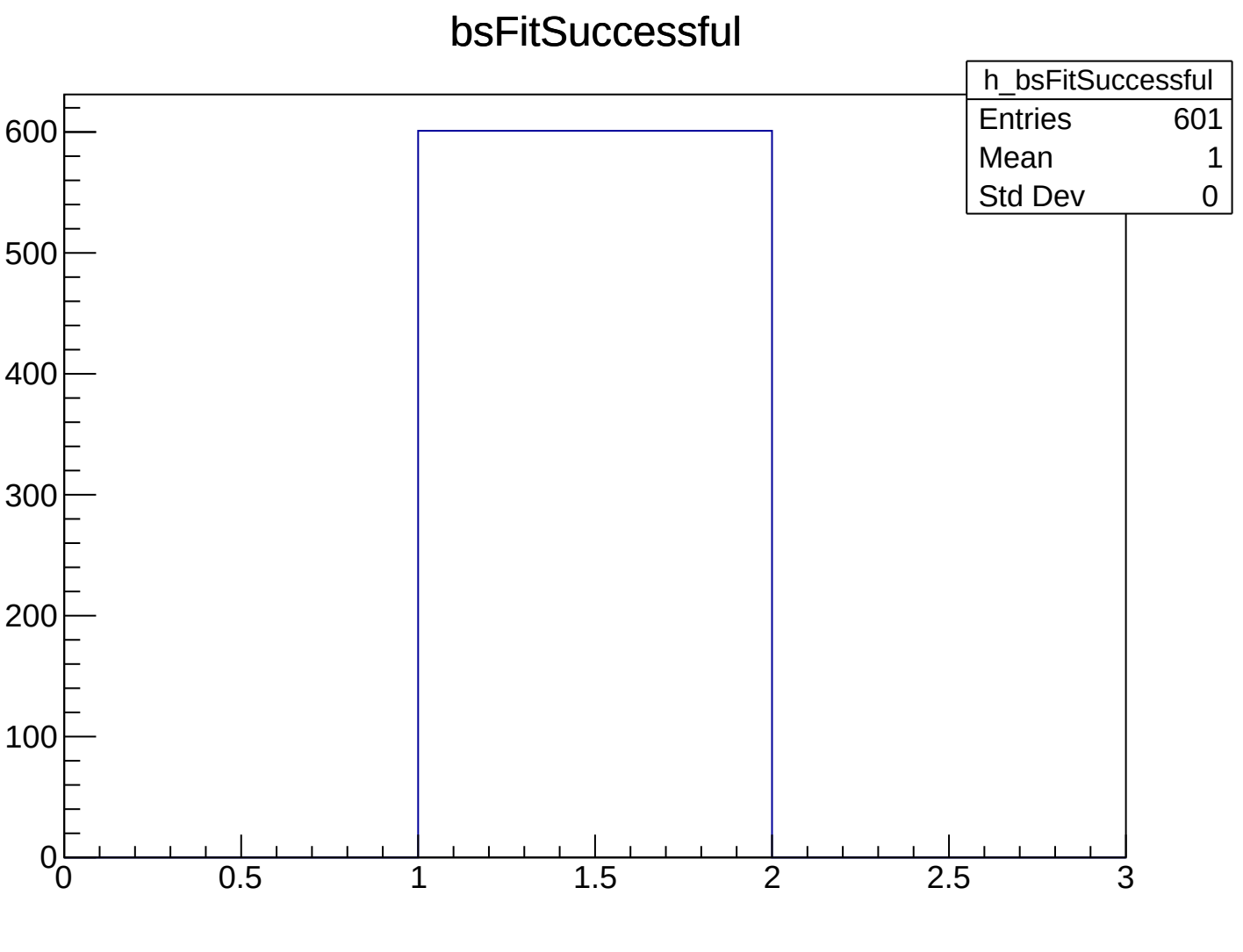

bsMass
h_bsMass
Entries 601
Mean 5.382
Std Dev 0.211
60
50
40
30
20
10
0
3.5
4
4.5
5
5.5
6


bsEnergy
h_bsEnergy
Entries 601
Mean 44.92
Std Dev 25.51
0
10
20
30
40
50
0
50
100
150
200
250


bsPt
h_bsPt
Entries 601
Mean 30.81
Std Dev 16.26
0
10
20
30
40
50
60
70
0
20
40
60
80
100
120
140
160
180
200


bsEta
h_bsEta
Entries 601
Mean −0.004655
Std Dev 0.9219
14
12
10
8
6
4
2
0
−1.5
−1
−0.5
0
0.5
1
1.5


bsPhi
h_bsPhi
Entries 601
Mean −0.05749
Std Dev 1.806
16
14
12
10
8
6
4
2
0
−3
−2
−1
0
1
2
3


bsFitSuccessful
h_bsFitSuccessful
Entries 601
Mean 1
Std Dev 0
600
500
400
300
200
100
0
0
0.5
1
1.5
2
2.5
3